

Krishna's
TEXT BOOK on
 VECTOR SPACES
AND MATRICES

(For B.A. and B.Sc. IVth Semester students of Kumaun University)

Kumaun University Semester Syllabus w.e.f. 2017-18

By

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Dedicated
to
Lord
Krishna

Authors & Publishers

Preface

This book on **Vector Spaces and Matrices** has been specially written according to the latest **Syllabus** to meet the requirements of **B.A. and B.Sc. Semester-IV Students** of all colleges affiliated to Kumaun University.

The subject matter has been discussed in such a simple way that the students will find no difficulty to understand it. The proofs of various theorems and examples have been given with minute details. Each chapter of this book contains complete theory and a fairly large number of solved examples. Sufficient problems have also been selected from various university examination papers. At the end of each chapter an exercise containing objective questions has been given.

We have tried our best to keep the book free from misprints. The authors shall be grateful to the readers who point out errors and omissions which, inspite of all care, might have been there.

The authors, in general, hope that the present book will be warmly received by the students and teachers. We shall **indeed** be very thankful to our colleagues for their recommending this book to their students.

The authors wish to express their thanks to Mr. S.K. Rastogi, *M.D.*, Mr. Sugam Rastogi, *Executive Director*, Mrs. Kanupriya Rastogi, *Director* and entire team of **KRISHNA Prakashan Media (P) Ltd., Meerut** for bringing out this book in the present nice form.

The authors will feel amply rewarded if the book serves the purpose for which it is meant. Suggestions for the improvement of the book are always welcome.

— **Authors**

Syllabus



VECTOR SPACES AND MATRICES



B.A./B.Sc. IV Semester

Kumaun University

Fourth Semester – First Paper

B.A./B.Sc. Paper-I

M.M.-60

Vector spaces: Vector space, sub spaces, Linear combinations, linear spans, Sums and direct sums.

Bases and Dimensions: Linear dependence and independence, Bases and dimensions, Dimensions and subspaces, Coordinates and change of bases.

Matrices: Idempotent, nilpotent, involutory, orthogonal and unitary matrices, singular and nonsingular matrices, negative integral powers of a nonsingular matrix; Trace of a matrix.

Rank of a matrix: Rank of a matrix, linear dependence of rows and columns of a matrix, row rank, column rank, equivalence of row rank and column rank, elementary transformations of a matrix and invariance of rank through elementary transformations, normal form of a matrix, elementary matrices, rank of the sum and product of two matrices, inverse of a non-singular matrix through elementary row transformations; equivalence of matrices.

Applications of Matrices: Solutions of a system of linear homogeneous equations, condition of consistency and nature of the general solution of a system of linear non-homogeneous equations, matrices of rotation and reflection.

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Krishna's

VECTOR SPACES AND MATRICES

Chapters



1. Vector Spaces

2. Linear Dependence of Vectors

3. Basis and Dimensions

4. Homomorphism of Vector Spaces



5. Matrices

6. Rank of a Matrix

7. Applications of Matrices

Chapter

1



Vector Spaces

So far we have studied groups and rings. Now we shall study another important algebraic structure known as **vector space**. Before giving the definition of a vector space we shall make a distinction between **internal** and **external compositions**.

Let A be any set. If $a * b \in A \forall a, b \in A$, and $a * b$ is unique then $*$ is said to be an **internal composition** in the set A . Here a and b are both elements of the set A .

Let V and F be any two sets. If $a \circ \alpha \in V$ for all $a \in F$ and for all $\alpha \in V$ and $a \circ \alpha$ is unique, then $\circ$ is said to be an **external composition** in V over F . Here a is an element of the set F and α is an element of the set V and the resulting element $a \circ \alpha$ is an element of the set V .

1 Vector Space

(Kumaun 2002; Garhwal 10)

Definition: Let $(F, +, \cdot)$ be a field. The elements of F will be called **scalars**. Let V be a non-empty set whose elements will be called **vectors**. Then V is a vector space over the field F , if

1. There is defined an internal composition in V called **addition of vectors** and denoted by '+'. Also for this composition V is an abelian group.
-

2. There is an external composition in V over F called **scalar multiplication** and denoted multiplicatively i.e., $a\alpha \in V$ for all $a \in F$ and for all $\alpha \in V$. In other words V is closed with respect to scalar multiplication.
3. The two compositions i.e., scalar multiplication and addition of vectors satisfy the following postulates :
 - (i) $a(\alpha + \beta) = a\alpha + a\beta \forall a \in F$ and $\forall \alpha, \beta \in V$.
 - (ii) $(a + b)\alpha = a\alpha + b\alpha \forall a, b \in F$ and $\forall \alpha \in V$.
 - (iii) $(ab)\alpha = a(b\alpha) \forall a, b \in F$ and $\forall \alpha \in V$.
 - (iv) $1\alpha = \alpha \forall \alpha \in V$ and 1 is the unity element of the field F .

When V is a vector space over the field F , we shall say that $V(F)$ is a vector space. If the field F is understood we can simply say that V is a vector space. If F is the field $\mathbf{R}$ of real numbers, V is called a *real vector space* ; similarly if F is $\mathbf{Q}$ or F is $\mathbf{C}$, we speak of *rational vector spaces* or *complex vector spaces*.

In the above definition of a vector space V over the field F , we have denoted the addition of vectors by the symbol '+'. This symbol also denotes the addition composition of the field F , i.e., addition of scalars. There should be no confusion about the two compositions though we have used the same symbol to denote each of them. If $\alpha, \beta \in V$, then $\alpha + \beta$ represents addition in V i.e., addition of vectors. If $a, b \in F$ then $a + b$ represents addition of scalars i.e., addition in the field F . Similarly there should be no confusion in multiplication of scalars i.e., multiplication of the elements of F and in scalar multiplication i.e., multiplication of an element of V by an element of F . If $a, b \in F$, then ab represents multiplication of F and $ab \in F$. If $a \in F$, and $\alpha \in V$, then $a\alpha$ represents scalar multiplication and $a\alpha \in V$. Since $1 \in F$ and $\alpha \in V$, therefore 1α represents scalar multiplication. Again $a\alpha \in V$, $a\beta \in V$, therefore $a\alpha + a\beta$ represents addition of vectors and thus $a\alpha + a\beta$ is an element of V . Further $a \in F$ and $\alpha + \beta \in V$, therefore $a(\alpha + \beta)$ represents scalar multiplication and we have $a(\alpha + \beta) \in V$.

Note 1: For V to be an abelian group with respect to addition of vectors, we must have the following conditions satisfied :

- (i) $\alpha + \beta \in V$ for all $\alpha, \beta \in V$.
- (ii) $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in V$.
- (iii) $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$ for all $\alpha, \beta, \gamma \in V$.
- (iv) There exists an element $\mathbf{0} \in V$ such that $\mathbf{0} + \alpha = \alpha \forall \alpha \in V$.
The element $\mathbf{0} \in V$ will be called the **zero vector**. It is the additive identity in V .
- (v) To every vector $\alpha \in V$ there exists a vector $-\alpha \in V$ such that $-\alpha + \alpha = \mathbf{0}$. Thus each vector should possess additive inverse. The vector $-\alpha$ is called the negative of the vector α .

Note 2: Since $(V, +)$ is an abelian group therefore all the properties of an abelian group will hold in V . A few of them are as follows :

- (i) $\alpha + \beta = \alpha + \gamma \Rightarrow \beta = \gamma$ (left cancellation law)
- (ii) $\beta + \alpha = \gamma + \alpha \Rightarrow \beta = \gamma$ (right cancellation law)
- (iii) $\alpha + \beta = \mathbf{0} \Rightarrow \alpha = -\beta$ and $\beta = -\alpha$.
- (iv) $-(\alpha + \beta) = -\alpha - \beta$ where by $\alpha - \beta$ we mean $\alpha + (-\beta)$.
- (v) The additive identity $\mathbf{0}$ will be unique.
- (vi) The additive inverse of each vector will be unique.
- (vii) If $\alpha + \beta = \gamma$, then $\alpha + \beta - \gamma = \mathbf{0}$.

Note 3: There should also be no confusion about the use of the word vector. Here by vector we do not mean the vector quantity which we have defined in vector algebra as a directed line segment. Here we shall call the elements of the set V as vectors.

Note 4: In a vector space we shall be dealing with two types of zero elements. One is the zero vector and the other is the zero element of the field F i.e., the 0 scalar. To distinguish between the two, we shall use the zero letter in bold type to represent the zero vector. However the students may use the same symbol 0 to denote the zero vector as well as the zero scalar. There will be no confusion in this use. The use of 0 will itself tell whether it stands for zero vector or for zero scalar.

Note 5: We shall usually use the lower case Greek letters α, β, γ etc. to denote vectors i.e., the elements of V and the lower case Latin letters, a, b, c etc. to denote the scalars i.e., the elements of the field F .

Illustrative Examples

Example 1: A field K can be regarded as a vector space over any subfield F of K .

(Garhwal 2007)

Solution: Here K is the set of vectors. Addition of vectors is the addition composition in the field K . Since K is a field, therefore $(K, +)$ is an abelian group. Further the elements of the subfield F constitute the set of scalars. The composition of scalar multiplication is the multiplication composition in the field K . K is a field, therefore $a\alpha \in K \forall a \in F$ and $\forall \alpha \in K$ because both a and α are elements of K . If 1 is the unity element of K , then 1 is also the unity element of the subfield F . We make the following observations.

- (i) $a(\alpha + \beta) = a\alpha + a\beta \forall a \in F$ and $\forall \alpha, \beta \in K$. This result follows from the left distributive law in K .
- (ii) $(a + b)\alpha = a\alpha + b\alpha \forall a, b \in F$ and $\forall \alpha \in K$. This result is a consequence of the right distributive law in K .
- (iii) $(ab)\alpha = a(b\alpha) \forall a, b \in F$ and $\forall \alpha \in K$. This result is a consequence of associativity of multiplication in K .
- (iv) $1\alpha = \alpha \forall \alpha \in K$ and 1 is the unity element of the subfield F . Since 1 is also the unity element of the field K , therefore $1\alpha = \alpha \forall \alpha \in K$.

Hence $K(F)$ is a vector space.

Note 1: If F is any field, then F itself is a vector space over the field F .

Note 2: If $\mathbf{C}$ is the field of complex numbers and $\mathbf{R}$ is the field of real numbers, then $\mathbf{C}$ is a vector space over $\mathbf{R}$ because $\mathbf{R}$ is a subfield of $\mathbf{C}$. But $\mathbf{R}$ is not a vector space over $\mathbf{C}$. Here $\mathbf{R}$ is not closed with respect to scalar multiplication. For example, $2 \in \mathbf{R}$ and $3 + 4i \in \mathbf{C}$ and $(3 + 4i)2 \notin \mathbf{R}$.

Example 2: The set V of all $m \times n$ matrices with their elements as real numbers is a vector space over the field F of real numbers with respect to addition of matrices as addition of vectors and multiplication of a matrix by a scalar as scalar multiplication.

(Garhwal 2007)

Solution: As in groups, we can easily prove that V is an abelian group with respect to addition of matrices. The null matrix O of the type $m \times n$ is the additive identity of this abelian group.

If $a \in F$ and $\alpha \in V$ (i.e., α is a matrix of the type $m \times n$ with elements as real numbers), then $a\alpha \in V$ because $a\alpha$ is also a matrix of the type $m \times n$ with elements as real numbers. Therefore V is closed with respect to scalar multiplication. Also from our study of matrices we observe that

- (i) $a(\alpha + \beta) = a\alpha + a\beta \quad \forall a \in F \text{ and } \forall \alpha, \beta \in V.$
- (ii) $(a + b)\alpha = a\alpha + b\alpha \quad \forall a, b \in F \text{ and } \forall \alpha \in V.$
- (iii) $(ab)\alpha = a(b\alpha) \quad \forall a, b \in F \text{ and } \forall \alpha \in V.$
- (iv) $1\alpha = \alpha \quad \forall \alpha \in V$ where 1 is the unity element of the field F of real numbers.

Hence $V(F)$ is a vector space.

Note: If V is the set of all $m \times n$ matrices with their elements as rational numbers and F is the field of real numbers, then V will not be closed with respect to scalar multiplication. For $\sqrt{7} \in F$ and if $\alpha \in V$, then $\sqrt{7}\alpha \notin V$ because the elements of the matrix $\sqrt{7}\alpha$ will not be rational numbers. Therefore $V(F)$ will not be a vector space.

Example 3: The vector space of all ordered n -tuples over a field F .

Solution: Let F be a field. An ordered set $\alpha = (a_1, a_2, a_3, \dots, a_n)$ of n elements of F is called an n -tuple over F . Let V be the totality of all ordered n -tuples over F i.e., let

$$V = \{(a_1, a_2, \dots, a_n) : a_1, a_2, a_3, \dots, a_n \in F\}.$$

Now we shall give a vector space structure to V over the field F . For this we define equality of two n -tuples, addition of two n -tuples and multiplication of an n -tuple by a scalar as follows :

Equality of two n -tuples. Two elements

$$\alpha = (a_1, a_2, \dots, a_n)$$

and

$$\beta = (b_1, b_2, \dots, b_n)$$

of V are said to be equal if and only if $a_i = b_i$ for each $i = 1, 2, \dots, n$.

Addition composition in V . We define

$$\alpha + \beta = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$\forall \alpha = (a_1, a_2, \dots, a_n), \beta = (b_1, b_2, \dots, b_n) \in V.$$

Since $a_1 + b_1, a_2 + b_2, \dots, a_n + b_n$ are all elements of F , therefore $\alpha + \beta \in V$ and thus V is closed with respect to addition of n -tuples.

Scalar multiplication in V over F . We define

$$a\alpha = (aa_1, aa_2, \dots, aa_n) \quad \forall a \in F, \forall \alpha = (a_1, a_2, \dots, a_n) \in V.$$

Since $aa_1, aa_2, \dots, aa_n$ are all elements of F , therefore $a\alpha \in V$ and thus V is closed with respect to scalar multiplication.

Now we shall see that V is a vector space for these two compositions.

Associativity of addition in V . We have

$$\begin{aligned} & (a_1, a_2, \dots, a_n) + [(b_1, b_2, \dots, b_n) + (c_1, c_2, \dots, c_n)] \\ &= (a_1, a_2, \dots, a_n) + (b_1 + c_1, b_2 + c_2, \dots, b_n + c_n) \\ &= (a_1 + [b_1 + c_1], a_2 + [b_2 + c_2], \dots, a_n + [b_n + c_n]) \end{aligned}$$

$$\begin{aligned}
&= ([a_1 + b_1] + c_1, [a_2 + b_2] + c_2, \dots, [a_n + b_n] + c_n) \\
&= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) + (c_1, c_2, \dots, c_n) \\
&= [(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)] + (c_1, c_2, \dots, c_n).
\end{aligned}$$

Commutativity of addition in V . We have

$$\begin{aligned}
(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) &= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \\
&= (b_1 + a_1, b_2 + a_2, \dots, b_n + a_n) \\
&= (b_1, b_2, \dots, b_n) + (a_1, a_2, \dots, a_n).
\end{aligned}$$

Existence of additive identity in V . We have

$$\begin{aligned}
(0, 0, \dots, 0) &\in V. \text{ Also if } (a_1, a_2, \dots, a_n) \in V, \text{ then} \\
(a_1, a_2, \dots, a_n) + (0, 0, \dots, 0) &= (a_1 + 0, a_2 + 0, \dots, a_n + 0) \\
&= (a_1, a_2, \dots, a_n).
\end{aligned}$$

$\therefore (0, 0, \dots, 0)$ is the additive identity in V .

Existence of additive inverse of each element of V . If

$$(a_1, a_2, \dots, a_n) \in V, \text{ then } (-a_1, -a_2, \dots, -a_n) \in V.$$

Also we have $(-a_1, -a_2, \dots, -a_n) + (a_1, a_2, \dots, a_n)$

$$= (-a_1 + a_1, -a_2 + a_2, \dots, -a_n + a_n) = (0, 0, \dots, 0).$$

$\therefore (-a_1, -a_2, \dots, -a_n)$ is the additive inverse of $(a_1, a_2, \dots, a_n)$.

Thus V is an abelian group with respect to addition. Further we observe that

1. If $a \in F$ and $\alpha = (a_1, a_2, \dots, a_n), \beta = (b_1, b_2, \dots, b_n) \in V$, then

$$\begin{aligned}
a(\alpha + \beta) &= a(a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \\
&= (a[a_1 + b_1], a[a_2 + b_2], \dots, a[a_n + b_n]) \\
&= (aa_1 + ab_1, aa_2 + ab_2, \dots, aa_n + ab_n) \\
&= (aa_1, aa_2, \dots, aa_n) + (ab_1, ab_2, \dots, ab_n) \\
&= a(a_1, a_2, \dots, a_n) + a(b_1, b_2, \dots, b_n) = a\alpha + a\beta.
\end{aligned}$$

2. If $a, b \in F$ and $\alpha = (a_1, a_2, \dots, a_n) \in V$, then

$$\begin{aligned}
(a + b)\alpha &= ([a + b]a_1, [a + b]a_2, \dots, [a + b]a_n) \\
&= (aa_1 + ba_1, aa_2 + ba_2, \dots, aa_n + ba_n) \\
&= (aa_1, aa_2, \dots, aa_n) + (ba_1, ba_2, \dots, ba_n) \\
&= a(a_1, a_2, \dots, a_n) + b(a_1, a_2, \dots, a_n) = a\alpha + b\alpha.
\end{aligned}$$

3. If $a, b \in F$ and $\alpha = (a_1, a_2, \dots, a_n) \in V$, then

$$\begin{aligned}
(ab)\alpha &= ([ab]a_1, [ab]a_2, \dots, [ab]a_n) \\
&= (a[ba_1], a[ba_2], \dots, a[ba_n]) \\
&= a(ba_1, ba_2, \dots, ba_n) = a[b(a_1, a_2, \dots, a_n)] = a(b\alpha).
\end{aligned}$$

4. If 1 is the unity element of F and $\alpha = (a_1, a_2, \dots, a_n) \in V$, then

$$1\alpha = (1a_1, 1a_2, \dots, 1a_n) = (a_1, a_2, \dots, a_n) = \alpha.$$

Hence V is a vector space over F . The vector space of all ordered n -tuples over F will be denoted by $V_n(F)$. Sometimes we also denote it by F^n or $F^{(n)}$ or $F^n(F)$. Here the zero vector i.e., $\mathbf{0}$ is the n -tuple $(0, 0, \dots, 0)$.

Note: $V_2(F) = \{(a_1, a_2) : a_1, a_2 \in F\}$ is the vector space of all **ordered pairs** over F . Similarly $V_3(F) = \{(a_1, a_2, a_3) : a_1, a_2, a_3 \in F\}$ is the vector space of all **ordered triads** over F .

Example 4: The vector space of all polynomials over a field F .

Solution: Let $F[x]$ denote the set of all polynomials in an indeterminate x over a field F . Then $F[x]$ is a vector space over the field F with respect to addition of two polynomials as addition of vectors and the product of a polynomial by a constant polynomial (i.e., by an element of F) as scalar multiplication.

Let
$$f(x) = \sum a_i x^i = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots,$$

$$g(x) = \sum b_i x^i = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \dots$$

and
$$h(x) = \sum c_i x^i = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

be any arbitrary members of $F[x]$.

Equality of two polynomials. We define $f(x) = g(x)$ if and only if $a_i = b_i$ for each $i = 0, 1, 2, \dots$.

Addition composition in $F[x]$. We define

$$\begin{aligned} f(x) + g(x) &= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \dots \\ &= \sum (a_i + b_i)x^i. \end{aligned}$$

Since $a_0 + b_0, a_1 + b_1, a_2 + b_2, \dots$ are all elements of F , therefore

$$f(x) + g(x) \in F[x]$$

and thus $F[x]$ is closed with respect to addition of polynomials.

Scalar multiplication in $F[x]$ over F . If k is any scalar i.e., $k \in F$, we define

$$\begin{aligned} kf(x) &= ka_0 + (ka_1)x + (ka_2)x^2 + (ka_3)x^3 + \dots \\ &= \sum (ka_i)x^i. \end{aligned}$$

Since $ka_0, ka_1, ka_2, \dots$ are all elements of F , therefore $kf(x) \in F[x]$ and thus $F[x]$ is closed with respect to scalar multiplication.

Now we shall show that $F[x]$ is a vector space for these two compositions.

Commutativity of addition in $F[x]$. We have

$$\begin{aligned} f(x) + g(x) &= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \dots \\ &= (b_0 + a_0) + (b_1 + a_1)x + (b_2 + a_2)x^2 + \dots \\ &\quad [\because \text{addition in the field } F \text{ is commutative}] \\ &= g(x) + f(x). \end{aligned}$$

Associativity of addition in $F[x]$. We have

$$\begin{aligned} [f(x) + g(x)] + h(x) &= \sum (a_i + b_i)x^i + \sum c_i x^i \\ &= \sum [(a_i + b_i) + c_i]x^i = \sum [a_i + (b_i + c_i)]x^i \\ &= \sum a_i x^i + \sum (b_i + c_i)x^i \\ &= f(x) + [g(x) + h(x)]. \end{aligned}$$

Existence of additive identity in $F[x]$. Let 0 denote the zero polynomial over the field F i.e., $0 = 0 + 0x + 0x^2 + 0x^3 + \dots$

Then $0 \in F[x]$ and $0 + f(x) = f(x)$.

$\therefore$ the zero polynomial 0 is the additive identity.

Existence of additive inverse of each member of $F[x]$. Let $-f(x)$ be the polynomial over the field F defined as

$$\begin{aligned} -f(x) &= -a_0 + (-a_1)x + (-a_2)x^2 + \dots \\ &= -a_0 - a_1x - a_2x^2 - \dots \end{aligned}$$

Then $-f(x) \in F[x]$ and we have

$$-f(x) + f(x) = 0 \text{ i.e., the zero polynomial.}$$

$\therefore -f(x)$ is the additive inverse of $f(x)$.

Thus $F[x]$ is an abelian group with respect to addition of polynomials.

Now for the operation of scalar multiplication we make the following observations.

1. If $k \in F$, then

$$\begin{aligned} k[f(x) + g(x)] &= k[(a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \dots] \\ &= k(a_0 + b_0) + k(a_1 + b_1)x + k(a_2 + b_2)x^2 + \dots \\ &= (ka_0 + kb_0) + (ka_1 + kb_1)x + (ka_2 + kb_2)x^2 + \dots \\ &= [ka_0 + (ka_1)x + (ka_2)x^2 + \dots] + [kb_0 + (kb_1)x + (kb_2)x^2 + \dots] \\ &= k(a_0 + a_1x + a_2x^2 + \dots) + k(b_0 + b_1x + b_2x^2 + \dots) \\ &= kf(x) + kg(x). \end{aligned}$$

2. If $k_1, k_2 \in F$, then

$$\begin{aligned} (k_1 + k_2)f(x) &= (k_1 + k_2)a_0 + [(k_1 + k_2)a_1]x + [(k_1 + k_2)a_2]x^2 + \dots \\ &= (k_1a_0 + k_2a_0) + (k_1a_1 + k_2a_1)x + (k_1a_2 + k_2a_2)x^2 + \dots \\ &= [k_1a_0 + (k_1a_1)x + (k_1a_2)x^2 + \dots] + [k_2a_0 + (k_2a_1)x + (k_2a_2)x^2 + \dots] \\ &= k_1(a_0 + a_1x + a_2x^2 + \dots) + k_2(a_0 + a_1x + a_2x^2 + \dots) \\ &= k_1f(x) + k_2f(x). \end{aligned}$$

3. If $k_1, k_2 \in F$, then

$$\begin{aligned} (k_1 k_2)f(x) &= (k_1 k_2)a_0 + [(k_1 k_2)a_1]x + [(k_1 k_2)a_2]x^2 + \dots \\ &= k_1(k_2a_0) + [k_1(k_2a_1)]x + [k_1(k_2a_2)]x^2 + \dots \\ &= k_1[k_2a_0 + (k_2a_1)x + (k_2a_2)x^2 + \dots] = k_1[k_2f(x)]. \end{aligned}$$

4. If 1 is the unity element of the field F , then

$$\begin{aligned} 1f(x) &= (1a_0) + (1a_1)x + (1a_2)x^2 + \dots \\ &= a_0 + a_1x + a_2x^2 + \dots = f(x). \end{aligned}$$

Hence $F[x]$ is a vector space over the field F .

Example 5: The set of all convergent real sequences is a vector space over the field of real numbers.

Solution: Let V denote the set of all convergent sequences over the field of real numbers.

Let $\alpha = \langle \alpha_1, \alpha_2, \dots, \alpha_n, \dots \rangle = \langle \alpha_n \rangle$, $\beta = \langle \beta_1, \beta_2, \dots, \beta_n, \dots \rangle = \langle \beta_n \rangle$

and $\gamma = \langle \gamma_1, \gamma_2, \dots, \gamma_n, \dots \rangle = \langle \gamma_n \rangle$

be any three convergent sequences.

1. $(V, +)$ is an abelian group.

(i) We have $\alpha + \beta = \langle \alpha_n \rangle + \langle \beta_n \rangle = \langle \alpha_n + \beta_n \rangle$ which is also a convergent sequence. Therefore V is closed for addition of sequences.

(ii) **Commutativity of addition.** We have

$$\begin{aligned}\alpha + \beta &= \langle \alpha_n \rangle + \langle \beta_n \rangle = \langle \alpha_n + \beta_n \rangle = \langle \beta_n + \alpha_n \rangle \\ &= \langle \beta_n \rangle + \langle \alpha_n \rangle = \beta + \alpha\end{aligned}$$

(iii) **Associativity of addition.** We have

$$\begin{aligned}\alpha + (\beta + \gamma) &= \langle \alpha_n \rangle + [\langle \beta_n \rangle + \langle \gamma_n \rangle] = \langle \alpha_n \rangle + \langle \beta_n + \gamma_n \rangle \\ &= \langle \alpha_n + (\beta_n + \gamma_n) \rangle = \langle (\alpha_n + \beta_n) + \gamma_n \rangle = \langle \alpha_n + \beta_n \rangle + \langle \gamma_n \rangle \\ &= [\langle \alpha_n \rangle + \langle \beta_n \rangle] + \langle \gamma_n \rangle = (\alpha + \beta) + \gamma.\end{aligned}$$

(iv) **Existence of additive identity.** The zero sequence $\langle 0 \rangle = \langle 0, 0, 0, \dots \rangle$ is the additive identity.

(v) **Existence of additive inverse.** For every sequence $\langle \alpha_n \rangle$ there exists a sequence $\langle -\alpha_n \rangle$ such that

$$\langle \alpha_n \rangle + \langle -\alpha_n \rangle = \langle \alpha_n - \alpha_n \rangle = \langle 0 \rangle = \text{the additive identity.}$$

$\therefore (V, +)$ is an abelian group.

2. V is closed for scalar multiplication.

Let a be any scalar i.e., a be any real number. Then

$a\alpha = a\langle \alpha_n \rangle = \langle a\alpha_n \rangle$ which is also a convergent sequence because

$$\lim_{n \rightarrow \infty} a\alpha_n = a \lim_{n \rightarrow \infty} \alpha_n.$$

Thus V is closed for scalar multiplication.

3. Laws of scalar multiplication. Let $a, b \in \mathbf{R}$. We have

(i) $a(\alpha + \beta) = a[\langle \alpha_n \rangle + \langle \beta_n \rangle] = a\langle \alpha_n + \beta_n \rangle$

$$= \langle a(\alpha_n + \beta_n) \rangle = \langle a\alpha_n + a\beta_n \rangle = \langle a\alpha_n \rangle + \langle a\beta_n \rangle$$

$$= a\langle \alpha_n \rangle + a\langle \beta_n \rangle = a\alpha + a\beta.$$

(ii) $(a + b)\alpha = (a + b)\langle \alpha_n \rangle = \langle (a + b)\alpha_n \rangle$

$$= \langle a\alpha_n + b\alpha_n \rangle = \langle a\alpha_n \rangle + \langle b\alpha_n \rangle$$

$$= a\langle \alpha_n \rangle + b\langle \alpha_n \rangle = a\alpha + b\alpha.$$

(iii) $(ab)\alpha = (ab)\langle \alpha_n \rangle = \langle (ab)\alpha_n \rangle = \langle a(b\alpha_n) \rangle$

$$= a\langle b\alpha_n \rangle = a[b\langle \alpha_n \rangle] = a(b\alpha).$$

(iv) $1\alpha = 1\langle \alpha_n \rangle = \langle 1\alpha_n \rangle = \langle \alpha_n \rangle = \alpha.$

Thus all the postulates of a vector space are satisfied. Hence V is a vector space over the field of real numbers.

Example 6: Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Define

$$(x, y) + (x_1, y_1) = (x + x_1, 0)$$

$$c(x, y) = (cx, 0).$$

Is V , with these operations, a vector space over the field of real numbers? (Meerut 2002)

Solution: If any of the postulates of a vector space is not satisfied, then V will not be a vector space. We shall show that for the operation of addition of vectors as defined in this problem the identity element does not exist. Suppose the ordered pair (x_1, y_1) is to be the identity element for the operation of addition of vectors. Then we must have

$$(x, y) + (x_1, y_1) = (x, y) \quad \forall x, y \in \mathbf{R}$$

$$\Rightarrow (x + x_1, 0) = (x, y) \quad \forall x, y \in \mathbf{R}.$$

But if $y \neq 0$, then we cannot have $(x + x_1, 0) = (x, y)$. Thus there exists no element (x_1, y_1) of V such that

$$(x, y) + (x_1, y_1) = (x, y) \quad \forall (x, y) \in V.$$

Therefore the identity element does not exist and V is not a vector space over the field $\mathbf{R}$.

Example 7: Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Define

$$(x, y) + (x_1, y_1) = (3y + 3y_1, -x - x_1)$$

$$c(x, y) = (3cy, -cx).$$

Verify that V , with these operations, is not a vector space over the field of real numbers.

Solution: Take $\alpha = (1, 2), \beta = (2, 3), \gamma = (1, 3)$

$$\begin{aligned} \text{Then } (\alpha + \beta) + \gamma &= \{(1, 2) + (2, 3)\} + (1, 3) \\ &= (15, -3) + (1, 3) = (0, -16) \end{aligned}$$

$$\begin{aligned} \text{and } \alpha + (\beta + \gamma) &= (1, 2) + \{(2, 3) + (1, 3)\} \\ &= (1, 2) + (18, -3) = (-3, -19). \end{aligned}$$

Thus we conclude that in general $(\alpha + \beta) + \gamma \neq \alpha + (\beta + \gamma)$. Hence V is not a vector space over the field of real numbers.

Example 8: How many elements are there in the vector space of polynomials of degree at most n in which the coefficients are the elements of the field $\mathbf{I}(p)$ over the field $\mathbf{I}(p)$, p being a prime number?

Solution: The field $\mathbf{I}(p)$ is the field $(\{0, 1, 2, \dots, p-1\}, +_p, \times_p)$.

The number of distinct elements in the field $\mathbf{I}(p)$ is p .

If $f(x)$ is a polynomial of degree at most n over the field $\mathbf{I}(p)$, then

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n, \text{ where } a_0, a_1, a_2, \dots, a_n \in \mathbf{I}(p).$$

Now in the polynomial $f(x)$, the coefficient of each of the $n+1$ terms $a_0, a_1 x, a_2 x^2, \dots, a_n x^n$, can be filled in p ways because any of the p elements of the field $\mathbf{I}(p)$ can be filled there.

Thus we can have $p \times p \times p \times \dots$ upto $(n+1)$ times i.e., p^{n+1} distinct polynomials of degree at most n over the field $\mathbf{I}(p)$. Hence if P_n is the vector space of polynomials of degree at most n in which the coefficients are the elements of the field $\mathbf{I}(p)$ over the field $\mathbf{I}(p)$, then P_n has p^{n+1} distinct elements.

Comprehensive Exercise 1

1. Show that a field F may be considered as a vector space over F if scalar multiplication is identified with field multiplication.
2. Show that the complex field C is a vector space over the real field R .

(Garhwal 2007)

3. Let V be the set of ordered pairs (z_1, z_2) of complex numbers. Show that V is a vector space over the real field R with addition in V and scalar multiplication on V defined by $(z_1, z_2) + (w_1, w_2) = (z_1 + w_1, z_2 + w_2)$ and $a(z_1, z_2) = (az_1, az_2)$ where $z_1, z_2, w_1, w_2 \in C$ and $a \in R$.
4. Let F be the field $I(p)$, p being a prime number. Show that $V_n(F)$ is a vector space. How many vectors are there in this vector space?
5. Let S be any non-empty set and let F be any field. Let V be the set of all functions from S to F i.e., let

$$V = \{f : f : S \rightarrow F\}$$

Let us define sum of two elements f and g in V as follows :

$$(f + g)(x) = f(x) + g(x) \quad \forall x \in S.$$

Also let us define scalar multiplication of an element f in V by an element c in F as follows :

$$(cf)(x) = cf(x) \quad \forall x \in S.$$

Then $V(F)$ is a vector space.

6. The vector space of all real valued continuous functions defined in some interval $[0, 1]$.
7. Prove that the set of all vectors in a plane over the field of real numbers is a vector space.
8. Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Examine in each of the following cases whether V is a vector space over the field of real numbers or not :
 - (i) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$
 $c(x, y) = (|c|x, |c|y)$
 - (ii) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$
 $c(x, y) = (0, cy)$
 - (iii) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$
 $c(x, y) = (c^2x, c^2y)$
 - (iv) $(x, y) + (x_1, y_1) = (x + y_1, y + x_1)$
 $c(x, y) = (cx, cy)$
9. Show that the axiom $1\alpha = \alpha \quad \forall \alpha \in V$ and $1 \in F$, defining the vector space is not a consequence of the remaining axioms.
10. Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Define

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$$

$$c(x, y) = (cx, y).$$

Show that with these operations V is not a vector space over the field of real numbers.

11. Let V be the set of ordered pairs (x, y) of real numbers, Show that V is not a vector space over $\mathbf{R}$ with addition in V and scalar multiplication on V defined by :
- (i) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$ and $c(x, y) = (x, y)$;
 - (ii) $(x, y) + (x_1, y_1) = (x, y)$ and $c(x, y) = (cx, cy)$;
 - (iii) $(x, y) + (x_1, y_1) = (0, 0)$ and $c(x, y) = (cx, cy)$;
 - (iv) $(x, y) + (x_1, y_1) = (xx_1, yy_1)$ and $c(x, y) = (cx, cy)$.
12. Let V be the set of all pairs (x, y) of real numbers and let F be the field of real numbers. Define

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$$

$$c(x, y) = (cx, 0).$$

Is V a vector space over the field of real numbers with these operations ?

13. Let $\mathbf{R}$ be the field of real numbers and let P_n be the set of all polynomials (of degree at most n) over the field $\mathbf{R}$. Prove that P_n is a vector space over the field $\mathbf{R}$.
14. Let U and W be vector spaces over a field F . Let V be the set of ordered pairs *i.e.*,

$$V = \{(u, w) : u \in U, w \in W\}.$$

Show that V is a vector space over F with addition in V and scalar multiplication on V defined by

$$(u, w) + (u_1, w_1) = (u + u_1, w + w_1)$$

and $k(u, w) = (ku, kw)$ where $u, u_1 \in U$ and $w, w_1 \in W$ and $k \in F$.

(Kumaun 2003)

Answers 1

4. p^n
8. (i) not a vector space (ii) not a vector space
(iii) not a vector space (iv) not a vector space
12. not a vector space

2 General Properties of Vector Spaces

Theorem 1: Let $V(F)$ be a vector space and $\mathbf{0}$ be the zero vector of V . Then

- (i) $a\mathbf{0} = \mathbf{0} \forall a \in F$. (Gorakhpur 2015)
- (ii) $0\alpha = \mathbf{0} \forall \alpha \in V$. (Kumaun 2006, 08; Gorakhpur 15)
- (iii) $a(-\alpha) = -(a\alpha) \forall a \in F, \forall \alpha \in V$. (Garhwal 2007)
- (iv) $(-a)\alpha = -(a\alpha) \forall a \in F, \forall \alpha \in V$. (Garhwal 2007)
- (v) $a(\alpha - \beta) = a\alpha - a\beta \forall a \in F$ and $\forall \alpha, \beta \in V$.
- (vi) $a\alpha = \mathbf{0} \Rightarrow a = 0$ or $\alpha = \mathbf{0}$. (Kumaun 2006)

Proof: (i) We have $a\mathbf{0} = a(\mathbf{0} + \mathbf{0})$
 $= a\mathbf{0} + a\mathbf{0}.$

$$[\because \mathbf{0} = \mathbf{0} + \mathbf{0}]$$

$$\therefore \mathbf{0} + a\mathbf{0} = a\mathbf{0} + a\mathbf{0}$$

$$[\because a\mathbf{0} \in V \text{ and } \mathbf{0} + a\mathbf{0} = a\mathbf{0}]$$

Now V is an abelian group with respect to addition.

Therefore by right cancellation law in V , we get $\mathbf{0} = a\mathbf{0}.$

$$(ii) \text{ We have } 0\alpha = (0 + 0)\alpha$$

$$= 0\alpha + 0\alpha.$$

$$[\because 0 = 0 + 0]$$

$$\therefore \mathbf{0} + 0\alpha = 0\alpha + 0\alpha.$$

$$[\because 0\alpha \in V \text{ and } \mathbf{0} + 0\alpha = 0\alpha]$$

Now V is an abelian group with respect to addition of vectors. Therefore by right cancellation law in V , we get $\mathbf{0} = 0\alpha.$

$$(iii) \text{ We have } a[\alpha + (-\alpha)] = a\alpha + a(-\alpha) \Rightarrow a\mathbf{0} = a\alpha + a(-\alpha)$$

$$\Rightarrow \mathbf{0} = a\alpha + a(-\alpha)$$

$$[\because a\mathbf{0} = \mathbf{0}]$$

$$\Rightarrow a(-\alpha) \text{ is the additive inverse of } a\alpha \Rightarrow a(-\alpha) = -(a\alpha).$$

$$(iv) \text{ We have } [a + (-a)]\alpha = a\alpha + (-a)\alpha \Rightarrow 0\alpha = a\alpha + (-a)\alpha$$

$$\Rightarrow \mathbf{0} = a\alpha + (-a)\alpha \Rightarrow (-a)\alpha \text{ is the additive inverse of } a\alpha$$

$$\Rightarrow (-a)\alpha = -(a\alpha).$$

$$(v) \text{ We have } a(\alpha - \beta) = a[\alpha + (-\beta)] = a\alpha + a(-\beta)$$

$$= a\alpha + [-(a\beta)]$$

$$[\because a(-\beta) = -(a\beta)]$$

$$= a\alpha - a\beta.$$

(vi) Let $a\alpha = \mathbf{0}$ and $a \neq 0$. Then a^{-1} exists because a is a non-zero element of the field F .

$$\therefore a\alpha = \mathbf{0} \Rightarrow a^{-1}(a\alpha) = a^{-1}\mathbf{0} \Rightarrow (a^{-1}a)\alpha = \mathbf{0} \Rightarrow 1\alpha = \mathbf{0} \Rightarrow \alpha = \mathbf{0}.$$

Again let $a\alpha = \mathbf{0}$ and $\alpha \neq \mathbf{0}$. Then to prove that $a = 0$. Suppose $a \neq 0$. Then a^{-1} exists.

$$\therefore a\alpha = \mathbf{0} \Rightarrow a^{-1}(a\alpha) = a^{-1}\mathbf{0} \Rightarrow (a^{-1}a)\alpha = \mathbf{0} \Rightarrow 1\alpha = \mathbf{0} \Rightarrow \alpha = \mathbf{0}.$$

Thus we get a contradiction that α must be a zero vector. Therefore a must be equal to 0. Hence $\alpha \neq \mathbf{0}$ and $a\alpha = \mathbf{0} \Rightarrow a = 0$.

Theorem 2: Let $V(F)$ be a vector space. Then

(i) If $a, b \in F$ and α is a non-zero element of V , we have $a\alpha = b\alpha \Rightarrow a = b$.

(ii) If $\alpha, \beta \in V$ and a is a non-zero element of F , we have $a\alpha = a\beta \Rightarrow \alpha = \beta$.

Proof: (i) We have $a\alpha = b\alpha \Rightarrow a\alpha - b\alpha = \mathbf{0} \Rightarrow (a - b)\alpha = \mathbf{0}.$

But $\alpha \neq \mathbf{0}$. Therefore $(a - b)\alpha = \mathbf{0} \Rightarrow a - b = 0 \Rightarrow a = b$.

(ii) We have $a\alpha = a\beta \Rightarrow a\alpha - a\beta = \mathbf{0} \Rightarrow a(\alpha - \beta) = \mathbf{0}.$

But $a \neq 0$. Therefore $a(\alpha - \beta) = \mathbf{0} \Rightarrow \alpha - \beta = \mathbf{0} \Rightarrow \alpha = \beta$.

3 Vector Subspaces

Definition: Let V be a vector space over the field F and $W \subseteq V$. Then W is called a subspace of V if W itself is a vector space over F with respect to the operations of vector addition and scalar multiplication in V . (Garhwal 2010)

Remark: Let $V(F)$ be any vector space. Then V itself and the subset of V consisting of zero vector only are always subspaces of V . These two are called **improper subspaces**. If V has any other subspace, then it is called a **proper subspace**. The subspace of V consisting of zero vector only is called the **zero subspace**.

Theorem 1: *The necessary and sufficient condition for a non-empty subset W of a vector space $V(F)$ to be a subspace of V is that W is closed under vector addition and scalar multiplication in V .* (Kumaun 2003, 10, 13)

Proof: If W itself is a vector space over F with respect to vector addition and scalar multiplication in V , then W must be closed with respect to these two compositions. Hence the condition is necessary.

The condition is sufficient. Now suppose that W is a non-empty subset of V and W is closed under vector addition and scalar multiplication in V .

Let $\alpha \in W$. If 1 is the unity element of F , then $-1 \in F$. Now W is closed under scalar multiplication. Therefore

$$-1 \in F, \alpha \in W \Rightarrow (-1)\alpha \in W \Rightarrow -(1\alpha) \in W \Rightarrow -\alpha \in W.$$

$$[\because \alpha \in W \Rightarrow \alpha \in V \text{ and } 1\alpha = \alpha \text{ in } V.]$$

Thus the additive inverse of each element of W is also in W .

Now W is closed under vector addition.

Therefore $\alpha \in W, -\alpha \in W \Rightarrow \alpha + (-\alpha) \in W \Rightarrow \mathbf{0} \in W$,

where $\mathbf{0}$ is the zero vector of V .

Hence the zero vector of V is also the zero vector of W . Since the elements of W are also the elements of V , therefore vector addition will be commutative as well as associative in W . Hence W is an abelian group with respect to vector addition. Also it is given that W is closed under scalar multiplication. The remaining postulates of a vector space will hold in W since they hold in V of which W is a subset.

Hence W itself is a vector space for the two compositions.

$\therefore W$ is a subspace of V .

Theorem 2: *The necessary and sufficient conditions for a non-empty subset W of a vector space $V(F)$ to be a subspace of V are*

- (i) $\alpha \in W, \beta \in W \Rightarrow \alpha - \beta \in W$. (Garhwal 2013)
- (ii) $a \in F, \alpha \in W \Rightarrow a\alpha \in W$.

Proof: The conditions are necessary. If W is a subspace of V , then W is an abelian group with respect to vector addition. Therefore $\alpha \in W, \beta \in W \Rightarrow \alpha - \beta \in W$. Also W must be closed under scalar multiplication. Therefore the condition (ii) is also necessary.

The conditions are sufficient: Now suppose W is a non-empty subset of V satisfying the two given conditions. From condition (i), we have $\alpha \in W, \alpha \in W \Rightarrow \alpha - \alpha \in W \Rightarrow \mathbf{0} \in W$.

Thus the zero vector of V belongs to W and it will also be the zero vector of W .

Now $\mathbf{0} \in W, \alpha \in W \Rightarrow \mathbf{0} - \alpha \in W \Rightarrow -\alpha \in W$.

Thus the additive inverse of each element of W is also in W .

Again $\alpha \in W, \beta \in W \Rightarrow \alpha \in W, -\beta \in W$
 $\Rightarrow \alpha - (-\beta) \in W \Rightarrow \alpha + \beta \in W.$

Thus W is closed with respect to vector addition.

Since the elements of W are also the elements of V , therefore vector addition will be commutative as well as associative in W . Hence W is an abelian group under vector addition. Also from condition (ii), W is closed under scalar multiplication. The remaining postulates of a vector space will hold in W since they hold in V which is a superset of W . Hence W is a subspace of V .

Theorem 3: *The necessary and sufficient condition for a non-empty subset W of a vector space V (F) to be a subspace of V is*

$$a, b \in F \text{ and } \alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W. \\ \text{(Kumaun 2007,14 ; Garhwal 10 ; Gorakhpur 10)}$$

Proof: **The condition is necessary.** If W is a subspace of V , then W must be closed under scalar multiplication and vector addition.

Therefore $a \in F, \alpha \in W \Rightarrow a\alpha \in W$

and $b \in F, \beta \in W \Rightarrow b\beta \in W.$

Now $a\alpha \in W, b\beta \in W \Rightarrow a\alpha + b\beta \in W$. Hence the condition is necessary.

The condition is sufficient. Now suppose W is a non-empty subset of V satisfying the given condition i.e.,

$$a, b \in F \text{ and } \alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W.$$

Taking $a = 1, b = 1$, we see that if $\alpha, \beta \in W$ then $1\alpha + 1\beta \in W$

$$\Rightarrow \alpha + \beta \in W. \quad [\because \alpha \in W \Rightarrow \alpha \in V \text{ and } 1\alpha = \alpha \text{ in } V]$$

Thus W is closed under vector addition.

Now taking $a = -1, b = 0$, we see that if $\alpha \in W$, then

$$(-1)\alpha + 0\alpha \in W \quad [\text{in place of } \beta \text{ we have taken } \alpha]$$

$$\Rightarrow -(1\alpha) + 0\alpha \in W \Rightarrow -\alpha \in W.$$

Thus the additive inverse of each element of W is also in W .

Taking $a = 0, b = 0$, we see that if $\alpha \in W$, then

$$0\alpha + 0\alpha \in W \Rightarrow 0 + 0 \in W \Rightarrow 0 \in W.$$

Thus the zero vector of V belongs to W . It will also be the zero vector of W .

Since the elements of W are also the elements of V , therefore vector addition will be associative as well as commutative in W . Thus W is an abelian group with respect to vector addition.

Now taking $\beta = 0$, we see that if $a, b \in F$

and $\alpha \in W$, then

$$a\alpha + b0 \in W \text{ i.e., } a\alpha + 0 \in W \text{ i.e., } a\alpha \in W.$$

Thus W is closed under scalar multiplication.

The remaining postulates of a vector space will hold in W since they hold in V of which W is a subset. Hence W (F) is a subspace of V (F).

Theorem 4: *A non-empty subset W of a vector space V (F) is a subspace of V if and only if for each pair of vectors α, β in W and each scalar a in F the vector $a\alpha + \beta$ is again in W .*

Proof: The condition is necessary. If W is a subset of V , then W must be closed with respect to scalar multiplication and as well as with respect to vector addition. Therefore

$$a \in F, \alpha \in W \Rightarrow a\alpha \in W.$$

$$\text{Further } a\alpha \in W, \beta \in W \Rightarrow a\alpha + \beta \in W.$$

Hence the condition is necessary.

The condition is sufficient. It is given that W is a non-empty subset of V and $a \in F, \alpha, \beta \in W \Rightarrow a\alpha + \beta \in W$. We are to prove that W is a subspace of V .

(i) Since W is non-empty, therefore there is at least one vector in W , say γ . Now $1 \in F \Rightarrow -1 \in F$. Therefore taking $a = -1, \alpha = \gamma, \beta = \gamma$, we get from the given condition that

$$(-1)\gamma + \gamma = -(1\gamma) + \gamma = -\gamma + \gamma = \mathbf{0} \text{ is in } W.$$

(ii) Now let $a \in F, \alpha \in W$. Since $\mathbf{0}$ is in W , therefore taking $\beta = \mathbf{0}$ in the given condition, we get

$$a\alpha + \mathbf{0} = a\alpha \text{ is in } W.$$

Thus W is closed with respect to scalar multiplication.

(iii) Let $\alpha \in W$. Since $-1 \in F$ and W is closed with respect to scalar multiplication, therefore $(-1)\alpha = -(1\alpha) = -\alpha$ is in W .

(iv) We have $1 \in F$. If $\alpha, \beta \in W$, then $1\alpha + \beta = \alpha + \beta$ is in W . Thus W is closed with respect to vector addition.

The remaining postulates of a vector space will hold in W since they hold in V of which W is a subset.

Hence W is a subspace of V .

Note: If we are to prove that a non-empty subset W of a vector space V is a subspace of V , then either it is sufficient to prove that

$$a, b \in F$$

$$\text{and } \alpha, \beta \in W$$

$$\Rightarrow a\alpha + b\beta \in W$$

or it is sufficient to prove that

$$a \in F, \text{ and } \alpha, \beta \in W \Rightarrow a\alpha + \beta \in W.$$

Illustrative Examples

Example 9: The set W of ordered triads $(a_1, a_2, 0)$ where $a_1, a_2 \in F$ is a subspace of $V_3(F)$.

Solution: W is non-empty since $\mathbf{0} = (0, 0, 0) \in W$. Let $\alpha = (a_1, a_2, 0)$ and $\beta = (b_1, b_2, 0)$ be any two elements of W . Then $a_1, a_2, b_1, b_2 \in F$. If a, b be any two elements of F , we have

$$\begin{aligned} a\alpha + b\beta &= a(a_1, a_2, 0) + b(b_1, b_2, 0) \\ &= (aa_1, aa_2, 0) + (bb_1, bb_2, 0) \\ &= (aa_1 + bb_1, aa_2 + bb_2, 0) \in W \end{aligned}$$

since $aa_1 + bb_1, aa_2 + bb_2 \in F$ and the last co-ordinate of this triad is zero.

Hence W is a subspace of $V_3(F)$.

Example 10: Let V be the vector space of all polynomials in an indeterminate x over a field F . Let W be a subset of V consisting of all polynomials of degree $\leq n$. Then W is a subspace of V .

Solution: Let α and β be any two elements of W . Then α, β are polynomials over F of degree $\leq n$. If a, b are any two elements of F , then $a\alpha + b\beta$ will also be a polynomial of degree $\leq n$.

Therefore $a\alpha + b\beta \in W$. Hence W is a subspace of V .

Example 11: If a_1, a_2, a_3 are fixed elements of a field F , then the set W of all ordered triads (x_1, x_2, x_3) of elements of F , such that $a_1 x_1 + a_2 x_2 + a_3 x_3 = 0$, is a subspace of $V_3(F)$.
(Kumaun 2007)

Solution: W is non-empty since $(0, 0, 0) \in W$. Let $\alpha = (x_1, x_2, x_3)$ and $\beta = (y_1, y_2, y_3)$ be any two elements of W . Then $x_1, x_2, x_3, y_1, y_2, y_3$ are elements of F and are such that

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = 0 \quad \dots(1)$$

$$\text{and} \quad a_1 y_1 + a_2 y_2 + a_3 y_3 = 0. \quad \dots(2)$$

If a, b be any two elements of F , we have

$$\begin{aligned} a\alpha + b\beta &= a(x_1, x_2, x_3) + b(y_1, y_2, y_3) \\ &= (ax_1, ax_2, ax_3) + (by_1, by_2, by_3) \\ &= (ax_1 + by_1, ax_2 + by_2, ax_3 + by_3). \end{aligned}$$

$$\begin{aligned} \text{Now } a_1(ax_1 + by_1) + a_2(ax_2 + by_2) + a_3(ax_3 + by_3) \\ &= a(a_1 x_1 + a_2 x_2 + a_3 x_3) + b(a_1 y_1 + a_2 y_2 + a_3 y_3) \\ &= a \cdot 0 + b \cdot 0 \quad [\text{by (1) and (2)}] \\ &= 0. \end{aligned}$$

$$\therefore a\alpha + b\beta = (ax_1 + by_1, ax_2 + by_2, ax_3 + by_3) \in W.$$

Hence W is a subspace of $V_3(F)$.

Example 12: Which of the following sets of vectors $\alpha = (a_1, a_2, \dots, a_n)$ in $\mathbf{R}^n$ are subspaces of $\mathbf{R}^n$ ($n \geq 3$)?

- (i) all α such that $a_1 \leq 0$;
- (ii) all α such that a_3 is an integer;
- (iii) all α such that $a_2 + 4a_3 = 0$;
- (iv) all α such that $a_1 + a_2 + \dots + a_n = k$ (k a given constant).

Solution: (i) Let $W = \{\alpha : \alpha \in \mathbf{R}^n \text{ and } a_1 \leq 0\}$.

If we take $a_1 = -3$, then $a_1 < 0$ and so

$$\alpha = (-3, a_2, \dots, a_n) \in W.$$

Now if we take $a = -2$, then

$$a\alpha = (6, -2a_2, \dots, -2a_n).$$

Since the first coordinate of $a\alpha$ is 6 which is > 0 , therefore $a\alpha \notin W$.

Thus $\alpha \in W, a \in \mathbf{R}$ but $a\alpha \notin W$. Therefore W is not closed for scalar multiplication and so W is not a subspace of $\mathbf{R}^n$.

(ii) Let $W = \{\alpha : \alpha \in \mathbf{R}^n \text{ and } a_3 \text{ is an integer}\}$.

If we take $a_3 = 5$, then a_3 is an integer and so

$$\alpha = (a_1, a_2, 5, \dots, a_n) \in W.$$

Now if we take $a = \frac{1}{2}$, then

$$a\alpha = \left(\frac{1}{2} a_1, \frac{1}{2} a_2, \frac{5}{2}, \dots, \frac{1}{2} a_n \right).$$

Since the third coordinate of $a\alpha$ is $5/2$ which is not an integer therefore $a\alpha \notin W$.
Thus $\alpha \in W, a \in \mathbf{R}$ but $a\alpha \notin W$. Therefore W is not closed for scalar multiplication and so W is not a subspace of $\mathbf{R}^n$.

(iii) Let $W = \{\alpha : \alpha \in \mathbf{R}^n \text{ and } a_2 + 4a_3 = 0\}$.

W is non-empty since $\mathbf{0} = (0, 0, \dots, 0) \in W$.

Let $\alpha = (a_1, \dots, a_n)$ and $\beta = (b_1, \dots, b_n)$ be any two members of W .

Then $a_2 + 4a_3 = 0$ and $b_2 + 4b_3 = 0$.

If $a, b \in \mathbf{R}$, then $a\alpha + b\beta = (aa_1 + bb_1, \dots, aa_n + bb_n)$.

We have $(aa_2 + bb_2) + 4(aa_3 + bb_3) = a(a_2 + 4a_3) + b(b_2 + 4b_3) = a \cdot 0 + b \cdot 0 = 0$.

Thus according to the definition of W , $a\alpha + b\beta \in W$.

In this way $\alpha, \beta \in W$ and $a, b \in \mathbf{R} \Rightarrow a\alpha + b\beta \in W$. Hence W is a subspace of $\mathbf{R}^n$.

(iv) A subspace if $k = 0$ and not a subspace if $k \neq 0$.

Example 13: Let $\mathbf{R}$ be the field of real numbers. Which of the following are subspaces of $V_3(\mathbf{R})$?

(i) $\{(x, 2y, 3z) : x, y, z \in \mathbf{R}\}$.

(ii) $\{(x, x, x) : x \in \mathbf{R}\}$.

(iii) $\{(x, y, z) : x, y, z \text{ are rational numbers}\}$.

Solution: (i) Let $W = \{(x, 2y, 3z) : x, y, z \in \mathbf{R}\}$.

Let $\alpha = (x_1, 2y_1, 3z_1)$ and $\beta = (x_2, 2y_2, 3z_2)$ be any two elements of W . Then $x_1, y_1, z_1, x_2, y_2, z_2$ are real numbers. If a, b are any two real numbers, then

$$\begin{aligned} a\alpha + b\beta &= a(x_1, 2y_1, 3z_1) + b(x_2, 2y_2, 3z_2) \\ &= (ax_1 + bx_2, 2ay_1 + 2by_2, 3az_1 + 3bz_2) \\ &= (ax_1 + bx_2, 2[ay_1 + by_2], 3[az_1 + bz_2]) \\ &\in W, \text{ since } ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2 \end{aligned}$$

are real numbers.

Thus $a, b \in \mathbf{R}$ and $\alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W$.

$\therefore W$ is a subspace of $V_3(\mathbf{R})$.

(ii) Let $W = \{(x, x, x) : x \in \mathbf{R}\}$.

Let $\alpha = (x_1, x_1, x_1)$ and $\beta = (x_2, x_2, x_2)$ be any two elements of W . Then x_1, x_2 are real numbers. If a, b are any real numbers, then

$$\begin{aligned} a\alpha + b\beta &= a(x_1, x_1, x_1) + b(x_2, x_2, x_2) \\ &= (ax_1 + bx_2, ax_1 + bx_2, ax_1 + bx_2) \in W, \text{ since } ax_1 + bx_2 \in \mathbf{R}. \end{aligned}$$

Thus W is a subspace of $V_3(\mathbf{R})$.

(iii) Let $W = \{(x, y, z) : x, y, z \text{ are rational numbers}\}$.

Now $\alpha = (3, 4, 5)$ is an element of W . Also $a = \sqrt{7}$ is an element of $\mathbf{R}$. But

$$a\alpha = \sqrt{7}(3, 4, 5) = (3\sqrt{7}, 4\sqrt{7}, 5\sqrt{7}) \notin W \text{ since } 3\sqrt{7}, 4\sqrt{7}, 5\sqrt{7}$$

are not rational numbers.

Therefore W is not closed under scalar multiplication. Hence W is not a subspace of $V_3(\mathbf{R})$.

Example 14: Let V be the vector space of all 2×2 matrices over the field $\mathbf{R}$. Show that W is not a subspace of V , where W contains all 2×2 matrices with zero determinants.

Solution: Let $A = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & b \end{bmatrix}$, where $a, b \in \mathbf{R}$ and $a \neq 0, b \neq 0$.

We have $|A| = 0, |B| = 0$ so $A, B \in W$.

Now $A + B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$. We have $|A + B| = ab \neq 0$.

$\therefore A + B \notin W$.

Thus $A \in W, B \in W$ but $A + B \notin W$.

Hence W is not a subspace of V .

Example 15: If a vector space V is the set of all real valued continuous functions over the field of real numbers $\mathbf{R}$, then show that the set W of solutions of the differential equation.

$$2 \frac{d^2 y}{dx^2} - 9 \frac{dy}{dx} + 2y = 0$$

is a subspace of V .

Solution: We have $W = \left\{ y : 2 \frac{d^2 y}{dx^2} - 9 \frac{dy}{dx} + 2y = 0 \right\}$, where $y = f(x)$.

Obviously $y = 0$ satisfies the given differential equation and as such it belongs to W and thus $W \neq \emptyset$.

Now let $y_1, y_2 \in W$. Then

$$2 \frac{d^2 y_1}{dx^2} - 9 \frac{dy_1}{dx} + 2y_1 = 0 \quad \dots(1)$$

$$\text{and} \quad 2 \frac{d^2 y_2}{dx^2} - 9 \frac{dy_2}{dx} + 2y_2 = 0. \quad \dots(2)$$

Let $a, b \in \mathbf{R}$. If W is to be a subspace then we should show that $ay_1 + by_2$ also belongs to W i.e., it is a solution of the given differential equation.

$$\begin{aligned} \text{We have} \quad & 2 \frac{d^2}{dx^2} (ay_1 + by_2) - 9 \frac{d}{dx} (ay_1 + by_2) + 2 (ay_1 + by_2) \\ &= 2a \frac{d^2 y_1}{dx^2} + 2b \frac{d^2 y_2}{dx^2} - 9a \frac{dy_1}{dx} - 9b \frac{dy_2}{dx} + 2ay_1 + 2by_2 \\ &= a \left(2 \frac{d^2 y_1}{dx^2} - 9 \frac{dy_1}{dx} + 2y_1 \right) + b \left(2 \frac{d^2 y_2}{dx^2} - 9 \frac{dy_2}{dx} + 2y_2 \right) \\ &= a \cdot 0 + b \cdot 0, \text{ by (1) and (2)} \\ &= 0. \end{aligned}$$

Thus $ay_1 + by_2$ is a solution of the given differential equation and so it belongs to W .

Hence W is a subspace of V .

Example 16: Find the solution of system of homogeneous linear equations. Let $V(F)$ be the vector space of all $n \times 1$ matrices over the field F . Let A be an $m \times n$ matrix over F . Then the set W of all $n \times 1$ matrices X over F such that $AX = O$ is a subspace of V . Here O is a null matrix of the type $m \times 1$.

Solution: W is non-empty since $O_{n \times 1} \in W$. Let $X, Y \in W$. Then X and Y are $n \times 1$ matrices over F such that $AX = O, AY = O$.

Let $a \in F$. Then $aX + Y$ is also an $n \times 1$ matrix over F .

We have $A(aX + Y) = A(aX) + AY = a(AX) + AY = aO + O = O + O = O$

Therefore $aX + Y \in W$. Thus

$$a \in F, X, Y \in W \Rightarrow aX + Y \in W.$$

Hence W is a subspace of V .

Comprehensive Exercise 2

1. Show that the subset $\{(a, b, c) : a + b + c = 0\}$ of $\mathbf{R}^3$ is a subspace of $\mathbf{R}^3$.
(Gorakhpur 2014)
2. Prove that the set of all solutions (a, b, c) of the equation $a + b + 2c = 0$ is a subspace of the vector space $V_3(\mathbf{R})$.
3. Show that the set W of the elements of the vector space $V_3(\mathbf{R})$ of the form $(x + 2y, y, -x + 3y)$ where $x, y \in \mathbf{R}$ is a subspace of $V_3(\mathbf{R})$. (Kumaun 2013)
4. Let V be the vector space of all polynomials over the field $\mathbf{R}$. Determine whether or not W is a subspace of V where (i) W contains all polynomials with integral coefficients (ii) W consists of all polynomials $b_0 + b_1x^2 + b_2x^4 + \dots + b_nx^{2n}$ i.e., polynomials with only even powers of x .
5. Which of the following sets of vectors $\alpha = (a_1, \dots, a_n)$ in $\mathbf{R}^n$ are subspaces of $\mathbf{R}^n$? ($n \geq 3$).
 (i) all α such that $a_1 \geq 0$;
 (ii) all α such that $a_1 + 3a_2 = a_3$;
 (iii) all α such that $a_2 = a_1^2$;
 (iv) all α such that $a_1 a_2 = 0$;
 (v) all α such that a_2 is rational.
 (Kumaun 2011)
6. Let F be the field of integers modulo 2, V be the set of all 2×2 matrices over F . Show that $V(F)$ is a finite vector space. Give two non-trivial subspaces of this vector space.
7. Let V be the vector space of all $n \times n$ square matrices over a field F . Show that W is a subspace of V if W consists of all matrices which are
 (i) symmetric i.e., all matrices $A = [a_{ij}]$ for which $a_{ji} = a_{ij}$.
 (ii) skew-symmetric.
 (iii) upper triangular.
 (iv) diagonal.
 (v) scalar.

8. Let V be a vector space of all real $n \times n$ matrices. Prove that the set W consisting of all $n \times n$ real matrices which commute with a given matrix T of V forms a subspace of V .
9. Let V be the vector space of all 2×2 matrices over the real field $\mathbf{R}$. Show that the subset of V consisting of all matrices A for which $A^2 = A$ is not a subspace of V .
10. Determine whether or not W is a subspace of $\mathbf{R}^3$ if W consists of those vectors $(a, b, c) \in \mathbf{R}^3$ for which :
 - (i) $a = 2b$ (ii) $ab = 0$
 - (iii) $a \leq b \leq c$ (iv) $a = b^2$
 - (v) $a^2 + b^2 + c^2 \leq 1$ (vi) $k_1 a + k_2 b + k_3 c = 0, k_i \in \mathbf{R}$.
11. Let V be the (real) vector space of all functions f from $\mathbf{R}$ into $\mathbf{R}$. Which of the following sets of functions are sub-spaces of V ?
 - (i) all f such that $f(x^2) = [f(x)]^2$;
 - (ii) all f such that $f(0) = f(1)$;
 - (iii) all f such that $f(3) = 1 + f(-5)$;
 - (iv) all f such that $f(-1) = 0$;
 - (v) all f which are continuous.
12. Let V be the vector space of all functions from the real field $\mathbf{R}$ into $\mathbf{R}$. Check whether W is a subspace of V where
 - (i) $W = \{f : f(7) = 2 + f(1)\}$;
 - (ii) W consists of all bounded functions;
 - (iii) W consists of all integrable functions in the interval $[0, 1]$.
13. Let V be the vector space of infinite sequences $\langle a_1, a_2, \dots, a_n, \dots \rangle$ in a field F . Show that W is a subspace of V , if
 - (i) W consists of all sequences with 0 as the first component ;
 - (ii) W consists of all sequences with only a finite number of non-zero components.
14. Let $AX = B$ be a non-homogeneous system of linear equations in n unknowns over a field F . Show that the solution set W of the system is not a subspace of F^n .

Answers 2

4. (i) not a subspace (ii) a subspace
5. (i) not a subspace (ii) a subspace (iii) not a subspace;
- (iv) not a subspace (v) not a subspace
10. (i) a subspace (ii) not a subspace
- (iii) not a subspace (iv) not a subspace
- (v) not a subspace (vi) a subspace
11. (i) not a subspace (ii) a subspace (iii) not a subspace
- (iv) a subspace (v) a subspace
12. (i) not a subspace (ii) a subspace (iii) a subspace

4 Algebra of Subspaces

Theorem 1: *The intersection of any two subspaces W_1 and W_2 of a vector space $V(F)$ is also a subspace of $V(F)$.*

(Kumaun 2002, 07, 09; Garhwal 07, 12; Gorakhpur 10, 13, 15)

Proof: Since $0 \in W_1$ and W_2 both, therefore $W_1 \cap W_2$ is not empty.

Let $\alpha, \beta \in W_1 \cap W_2$ and $a, b \in F$.

Now $\alpha \in W_1 \cap W_2 \Rightarrow \alpha \in W_1$ and $\alpha \in W_2$

and $\beta \in W_1 \cap W_2 \Rightarrow \beta \in W_1$ and $\beta \in W_2$.

Since W_1 is a subspace, therefore $a, b \in F$ and $\alpha, \beta \in W_1 \Rightarrow a\alpha + b\beta \in W_1$.

Similarly $a, b \in F$ and $\alpha, \beta \in W_2 \Rightarrow a\alpha + b\beta \in W_2$.

Now $a\alpha + b\beta \in W_1, a\alpha + b\beta \in W_2 \Rightarrow a\alpha + b\beta \in W_1 \cap W_2$.

Thus $a, b \in F$ and $\alpha, \beta \in W_1 \cap W_2 \Rightarrow a\alpha + b\beta \in W_1 \cap W_2$.

Hence $W_1 \cap W_2$ is a subspace of $V(F)$.

Note: The union of two subspaces of $V(F)$ may not be a subspace of $V(F)$. For example if $\mathbf{R}$ be the field of real numbers, then $W_1 = \{(0, 0, z) : z \in \mathbf{R}\}$ and $W_2 = \{(0, y, 0) : y \in \mathbf{R}\}$ are two subspaces of $V_3(\mathbf{R})$.

We have $(0, 0, 3) \in W_1$ and $(0, 5, 0) \in W_2$.

But $(0, 0, 3) + (0, 5, 0) = (0, 5, 3) \notin W_1 \cup W_2$ since neither $(0, 5, 3) \in W_1$ nor $(0, 5, 3) \in W_2$. Thus $W_1 \cup W_2$ is not closed under vector addition. Hence $W_1 \cup W_2$ is not a subspace of $V_3(\mathbf{R})$.

(Gorakhpur 2015)

Theorem 2: *The union of two subspaces is a subspace if and only if one is contained in the other.*

(Kumaun 2003, 07, 11; Garhwal 06, 07, 13)

Proof: Suppose W_1 and W_2 are two subspaces of a vector space V .

Let $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. Then $W_1 \cup W_2 = W_2$ or W_1 . But W_1, W_2 are subspaces and therefore, $W_1 \cup W_2$ is also a subspace.

Conversely, suppose $W_1 \cup W_2$ is a subspace.

To prove that $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Let us assume that W_1 is not a subset of W_2 and W_2 is also not a subset of W_1 .

Now W_1 is not a subset of $W_2 \Rightarrow \exists \alpha \in W_1$ and $\alpha \notin W_2$... (1)

and W_2 is not a subset of $W_1 \Rightarrow \exists \beta \in W_2$ and $\beta \notin W_1$ (2)

From (1) and (2), we have $\alpha \in W_1 \cup W_2$ and $\beta \in W_1 \cup W_2$.

Since $W_1 \cup W_2$ is a subspace, therefore $\alpha + \beta$ is also in $W_1 \cup W_2$.

But $\alpha + \beta \in W_1 \cup W_2 \Rightarrow \alpha + \beta \in W_1$ or W_2 .

Suppose $\alpha + \beta \in W_1$. Since $\alpha \in W_1$ and W_1 is a subspace, therefore $(\alpha + \beta) - \alpha = \beta$ is in W_1 .

But from (2), we have $\beta \notin W_1$. Thus we get a contradiction. Again suppose that $\alpha + \beta \in W_2$. Since $\beta \in W_2$ and W_2 is a subspace, therefore $(\alpha + \beta) - \beta = \alpha$ is in W_2 . But from (1), we have $\alpha \notin W_2$. Thus here also we get a contradiction. Hence either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Theorem 3: *Arbitrary intersection of subspaces i.e., the intersection of any family of subspaces of a vector space is a subspace.* (Kumaun 2010, 14)

Proof: Let $V(F)$ be a vector space and let $\{W_t : t \in T\}$ be any family of subspaces of V . Here T is an index set and is such that $\forall t \in T, W_t$ is a subspace of V .

$$\text{Let } U = \bigcap_{t \in T} W_t = \{x \in V : x \in W_t \forall t \in T\}$$

be the intersection of this family of subspaces of V . Then to prove that U is also a subspace of V .

Obviously $U \neq \emptyset$, since at least the zero vector $\mathbf{0}$ of V is in $W_t \forall t \in T$.

Now let $a, b \in F$ and α, β be any two elements of $\bigcap_{t \in T} W_t$.

Then $\alpha, \beta \in W_t \forall t \in T$. Since each W_t is a subspace of V , therefore

$$a\alpha + b\beta \in W_t \forall t \in T.$$

Thus $a\alpha + b\beta \in \bigcap_{t \in T} W_t$.

Thus $a, b \in F$ and $\alpha, \beta \in \bigcap_{t \in T} W_t \Rightarrow a\alpha + b\beta \in \bigcap_{t \in T} W_t$.

Hence $\bigcap_{t \in T} W_t$ is a subspace of $V(F)$.

Note: In the above theorem we cannot replace intersection by union.

5 Smallest Subspace Containing any Subset of $V(F)$

Let $V(F)$ be a vector space and S be any subset of V . If U is a subspace of V containing S and is itself contained in every subspace of V containing S , then U is called the **smallest subspace** of V containing S . The smallest subspace of V containing S is also called the subspace of V generated or spanned by S and we shall denote it by the symbol $\{S\}$ or by (S) . It can be easily seen that the intersection of all the subspaces of $V(F)$ containing S is the subspace of $V(F)$ generated by S . If $\{S\} = V$, then we say that V is spanned by S .

6 Linear Combination of Vectors and Linear Span of a Set

Linear combination:

Definition: Let $V(F)$ be a vector space. If $\alpha_1, \alpha_2, \dots, \alpha_n \in V$, then any vector

$$\alpha = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n \text{ where } a_1, a_2, \dots, a_n \in F$$

is called a linear combination of the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$.

Linear span:

Definition: Let $V(F)$ be a vector space and S be any non-empty subset of V . Then the linear span of S is the set of all linear combinations of finite sets of elements of S and is denoted by $L(S)$.

Thus we have

$$L(S) = \{a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n : \alpha_1, \alpha_2, \dots, \alpha_n$$

is any arbitrary finite subset of S and $a_1, a_2, \dots, a_n$

is any arbitrary finite subset of $F\}$.

(Kumaun 2009; Garhwal 14)

Theorem 1: The linear span $L(S)$ of any subset S of a vector space $V(F)$ is a subspace of V generated by S i.e., $L(S) = \{S\}$. (Kumaun 2009)

Proof: Let α, β be any two elements of $L(S)$.

$$\text{Then } \alpha = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m$$

$$\text{and } \beta = b_1 \beta_1 + b_2 \beta_2 + \dots + b_n \beta_n$$

where the a 's and b 's are elements of F and the α 's and β 's are elements of S .

If a, b be any two elements of F , then

$$\begin{aligned} a\alpha + b\beta &= a(a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_m \alpha_m) \\ &\quad + b(b_1 \beta_1 + b_2 \beta_2 + \dots + b_n \beta_n) \\ &= a(a_1 \alpha_1) + a(a_2 \alpha_2) + \dots + a(a_m \alpha_m) + b(b_1 \beta_1) + b(b_2 \beta_2) + \dots \\ &\quad \dots + b(b_n \beta_n) \\ &= (aa_1) \alpha_1 + (aa_2) \alpha_2 + \dots + (aa_m) \alpha_m + (bb_1) \beta_1 + (bb_2) \beta_2 + \dots \\ &\quad \dots + (bb_n) \beta_n. \end{aligned}$$

Thus $a\alpha + b\beta$ has been expressed as a linear combination of a finite set $\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_n$ of the elements of S . Consequently $a\alpha + b\beta \in L(S)$.

Thus $a, b \in F$ and $\alpha, \beta \in L(S) \Rightarrow a\alpha + b\beta \in L(S)$.

Hence $L(S)$ is a subspace of $V(F)$.

Also each element of S belongs to $L(S)$ because if $\alpha_r \in S$, then $\alpha_r = 1\alpha_r$ and this implies that $\alpha_r \in L(S)$. Thus $L(S)$ is a subspace of V and S is contained in $L(S)$.

Now if W is any subspace of V containing S , then each element of $L(S)$ must be in W because W is to be closed under vector addition and scalar multiplication. Therefore $L(S)$ will be contained in W .

Hence $L(S) = \{S\}$ i.e., $L(S)$ is the smallest subspace of V containing S .

Note 1: Important. Suppose S is a non-empty subset of a vector space $V(F)$. Then a vector $\alpha \in V$ will be in the subspace of V generated by S if it can be expressed as a linear combination over F of a finite number of vectors belonging to S .

Note 2: If in any case we are to prove that $L(S) = V$, then we should prove that $V \subseteq L(S)$ because $L(S) \subseteq V$ since $L(S)$ is a subspace of V . In order to prove that $V \subseteq L(S)$, we should prove that each element of V can be expressed as a linear combination of a finite number of elements of S . Then each element of V will also be an element of $L(S)$ and we shall have $V \subseteq L(S)$.

Finally $V \subseteq L(S)$ and $L(S) \subseteq V \Rightarrow L(S) = V$.

Illustration 1: The subset containing a single element $(1, 0, 0)$ of the vector space $V_3(F)$ generates the subspace which is the totality of elements of the form $(a, 0, 0)$.

Illustration 2: The subset $\{(1, 0, 0), (0, 1, 0)\}$ of $V_3(F)$ generates the subspace which is the totality of the elements of the form $(a, b, 0)$.

Illustration 3: The subset $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ of $V_3(F)$ generates or spans the entire vector space $V_3(F)$ i.e., $L(S) = V$.

If (a, b, c) be any element of V , then

$$(a, b, c) = a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1).$$

Thus $(a, b, c) \in L(S)$.

Hence $V \subseteq L(S)$. Also $L(S) \subseteq V$. Hence $L(S) = V$.

Illustration 4: Let V be the vector space of all polynomials over the field F . Let S be the subset of V consisting of the polynomials $f_0, f_1, f_2, \dots$, defined by $f_n = x^n, n = 0, 1, 2, \dots$

Then $V = L(S)$.

7 Linear Sum of Two Subspaces

Definition: Let W_1 and W_2 be two subspaces of the vector space $V(F)$. Then the linear sum of the subspaces W_1 and W_2 , denoted by $W_1 + W_2$, is the set of all sums $\alpha_1 + \alpha_2$ such that $\alpha_1 \in W_1, \alpha_2 \in W_2$.

Thus $W_1 + W_2 = \{\alpha_1 + \alpha_2 : \alpha_1 \in W_1, \alpha_2 \in W_2\}$.

For any subspace W of a vector space V , we have $W + W = W$.

Theorem 1: If W_1 and W_2 are subspaces of the vector space $V(F)$, then

(i) $W_1 + W_2$ is a subspace of $V(F)$.

(Kumaun 2004, 12; Garhwal 10; Gorakhpur 12, 14)

(ii) $W_1 + W_2 = \{W_1 \cup W_2\}$ i.e., $L(W_1 \cup W_2) = W_1 + W_2$.

(Kumaun 2002; Gorakhpur 11, 14)

Proof: (i) Let α, β be any two elements of $W_1 + W_2$.

Then $\alpha = \alpha_1 + \alpha_2$ and $\beta = \beta_1 + \beta_2$ where $\alpha_1, \beta_1 \in W_1$ and $\alpha_2, \beta_2 \in W_2$. If $a, b \in F$, we have

$$a\alpha + b\beta = a(\alpha_1 + \alpha_2) + b(\beta_1 + \beta_2) = (a\alpha_1 + b\beta_1) + (a\alpha_2 + b\beta_2).$$

Since W_1 is a subspace of V , therefore $a, b \in F$

and $\alpha_1, \beta_1 \in W_1 \Rightarrow a\alpha_1 + b\beta_1 \in W_1$.

Similarly $a\alpha_2 + b\beta_2 \in W_2$.

Consequently $a\alpha + b\beta = (a\alpha_1 + b\beta_1) + (a\alpha_2 + b\beta_2) \in W_1 + W_2$.

Thus $a, b \in F$ and $\alpha, \beta \in W_1 + W_2 \Rightarrow a\alpha + b\beta \in W_1 + W_2$.

Hence $W_1 + W_2$ is a subspace of $V(F)$.

(ii) Since W_2 contains the zero vector, therefore if $\alpha_1 \in W_1$, then we can write

$$\alpha_1 = \alpha_1 + \mathbf{0} \in W_1 + W_2.$$

Thus $W_1 \subseteq W_1 + W_2$.

Similarly $W_2 \subseteq W_1 + W_2$.

Hence $W_1 \cup W_2 \subseteq W_1 + W_2$.

Therefore $W_1 + W_2$ is a subspace of $V(F)$ containing $W_1 \cup W_2$.

Now to prove that $W_1 + W_2 = \{W_1 \cup W_2\}$ we should prove that

$$W_1 + W_2 \subseteq L(W_1 \cup W_2) \text{ and } L(W_1 \cup W_2) \subseteq W_1 + W_2.$$

Let $\alpha = \alpha_1 + \beta_1$ be any element of $W_1 + W_2$. Then $\alpha_1 \in W_1$ and $\beta_1 \in W_2$. Therefore $\alpha_1, \beta_1 \in W_1 \cup W_2$. We can write

$$\alpha_1 + \beta_1 = 1\alpha_1 + 1\beta_1.$$

Thus $\alpha_1 + \beta_1$ is a linear combination of a finite number of elements

$$\alpha_1, \beta_1 \in W_1 \cup W_2.$$

Therefore $\alpha_1 + \beta_1 \in L(W_1 \cup W_2)$.

$$\therefore W_1 + W_2 \subseteq L(W_1 \cup W_2).$$

Also $L(W_1 \cup W_2)$ is the smallest subspace containing $W_1 \cup W_2$ and $W_1 + W_2$ is a subspace containing $W_1 \cup W_2$. Therefore $L(W_1 \cup W_2)$ must be contained in $W_1 + W_2$. Consequently

$$L(W_1 \cup W_2) \subseteq W_1 + W_2.$$

Hence $W_1 + W_2 = L(W_1 \cup W_2) = \{W_1 \cup W_2\}$.

Note: If $W_1, W_2, \dots, W_k$ are subspaces of the vector space V , then their linear sum, denoted by $W_1 + W_2 + \dots + W_k$, is the set of all sums $\alpha_1 + \alpha_2 + \dots + \alpha_k$ such that $\alpha_i \in W_i$. It can be proved that $W_1 + W_2 + \dots + W_k$ is a subspace of V which contains each of the subspaces W_i and which is spanned by the union of $W_1, W_2, \dots, W_k$.
(Kumaun 2006)

Illustration: Let V be the vector space of all 2×2 matrices over $\mathbf{R}$. If W_1 consists of those matrices in V whose second row is zero and W_2 consists of those matrices in V whose second column is zero i.e.,

$$W_1 = \left\{ \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} : a, b \in \mathbf{R} \right\}, W_2 = \left\{ \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} : a, c \in \mathbf{R} \right\}$$

then W_1 and W_2 are subspaces of V . We have

$$W_1 + W_2 = \left\{ \begin{bmatrix} a & b \\ c & 0 \end{bmatrix} : a, b, c \in \mathbf{R} \right\}, W_1 \cap W_2 = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} : a \in \mathbf{R} \right\}.$$

Illustrative Examples

Example 17: If S, T are subsets of $V(F)$, then

- (i) $S \subseteq T \Rightarrow L(S) \subseteq L(T)$.
- (ii) $L(S \cup T) = L(S) + L(T)$.
- (iii) S is a subspace of $V \Leftrightarrow L(S) = S$.
- (iv) $L(L(S)) = L(S)$.

Solution: (i) Let $\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n \in L(S)$ where $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a finite subset of S . Since $S \subseteq T$, therefore $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is also a finite subset of T . So $\alpha \in L(T)$.

Thus $\alpha \in L(S) \Rightarrow \alpha \in L(T)$.

$$\therefore L(S) \subseteq L(T).$$

(ii) Let α be any element of $L(S \cup T)$. Then

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m + b_1\beta_1 + b_2\beta_2 + \dots + b_p\beta_p$$

where $\{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_p\}$ is a finite subset of $S \cup T$ such that

$$\{\alpha_1, \alpha_2, \dots, \alpha_m\} \subseteq S \text{ and } \{\beta_1, \beta_2, \dots, \beta_p\} \subseteq T.$$

Now $a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m \in L(S)$

and $b_1\beta_1 + b_2\beta_2 + \dots + b_p\beta_p \in L(T)$.

Therefore $\alpha \in L(S) + L(T)$.

Consequently $L(S \cup T) \subseteq L(S) + L(T)$.

Now let γ be any element of $L(S) + L(T)$. Then $\gamma = \beta + \delta$ where $\beta \in L(S)$ and $\delta \in L(T)$. Now β will be a linear combination of a finite number of elements of S and δ will be a linear combination of a finite number of elements of T . Therefore $\beta + \delta$ will be a linear combination of a finite number of elements of $S \cup T$.

Thus $\beta + \delta \in L(S \cup T)$. Consequently

$$L(S) + L(T) \subseteq L(S \cup T).$$

Hence $L(S \cup T) = L(S) + L(T)$.

(iii) Suppose S is a subspace of V . Then we are to prove that $L(S) = S$.

Let $\alpha \in L(S)$. Then $\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$ where $a_1, \dots, a_n \in F$ and $\alpha_1, \dots, \alpha_n \in S$. But S is a subspace of V . Therefore it is closed with respect to scalar multiplication and vector addition. Hence $\alpha = a_1\alpha_1 + \dots + a_n\alpha_n \in S$. Thus

$$\alpha \in L(S) \Rightarrow \alpha \in S.$$

Therefore $L(S) \subseteq S$. Also $S \subseteq L(S)$. Therefore $L(S) = S$.

Converse. Suppose $L(S) = S$. Then to prove that S is a subspace of V . We know that $L(S)$ is a subspace of V . Since $S = L(S)$, therefore S is also a subspace of V .

(iv) $L(L(S))$ is the smallest subspace of V containing $L(S)$. But $L(S)$ is a subspace of V . Therefore the smallest subspace of V containing $L(S)$ is $L(S)$ itself.

Hence $L(L(S)) = L(S)$.

Example 18: In the vector space $\mathbf{R}^3$ express the vector $(1, -2, 5)$ as a linear combination of the vectors $(1, 1, 1)$, $(1, 2, 3)$ and $(2, -1, 1)$. (Kumaun 2008)

Solution: Let $\alpha = (1, -2, 5)$, $\alpha_1 = (1, 1, 1)$, $\alpha_2 = (1, 2, 3)$, $\alpha_3 = (2, -1, 1)$.

We wish to express α as a linear combination of $\alpha_1, \alpha_2, \alpha_3$.

Let $\alpha = a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$ where $a_1, a_2, a_3 \in \mathbf{R}$.

Then $(1, -2, 5) = a_1(1, 1, 1) + a_2(1, 2, 3) + a_3(2, -1, 1)$

$$= (a_1, a_1, a_1) + (a_2, 2a_2, 3a_2) + (2a_3, -a_3, a_3)$$

$$= (a_1 + a_2 + 2a_3, a_1 + 2a_2 - a_3, a_1 + 3a_2 + a_3)$$

$$\therefore \begin{aligned} a_1 + a_2 + 2a_3 &= 1 & \dots(1) \\ a_1 + 2a_2 - a_3 &= -2 & \dots(2) \\ a_1 + 3a_2 + a_3 &= 5 & \dots(3) \end{aligned}$$

Method 1. The augmented matrix of the system of equations (1), (2) and (3) is

$$[A:B] = \left[\begin{array}{cccc|c} 1 & 1 & 2 & : & 1 \\ 1 & 2 & -1 & : & -2 \\ 1 & 3 & 1 & : & 5 \end{array} \right].$$

We shall reduce the matrix $[A:B]$ to Echelon form by applying the row transformations only.

We have $[A : B] \sim \begin{bmatrix} 1 & 1 & 2 & \vdots & 1 \\ 0 & 1 & -3 & \vdots & -3 \\ 0 & 2 & -1 & \vdots & 4 \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 1 & 2 & \vdots & 1 \\ 0 & 1 & -3 & \vdots & -3 \\ 0 & 0 & 5 & \vdots & 10 \end{bmatrix}.$$

Thus we have $\text{rank } A = \text{rank } [A : B]$ i.e., the system of equations (1), (2), (3) is consistent and so has a solution i.e., α can be expressed as a linear combination of $\alpha_1, \alpha_2, \alpha_3$. The equations (1), (2) and (3) are equivalent to equations

$$a_1 + a_2 + 2a_3 = 1, \quad a_2 - 3a_3 = -3, \quad 5a_3 = 10.$$

Solving for unknowns, we get $a_3 = 2, \quad a_2 = 3, \quad a_1 = -6$.

Hence $(1, -2, 5) = -6(1, 1, 1) + 3(1, 2, 3) + 2(2, -1, 1)$.

Method 2. Eliminating a_3 between (1) and (2), we get

$$3a_1 + 5a_2 = -3 \quad \dots(4)$$

Eliminating a_3 between (2) and (3), we get

$$2a_1 + 5a_2 = 3 \quad \dots(5)$$

Solving (4) and (5), we get $a_1 = -6, \quad a_2 = 3$.

Putting $a_1 = -6, \quad a_2 = 3$ in (1), we get $a_3 = 2$.

Hence $(1, -2, 5) = -6(1, 1, 1) + 3(1, 2, 3) + 2(2, -1, 1)$.

Example 19: For what value of m , the vector $(m, 3, 1)$ is a linear combination of the vectors $(3, 2, 1)$ and $(2, 1, 0)$?

Solution: Let $\alpha_1 = (3, 2, 1)$ and $\alpha_2 = (2, 1, 0)$.

Let $(m, 3, 1) = a\alpha_1 + b\alpha_2$ where $a, b \in \mathbf{R}$.

$$\begin{aligned} \text{Then } (m, 3, 1) &= a(3, 2, 1) + b(2, 1, 0) = (3a, 2a, a) + (2b, b, 0) \\ &= (3a + 2b, 2a + b, a) \end{aligned}$$

$$\therefore 3a + 2b = m, \quad 2a + b = 3, \quad a = 1.$$

These give $a = 1, b = 1, m = 5$.

Hence the required value of m is 5.

Example 20: In the vector space $\mathbf{R}^4$ determine whether or not the vector $(3, 9, -4, -2)$ is a linear combination of the vectors $(1, -2, 0, 3), (2, 3, 0, -1)$ and $(2, -1, 2, 1)$.

Solution: Let $\alpha = (3, 9, -4, -2), \alpha_1 = (1, -2, 0, 3),$

$$\alpha_2 = (2, 3, 0, -1), \alpha_3 = (2, -1, 2, 1).$$

Let $\alpha = a\alpha_1 + b\alpha_2 + c\alpha_3$ where $a, b, c \in \mathbf{R}$.

$$\text{Then } (3, 9, -4, -2) = a(1, -2, 0, 3) + b(2, 3, 0, -1) + c(2, -1, 2, 1)$$

$$\text{or } (3, 9, -4, -2) = (a + 2b + 2c, -2a + 3b - c, 2c, 3a - b + c)$$

$$\therefore \begin{aligned} a + 2b + 2c &= 3 & \dots(1) \\ -2a + 3b - c &= 9 & \dots(2) \\ 2c &= -4 & \dots(3) \\ 3a - b + c &= -2 & \dots(4) \end{aligned}$$

Solving (1), (2) and (3), we get $a = 1, b = 3, c = -2$. These values satisfy (4). Thus the system of equations (1), (2), (3) and (4) has a solution *i.e.*, α can be written as a linear combination of α_1, α_2 and α_3 .

Also we have $\alpha = \alpha_1 + 3\alpha_2 - 2\alpha_3$.

Example 21: In the vector space $\mathbf{R}^3$, let $\alpha = (1, 2, 1), \beta = (3, 1, 5), \gamma = (3, -4, 7)$. Show that the subspaces spanned by $S = \{\alpha, \beta\}$ and $T = \{\alpha, \beta, \gamma\}$ are the same. (Kumaun 2008, 11)

Solution: First we shall show that the vector γ can be expressed as a linear combination of the vectors α and β . Let

$$(3, -4, 7) = a(1, 2, 1) + b(3, 1, 5).$$

Then $a + 3b = 3, 2a + b = -4, a + 5b = 7$. Solving the first two equations, we get $a = -3, b = 2$ and these values satisfy the third equation also. Therefore we can write $\gamma = -3\alpha + 2\beta$.

Now $S \subseteq T \Rightarrow L(S) \subseteq L(T)$.

Further let $\delta \in L(T)$. Then δ can be expressed as a linear combination of the vectors α, β and γ . In this linear combination the vector γ can be replaced by $-3\alpha + 2\beta$. Thus δ can be expressed as a linear combination of the vectors α and β . Therefore $\delta \in L(S)$. Thus $\delta \in L(T) \Rightarrow \delta \in L(S)$. Therefore $L(T) \subseteq L(S)$.

Hence $L(T) = L(S)$.

Example 22: Show that $(1, 1, 1), (0, 1, 1)$ and $(0, 1, -1)$ generate $\mathbf{R}^3$.

Solution: We shall show that any vector $(a, b, c) \in \mathbf{R}^3$ is a linear combination of the given vectors.

Let $(a, b, c) = x(1, 1, 1) + y(0, 1, 1) + z(0, 1, -1)$, where $x, y, z \in \mathbf{R}$.

Then $(a, b, c) = (x, x + y + z, x + y - z)$.

$\therefore x = a, x + y + z = b, x + y - z = c$.

These give $x = a, y = \frac{b + c - 2a}{2}, z = \frac{b - c}{2}$ *i.e.*, the above system has a solution. Thus the given vectors generate $\mathbf{R}^3$.

Example 23: Let V be the vector space of all functions from $\mathbf{R}$ into $\mathbf{R}$; let V_e be the subset of even functions, $f(-x) = f(x)$; let V_o be the subset of odd functions, $f(-x) = -f(x)$.

(i) Prove that V_e and V_o are subspaces of V .

(ii) Prove that $V_e + V_o = V$.

(iii) Prove that $V_e \cap V_o = \{0\}$.

Solution: (i) Suppose f_e and $g_e \in V_e$ and a is any scalar *i.e.*, $a \in \mathbf{R}$. Then

$$\begin{aligned} (af_e + g_e)(-x) &= af_e(-x) + g_e(-x) \\ &= af_e(x) + g_e(x) = (af_e + g_e)(x). \end{aligned}$$

Therefore $af_e + g_e$ is an even function.

Thus $a \in \mathbf{R}$ and $f_e, g_e \in V_e \Rightarrow af_e + g_e \in V_e$.

Hence V_e is a subspace of V .

Again suppose f_o and $g_o \in V_o$ and a is any scalar. Then

$$\begin{aligned}(af_o + g_o)(-x) &= af_o(-x) + g_o(-x) = a[-f_o(x)] - g_o(x) \\ &= -[af_o(x) + g_o(x)] = -(af_o + g_o)(x).\end{aligned}$$

Therefore $af_o + g_o$ is an odd function.

Thus $a \in \mathbf{R}$ and $f_o, g_o \in V_o \Rightarrow af_o + g_o \in V_o$.

Hence V_o is a subspace of V .

(ii) Since V_e and V_o are subspaces of V , therefore $V_e + V_o$ is also a subspace of V and consequently $V_e + V_o \subseteq V$.

Now let $f \in V$. We shall show that f can be expressed as the sum of an even and an odd function.

Let
$$f_{e_1}(x) = \frac{1}{2}[f(x) + f(-x)] \text{ and } f_{o_1}(x) = \frac{1}{2}[f(x) - f(-x)].$$

Then obviously f_{e_1} is an even function and f_{o_1} is an odd function. We can easily see that $f_{e_1}(-x) = f_{e_1}(x)$ and $f_{o_1}(-x) = -f_{o_1}(x)$.

Now
$$\begin{aligned}f(x) &= \frac{1}{2}[f(x) + f(-x)] + \frac{1}{2}[f(x) - f(-x)] \\ &= f_{e_1}(x) + f_{o_1}(x) = (f_{e_1} + f_{o_1})(x).\end{aligned}$$

Therefore $f = f_{e_1} + f_{o_1}$ where $f_{e_1} \in V_e$, $f_{o_1} \in V_o$.

Thus $f \in V \Rightarrow f \in V_e + V_o$. Therefore $V \subseteq V_e + V_o$.

Hence $V = V_e + V_o$.

(iii) Let $\mathbf{0}$ denote the zero function i.e., $\mathbf{0}(x) = 0 \forall x \in \mathbf{R}$.

Then $\mathbf{0} \in V_e$ and also $\mathbf{0} \in V_o$.

Let $f \in V_e$ and also $f \in V_o$.

Then $f(-x) = f(x) = -f(x)$.

$\therefore 2f(x) = 0 \Rightarrow f(x) = 0 = \mathbf{0}(x)$.

Therefore $f = \mathbf{0}$ (zero function).

Thus $f \in V_e \cap V_o \Leftrightarrow f = \mathbf{0}$.

Hence $V_e \cap V_o = \{\mathbf{0}\}$.

Comprehensive Exercise 3

- For which value of k will the vector $(1, -2, k)$ in $\mathbf{R}^3$ be a linear combination of the vectors $(3, 0, -2)$ and $(2, -1, -5)$?
- In the vector space $\mathbf{R}^4$ determine whether or not the vector $(3, 9, -4, 2)$ is a linear combination of the vectors $(1, -2, 0, 3)$, $(2, 3, 0, -1)$ and $(2, -1, 2, 1)$.
- Is the vector $(3, -1, 0, -1)$ in the subspace of $\mathbf{R}^4$ spanned by the vectors $(2, -1, 3, 2)$, $(-1, 1, 1, -3)$ and $(1, 1, 9, -5)$?
- Is the vector $(2, -5, 3)$ in the subspace of $\mathbf{R}^3$ spanned by the vectors $(1, -3, 2)$, $(2, -4, -1)$, $(1, -5, 7)$?

(Kumaun 2009; Garhwal 13)

5. Express the matrix $E = \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}$ as a linear combination of the matrices
- $$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 0 & 2 \\ 0 & -1 \end{bmatrix}.$$
6. Write E as a linear combination of
- $$A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$
- where (i) $E = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ (ii) $E = \begin{bmatrix} 2 & 1 \\ -1 & -2 \end{bmatrix}$.
7. Express the polynomial $f = x^2 + 4x - 3$ over $\mathbf{R}$ as a linear combination of the polynomials
- $$f_1 = x^2 - 2x + 5, f_2 = 2x^2 - 3x \text{ and } f_3 = x + 3.$$
8. Express f as a linear combination of the polynomials
- $$f_1 = 2x^2 + 3x - 4, f_2 = x^2 - 2x - 3$$
- where (i) $f = 3x^2 + 8x - 5$ (ii) $f = 4x^2 - 6x - 1$.
9. Find the condition on a, b and c so that $(a, b, c) \in \mathbf{R}^3$ is in the space generated by $(2, 1, 0), (1, -1, 2)$ and $(0, 3, -4)$.
10. Find one vector in $\mathbf{R}^3$ which generates the intersection of W_1 and W_2 where $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$ and W_2 is the space generated by the vectors $(1, 2, 3)$ and $(1, -1, 1)$.
11. If α, β and γ are vectors such that $\alpha + \beta + \gamma = \mathbf{0}$, then α and β span the same subspace as β and γ .
12. Give an example of a subset W of a vector space V which is not a subspace of V but for which
- (i) $W + W = W$. (ii) $W + W \subset W$.
13. Let V be the vector space of $n \times n$ matrices over a field F . Let W_1 and W_2 be the subspaces of upper triangular matrices and lower triangular matrices respectively. Find
- (i) $W_1 + W_2$ (ii) $W_1 \cap W_2$.
14. Show that the space U generated by the vectors $u_1 = (1, 2, -1, 3), u_2 = (2, 4, 1, -2)$ and $u_3 = (3, 6, 3, -7)$ and the space V generated by the vectors $v_1 = (1, 2, -4, 1)$ and $v_2 = (2, 4, -5, 1)$ are equal.

Answers 3

- | | | |
|------------------------------|---------------------------------|-------------------|
| 1. $k = -8$ | 2. No | 3. No |
| 4. No | 5. $E = 3A - 2B - C$ | |
| 6. (i) $E = 2A - B + 2C$ | (ii) Not possible | |
| 7. $f = -3f_1 + 2f_2 + 4f_3$ | 8. (i) $f = 2f_1 - f_2$ | (ii) Not possible |
| 9. $2a - 4b - 3c = 0$ | 10. $(-2, 5, 0)$ | |
| 13. (i) $W_1 + W_2 = V$ | (ii) space of diagonal matrices | |

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The zero vector in the vector space $\mathbf{R}^4$ is
 (a) $(0, 0)$ (b) $(0, 0, 0)$
 (c) $(0, 0, 0, 0)$ (d) None of these.
- The number of elements in the vector space of polynomials of degree at most n in which the coefficients are the elements of the field $\mathbf{I}(p)$ over the field $\mathbf{I}(p)$, p being a prime number, is
 (a) p^{n+1} (b) p^n
 (c) p^{n-1} (d) None of these.
- If $(m, 3, 1)$ is a linear combination of vectors $(3, 2, 1)$ and $(2, 1, 0)$ in $\mathbf{R}^3$, then the value of m is
 (a) 2 (b) 3
 (c) 5 (d) None of these.
- A subspace of a vector space V over a field F other than V is called.
 (a) zero subspace of V (b) trivial subspace of V
 (c) proper subspace of V (d) none of these (Garhwal 2007)
- Let V be a vector space over the field of F . The intersection of any collection of subspaces of V is
 (a) zero subspace of V (b) trivial subspace of V
 (c) a subspace of V (d) none of these (Garhwal 2008)
- If $\alpha_1 = (1, 1, 1)$, $\alpha_2 = (1, 2, 3)$ and $\alpha_3 = (2, -1, 1)$ be three vectors in $\mathbf{R}^3$, the vector $(1, -2, 5) \in \mathbf{R}^3$ as a linear combination of $\alpha_1, \alpha_2, \alpha_3$ will be
 (a) $-6\alpha_1 + 3\alpha_2 + 2\alpha_3$ (b) $6\alpha_1 - 3\alpha_2 + 2\alpha_3$
 (c) $6\alpha_1 + 3\alpha_2 - 2\alpha_3$ (d) $6\alpha_1 + 3\alpha_2 + 2\alpha_3$
- If W_1 and W_2 be two subspaces of the vector space V over a field F , then which one will always be the subspace of V
 (a) $W_1 \cup W_2$ (b) $W_1 \cap W_2$
 (c) $W_1 - W_2$ (d) none of these

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- The necessary and sufficient condition for a non-empty subset W of a vector space $V(F)$ to be a subspace of V is
 $a, b \in F$ and $\alpha, \beta \in W \Rightarrow \dots \in W$.
- The subset containing a single element $(1, 0, 0)$ of the vector space $V_3(F)$ generates the subspace which is the totality of the elements of the form

3. If W_1 and W_2 be two subspaces of the vector space $V(F)$,
then $L(W_1 \cup W_2) = \dots$. (Garhwal 2006)

True or False

Write 'T' for true and 'F' for false statement.

1. A subspace of $V_3(\mathbf{R})$, where $\mathbf{R}$ is the real field, must always contain the origin.
2. The set of vectors $\alpha = (x, y) \in V_2(\mathbf{R})$ for which $x^2 = y^2$ is a subspace of $V_2(\mathbf{R})$.
3. The set of ordered triads (x, y, z) of real numbers with $x > 0$ is a subspace of $V_3(\mathbf{R})$.
4. The set of ordered triads (x, y, z) of real numbers with $x + y = 0$ is a subspace of $V_3(\mathbf{R})$.
5. The union of any two subspaces W_1 and W_2 of a vector space $V(F)$ is always a subspace of $V(F)$.
6. The intersection of two subspaces of a vector space V is also a subspace of V .
7. If A and B are subsets of a vector space, then $A \neq B \Rightarrow L(A) \neq L(B)$.

Answers

Multiple Choice Questions

1. (c)
2. (a)
3. (c)
4. (c)
5. (c)
6. (a)
7. (b)

Fill in the Blank(s)

1. $a\alpha + b\beta$
2. $(a, 0, 0)$
3. $W_1 + W_2$

True or False

1. T
2. F
3. F
4. T
5. F
6. T
7. F



Chapter

2



Linear Dependence of Vectors

1 Linear Dependence and Linear Independence of Vectors

Linear dependence:

Definition: Let $V(F)$ be a vector space. A finite set $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of vectors of V is said to be linearly dependent if there exist scalars $a_1, a_2, \dots, a_n \in F$ not all of them 0 (some of them may be zero) such that

$$a_1 \alpha_1 + a_2 \alpha_2 + a_3 \alpha_3 + \dots + a_n \alpha_n = \mathbf{0}.$$

Linear independence:

Definition: Let $V(F)$ be a vector space. A finite set $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of vectors of V is said to be linearly independent if every relation of the form

$$a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n = \mathbf{0}, a_i \in F, 1 \leq i \leq n$$

$\Rightarrow a_i = 0$ for each $1 \leq i \leq n$.

Any infinite set of vectors of V is said to be linearly independent if its every finite subset is linearly independent, otherwise it is linearly dependent.

Note: The empty set $\emptyset$ is defined as a linearly independent set.

2 Some Theorems on Linear Dependence and Linear Independence

Theorem 1: Let $V(F)$ be a vector space. If $\alpha_1, \alpha_2, \dots, \alpha_n$ are non-zero vectors $\in V$ then either they are linearly independent or some $\alpha_k, 2 \leq k \leq n$, is a linear combination of the preceding ones $\alpha_1, \alpha_2, \dots, \alpha_{k-1}$. (Garhwal 2004; Gorakhpur 11; Kumaun 13)

Proof: If $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly independent we are nothing to prove. So let $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly dependent. Then there exists a relation of the form

$$a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n = \mathbf{0}, \quad \dots(1)$$

where, not all the scalar coefficients $a_1, a_2, \dots, a_n$ are 0. Let k be the largest integer for which $a_k \neq 0$, i.e., $a_{k+1} = 0, a_{k+2} = 0, \dots, a_n = 0$ and $a_k \neq 0$. There is no harm in this assumption because at the most if $a_n \neq 0$ then $k = n$.

Also $2 \leq k$. Because if $a_2 = 0, a_3 = 0, \dots, a_n = 0$ then $a_1 \alpha_1 = \mathbf{0}$ and $\alpha_1 \neq \mathbf{0} \Rightarrow a_1 = 0$. This contradicts the fact that not all the a 's are 0.

Now the relation (1) reduces to

$$a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_k \alpha_k = \mathbf{0}, \text{ where } a_k \neq 0$$

$$\text{or } a_k \alpha_k = -a_1 \alpha_1 - a_2 \alpha_2 - \dots - a_{k-1} \alpha_{k-1}$$

$$\text{or } a_k^{-1} (a_k \alpha_k) = a_k^{-1} (-a_1 \alpha_1 - a_2 \alpha_2 - \dots - a_{k-1} \alpha_{k-1})$$

$$\text{or } \alpha_k = (-a_k^{-1} a_1) \alpha_1 + (-a_k^{-1} a_2) \alpha_2 + \dots + (-a_k^{-1} a_{k-1}) \alpha_{k-1}.$$

Thus α_k is a linear combination of its preceding vectors.

Theorem 2: The set of non-zero vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ of $V(F)$ is linearly dependent if some $\alpha_k, 2 \leq k \leq n$, is a linear combination of the preceding ones. (Gorakhpur 2012)

Proof: If some $\alpha_k, 2 \leq k \leq n$, is a linear combination of the preceding ones $\alpha_1, \alpha_2, \dots, \alpha_{k-1}$ then $\exists$ scalars $a_1, a_2, \dots, a_{k-1}$ such that

$$\alpha_k = a_1 \alpha_1 + \dots + a_{k-1} \alpha_{k-1}$$

$$\Rightarrow \alpha_k - a_1 \alpha_1 - a_2 \alpha_2 - \dots - a_{k-1} \alpha_{k-1} = \mathbf{0}$$

$$\Rightarrow \text{the set } \{\alpha_1, \alpha_2, \dots, \alpha_k\} \text{ is linearly dependent.}$$

Hence the set $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of which $\{\alpha_1, \alpha_2, \dots, \alpha_k\}$ is a subset, must be linearly dependent.

Theorem 3: If in a vector space $V(F)$, a vector β is a linear combination of the set of vectors $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$, then the set of vectors $\beta, \alpha_1, \alpha_2, \dots, \alpha_n$ is linearly dependent.

Proof: Since β is a linear combination of $\alpha_1, \alpha_2, \dots, \alpha_n$, therefore there exist scalars $a_1, a_2, \dots, a_n$ such that

$$\beta = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n$$

$$\Rightarrow 1\beta - a_1 \alpha_1 - a_2 \alpha_2 - \dots - a_n \alpha_n = \mathbf{0}. \quad \dots(1)$$

In the relation (1) the scalar coefficient of β is 1 which is $\neq 0$. Hence in the relation (1) not all the scalar coefficients are 0. Therefore the set $\beta, \alpha_1, \dots, \alpha_n$ is linearly dependent.

Theorem 4: Let S be a linearly independent subset of a vector space V . Suppose β is a vector in V which is not in the subspace spanned by S . Then the set obtained by adjoining β to S is linearly independent. (Kumaun 2006)

Proof: Suppose $\alpha_1, \alpha_2, \dots, \alpha_m$ are distinct vectors in S . Let

$$c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_m \alpha_m + b\beta = \mathbf{0}. \quad \dots(1)$$

Then b must be zero, for otherwise

$$\beta = \left(-\frac{c_1}{b}\right)\alpha_1 + \dots + \left(-\frac{c_m}{b}\right)\alpha_m$$

and consequently β is in the subspace spanned by S which is a contradiction.

Putting $b = 0$ in (1), we get

$$c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_m \alpha_m = \mathbf{0} \Rightarrow c_1 = 0, c_2 = 0, \dots, c_m = 0 \text{ because}$$

the set $\{\alpha_1, \alpha_2, \dots, \alpha_m\}$ is linearly independent since it is a subset of a linearly independent set S .

Thus the relation (1) implies $c_1 = 0, c_2 = 0, \dots, c_m = 0, b = 0$.

Therefore the set $\{\alpha_1, \alpha_2, \dots, \alpha_m, \beta\}$ is linearly independent. If S' is the set obtained by adjoining β to S , then we have proved that every finite subset of S' is linearly independent. Hence S' is linearly independent.

Theorem 5: *The set of non-zero vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ of $V(F)$ is linearly dependent iff one of these vectors is a linear combination of the remaining $(n-1)$ vectors.*

Proof: This theorem can be easily proved.

Illustrative Examples

Example 1: *Prove that if two vectors are linearly dependent, one of them is a scalar multiple of the other.* (Gorakhpur 2011; Kumaun 13)

Solution: Let α, β be two linearly dependent vectors of the vector space V . Then $\exists$ scalars a, b not both zero, such that

$$a\alpha + b\beta = \mathbf{0}.$$

If $a \neq 0$, then we get $a\alpha = -b\beta$

$$\Rightarrow \alpha = \left(-\frac{b}{a}\right)\beta \Rightarrow \alpha \text{ is a scalar multiple of } \beta.$$

If $b \neq 0$, then we get $b\beta = -a\alpha$

$$\Rightarrow \beta = \left(-\frac{a}{b}\right)\alpha \Rightarrow \beta \text{ is a scalar multiple of } \alpha.$$

Thus one of the vectors α and β is a scalar multiple of the other.

Example 2: *In the vector space $V_n(F)$, the system of n vectors*

$$e_1 = (1, 0, 0, \dots, 0), e_2 = (0, 1, 0, \dots, 0), \dots, e_n = (0, 0, \dots, 0, 1)$$

is linearly independent where 1 denotes the unity of the field F .

Solution: If $a_1, a_2, a_3, \dots, a_n$ be any scalars, then

$$a_1 e_1 + a_2 e_2 + \dots + a_n e_n = \mathbf{0}$$

$$\Rightarrow a_1 (1, 0, 0, \dots, 0) + a_2 (0, 1, 0, \dots, 0) + \dots + a_n (0, 0, \dots, 0, 1) = \mathbf{0}$$

$$\Rightarrow (a_1, a_2, \dots, a_n) = (0, 0, \dots, 0)$$

$$\Rightarrow a_1 = 0, a_2 = 0, \dots, a_n = 0.$$

Therefore the given set of n vectors is linearly independent.

In particular $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a linearly independent subset of $V_3(F)$.

Example 3: *If the set $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of vectors of $V(F)$ is linearly independent, then none of the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ can be zero vector.*

Solution: Let α_r be equal to zero vector where $1 \leq r \leq n$.

Then $0 \alpha_1 + 0 \alpha_2 + \dots + a \alpha_r + 0 \alpha_{r+1} + \dots + 0 \alpha_n = \mathbf{0}$ for any $a \neq 0$ in F .

Since $a \neq 0$, therefore from this relation we conclude that S is linearly dependent.

Thus we get a contradiction because it is given that S is linearly independent. Hence none of the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ can be zero vector. We also conclude that *a set of vectors which contains the zero vector is necessarily linearly dependent.*

Example 4: Every superset of a linearly dependent set of vectors is linearly dependent.

Solution: Let $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a linearly dependent set of vectors. Then there exist scalars $a_1, a_2, \dots, a_n$ not all zero such that

$$a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n = \mathbf{0}. \quad \dots(1)$$

Now let $S' = \{\alpha_1, \alpha_2, \dots, \alpha_n, \beta_1, \beta_2, \dots, \beta_m\}$ be a superset of S .

Then we have from (1)

$$a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n + 0 \beta_1 + 0 \beta_2 + \dots + 0 \beta_m = \mathbf{0}. \quad \dots(2)$$

Since in the relation (2) the scalar coefficients are not all 0, therefore S' is linearly dependent.

From this we also conclude that *any subset of a linearly independent set of vectors is also linearly independent.*

Example 5: A system consisting of a single non-zero vector is always linearly independent.

Solution: Let $S = \{\alpha\}$ be a subset of a vector space V and let α be not equal to zero vector. If a is any scalar, then

$$a \alpha = \mathbf{0}$$

$$\Rightarrow a = 0. \quad [\text{Since } \alpha \text{ is not-zero vector}]$$

$\therefore$ the set S is linearly independent.

Example 6: Show that $S = \{(1, 2, 4), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a linearly dependent subset of the vector space $V_3(\mathbf{R})$ where $\mathbf{R}$ is the field of real numbers.

Solution: We have

$$\begin{aligned} & 1(1, 2, 4) + (-1)(1, 0, 0) + (-2)(0, 1, 0) + (-4)(0, 0, 1) \\ &= (1, 2, 4) + (-1, 0, 0) + (0, -2, 0) + (0, 0, -4) = (0, 0, 0) \end{aligned}$$

i.e., the zero vector.

Since in this relation the scalar coefficients $1, -1, -2, -4$ are not all zero, therefore the given system S is linearly dependent.

Example 7: Show that the system of three vectors $(1, 3, 2), (1, -7, -8), (2, 1, -1)$ of $V_3(\mathbf{R})$ is linearly dependent. (Garhwal 2010; Kumaun 10)

Solution: Let a, b, c be scalars *i.e.,* real numbers such that

$$a(1, 3, 2) + b(1, -7, -8) + c(2, 1, -1) = (0, 0, 0)$$

$$\text{i.e., } (a + b + 2c, 3a - 7b + c, 2a - 8b - c) = (0, 0, 0).$$

$$\text{Then } a + b + 2c = 0, \quad \dots(1)$$

$$3a - 7b + c = 0, \quad \dots(2)$$

$$\text{and } 2a - 8b - c = 0. \quad \dots(3)$$

If the equations (1), (2), (3) possess a non-zero solution, then the given vectors are linearly dependent.

Adding (2) and (3), we get

$$5a - 15b = 0 \quad \text{or} \quad a - 3b = 0. \quad \dots(4)$$

Multiplying (3) by 2 and adding to (1), we get

$$5a - 15b = 0 \quad \text{or} \quad a - 3b = 0. \quad \dots(5)$$

The equations (4) and (5) are the same and give $a = 3b$.

Putting $a = 3b$ in (1), we get $2c + 4b = 0$ or $c = -2b$.

If we take $b = 1$, we get $a = 3, c = -2$.

Thus $a = 3, b = 1, c = -2$ is a non-zero solution of the equations (1), (2) and (3). Hence the given set of vectors is linearly dependent.

Alternative method. The coefficient matrix A of the system of equations (1), (2) and (3) is

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & -7 & 1 \\ 2 & -8 & -1 \end{bmatrix}.$$

$$\begin{aligned} \text{We have } \det A &= \begin{vmatrix} 1 & 1 & 2 \\ 3 & -7 & 1 \\ 2 & -8 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 3 & -10 & -5 \\ 2 & -10 & -5 \end{vmatrix} \text{ by } C_2 - C_1 \text{ and } C_3 - 2C_1 \\ &= 50 - 50 = 0. \end{aligned}$$

$\therefore$ rank $A < 3$ i.e., rank $A <$ the number of unknowns a, b, c in the equations (1), (2) and (3).

Therefore the equations (1), (2) and (3) must possess a non-zero solution. Hence the given vectors are linearly dependent.

Example 8: In $V_3(\mathbf{R})$, where $\mathbf{R}$ is the field of real numbers, examine each of the following sets of vectors for linear dependence :

(i) $\{(2, 1, 2), (8, 4, 8)\}$

(ii) $\{(1, 2, 0), (0, 3, 1), (-1, 0, 1)\}$

(iii) $\{(-1, 2, 1), (3, 0, -1), (-5, 4, 3)\}$

(Gorakhpur 2010)

(iv) $\{(2, 3, 5), (4, 9, 25)\}$

(v) $\{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$.

Solution: (i) We have $4(2, 1, 2) + (-1)(8, 4, 8) = (8, 4, 8) + (-8, -4, -8) = (0, 0, 0)$ i.e., the zero vector.

Since in this relation the scalar coefficients $4, -1$ are not both zero, therefore the given set is linearly dependent.

(ii) Let a, b, c be scalars i.e., real numbers such that

$$a(1, 2, 0) + b(0, 3, 1) + c(-1, 0, 1) = (0, 0, 0)$$

$$\text{i.e., } (a - c, 2a + 3b, b + c) = (0, 0, 0)$$

$$\text{i.e., } a + 0b - c = 0, 2a + 3b + 0c = 0, 0a + b + c = 0.$$

These equations will have a non-zero solution i.e., a solution in which a, b, c are not all zero if the rank of the coefficient matrix is less than three i.e., the number of unknowns a, b, c . If the rank is 3, then the zero solution $a = 0, b = 0, c = 0$ will be the only solution.

$$\text{Coefficient matrix } A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$\text{We have } |A| = 1(3 - 0) - 2(0 + 1) = 1 \neq 0.$$

$\therefore$ Rank $A = 3$. Hence $a = 0, b = 0, c = 0$ is the only solution. Therefore the given system is linearly independent.

(iii) Let a, b, c be scalars such that

$$a(-1, 2, 1) + b(3, 0, -1) + c(-5, 4, 3) = (0, 0, 0)$$

$$\text{i.e., } (-a + 3b - 5c, 2a + 0b + 4c, a - b + 3c) = (0, 0, 0)$$

$$\text{i.e., } -a + 3b - 5c = 0, 2a + 0b + 4c = 0, a - b + 3c = 0.$$

The coefficient matrix of these equations is

$$A = \begin{bmatrix} -1 & 3 & -5 \\ 2 & 0 & 4 \\ 1 & -1 & 3 \end{bmatrix}.$$

$$\text{We have } |A| = -1(0 + 4) - 2(9 - 5) + 1(12 - 0) = 0.$$

$\therefore$ Rank A is < 3 i.e., the number of unknowns a, b, c . Therefore the given system of equations will possess a non-zero solution. For example $a = -2, b = 1, c = 1$ is a non-zero solution. Hence the given system of vectors is linearly dependent.

(iv) Let a, b be scalars i.e., real numbers such that

$$a(2, 3, 5) + b(4, 9, 25) = (0, 0, 0)$$

$$\text{i.e., } (2a + 4b, 3a + 9b, 5a + 25b) = (0, 0, 0)$$

$$\text{i.e., } 2a + 4b = 0, 3a + 9b = 0, 5a + 25b = 0.$$

$$\text{The coefficient matrix of these equations is } A = \begin{bmatrix} 2 & 4 \\ 3 & 9 \\ 5 & 25 \end{bmatrix}.$$

Obviously rank $A = 2$ i.e., equal to the number of unknowns a and b .

Therefore these equations have the only solution $a = 0, b = 0$.

Hence the given set of vectors is linearly independent.

(v) Let a, b, c be scalars i.e., real numbers such that

$$a(1, 2, 1) + b(3, 1, 5) + c(3, -4, 7) = (0, 0, 0)$$

$$\text{i.e., } (a + 3b + 3c, 2a + b - 4c, a + 5b + 7c) = (0, 0, 0)$$

$$\text{i.e., } a + 3b + 3c = 0, \quad \dots(1)$$

$$2a + b - 4c = 0, \quad \dots(2)$$

$$a + 5b + 7c = 0. \quad \dots(3)$$

Multiplying (1) by 2, we get

$$2a + 6b + 6c = 0. \quad \dots(4)$$

Subtracting (4) from (2), we get

$$-5b - 10c = 0$$

$$\text{or } b + 2c = 0. \quad \dots(5)$$

Again subtracting (3) from (1), we get

$$-2b - 4c = 0 \quad \text{or} \quad b + 2c = 0. \quad \dots(6)$$

The equations (5) and (6) are the same and give $b = -2c$.

Putting $b = -2c$ in (1), we get $a = 3c$. If we take $c = 1$, we get $b = -2$ and $a = 3$. Thus $a = 3, b = -2, c = 1$ is a non-zero solution of the equations (1), (2) and (3). Hence the given set of vectors is linearly dependent.

Example 9: If F is the field of complex numbers, prove that the vectors (a_1, a_2) and (b_1, b_2) in $V_2(F)$ are linearly dependent iff $a_1 b_2 - a_2 b_1 = 0$. (Meerut 2003)

Solution: Let $x, y \in F$. Then

$$\begin{aligned} & x(a_1, a_2) + y(b_1, b_2) = (0, 0) \Rightarrow (xa_1 + yb_1, xa_2 + yb_2) = (0, 0). \\ \text{Therefore, } & \left. \begin{aligned} a_1 x + b_1 y &= 0 \\ \text{and } a_2 x + b_2 y &= 0 \end{aligned} \right\}. \end{aligned}$$

The necessary and sufficient condition for these equations to possess a non-zero solution is that

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 0 \text{ i.e., } a_1 b_2 - a_2 b_1 = 0.$$

Hence the given system is linearly dependent iff $a_1 b_2 - a_2 b_1 = 0$.

Example 10: If α_1 and α_2 are vectors of $V(F)$, and $a, b \in F$, show that the set $\{\alpha_1, \alpha_2, a\alpha_1 + b\alpha_2\}$ is linearly dependent.

Solution: We have $(-a)\alpha_1 + (-b)\alpha_2 + 1(a\alpha_1 + b\alpha_2)$
 $= (-a + a)\alpha_1 + (-b + b)\alpha_2$
 $= 0\alpha_1 + 0\alpha_2 = \mathbf{0}$ i.e., zero vector.

In the above linear combination the scalar coefficient $1 \neq 0$. Therefore whatever may be the scalars $-a$ and $-b$, the given set of vectors is linearly dependent.

Example 11: If α, β, γ are linearly independent vectors of $V(F)$ where F is any subfield of the field of complex numbers, then so also are $\alpha + \beta, \beta + \gamma, \gamma + \alpha$.

(Kumaun 2007, 12 ; Garhwal 13)

Solution: Let a, b, c be scalars such that

$$\begin{aligned} & a(\alpha + \beta) + b(\beta + \gamma) + c(\gamma + \alpha) = \mathbf{0} \\ \text{i.e., } & (a + c)\alpha + (a + b)\beta + (b + c)\gamma = \mathbf{0}. \end{aligned} \quad \dots(1)$$

But α, β, γ are linearly independent. Therefore (1) implies

$$a + 0b + c = 0, a + b + 0c = 0, 0a + b + c = 0.$$

The coefficient matrix A of these equations is

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

We have $\text{rank } A = 3$ i.e., the number of unknowns a, b, c . Therefore $a = 0, b = 0, c = 0$ is the only solution of the given equations.

Hence $\alpha + \beta, \beta + \gamma, \gamma + \alpha$ are also linearly independent.

Example 12: Show that the set $\{1, x, 1 + x + x^2\}$ is a linearly independent set of vectors in the vector space of all polynomials over the real number field.

Solution: Let a, b, c be scalars (real numbers) such that

$$a(1) + bx + c(1 + x + x^2) = 0.$$

$$\text{We have } a(1) + bx + c(1 + x + x^2) = 0$$

$$\Rightarrow (a + c) + (b + c)x + cx^2 = 0$$

$$\Rightarrow a + c = 0, b + c = 0, c = 0$$

$$\Rightarrow c = 0, b = 0, a = 0.$$

$\therefore$ the vectors $1, x, 1 + x + x^2$ are linearly independent over the field of real numbers.

Example 13: In the vector space $F[x]$ of all polynomials over the field F the infinite set $S = \{1, x, x^2, x^3, \dots\}$ is linearly independent.

Solution: Let $S' = \{x^{m_1}, x^{m_2}, \dots, x^{m_n}\}$ be any finite subset of S having n vectors. Here $m_1, m_2, \dots, m_n$ are some non-negative integers. Let $a_1, a_2, \dots, a_n$ be scalars such that

$$a_1 x^{m_1} + a_2 x^{m_2} + \dots + a_n x^{m_n} = \mathbf{0}. \quad (\text{i.e., zero polynomial}) \dots(1)$$

By the definition of equality of two polynomials we have from (1)

$$a_1 = 0, a_2 = 0, \dots, a_n = 0.$$

Thus every finite subset of S is linearly independent.

Therefore S is linearly independent.

Example 14: Show that the three vectors $(1, 1, -1)$, $(2, -3, 5)$ and $(-2, 1, 4)$ of $\mathbf{R}^3$ are linearly independent. (Kumaun 2010, 13)

Solution: Let a, b, c be scalars i.e., real numbers such that

$$a(1, 1, -1) + b(2, -3, 5) + c(-2, 1, 4) = (0, 0, 0)$$

$$\text{i.e.,} \quad (a + 2b - 2c, a - 3b + c, -a + 5b + 4c) = (0, 0, 0)$$

$$\text{i.e.,} \quad a + 2b - 2c = 0 \quad \dots(1)$$

$$a - 3b + c = 0 \quad \dots(2)$$

$$-a + 5b + 4c = 0 \quad \dots(3)$$

Now we shall solve the simultaneous equations (1), (2) and (3).

Multiplying (2) by 2 and adding to (1), we get

$$3a - 4b = 0. \quad \dots(4)$$

Again multiplying (1) by 2 and adding to (3), we get

$$a + 9b = 0. \quad \dots(5)$$

Multiplying (5) by 3 and subtracting from (4), we get

$$-31b = 0 \quad \text{or} \quad b = 0.$$

Putting $b = 0$ in (5), we get $a = 0$.

Now putting $a = 0, b = 0$ in (1), we get $c = 0$.

Thus $a = 0, b = 0, c = 0$ is the only solution of the equations (1), (2) and (3).

$$\therefore a(1, 1, -1) + b(2, -3, 5) + c(-2, 1, 4) = (0, 0, 0)$$

$$\Rightarrow a = 0, b = 0, c = 0.$$

Hence the vectors $(1, 1, -1)$, $(2, -3, 5)$, $(-2, 1, 4)$ of $\mathbf{R}^3$ are linearly independent.

Example 15: Find a maximal linearly independent subsystem of the system of vectors

$$\alpha_1 = (2, -2, -4), \alpha_2 = (1, 9, 3), \alpha_3 = (-2, -4, 1)$$

$$\text{and} \quad \alpha_4 = (3, 7, -1).$$

Solution: Let A denote the matrix $\begin{bmatrix} 2 & -2 & -4 \\ 1 & 9 & 3 \\ -2 & -4 & 1 \\ 3 & 7 & -1 \end{bmatrix}$ whose rows consist of the vectors

$\alpha_1, \alpha_2, \alpha_3$ and α_4 .

We shall reduce the matrix A to Echelon form by applying the row transformations.

We have

$$A \sim \begin{bmatrix} 1 & -1 & -2 \\ 1 & 9 & 3 \\ -2 & -4 & 1 \\ 3 & 7 & -1 \end{bmatrix}, \text{ applying } R_1 \rightarrow \frac{1}{2} R_1$$

$$\sim \begin{bmatrix} 1 & -1 & -2 \\ 0 & 10 & 5 \\ 0 & -6 & -3 \\ 0 & 10 & 5 \end{bmatrix}, \quad \begin{array}{l} \text{by } R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 + 2R_1 \\ R_4 \rightarrow R_4 - 3R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \end{bmatrix}, \quad \begin{array}{l} \text{by } R_2 \rightarrow \frac{1}{10} R_2 \\ R_3 \rightarrow -\frac{1}{6} R_3 \\ R_4 \rightarrow \frac{1}{10} R_4 \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{array}{l} \text{by } R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array}$$

which is in Echelon form.

We have $\text{rank } A = \text{the number of non-zero rows in its Echelon form} = 2$.

$\therefore$ the maximum number of linearly independent row vectors in the matrix A
 = the rank $A = 2$.

The vectors α_1 and α_2 are linearly independent and so $\{\alpha_1, \alpha_2\}$ is a maximal linearly independent subsystem of the given system of vectors. We observe that none of the given four vectors is a scalar multiple of any of the remaining three vectors. So any two of the given four vectors form a maximal linearly independent subsystem of the given system of vectors.

Comprehensive Exercise 1

1. Show that the vectors $(1, 1, 2, 4), (2, -1, -5, 2), (1, -1, -4, 0)$ and $(2, 1, 1, 6)$ are linearly dependent in $\mathbf{R}^4$.
2. Show that the vectors $(1, 1, 0, 0), (0, 1, -1, 0), (0, 0, 0, 3)$ in $\mathbf{R}^4$ are linearly independent.
3. Determine whether the following set of vectors in $V_3(\mathbf{Q})$ is linearly dependent or independent, $\mathbf{Q}$ being the field of rational numbers :
 $\{(-1, 2, 1), (3, 1, -2)\}$.
4. Find a linearly independent subset T of the set $S = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ where $\alpha_1 = (1, 2, -1), \alpha_2 = (-3, -6, 3), \alpha_3 = (2, 1, 3), \alpha_4 = (8, 7, 7) \in \mathbf{R}^3$ which spans the same space as S .
5. If the vectors $(0, 1, x), (x, 1, 0), (1, x, 1)$ of the vector space $\mathbf{R}^3$ are linearly dependent, then find the value of x . (Kumaun 2012, 14)
6. Show that the vectors $(0, 2, -4), (1, -2, -1), (1, -4, 3)$ in $\mathbf{R}^3$ are linearly dependent. Also express $(0, 2, -4)$ as a linear combination of $(1, -2, -1)$ and $(1, -4, 3)$.

7. Determine if the vectors $(1, -2, 1), (2, 1, -1), (7, -4, 1)$ in $\mathbf{R}^3$ are linearly independent or linearly dependent.
8. Let $\alpha_1, \alpha_2, \alpha_3$ be the vectors of $V(F)$, $a, b \in F$. Show that the set $\{\alpha_1, \alpha_2, \alpha_3\}$ is linearly dependent if the set $\{\alpha_1 + a\alpha_2 + b\alpha_3, \alpha_2, \alpha_3\}$ is linearly dependent.
9. If α, β, γ are linearly independent vectors of $V(F)$ where F is the field of complex numbers, then so also are $\alpha + \beta, \alpha - \beta, \alpha - 2\beta + \gamma$.
10. Show that the set $\{1, x, x(1-x)\}$ is a linearly independent set of vectors in the space of all polynomials over the real number field.
11. Find whether the vectors $2x^3 + x^2 + x + 1, x^3 + 3x^2 + x - 2$ and $x^3 + 2x^2 - x + 3$ of $\mathbf{R}[x]$, the vector space of all polynomials over the real number field, are linearly independent or not.
12. Prove that a set of vectors which contains the zero vector is linearly dependent.
13. Let V be the vector space of 2×2 matrices over $\mathbf{R}$. Determine whether the matrices $A, B, C \in V$ are linearly dependent, where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 3 & -1 \\ 2 & 2 \end{bmatrix}, C = \begin{bmatrix} 1 & -5 \\ -4 & 0 \end{bmatrix}.$$
(Garhwal 2013)
14. Prove that the three non-zero non-coplanar vectors are linearly independent.
15. Show that any three non-zero coplanar vectors are linearly dependent in $\mathbf{R}^3$.
16. Show that the set of functions $\{x, |x|\}$ is L.I. in vector space of continuous functions defined in $(-1, 1)$.
17. Show that the vectors $\alpha = (1 + i, 2i), \beta = (1, 1 + i)$ in $\mathbf{C}^2(C)$ are L.D. but in $\mathbf{C}^2(\mathbf{R})$ are L.I.
18. If $V(F)$ is a vector space and $\alpha_1, \alpha_2, \dots, \alpha_n \in V$, $\lambda_2, \lambda_3, \dots, \lambda_n \in F$ such that $\{\alpha_2 + \lambda_2 \alpha_1, \alpha_3 + \lambda_3 \alpha_1, \dots, \alpha_n + \lambda_n \alpha_1\}$ is L.D. then show that $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is L.D.
19. Let V be the vector space of functions from $\mathbf{R}$ into $\mathbf{R}$. Show that $f, g, h \in V$ are linearly independent where
 - (i) $f(x) = e^{2x}, g(x) = x^2, h(x) = x$
 - (ii) $f(x) = \sin x, g(x) = \cos x, h(x) = x$.
20. Test the vectors $(0, 1, 0, 1, 1), (1, 0, 1, 0, 1), (0, 1, 0, 1, 1), (1, 1, 1, 1, 1)$ in V_5 over the field of rational numbers for linear independence.
21. Prove that the four vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$ in $V_3(\mathbf{C})$ form a linearly dependent set but any three of them are linearly independent.

Answers 1

- | | | |
|--|-----------------------------|------------------------|
| 3. linearly independent | 4. $\{\alpha_1, \alpha_3\}$ | 5. $0, \pm \sqrt{2}$ |
| 6. $(0, 2, -4) = (1, -2, -1) - (1, -4, 3)$ | | 7. linearly dependent |
| 11. linearly independent | 13. Yes | 20. linearly dependent |

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The singleton set $\{x\}$ is linearly independent if and only if
 - $x = 0$
 - $x \neq 0$
 - may or may not be zero
 - none of these(Garhwal 2006, 10)
- The set of vectors $\{(1, 2, 0), (0, 3, 1), (-1, 0, 1)\}$ are
 - linearly independent
 - linearly dependent
 - neither linearly independent nor dependent
 - none of these
- The vectors (a_1, a_2) and (b_1, b_2) are linearly dependent if
 - $a_1b_1 + a_2b_2 = 0$
 - $a_1b_2 + a_2b_1 = 0$
 - $a_1b_2 - a_2b_1 = 0$
 - $a_1b_1 - a_2b_2 = 0$
- Every superset of a linearly dependent set of vectors is
 - linearly dependent
 - linearly independent
 - neither linearly dependent nor independent
 - none of these
- Three non-zero coplanar vectors are
 - linearly independent
 - linearly dependent
 - neither dependent nor independent
 - none of these

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- Any set of vectors containing the zero vector as a member is linearly
- Intersection of two linearly independent subsets of a vector space will be linearly
- The subset $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ of the vector space $\mathbf{R}^3$ is linearly
- A system consisting of a single non-zero vector is always linearly
- In the vector space $\mathbf{R}^3$ the vectors $(1, 0, 1)$, $(3, 7, 0)$ and $(-1, 0, -1)$ are linearly

True or False

Write 'T' for true and 'F' for false statement.

- A set containing a linearly independent set of vectors is itself linearly independent.
- Union of two linearly independent subsets of a vector space is linearly independent.
- The vectors $(2, -3, 7)$, $(0, 0, 0)$, $(3, -1, -4)$ are linearly independent in $\mathbf{R}^3$.
- The vectors $(1, -3, 7)$, $(2, 0, -6)$, $(3, -1, -1)$, $(2, 4, -5)$ are linearly dependent in $\mathbf{R}^3$.

Answers

Multiple Choice Questions

- | | | | |
|--------|--------|--------|--------|
| 1. (b) | 2. (a) | 3. (c) | 4. (a) |
| 5. (b) | | | |

Fill in the Blank(s)

- | | |
|----------------|----------------|
| 1. dependent | 2. independent |
| 3. independent | 4. independent |
| 5. dependent | |

True or False

- | | | | |
|------|------|------|------|
| 1. F | 2. F | 3. F | 4. T |
|------|------|------|------|



Chapter

3



Basis and Dimensions

1 Basis of A Vector Space

Definition: A subset S of a vector space $V(F)$ is said to be a basis of $V(F)$, if

- (i) S consists of linearly independent vectors,
- (ii) S generates $V(F)$ i.e., $L(S) = V$ i.e., each vector in V is a linear combination of a finite number of elements of S . (Garhwal 2010; Gorakhpur 10)

Illustrative Examples

Example 1: A system S consisting of n vectors,

$$e_1 = (1, 0, 0, \dots, 0), e_2 = (0, 1, 0, \dots, 0), \dots, e_n = (0, 0, \dots, 0, 1)$$

is a basis of $V_n(F)$.

(Kumaun 2009)

Solution: First we should show that S is a linearly independent set of vectors. We have proved it in one of the previous examples.

Now we should prove that $L(S) = V_n(F)$. We have always $L(S) \subseteq V_n(F)$. So we should prove that $V_n(F) \subseteq L(S)$ i.e., each vector in $V_n(F)$ is a linear combination of elements of S .

Let $\alpha = (a_1, a_2, \dots, a_n)$ be any vector in $V_n(F)$.

We can write

$$(a_1, a_2, \dots, a_n) = a_1 (1, 0, \dots, 0) + a_2 (0, 1, \dots, 0) + \dots + a_n (0, 0, \dots, 1)$$

$$\text{i.e., } \alpha = a_1 e_1 + a_2 e_2 + \dots + a_n e_n.$$

Hence S is a basis of $V_n(F)$. We shall call this particular basis the **standard basis** of $V_n(F)$.

Note: The set $\{(1, 0), (0, 1)\}$ is a basis of $V_2(F)$. The set $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a basis of $V_3(F)$. As a particular case a basis of $F(F)$ is the set consisting of only the unity element of F .

Example 2: Show that the infinite set $S = \{1, x, x^2, \dots, x^n, \dots\}$ is a basis of the vector space $F[x]$ of polynomials over the field F .

Solution: First we should prove that S is a linearly independent set of vectors. For proof refer some previous example.

Now we should show that S spans $F[x]$ i.e., each polynomial in $F[x]$ can be expressed as a linear combination of a finite number of elements of S .

Let $f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_t x^t$ be a polynomial of degree t .

Then $f(x) = (a_0)1 + a_1 x + a_2 x^2 + \dots + a_t x^t$. Hence S is a basis of $F[x]$.

Note: The vector space $F[x]$ has no finite basis. If we take any finite set W of polynomials, we can find a polynomial of degree greater than that of each of them. Such a polynomial cannot at any cost be expressed as a linear combination of the elements of W .

2 Finite Dimensional Vector Space

(Garhwal 2003, 07)

Definition: The vector space $V(F)$ is said to be **finite dimensional** or **finitely generated** if there exists a **finite** subset S of V such that $V = L(S)$.

The vector space $V_n(F)$ of n -tuples is a finite dimensional vector space.

The vector space $F[x]$ of all polynomials over a field F is not finite dimensional. There exists no finite subset S of $F[x]$ which spans $F[x]$. A vector space which is not finitely generated may be referred to as an **infinite dimensional space**. Thus the vector space $F[x]$ of all polynomials over a field F is infinite dimensional.

Note: The dimension of the zero space $\{0\}$ is defined to be zero. It is generated by $\emptyset$.

3 Existence of Basis of a Finite Dimensional Vector Space

Theorem: There exists a basis for each finite dimensional vector space.

(Garhwal 2006, 10, 12, 14; Kumaun 06; Gorakhpur 10)

Proof: Let $V(F)$ be a finitely generated vector space.

Let $S = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$

be a finite subset of V such that $L(S) = V$. We may suppose that no member of S is 0 .

If S is linearly independent, then S itself is a basis of V .

If S is linearly dependent, then there exists a vector $\alpha_i \in S$ which can be expressed as a linear combination of the preceding vectors $\alpha_1, \alpha_2, \dots, \alpha_{i-1}$.

If we omit this vector α_i from S , then the remaining set S' of $m-1$ vectors

$$\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_m$$

also generates V i.e., $V = L(S')$. For if α is any element of V , then $L(S) = V$ implies that α can be written as a linear combination of $\alpha_1, \alpha_2, \dots, \alpha_m$. Let

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_{i-1}\alpha_{i-1} + a_i\alpha_i + a_{i+1}\alpha_{i+1} + \dots + a_m\alpha_m.$$

But α_i can be expressed as a linear combination of $\alpha_1, \dots, \alpha_{i-1}$. Let

$$\alpha_i = b_1\alpha_1 + \dots + b_{i-1}\alpha_{i-1}.$$

Putting this value of α_i in the expression for α , we get

$$\alpha = a_1\alpha_1 + \dots + a_{i-1}\alpha_{i-1} + a_i(b_1\alpha_1 + \dots + b_{i-1}\alpha_{i-1}) + a_{i+1}\alpha_{i+1} + \dots + a_m\alpha_m.$$

Thus α has been expressed as a linear combination of the vectors

$$\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_m.$$

In this way $\alpha \in V \Rightarrow \alpha$ can be expressed as a linear combination of the vectors belonging to the set S' .

Thus S' generates V i.e., $L(S') = V$.

If S' is linearly independent, then S' will be a basis of V . If S' is linearly dependent, then proceeding as above we shall get a new set of $n-2$ vectors which generates V . Continuing this process, we shall, after finite number of steps, obtain a linearly independent subset of S which generates V and which is therefore a basis of V .

At the most it may happen that we shall be left with a subset of S which contains only one non-zero vector and which spans V . We know that a set containing a single non-zero vector is definitely linearly independent and so it will form a basis of V .

Note: The above theorem may also be stated as below :

If a finite set S of vectors spans a finite dimensional vector space $V(F)$, there exists a subset of S which forms a basis of V .

4 Invariance of Number of Elements in the Basis of a Finite Dimensional Vector Space

Dimension theorem for vector spaces: *If $V(F)$ is a finite dimensional vector space, then any two bases of V have the same number of elements.*

(Garhwal 2004; Gorakhpur 11, 14)

Proof: Suppose $V(F)$ is a finite dimensional vector space. Then V definitely possesses a basis. Let

$$S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$$

$$\text{and } S_2 = \{\beta_1, \beta_2, \dots, \beta_n\}$$

be two bases of V . We shall prove that $m = n$.

Since $V = L(S_1)$ and $\beta_1 \in V$, therefore β_1 can be expressed as a linear combination of $\alpha_1, \alpha_2, \dots, \alpha_m$. Consequently the set $S_3 = \{\beta_1, \alpha_1, \alpha_2, \dots, \alpha_m\}$ which also obviously generates $V(F)$ is linearly dependent. Therefore there exists a member $\alpha_i \neq \beta_1$ of this set S_3 such that α_i is a linear combination of the preceding vectors

$$\beta_1, \alpha_1, \alpha_2, \dots, \alpha_{i-1}.$$

If we omit the vector α_i from S_3 then V is also generated by the remaining set

$$S_4 = \{\beta_1, \alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_m\}.$$

Since $V = L(S_4)$ and $\beta_2 \in V$, therefore β_2 can be expressed as a linear combination of the vectors belonging to S_4 . Consequently the set

$$S_5 = \{\beta_2, \beta_1, \alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_m\}$$

is linearly dependent. Therefore there exists a member α_j of this set S_5 such that α_j is a linear combination of the preceding vectors. Obviously α_j will be different from β_1 and β_2 since $\{\beta_1, \beta_2\}$ is a linearly independent set. If we exclude the vector α_j from S_5 , then the remaining set will generate $V(F)$.

We may continue to proceed in this manner. Here each step consists in the exclusion of an α and the inclusion of a β in the set S_1 .

Obviously the set S_1 of α 's cannot be exhausted before the set S_2 of β 's otherwise $V(F)$ will be a linear span of a proper subset of S_2 and thus S_2 will become linearly dependent. Therefore we must have

$$m \nless n.$$

Interchanging the roles of S_1 and S_2 , we shall get that

$$n \nless m.$$

Hence $n = m$.

Illustration: For the vector space $V_3(F)$, the sets

$$S_1 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

and

$$S_2 = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$$

are bases as can easily be seen. Both these bases contain the same number of elements i.e., 3.

5 Dimension of a Finitely Generated Vector Space

Definition: The number of elements in any basis of a finite dimensional vector space $V(F)$ is called the dimension of the vector space $V(F)$ and will be denoted by $\dim V$.

(Garhwal 2010)

The vector space $V_n(F)$ is of dimension n . The vector space $V_3(F)$ is of dimension 3. If a field F is regarded as a vector space over F , then F will be of dimension 1 and the set $S = \{1\}$ consisting of unity element of F alone is a basis of F . In fact every non-zero element of F will form a basis of F .

6 Some Properties of Finite Dimensional Vector Spaces

Theorem 1: Extension theorem. Every linearly independent subset of a finitely generated vector space $V(F)$ forms a part of a basis of V .

Or

Every linearly independent subset of a finitely generated vector space $V(F)$ is either a basis of V or can be extended to form a basis of V . (Meerut 2003)

Proof: Let $S = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$ be a linearly independent subset of a finite dimensional vector space $V(F)$. If $\dim V = n$, then V has a finite basis, say $\{\beta_1, \beta_2, \dots, \beta_n\}$. Consider the set

$$S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_n\}.$$

Obviously $L(S_1) = V$. Since the α 's can be expressed as linear combinations of the β 's therefore the set S_1 is linearly dependent.

Therefore there is some vector of S_1 which is a linear combination of its preceding vectors. This vector cannot be any of the α 's since the α 's are linearly independent. Therefore this vector must be some β , say β_i . Now omit the vector β_i from S_1 and consider the set

$$S_2 = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_{i-1}, \beta_{i+1}, \dots, \beta_n\}.$$

Obviously $L(S_2) = V$. If S_2 is linearly independent, then S_2 will be a basis of V and it is the required extended set which is a basis of V . If S_2 is not linearly independent, then repeating the above process a finite number of times, we shall get a linearly independent set containing $\alpha_1, \alpha_2, \dots, \alpha_m$ and spanning V . This set will be a basis of V and it will contain S . Since each basis of V contains the same number of elements, therefore exactly $n - m$ elements of set of β 's will be adjoined to S so as to form a basis of V .

Theorem 2: Each set of $(n + 1)$ or more vectors of a finite dimensional vector space $V(F)$ of dimension n is linearly dependent.

Proof: Let $V(F)$ be a finite dimensional vector space of dimension n . Let S be a linearly independent subset of V containing $(n + 1)$ or more vectors. Then S will form a part of basis of V . Thus we shall get a basis of V containing more than n vectors. But every basis of V will contain exactly n vectors. Hence our assumption is wrong. Therefore if S contains $(n + 1)$ or more vectors, then S must be linearly dependent.

Theorem 3: Let V be a vector space which is spanned by a finite set of vectors $\beta_1, \beta_2, \dots, \beta_m$. Then any linearly independent set of vectors in V is finite and contains no more than m vectors.

(Garhwal 2009; Kumaun 14)

Proof: Let $S = \{\beta_1, \beta_2, \dots, \beta_m\}$.

Since $L(S) = V$, therefore V has a finite basis and $\dim V \leq m$. Hence every subset S' of V which contains more than m vectors is linearly dependent. This proves the theorem.

Theorem 4: If $V(F)$ is a finite dimensional vector space of dimension n , then any set of n linearly independent vectors in V forms a basis of V .

Proof: Let $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a linearly independent subset of a finite dimensional vector space $V(F)$ of dimension n . If S is not a basis of V , then it can be extended to form a basis of V . Thus we shall get a basis of V containing more than n vectors. But every basis of V must contain exactly n vectors. Therefore our assumption is wrong and S must be a basis of V .

Theorem 5: If a set S of n vectors of a finite dimensional vector space $V(F)$ of dimension n generates $V(F)$, then S is a basis of V .

Proof: Let $V(F)$ be a finite dimensional vector space of dimension n . Let $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a subset of V such that $L(S) = V$. If S is linearly independent, then S will form a basis of V . If S is not linearly independent, then there will exist a proper subset of S which will form a basis of V . Thus we shall get a basis of V containing less than n elements. But every basis of V must contain exactly n elements. Hence S cannot be linearly dependent and so S must be a basis of V .

Note: If V is a finite dimensional vector space of dimension n , then V cannot be generated by fewer than n vectors.

Dimension of a Subspace:

Theorem 6: Each subspace W of a finite dimensional vector space V (F) of dimension n is a finite dimensional space with $\dim W \leq n$.

Also $V = W$ iff $\dim V = \dim W$.

(Garhwal 2008; Kumaun 10, 12)

Proof: Let V (F) be a finite dimensional vector space of $\dim n$. Let W be a subspace of V . Any subset of W containing $(n + 1)$ or more vectors is also a subset of V and any $(n + 1)$ vectors in V are linearly dependent. Therefore any linearly independent set of vectors in W can contain, at the most n vectors. Let

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$$

be a linearly independent subset of W with maximum number of elements. We claim that S is a basis of W . The proof is as follows :

(i) S is a linearly independent subset of W .

(ii) $L(S) = W$. Let α be any element of W .

Then the $(m + 1)$ vectors $\alpha, \alpha_1, \alpha_2, \dots, \alpha_m$ belonging to W are linearly dependent because we have supposed that the largest independent subset of W contains m vectors.

Now $\{\alpha_1, \alpha_2, \dots, \alpha_m, \alpha\}$ is a linearly dependent set. Therefore there exists a vector belonging to it which can be expressed as a linear combination of the preceding vectors. Since $\alpha_1, \alpha_2, \dots, \alpha_m$ are linearly independent, therefore this vector cannot be any of these m vectors. So it must be α itself. Thus α can be expressed as a linear combination of $\alpha_1, \alpha_2, \dots, \alpha_m$. Hence $L(S) = W$.

$\therefore S$ is a basis of W .

$\therefore \dim W = m$ and $m \leq n$.

Now if $V = W$, then every basis of V is also a basis of W .

Hence $\dim V = \dim W = n$.

Conversely let $\dim W = \dim V = n$. Then to prove that $W = V$.

Let S be a basis of W . Then $L(S) = W$ and S contains n vectors. Since S is also a subset of V and S contains n linearly independent vectors, therefore S will also be a basis of V . Therefore $L(S) = V$. Hence $W = V$. We thus conclude :

If W is a proper subspace of a finite-dimensional vector space V , then W is finite dimensional and $\dim W < \dim V$. (Garhwal 2008)

Theorem 7: If W is a subspace of a finite-dimensional vector space V , every linearly independent subset of W is finite and is part of a (finite) basis for W .

Proof: Let $\dim V = n$. Let W be a subspace of V . Let S_0 be a linearly independent subset of W . Let S be a linearly independent subset of W which contains S_0 and which has maximum number of elements. Then S is also a linearly independent subset of V . So S will have at the most n elements. Therefore S is finite and consequently S_0 is finite.

Now our claim is that S is a basis of W . The proof is as follows :

(i) S is a linearly independent subset of W .

(ii) $L(S) = W$. Because if $\beta \in W$, then β must be in the linear span of S . If β is not in the linear span of S , then the subset of W obtained by adjoining β to S will be linearly independent. Thus S will not remain the maximum linearly independent subset of W containing S_0 . Hence $\beta \in W \Rightarrow \beta \in L(S)$. Thus $W \subseteq L(S)$. Since S is a subset of the subspace W , therefore $L(S) \subseteq W$.

Hence $L(S) = W$.

Thus S is a finite basis of W and $S_0 \subseteq S$.

Theorem 8: Let $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a basis of a finite dimensional vector space $V(F)$ of dimension n . Then every element α of V can be **uniquely** expressed as

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n \quad \text{where } a_1, a_2, \dots, a_n \in F.$$

Proof: Since S is a basis of V , therefore $L(S) = V$. Therefore any vector $\alpha \in V$ can be expressed as

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n.$$

To show uniqueness let us suppose that

$$\alpha = b_1\alpha_1 + b_2\alpha_2 + \dots + b_n\alpha_n.$$

Then we must show that $a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$.

We have $a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = b_1\alpha_1 + b_2\alpha_2 + \dots + b_n\alpha_n$

$$\Rightarrow (a_1 - b_1)\alpha_1 + (a_2 - b_2)\alpha_2 + \dots + (a_n - b_n)\alpha_n = \mathbf{0}$$

$$\Rightarrow a_1 - b_1 = 0, a_2 - b_2 = 0, \dots, a_n - b_n = 0,$$

since $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly independent

$$\Rightarrow a_1 = b_1, a_2 = b_2, \dots, a_n = b_n.$$

Hence the theorem.

Theorem 9: If W_1, W_2 are two subspaces of a finite dimensional vector space $V(F)$, then $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$.

(Kumaun 2004, 13; Garhwal 08, 13; Gorakhpur 15)

Proof: Let $\dim(W_1 \cap W_2) = k$ and let the set

$$S = \{\gamma_1, \gamma_2, \gamma_3, \dots, \gamma_k\}$$

be a basis of $W_1 \cap W_2$. Then $S \subseteq W_1$ and $S \subseteq W_2$.

Since S is linearly independent and $S \subseteq W_1$, therefore S can be extended to form a basis of W_1 . Let

$$\{\gamma_1, \gamma_2, \dots, \gamma_k, \alpha_1, \alpha_2, \dots, \alpha_m\}$$

be a basis of W_1 . Then $\dim W_1 = k + m$. Similarly let

$$\{\gamma_1, \gamma_2, \dots, \gamma_k, \beta_1, \beta_2, \dots, \beta_t\}$$

be a basis of W_2 . Then $\dim W_2 = k + t$.

$$\therefore \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2) = (m + k) + (k + t) - k = k + m + t.$$

$\therefore$ to prove the theorem we must show that

$$\dim(W_1 + W_2) = k + m + t.$$

We claim that the set

$$S_1 = \{\gamma_1, \gamma_2, \dots, \gamma_k, \alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_t\}$$

is a basis of $W_1 + W_2$.

First we show that S_1 is linearly independent. Let

$$c_1\gamma_1 + c_2\gamma_2 + \dots + c_k\gamma_k + a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m + b_1\beta_1 + b_2\beta_2 + \dots + b_t\beta_t = \mathbf{0} \quad \dots(1)$$

$$\Rightarrow b_1\beta_1 + b_2\beta_2 + \dots + b_t\beta_t = -(c_1\gamma_1 + \dots + c_k\gamma_k + a_1\alpha_1 + \dots + a_m\alpha_m) \quad \dots(2)$$

Now $-(c_1\gamma_1 + \dots + c_k\gamma_k + a_1\alpha_1 + \dots + a_m\alpha_m) \in W_1$ since it is a linear combination of elements of a basis of W_1 . Again

$$b_1\beta_1 + b_2\beta_2 + \dots + b_t\beta_t \in W_2$$

since it is a linear combination of elements belonging to a basis of W_2 .

Also by virtue of the equality (2), $b_1\beta_1 + \dots + b_t\beta_t \in W_1$. Therefore

$$b_1\beta_1 + b_2\beta_2 + \dots + b_t\beta_t \in W_1 \cap W_2.$$

Therefore it can be expressed as a linear combination of elements of the basis of $W_1 \cap W_2$. Thus we have a relation of the form

$$b_1\beta_1 + b_2\beta_2 + \dots + b_t\beta_t = d_1\gamma_1 + \dots + d_k\gamma_k$$

$$\Rightarrow b_1\beta_1 + b_2\beta_2 + \dots + b_t\beta_t - d_1\gamma_1 - d_2\gamma_2 - \dots - d_k\gamma_k = \mathbf{0}.$$

But $\beta_1, \beta_2, \dots, \beta_t, \gamma_1, \dots, \gamma_k$ are linearly independent vectors.

Therefore we must have

$$b_1 = 0, b_2 = 0, \dots, b_t = 0.$$

Putting these values of b 's in (1), it reduces to

$$c_1\gamma_1 + c_2\gamma_2 + \dots + c_k\gamma_k + a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m = \mathbf{0}$$

$$\Rightarrow c_1 = 0, c_2 = 0, \dots, c_k = 0, a_1 = 0, a_2 = 0, \dots, a_m = 0$$

since the vectors $\gamma_1, \gamma_2, \dots, \gamma_k, \alpha_1, \alpha_2, \dots, \alpha_m$ are linearly independent.

Thus the relation (1) implies that

$$c_1 = 0, c_2 = 0, \dots, c_k = 0, a_1 = 0, \dots, a_m = 0, b_1 = 0, \dots, b_t = 0.$$

Therefore the set S_1 of vectors

$$\gamma_1, \dots, \gamma_k, \alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_t$$

is linearly independent.

Now to show that $L(S_1) = W_1 + W_2$.

Since $W_1 + W_2$ is a subspace of V and each element of S_1 belongs to $W_1 + W_2$, therefore $L(S_1) \subseteq W_1 + W_2$.

Again let α be any element of $W_1 + W_2$. Then

$$\alpha = \text{some element of } W_1 + \text{some element of } W_2$$

$$= \text{a linear combination of elements of basis of } W_1$$

$$+ \text{a linear combination of elements of basis of } W_2$$

$$= \text{a linear combination of elements of } S_1.$$

$$\therefore \alpha \in L(S_1). \text{ Hence } W_1 + W_2 \subseteq L(S_1).$$

$$\therefore L(S_1) = W_1 + W_2.$$

$$\therefore S_1 \text{ is a basis of } W_1 + W_2 \text{ and consequently}$$

$$\dim(W_1 + W_2) = k + m + t.$$

Hence the theorem.

Illustrative Examples

Example 3: Let V be the vector space of all 2×2 matrices over the field F . Prove that V has dimension 4 by exhibiting a basis for V which has 4 elements.

(Garhwal 2006; Kumaun 11)

Solution: Let $\alpha = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\beta = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\gamma = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ and $\delta = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

be four elements of V .

The subset $S = \{\alpha, \beta, \gamma, \delta\}$ of V is linearly independent because

$$\begin{aligned} & a\alpha + b\beta + c\gamma + d\delta = \mathbf{0} \\ \Rightarrow & a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ \Rightarrow & a = 0, b = 0, c = 0, d = 0. \end{aligned}$$

Also $L(S) = V$ because if $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is any vector in V , then we can write

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a\alpha + b\beta + c\gamma + d\delta.$$

Therefore S is a basis of V . Since the number of elements in S is 4, therefore $\dim V = 4$.

Example 4: Show that if $S = \{\alpha, \beta, \gamma\}$ is a basis of $\mathbf{C}^3(\mathbf{C})$, then the set

$$S' = \{\alpha + \beta, \beta + \gamma, \gamma + \alpha\}$$

is also a basis of $\mathbf{C}^3(\mathbf{C})$.

(Kumaun 2011)

Solution: The vector space $\mathbf{C}^3(\mathbf{C})$ is of dimension 3. Any subset of $\mathbf{C}^3$ having three linearly independent vectors will form a basis of $\mathbf{C}^3$. We have shown in one of the previous examples that if S is a linearly independent subset of $\mathbf{C}^3$, then S' is also a linearly independent subset of $\mathbf{C}^3$.

Therefore S' is also a basis of $\mathbf{C}^3$.

Example 5: Let V be the vector space of ordered pairs of complex numbers over the real field $\mathbf{R}$ i.e., let V be the vector space $\mathbf{C}^2(\mathbf{R})$. Show that the set

$$S = \{(1, 0), (i, 0), (0, 1), (0, i)\}$$

is a basis for V .

Solution: S is linearly independent. We have

$$\begin{aligned} & a(1, 0) + b(i, 0) + c(0, 1) + d(0, i) = (0, 0) \text{ where } a, b, c, d \in \mathbf{R} \\ \Rightarrow & (a + ib, c + id) = (0, 0) \\ \Rightarrow & a + ib = 0, c + id = 0 \\ \Rightarrow & a = 0, b = 0, c = 0, d = 0. \end{aligned}$$

Therefore S is linearly independent.

Now we shall show that $L(S) = V$. Let any ordered pair $(a + ib, c + id) \in V$ where $a, b, c, d \in \mathbf{R}$. Then as shown above we can write

$$(a + ib, c + id) = a(1, 0) + b(i, 0) + c(0, 1) + d(0, i).$$

Thus any vector in V is expressible as a linear combination of elements of S . Therefore $L(S) = V$ and so S is a basis for V .

Example 6: Show that the vectors $(1, 2, 1), (2, 1, 0), (1, -1, 2)$ form a basis of $\mathbf{R}^3$.

(Garhwal 2010; Kumaun 14)

Solution: We know that the set $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ forms a basis for $\mathbf{R}^3$.

Therefore $\dim \mathbf{R}^3 = 3$. If we show that the set $S = \{(1, 2, 1), (2, 1, 0), (1, -1, 2)\}$ is linearly independent, then this set will also form a basis for $\mathbf{R}^3$.

[See theorem 4 of article 3]

We have $a_1(1, 2, 1) + a_2(2, 1, 0) + a_3(1, -1, 2) = (0, 0, 0)$

$$\Rightarrow (a_1 + 2a_2 + a_3, 2a_1 + a_2 - a_3, a_1 + 2a_3) = (0, 0, 0).$$

$$\therefore a_1 + 2a_2 + a_3 = 0 \quad \dots(1)$$

$$2a_1 + a_2 - a_3 = 0 \quad \dots(2)$$

$$a_1 + 2a_3 = 0. \quad \dots(3)$$

Now we shall solve these equations to get the values of a_1, a_2, a_3 . Multiplying the equation (2) by 2, we get

$$4a_1 + 2a_2 - 2a_3 = 0. \quad \dots(4)$$

Subtracting (4) from (1), we get

$$-3a_1 + 3a_3 = 0$$

$$\text{or} \quad -a_1 + a_3 = 0. \quad \dots(5)$$

Adding (3) and (5), we get $3a_3 = 0$ or $a_3 = 0$. Putting $a_3 = 0$ in (3), we get $a_1 = 0$. Now putting $a_3 = 0$ and $a_1 = 0$ in (1), we get $a_2 = 0$.

Thus solving the equations (1), (2) and (3), we get $a_1 = 0, a_2 = 0, a_3 = 0$. Therefore the set S is linearly independent. Hence it forms a basis for $\mathbf{R}^3$.

Example 7: Determine whether or not the following vectors form a basis of $\mathbf{R}^3$:

$$(1, 1, 2), (1, 2, 5), (5, 3, 4). \quad \text{(Kumaun 2013)}$$

Solution: We know that $\dim \mathbf{R}^3 = 3$. If the given set of vectors is linearly independent, it will form a basis of $\mathbf{R}^3$ otherwise not. We have

$$a_1(1, 1, 2) + a_2(1, 2, 5) + a_3(5, 3, 4) = (0, 0, 0)$$

$$\Rightarrow (a_1 + a_2 + 5a_3, a_1 + 2a_2 + 3a_3, 2a_1 + 5a_2 + 4a_3) = (0, 0, 0).$$

$$a_1 + a_2 + 5a_3 = 0 \quad \dots(1)$$

$$\therefore a_1 + 2a_2 + 3a_3 = 0 \quad \dots(2)$$

$$2a_1 + 5a_2 + 4a_3 = 0 \quad \dots(3)$$

Now we shall solve these equations to get the values of a_1, a_2, a_3 . Subtracting (2) from (1), we get

$$-a_2 + 2a_3 = 0. \quad \dots(4)$$

Multiplying (1) by 2, we get

$$2a_1 + 2a_2 + 10a_3 = 0. \quad \dots(5)$$

Subtracting (5) from (3), we get

$$3a_2 - 6a_3 = 0$$

$$\text{or} \quad a_2 - 2a_3 = 0. \quad \dots(6)$$

We see that the equations (4) and (6) are the same and give $a_2 = 2a_3$. Putting $a_2 = 2a_3$ in (1), we get $a_1 = -7a_3$. If we put $a_3 = 1$, we get $a_2 = 2$ and $a_1 = -7$. Thus $a_1 = -7, a_2 = 2, a_3 = 1$ is a non-zero solution of the equations (1), (2) and (3). Hence the given set is linearly dependent so it does not form a basis of $\mathbf{R}^3$.

Example 8: Show that the vectors $\alpha_1 = (1, 0, -1)$, $\alpha_2 = (1, 2, 1)$, $\alpha_3 = (0, -3, 2)$ form a basis for $\mathbf{R}^3$. Express each of the standard basis vectors as a linear combination of $\alpha_1, \alpha_2, \alpha_3$.

(Gorakhpur 2013)

Solution: Let $S = \{\alpha_1, \alpha_2, \alpha_3\}$.

First we shall show that the set S is linearly independent. Let a, b, c be scalars i.e., real numbers such that

$$a\alpha_1 + b\alpha_2 + c\alpha_3 = \mathbf{0}$$

$$\text{i.e., } a(1, 0, -1) + b(1, 2, 1) + c(0, -3, 2) = (0, 0, 0)$$

$$\text{i.e., } (a + b + 0c, 0a + 2b - 3c, -a + b + 2c) = (0, 0, 0)$$

$$\text{i.e., } a + b = 0 \quad \dots(1)$$

$$2b - 3c = 0 \quad \dots(2)$$

$$-a + b + 2c = 0. \quad \dots(3)$$

Adding both sides of the equations (1) and (3), we get

$$2b + 2c = 0 \quad \dots(4)$$

Subtracting (2) from (4), we get

$$5c = 0 \quad \text{or} \quad c = 0.$$

Putting $c = 0$ in (2), we get $b = 0$ and then putting $b = 0$ in (1), we get $a = 0$.

Thus $a = 0, b = 0, c = 0$ is the only solution of the equations (1), (2) and (3) and so

$$a\alpha_1 + b\alpha_2 + c\alpha_3 = \mathbf{0}$$

$$\Rightarrow a = 0, b = 0, c = 0.$$

$\therefore$ The vectors $\alpha_1, \alpha_2, \alpha_3$ are linearly independent.

We have $\dim \mathbf{R}^3 = 3$. Therefore the vectors $\alpha_1, \alpha_2, \alpha_3$ form a basis of $\mathbf{R}^3$.

Now let $\gamma = (p, q, r)$ be any vector in $\mathbf{R}^3$ and let

$$\begin{aligned} \gamma &= (p, q, r) = x\alpha_1 + y\alpha_2 + z\alpha_3, \quad x, y, z \in \mathbf{R} \\ &= x(1, 0, -1) + y(1, 2, 1) + z(0, -3, 2). \end{aligned} \quad \dots(5)$$

$$\text{Then } x + y = p \quad \dots(6)$$

$$2y - 3z = q \quad \dots(7)$$

$$-x + y + 2z = r \quad \dots(8)$$

Adding both sides of equations (6) and (8), we get

$$2y + 2z = p + r. \quad \dots(9)$$

Subtracting (7) from (9), we get

$$5z = p + r - q$$

$$\text{or } z = \frac{1}{5}p - \frac{1}{5}q + \frac{1}{5}r.$$

Then from (9), we get

$$\begin{aligned} y &= -z + \frac{1}{2}p + \frac{1}{2}r \\ &= \frac{3}{10}p + \frac{1}{5}q + \frac{3}{10}r \end{aligned}$$

and from (6), we get

$$x = p - y$$

$$\begin{aligned}
 &= p - \frac{3}{10}p - \frac{1}{5}q - \frac{3}{10}r \\
 &= \frac{7}{10}p - \frac{1}{5}q - \frac{3}{10}r.
 \end{aligned}$$

Thus, the relation (5) expresses the vector $\gamma = (p, q, r)$ as a linear combination of α_1, α_2 and α_3 where $x, y, z \in \mathbf{R}$ are as found above.

The standard basis vectors are

$$e_1 = (1, 0, 0), e_2 = (0, 1, 0) \quad \text{and} \quad e_3 = (0, 0, 1).$$

If $\gamma = e_1$, then $p = 1, q = 0, r = 0$ and so

$$x = \frac{7}{10}, y = \frac{3}{10} \text{ and } z = \frac{1}{5}.$$

$$\therefore e_1 = \frac{7}{10}\alpha_1 + \frac{3}{10}\alpha_2 + \frac{1}{5}\alpha_3.$$

If $\gamma = e_2$, then $p = 0, q = 1, r = 0$ and so

$$x = -\frac{1}{5}, y = \frac{1}{5} \text{ and } z = -\frac{1}{5}.$$

$$\therefore e_2 = -\frac{1}{5}\alpha_1 + \frac{1}{5}\alpha_2 - \frac{1}{5}\alpha_3.$$

Finally if $\gamma = e_3$, then $p = 0, q = 0, r = 1$ and so

$$x = -\frac{3}{10}, y = \frac{3}{10} \text{ and } z = \frac{1}{5}.$$

$$\therefore e_3 = -\frac{3}{10}\alpha_1 + \frac{3}{10}\alpha_2 + \frac{1}{5}\alpha_3.$$

Example 9: Show that a system X consisting of the vectors

$$\alpha_1 = (1, 0, 0, 0), \alpha_2 = (0, 1, 0, 0), \alpha_3 = (0, 0, 1, 0) \text{ and } \alpha_4 = (0, 0, 0, 1)$$

is a basis set of $\mathbf{R}^4$.

Solution: First we show that the set X is a linearly independent set of vectors.

If a_1, a_2, a_3, a_4 be any scalars i.e., elements of the field $\mathbf{R}$, then

$$a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3 + a_4\alpha_4 = \text{zero vector}$$

$$\Rightarrow a_1(1, 0, 0, 0) + a_2(0, 1, 0, 0) + a_3(0, 0, 1, 0) + a_4(0, 0, 0, 1) = (0, 0, 0, 0)$$

$$\Rightarrow (a_1, a_2, a_3, a_4) = (0, 0, 0, 0)$$

$$\Rightarrow a_1 = 0, a_2 = 0, a_3 = 0, a_4 = 0.$$

Therefore the given set X of four vectors is linearly independent.

Now we shall show that X generates $\mathbf{R}^4$ i.e., each vector of $\mathbf{R}^4$ can be expressed as a linear combination of the vectors of X .

Let (a, b, c, d) be any vector in $\mathbf{R}^4$. We can write

$$\begin{aligned}
 (a, b, c, d) &= a(1, 0, 0, 0) + b(0, 1, 0, 0) + c(0, 0, 1, 0) + d(0, 0, 0, 1) \\
 &= a\alpha_1 + b\alpha_2 + c\alpha_3 + d\alpha_4.
 \end{aligned}$$

Thus (a, b, c, d) has been expressed as a linear combination of the vectors of X and so X generates $\mathbf{R}^4$.

Since X is a linearly independent subset of $\mathbf{R}^4$ and it also generates $\mathbf{R}^4$, therefore it is a basis of $\mathbf{R}^4$.

Example 10: Show that the set $S = \{1, x, x^2, \dots, x^n\}$ of $n+1$ polynomials in x is a basis of the vector space $P_n(\mathbf{R})$, of all polynomials in x (of degree at most n) over the field of real numbers.

Solution: $P_n(\mathbf{R})$ is the vector space of all polynomials in x (of degree at most n) over the field $\mathbf{R}$ of real numbers.

$S = \{1, x, x^2, \dots, x^n\}$ is a subset of P_n consisting of $n+1$ polynomials. To prove that S is a basis of the vector space $P_n(\mathbf{R})$.

First we show that the vectors in the set S are linearly independent over the field $\mathbf{R}$.

The zero vector of the vector space $P_n(\mathbf{R})$ is the zero polynomial. Let $a_0, a_1, a_2, \dots, a_n \in \mathbf{R}$ be such that

$$a_0(1) + a_1x + a_2x^2 + \dots + a_nx^n = 0 \quad \text{i.e., zero polynomial.}$$

Now by the definition of the equality of two polynomials, we have

$$a_0 + a_1x + a_2x^2 + \dots + a_nx^n = 0$$

$$\Rightarrow a_0 = 0, a_1 = 0, a_2 = 0, \dots, a_n = 0.$$

$\therefore$ the vectors $1, x, x^2, \dots, x^n$ of the vector space $P_n(\mathbf{R})$ are linearly independent.

Now we shall show that the set S generates the whole vector space $P_n(\mathbf{R})$.

Let $\alpha = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ be any arbitrary member of P_n , where $a_0, a_1, \dots, a_n \in \mathbf{R}$. Then α is a linear combination of the polynomials $1, x, x^2, \dots, x^n$ over the field $\mathbf{R}$. Therefore S generates $P_n(\mathbf{R})$.

Since S is a linearly independent subset of $P_n(\mathbf{R})$ and it also generates $P_n(\mathbf{R})$, therefore it is a basis of $P_n(\mathbf{R})$.

Example 11: Show that a finite subset W of a vector space $V(F)$ is linearly dependent if and only if some element of W can be expressed as a linear combination of the others.

Solution: Let $W = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a finite subset of a vector space $V(F)$.

First suppose that W is linearly dependent. Then to show that some vector of W can be expressed as a linear combination of the others.

Since the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly dependent, therefore there exist scalars $a_1, a_2, \dots, a_n$ not all zero such that

$$a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = \mathbf{0}. \quad \dots(1)$$

Suppose $a_r \neq 0, 1 \leq r \leq n$.

Then from (1), we have

$$a_r\alpha_r = -a_1\alpha_1 - a_2\alpha_2 - \dots - a_{r-1}\alpha_{r-1} - a_{r+1}\alpha_{r+1} - \dots - a_n\alpha_n$$

$$\text{or } a_r^{-1}(a_r\alpha_r) = a_r^{-1}(-a_1\alpha_1 - a_2\alpha_2 - \dots - a_{r-1}\alpha_{r-1} - a_{r+1}\alpha_{r+1} - \dots - a_n\alpha_n)$$

$$\text{or } \alpha_r = (-a_r^{-1}a_1)\alpha_1 + (-a_r^{-1}a_2)\alpha_2 + \dots + (-a_r^{-1}a_{r-1})\alpha_{r-1} + (-a_r^{-1}a_{r+1})\alpha_{r+1} + \dots + (-a_r^{-1}a_n)\alpha_n.$$

Thus α_r is a linear combination of the other vectors of the set W .

Conversely suppose that some vector of W is a linear combination of the others. Then to prove that W is linearly dependent.

Without loss of generality suppose α_1 is a linear combination of the vectors $\alpha_2, \dots, \alpha_n$.

Let $\alpha_1 = b_2\alpha_2 + b_3\alpha_3 + \dots + b_n\alpha_n$,

where $b_2, b_3, \dots, b_n \in F$.

Then $\alpha_1 - b_2\alpha_2 - b_3\alpha_3 - \dots - b_n\alpha_n = 0$(2)

Since in the linear relation (2) among the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$, the scalar coefficient of α_1 is 1 which is $\neq 0$, therefore the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly dependent.

Example 12: If n vectors span a vector space V containing r linearly independent vectors, then show that $n \geq r$. (Kumaun 2013)

Solution: Suppose a subset S of V containing n vectors spans V . Then there exists a linearly independent subset T of S which also spans V . This subset T of S will form a basis of V . Suppose the number of vectors in T is m . Then $\dim V = m \leq n$.

Since $\dim V = m$, therefore any subset of V containing more than m vectors will be linearly dependent.

Hence if S_1 is a linearly independent subset of V and S_1 contains r vectors, we must have

$$r \leq m \Rightarrow r \leq n. \quad [\because m \leq n]$$

Example 13: Let W_1 and W_2 be distinct subspaces of V and that $\dim W_1 = 4$, $\dim W_2 = 4$ and $\dim V = 6$. Find the possible dimensions of $W_1 \cap W_2$.

Solution: Since W_1 and W_2 are distinct, so $W_1 + W_2$ properly contains W_1 and W_2 i.e., $\dim(W_1 + W_2) > 4$. Also $\dim V = 6 \Rightarrow \dim(W_1 + W_2)$ cannot be greater than 6. Thus we have two possibilities i.e., dimension of $(W_1 + W_2)$ must be either 5 or 6.

Using theorem $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$, we have

(i) $5 = 4 + 4 - \dim(W_1 \cap W_2)$ or $\dim W_1 \cap W_2 = 3$

(ii) $6 = 4 + 4 - \dim(W_1 \cap W_2)$ or $\dim W_1 \cap W_2 = 2$.

Hence the possible dimensions of $W_1 \cap W_2$ are 2, 3.

Comprehensive Exercise 1

1. Give a basis for each of the following vector spaces over the indicated fields.

(i) $\mathbf{R}(\sqrt{2})$ over $\mathbf{R}$ (ii) $\mathbf{C}$ over $\mathbf{R}$ (iii) $\mathbf{Q}(2^{1/4})$ over $\mathbf{Q}$

where $\mathbf{Q}, \mathbf{R}, \mathbf{C}$ are fields of rational, real and complex numbers respectively.

2. Find the dimension of $\mathbf{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbf{Q}$.

3. Find bases for the subspaces

$$W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}, W_2 = \{(0, b, c) : b, c \in \mathbf{R}\} \text{ of } \mathbf{R}^3.$$

Find the dimensions of W_1, W_2 and $W_1 \cap W_2$.

4. Show that the real field $\mathbf{R}$ is a vector space of infinite dimension over the rational field $\mathbf{Q}$.
5. Let V be the vector space of all 2×2 symmetric matrices over F . Show that $\dim V = 3$.

6. Prove that the space of all $m \times n$ matrices over the field F has dimension mn , by exhibiting a basis for this space.
7. If $\{\alpha_1, \alpha_2, \alpha_3\}$ is a basis of $V_3(\mathbf{R})$, show that $\{\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1\}$ is also a basis of $V_3(\mathbf{R})$.
8. If W_1 and W_2 are finite-dimensional subspaces with the same dimension, and if $W_1 \subseteq W_2$, then $W_1 = W_2$.
9. In the vector space $\mathbf{R}^3$, let $\alpha = (1, 2, 1), \beta = (3, 1, 5), \gamma = (3, -4, 7)$. Show that there exists more than one basis for the subspace spanned by the set $S = \{\alpha, \beta, \gamma\}$.
10. For the 3-dimensional space $\mathbf{R}^3$ over the field of real numbers $\mathbf{R}$, determine if the set $\{(2, -1, 0), (3, 5, 1), (1, 1, 2)\}$ is a basis.
11. Show that the set $\{(1, i, 0), (2i, 1, 1), (0, 1 + i, 1 - i)\}$ is a basis for $V_3(\mathbf{C})$.
(Kumaun 2011)
12. Find three vectors in $\mathbf{R}^3$ which are linearly dependent, and are such that any two of them are linearly independent.
13. Select a basis, if any, of $\mathbf{R}^3(\mathbf{R})$ from the set $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$, where

$$\alpha_1 = (1, -3, 2), \alpha_2 = (2, 4, 1), \alpha_3 = (3, 1, 3), \alpha_4 = (1, 1, 1).$$
14. If W is a subspace of a finite dimensional vector space V , prove that any basis of W can be extended to form a basis of V .
15. Prove that any finite set S of vectors, not all the zero vectors, contains a linearly independent subset T which spans the same space as S .
16. Let V be a vector space. Let W be a subspace of V generated by the vectors $\alpha_1, \dots, \alpha_s$. Prove that W is spanned by a linearly independent subset of $\alpha_1, \dots, \alpha_s$.
17. If a vector space V is spanned by a finite set of m vectors, then show that any linearly independent set of vectors in V has at most m elements.
18. In a vector space V over the field F , let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ span V . Prove that the following two statements are equivalent :
 (i) B is linearly independent
 (ii) If $\alpha \in V$, then the expression $\alpha = \sum_{i=1}^n a_i \alpha_i$ with $a_i \in F$ is unique.
19. Determine a basis of the subspace spanned by the vectors
 $\alpha_1 = (1, 2, 3), \alpha_2 = (2, 1, -1), \alpha_3 = (1, -1, -4), \alpha_4 = (4, 2, -2).$
20. Let W be the subspace of $\mathbf{R}^4$ generated by the vectors $(1, -2, 5, -3), (2, 3, 1, -4)$ and $(3, 8, -3, -5)$. Find a basis and the dimension of W . Extend the basis of W to a basis of the whole space $\mathbf{R}^4$.
21. Let E be a subfield of a field F and F a subfield of a field K i.e., $E \subset F \subset K$. Let K be of dimension n over F and F is of dimension m over E . Show that K is of dimension mn over E .
22. Let W_1 and W_2 be the subspaces of $\mathbf{R}^4$:

$$W_1 = \{(a, b, c, d) : b + c + d = 0\}, W_2 = \{(a, b, c, d) : a + b = 0, c = 2d\}.$$
 Find the dimension of $W_1 \cap W_2$.

23. Let W_1 and W_2 be subspaces of V and that $\dim W_1 = 4$, $\dim W_2 = 5$ and $\dim V = 7$. Find the possible dimensions of $W_1 \cap W_2$.
24. Let W_1 and W_2 be the subspaces of $\mathbf{R}^4$ generated by the sets
 $B_1 = \{(1, 1, 0, -1), (1, 2, 3, 0), (2, 3, 3, -1)\}$
 and $B_2 = \{(1, 2, 2, -2), (2, 3, 2, -3), (1, 3, 4, -3)\}$ respectively.
 Find (i) $\dim(W_1 + W_2)$ (ii) $\dim(W_1 \cap W_2)$.

Answers 1

1. (i) $\{1, \sqrt{2}\}$ (ii) $\{1, i\}$ (iii) $\{1, 2^{1/4}\}$
2. 4
3. Basis for $W_1 = \{(1, 0, 0), (0, 1, 0)\}$; $\dim W_1 = 2$
 Basis for $W_2 = \{(0, 1, 0), (0, 0, 1)\}$; $\dim W_2 = 2$; $\dim(W_1 \cap W_2) = 1$
10. Yes 12. $(1, 0, 0), (0, 1, 0), (1, 1, 0)$
13. $\{\alpha_1, \alpha_2, \alpha_4\}$ 19. $\{\alpha_1, \alpha_2\}$
20. A basis of $W = \{(1, -2, 5, -3), (0, 7, -9, 2)\}$; $\dim W = 2$
 A basis of $\mathbf{R}^4$ is $\{(1, -2, 5, -3), (0, 7, -9, 2), (0, 0, 1, 0), (0, 0, 0, 1)\}$
22. 1 23. 2, 3 or 4
24. (i) 3 (ii) 1

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The vector space $V_3(\mathbf{R})$ is of dimension
 (a) 1 (b) 3
 (c) 2 (d) None of these.
2. Let V be the vector space of all 2×2 matrices over the field F . The dimension of V is
 (a) 2 (b) 3
 (c) 4 (d) None of these. (Garhwal 2007)
3. The dimension of the vector space $\mathbf{C}^2(\mathbf{R})$ is
 (a) 2 (b) 4
 (c) 1 (d) None of these.
4. The dimension of the vector space $\mathbf{C}(\mathbf{R})$ is
 (a) 1 (b) 2
 (c) 4 (d) None of these.

5. Let W_1 and W_2 be finite dimensional subspaces of a vector space V . If $\dim W_1 = 2$, $\dim W_2 = 2$, $\dim (W_1 + W_2) = 3$ then $\dim (W_1 \cap W_2)$ is
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4.
6. The dimension of the vector space spanned by $(1, -2, 3, -1)$ and $(1, 1, -2, 3)$ is
 - (a) 1
 - (b) 2
 - (c) 4
 - (d) None of these.
7. The dimension of the vector space spanned by $(3, -6, 3, -9)$ and $(-2, 4, -2, 6)$ is
 - (a) 1
 - (b) 2
 - (c) 4
 - (d) None of these.
8. In the vector space $\mathbf{R}^3$, the coordinate vector of $(3, 5, -2)$ relative to the ordered basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is
 - (a) $(5, 3, -2)$
 - (b) $(3, 5, -2)$
 - (c) $(-2, 5, 3)$
 - (d) $(-2, 3, 5)$.

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. Each set of $(n + 1)$ or more vectors of a finite dimensional vector space $V (F)$ of dimension is linearly dependent.
2. If W_1, W_2 are two subspaces of a finite dimensional vector space $V (F)$, then $\dim (W_1 + W_2) + \dim (W_1 \cap W_2) = \dots\dots\dots$
3. The set $S = \dots\dots\dots$ of $n + 1$ polynomials in x is a basis of the vector space $P_n(\mathbf{R})$, of all polynomials in x (of degree at most n) over the field of real numbers.
4. Let V be the vector space of $m \times n$ matrices over a field F . Then $\dim V = \dots\dots\dots$
5. If W is a proper subspace of a finite dimensional vector space V , then W is finite dimensional and
(Garhwal 2008)
6. The number of members in any basis of a vector space is called
(Garhwal 2010, 12)

True or False

Write 'T' for true and 'F' for false statement.

1. If a subset of an n -dimensional vector space V consists of n non-zero vectors, then it will be a basis of V .
2. If A and B are subspaces of a vector space, then $\dim A < \dim B \Rightarrow A \subset B$.
3. If A and B are subspaces of a vector space, then $A \neq B \Rightarrow \dim A \neq \dim B$.
4. If M and N are finite dimensional subspaces with the same dimension, and if $M \subset N$, then $M = N$.
5. In an n -dimensional space any subset consisting of n linearly independent vectors will form a basis.

Answers

Multiple Choice Questions

- | | | | |
|--------|--------|--------|--------|
| 1. (b) | 2. (c) | 3. (b) | 4. (b) |
| 5. (a) | 6. (b) | 7. (a) | 8. (b) |

Fill in the Blank(s)

- | | |
|--------------------------------|--------------------------|
| 1. n | 2. $\dim W_1 + \dim W_2$ |
| 3. $\{1, x, x^2, \dots, x^n\}$ | 4. mn |
| 5. $\dim W < \dim V$ | 6. dimension |

True or False

- | | | | |
|--------|--------|--------|--------|
| 1. F | 2. F | 3. F | 4. T |
| 5. T | | | |



Chapter

4

Homomorphism of Vector Spaces

1 Homomorphism of Vector Spaces or Linear Transformation

Definition: Let $U(F)$ and $V(F)$ be two vector spaces. Then a mapping $f: U \rightarrow V$

is called a homomorphism or a linear transformation of U into V , if

$$(i) \quad f(\alpha + \beta) = f(\alpha) + f(\beta), \forall \alpha, \beta \in U$$

$$\text{and } (ii) \quad f(a\alpha) = af(\alpha) \quad \forall a \in F, \quad \forall \alpha \in U.$$

The conditions (i) and (ii) can be combined into a single condition

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta) \quad \forall a, b \in F \quad \text{and} \quad \forall \alpha, \beta \in U.$$

If f is a homomorphism of U onto V , then V is called a homomorphic image of U .

Theorem: If f is a homomorphism of $U(F)$ into $V(F)$, then

$$(i) \quad f(\mathbf{0}) = \mathbf{0}' \quad \text{where } \mathbf{0} \text{ and } \mathbf{0}' \text{ are the zero vectors of } U \text{ and } V \text{ respectively.}$$

$$(ii) \quad f(-\alpha) = -f(\alpha) \quad \forall \alpha \in U.$$

Proof: (i) Let $\alpha \in U$. Then $f(\alpha) \in V$. Since $\mathbf{0}'$ is the zero vector of V , therefore

$$f(\alpha) + \mathbf{0}' = f(\alpha) = f(\alpha + \mathbf{0}) = f(\alpha) + f(\mathbf{0}).$$

Now V is an abelian group with respect to addition of vectors.

$$\therefore f(\alpha) + \mathbf{0}' = f(\alpha) + f(\mathbf{0})$$

$$\Rightarrow \mathbf{0}' = f(\mathbf{0}) \text{ by left cancellation law.}$$

(ii) If $\alpha \in U$, then $-\alpha \in U$. Also we have

$$\mathbf{0}' = f(\mathbf{0}) = f[\alpha + (-\alpha)] = f(\alpha) + f(-\alpha).$$

Now $f(\alpha) + f(-\alpha) = \mathbf{0}' \Rightarrow f(-\alpha) =$ the additive inverse of $f(\alpha)$

$$\Rightarrow f(-\alpha) = -f(\alpha).$$

Illustrative Examples

Example 1: Show that the mapping $f : V_3(F) \rightarrow V_2(F)$ defined by

$$f(a_1, a_2, a_3) = (a_1, a_2)$$

is a homomorphism of $V_3(F)$ onto $V_2(F)$.

(Kumaun 2011, 14)

Solution: Let $\alpha = (a_1, a_2, a_3)$ and $\beta = (b_1, b_2, b_3)$ be any two elements of $V_3(F)$. Also let a, b be any two elements of F . We have

$$\begin{aligned} f(a\alpha + b\beta) &= f[a(a_1, a_2, a_3) + b(b_1, b_2, b_3)] \\ &= f[(aa_1 + bb_1, aa_2 + bb_2, aa_3 + bb_3)] \\ &= (aa_1 + bb_1, aa_2 + bb_2) \\ &= a(a_1, a_2) + b(b_1, b_2) \\ &= af(a_1, a_2, a_3) + bf(b_1, b_2, b_3) \\ &= af(\alpha) + bf(\beta). \end{aligned}$$

$\therefore f$ is a linear transformation.

To show that f is onto $V_2(F)$. Let (a_1, a_2) be any element of $V_2(F)$. Then $(a_1, a_2, 0) \in V_3(F)$ and we have $f(a_1, a_2, 0) = (a_1, a_2)$. Therefore f is onto $V_2(F)$.

Therefore f is a homomorphism of $V_3(F)$ onto $V_2(F)$.

Comprehensive Exercise 1

1. Define linear transformation of a vector space $V(F)$ into a vector space $W(F)$. Show that the mapping

$$T : (a, b) \rightarrow (a + 2, b + 3)$$

of $V_2(\mathbb{R})$ into itself is not a linear transformation.

2. Let f be a linear transformation from a vector space U into a vector space V . If S is a subspace of U , prove that $f(S)$ will be a subspace of V .
3. If V is finite dimensional and f is a homomorphism of V into itself which is not onto prove that there is some $\alpha \neq \mathbf{0}$ in V such that $f(\alpha) = \mathbf{0}$.

2 Quotient Space

(Garhwal 2007, 10, 12; Gorakhpur 10)

Right and Left Cosets.

Let W be any subspace of a vector space V (F). Let α be any element of V . Then the set

$$W + \alpha = \{\gamma + \alpha : \gamma \in W\}$$

is called a **right coset** of W in V generated by α . Similarly the set

$$\alpha + W = \{\alpha + \gamma : \gamma \in W\}$$

is called a **left coset** of W in V generated by α .

Obviously $W + \alpha$ and $\alpha + W$ are both subsets of V . Since addition in V is commutative, therefore we have $W + \alpha = \alpha + W$. Hence we shall call $W + \alpha$ as simply a **coset** of W in V generated by α .

The following results about cosets are both to be remembered :

- (i) We have $\mathbf{0} \in V$ and $W + \mathbf{0} = W$. Therefore W itself is a coset of W in V .
- (ii) $\alpha \in W \Rightarrow W + \alpha = W$.

Proof: First we shall prove that $W + \alpha \subseteq W$.

Let $\gamma + \alpha$ be any arbitrary element of $W + \alpha$. Then $\gamma \in W$.

Now W is a subspace of V . Therefore

$$\gamma \in W, \alpha \in W$$

$$\Rightarrow \gamma + \alpha \in W.$$

Thus every element of $W + \alpha$ is also an element of W . Hence

$$W + \alpha \subseteq W.$$

Now we shall prove that $W \subseteq W + \alpha$.

Let $\beta \in W$. Since W is a subspace, therefore

$$\alpha \in W \Rightarrow -\alpha \in W.$$

Consequently $\beta \in W, -\alpha \in W \Rightarrow \beta - \alpha \in W$. Now we can write

$$\beta = (\beta - \alpha) + \alpha \in W + \alpha \text{ since } \beta - \alpha \in W.$$

Thus $\beta \in W \Rightarrow \beta \in W + \alpha$. Therefore $W \subseteq W + \alpha$.

Hence $W = W + \alpha$.

- (iii) If $W + \alpha$ and $W + \beta$ are two cosets of W in V , then

$$W + \alpha = W + \beta \Leftrightarrow \alpha - \beta \in W.$$

Proof: Since $\mathbf{0} \in W$, therefore $\mathbf{0} + \alpha \in W + \alpha$. Thus

$$\alpha \in W + \alpha.$$

Now $W + \alpha = W + \beta$

$$\Rightarrow \alpha \in W + \beta$$

$$\Rightarrow \alpha - \beta \in W + (\beta - \beta)$$

$$\Rightarrow \alpha - \beta \in W + \mathbf{0}$$

$$\Rightarrow \alpha - \beta \in W.$$

Conversely,

$$\alpha - \beta \in W$$

$$\Rightarrow W + (\alpha - \beta) = W$$

$$\Rightarrow W + [(\alpha - \beta) + \beta] = W + \beta$$

$$\Rightarrow W + \alpha = W + \beta.$$

Let V/W denote the set of all cosets of W in V i.e., let

$$V/W = \{W + \alpha : \alpha \in V\}.$$

We have just seen that if $\alpha - \beta \in W$, then $W + \alpha = W + \beta$. Thus a coset of W in V can have more than one representation.

Now if $V(F)$ is a vector space, then we shall give a vector space structure to the set V/W over the same field F . For this we shall have to define addition in V/W i.e., addition of cosets of W in V and multiplication of a coset by an element of F i.e., scalar multiplication.

Theorem: *If W is any subspace of a vector space $V(F)$, then the set V/W of all cosets $W + \alpha$ where α is any arbitrary element of V , is a vector space over F for the addition and scalar multiplication compositions defined as follows :*

$$(W + \alpha) + (W + \beta) = W + (\alpha + \beta) \quad \forall \alpha, \beta \in V$$

$$\text{and} \quad a(W + \alpha) = W + a\alpha; \quad a \in F, \alpha \in V.$$

Proof: We have $\alpha, \beta \in V \Rightarrow \alpha + \beta \in V$.

$$\text{Also} \quad a \in F, \alpha \in V \Rightarrow a\alpha \in V.$$

Therefore $W + (\alpha + \beta) \in V/W$ and also $W + a\alpha \in V/W$. Thus V/W is closed with respect to addition of cosets and scalar multiplication as defined above. Now first of all we shall show that these two compositions are well defined i.e., are independent of the particular representation chosen to denote a coset.

$$\text{Let} \quad W + \alpha = W + \alpha', \quad \alpha, \alpha' \in V$$

$$\text{and} \quad W + \beta = W + \beta', \quad \beta, \beta' \in V.$$

$$\text{We have} \quad W + \alpha = W + \alpha' \Rightarrow \alpha - \alpha' \in W$$

$$\text{and} \quad W + \beta = W + \beta' \Rightarrow \beta - \beta' \in W.$$

Now W is a subspace, therefore

$$\alpha - \alpha' \in W, \beta - \beta' \in W$$

$$\Rightarrow (\alpha - \alpha') + (\beta - \beta') \in W$$

$$\Rightarrow (\alpha + \beta) - (\alpha' + \beta') \in W$$

$$\Rightarrow W + (\alpha + \beta) = W + (\alpha' + \beta')$$

$$\Rightarrow (W + \alpha) + (W + \beta) = (W + \alpha') + (W + \beta').$$

Therefore addition in V/W is well defined.

$$\text{Again} \quad a \in F, \alpha - \alpha' \in W$$

$$\Rightarrow a(\alpha - \alpha') \in W$$

$$\Rightarrow a\alpha - a\alpha' \in W \Rightarrow W + a\alpha = W + a\alpha'.$$

$\therefore$ scalar multiplication in V/W is also well defined.

Commutativity of addition. Let $W + \alpha, W + \beta$ be any two elements of V/W . Then

$$\begin{aligned} (W + \alpha) + (W + \beta) &= W + (\alpha + \beta) \\ &= W + (\beta + \alpha) \\ &= (W + \beta) + (W + \alpha). \end{aligned}$$

Associativity of addition. Let $W + \alpha, W + \beta, W + \gamma$ be any three elements of V/W . Then

$$\begin{aligned}(W + \alpha) + [(W + \beta) + (W + \gamma)] &= (W + \alpha) + [W + (\beta + \gamma)] \\ &= W + [\alpha + (\beta + \gamma)] \\ &= W + [(\alpha + \beta) + \gamma] \\ &= [W + (\alpha + \beta)] + (W + \gamma) \\ &= [(W + \alpha) + (W + \beta)] + (W + \gamma).\end{aligned}$$

Existence of additive identity. If $\mathbf{0}$ is the zero vector of V , then

$$W + \mathbf{0} = W \in V/W.$$

If $W + \alpha$ is any element of V/W , then

$$(W + \mathbf{0}) + (W + \alpha) = W + (\mathbf{0} + \alpha) = W + \alpha.$$

$\therefore W + \mathbf{0} = W$ is the additive identity.

Existence of additive inverse. If $W + \alpha$ is any element of V/W , then

$$W + (-\alpha) = W - \alpha \in V/W.$$

Also we have

$$\begin{aligned}(W + \alpha) + (W - \alpha) &= W + (\alpha - \alpha) \\ &= W + \mathbf{0} \\ &= W.\end{aligned}$$

$\therefore W - \alpha$ is the additive inverse of $W + \alpha$.

Thus V/W is an abelian group with respect to addition composition. Further we observe that if

$$a, b \in F \text{ and } W + \alpha, W + \beta \in V/W, \text{ then}$$

1. $a [(W + \alpha) + (W + \beta)] = a [W + (\alpha + \beta)]$

$$\begin{aligned}&= W + a(\alpha + \beta) \\ &= W + (a\alpha + a\beta) \\ &= (W + a\alpha) + (W + a\beta) \\ &= a(W + \alpha) + a(W + \beta).\end{aligned}$$
2. $(a + b)(W + \alpha) = W + (a + b)\alpha$

$$\begin{aligned}&= W + (a\alpha + b\alpha) \\ &= (W + a\alpha) + (W + b\alpha) \\ &= a(W + \alpha) + b(W + \alpha).\end{aligned}$$
3. $(ab)(W + \alpha) = W + (ab)\alpha$

$$\begin{aligned}&= W + a(b\alpha) \\ &= a(W + b\alpha) \\ &= a[b(W + \alpha)].\end{aligned}$$
4. $1(W + \alpha) = W + 1\alpha$

$$= W + \alpha.$$

$\therefore V/W$ is a vector space over F for these two compositions. The vector space V/W is called the **Quotient Space** of V relative to W . The coset W is the zero vector of this vector space.

3 Dimension of a Quotient Space

Theorem: If W be a subspace of a finite dimensional vector space $V (F)$, then

$$\dim V/W = \dim V - \dim W.$$

(Garhwal 2006, 07, 08, 10, 14; Gorakhpur 10, 13)

Proof: Let m be the dimension of the subspace W of the vector space $V (F)$.
Let $S = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$

be a basis of W . Since S is a linearly independent subset of V , therefore it can be extended to form a basis of V . Let

$$S' = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l\}$$

be a basis of V . Then $\dim V = m + l$.

$$\therefore \dim V - \dim W = (m + l) - m = l.$$

So we should prove that $\dim V / W = l$.

We claim that the set of l cosets

$$S_1 = \{W + \beta_1, W + \beta_2, \dots, W + \beta_l\}$$

is a basis of V/W .

First we show that S_1 is linearly independent. The zero vector of V / W is W .

$$\text{Let } a_1 (W + \beta_1) + a_2 (W + \beta_2) + \dots + a_l (W + \beta_l) = W$$

$$\Rightarrow (W + a_1 \beta_1) + (W + a_2 \beta_2) + \dots + (W + a_l \beta_l) = W$$

$$\Rightarrow W + (a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l) = W + \mathbf{0}$$

$$\Rightarrow a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l \in W$$

$$\Rightarrow a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l = b_1 \alpha_1 + b_2 \alpha_2 + \dots + b_m \alpha_m$$

[$\because$ any vector in W can be expressed as a linear combination of its basis vectors]

$$\Rightarrow a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l - b_1 \alpha_1 - b_2 \alpha_2 - \dots - b_m \alpha_m = \mathbf{0}$$

$$\Rightarrow a_1 = 0, a_2 = 0, \dots, a_l = 0 \text{ since the vectors}$$

$$\beta_1, \beta_2, \dots, \beta_l, \alpha_1, \alpha_2, \dots, \alpha_m \text{ are linearly independent.}$$

$\therefore$ The set S_1 is linearly independent.

Now to show that $L(S_1) = V / W$. Let $W + \alpha$ be any element of V / W . The vector $\alpha \in V$ can be expressed as

$$\alpha = c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_m \alpha_m + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l$$

$$= \gamma + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l$$

$$\text{where } \gamma = c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_m \alpha_m \in W.$$

$$\text{So } W + \alpha = W + (\gamma + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l)$$

$$= (W + \gamma) + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l$$

$$= W + (d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l)$$

$$[\because \gamma \in W \Rightarrow W + \gamma = W]$$

$$= (W + d_1 \beta_1) + (W + d_2 \beta_2) + \dots + (W + d_l \beta_l)$$

$$= d_1 (W + \beta_1) + d_2 (W + \beta_2) + \dots + d_l (W + \beta_l).$$

Thus any element $W + \alpha$ of V / W can be expressed as a linear combination of elements of S_1 .

$$\therefore V/W = L(S_1).$$

$$\therefore S_1 \text{ is a basis of } V/W.$$

$$\therefore \dim V/W = l. \text{ Hence the theorem.}$$

4 Direct Sum of Spaces

(Garhwal 2006, 10)

Vector space as a direct sum of subspaces.

Definition: Let $V(F)$ be a vector space and let $W_1, W_2, \dots, W_m$ be subspaces of V . Then V is said to be the direct sum of $W_1, W_2, \dots, W_m$ if every element $\alpha \in V$ can be written in one and only one way as $\alpha = \alpha_1 + \alpha_2 + \dots + \alpha_m$ where

$$\alpha_1 \in W_1, \alpha_2 \in W_2, \dots, \alpha_m \in W_m.$$

If a vector space $V(F)$ is a direct sum of its two subspaces W_1 and W_2 then we should have not only $V = W_1 + W_2$ but also that each vector of V can be uniquely expressed as sum of an element of W_1 and an element of W_2 . Symbolically the direct sum is represented by the notation $V = W_1 \oplus W_2$.

Illustration 1: Let $V_2(F)$ be the vector space of all ordered pairs of F . Then $W_1 = \{(a, 0) : a \in F\}$ and $W_2 = \{(0, b) : b \in F\}$ are two subspaces of $V_2(F)$. Obviously any element $(x, y) \in V_2(F)$ can be uniquely expressed as the sum of an element of W_1 and an element of W_2 . The unique expression is $(x, y) = (x, 0) + (0, y)$. Thus $V_2(F)$ is the direct sum of W_1 and W_2 . Also we observe that the only element common to both W_1 and W_2 is the zero vector $(0, 0)$.

Illustration 2: Let $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$ and $W_2 = \{(0, 0, c) : c \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$.

Now any vector $(a, b, c) \in \mathbf{R}^3$ can be uniquely expressed as the sum of a vector in W_1 and a vector in W_2 : $(a, b, c) = (a, b, 0) + (0, 0, c)$.

Thus $\mathbf{R}^3$ is the direct sum of W_1 and W_2 .

Also we observe that the only element common to both W_1 and W_2 is the zero vector $(0, 0, 0)$.

Illustration 3: Let $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$ and $W_2 = \{(0, b, c) : b, c \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$. Then $\mathbf{R}^3 = W_1 + W_2$ since every vector in $\mathbf{R}^3$ is the sum of a vector in W_1 and a vector in W_2 .

However, $\mathbf{R}^3$ is not the direct sum of W_1 and W_2 since such sums are not unique.

For example, $(4, 6, 8) = (4, 5, 0) + (0, 1, 8)$.

Also $(4, 6, 8) = (4, -1, 0) + (0, 7, 8)$.

Here we observe that $W_1 \cap W_2 \neq \{0\}$.

5 Disjoint Subspaces

Definition: Two subspaces W_1 and W_2 of the vector space $V(F)$ are said to be disjoint if their intersection is the zero subspace i.e., if $W_1 \cap W_2 = \{0\}$.

Theorem: The necessary and sufficient conditions for a vector space $V(F)$ to be a direct sum of its two subspaces W_1 and W_2 are that

(i) $V = W_1 + W_2$

(ii) $W_1 \cap W_2 = \{0\}$ i.e., W_1 and W_2 are disjoint. (Garhwal 2010, 14; Gorakhpur 11)

Proof: The conditions are necessary.

Let V be direct sum of its two subspaces W_1 and W_2 . Then each element of V is expressible uniquely as sum of an element of W_1 and an element of W_2 . Therefore we have

$$V = W_1 + W_2.$$

Let, if possible $\mathbf{0} \neq \alpha \in W_1 \cap W_2$. Then $\alpha \in W_1, \alpha \in W_2$. Also $\alpha \in V$ and we can write

$$\alpha = \mathbf{0} + \alpha \text{ where } \mathbf{0} \in W_1, \alpha \in W_2$$

and $\alpha = \alpha + \mathbf{0}$ where $\alpha \in W_1, \mathbf{0} \in W_2$.

Thus $\alpha \in V$ can be expressed in at least two different ways as sum of an element of W_1 and an element of W_2 . This contradicts the fact that V is direct sum of W_1 and W_2 . Hence $\mathbf{0}$ is the only vector common to both W_1 and W_2 i.e., $W_1 \cap W_2 = \{\mathbf{0}\}$. Thus the conditions are necessary.

The conditions are sufficient.

Let $V = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$. Then to show that V is direct sum of W_1 and W_2 . $V = W_1 + W_2 \Rightarrow$ that each element of V can be expressed as sum of an element of W_1 and an element of W_2 . Now to show that this expression is unique.

Let, if possible,

$$\alpha = \alpha_1 + \alpha_2, \alpha \in V, \alpha_1 \in W_1, \alpha_2 \in W_2,$$

and $\alpha = \beta_1 + \beta_2, \beta_1 \in W_1, \beta_2 \in W_2$.

Then to show that $\alpha_1 = \beta_1$ and $\alpha_2 = \beta_2$.

We have $\alpha_1 + \alpha_2 = \beta_1 + \beta_2$

$$\Rightarrow \alpha_1 - \beta_1 = \beta_2 - \alpha_2.$$

Since W_1 is a subspace, therefore

$$\alpha_1 \in W_1, \beta_1 \in W_1 \Rightarrow \alpha_1 - \beta_1 \in W_1.$$

Similarly, $\beta_2 - \alpha_2 \in W_2$.

$$\therefore \alpha_1 - \beta_1 = \beta_2 - \alpha_2 \in W_1 \cap W_2.$$

But $\mathbf{0}$ is the only vector which belongs to $W_1 \cap W_2$. Therefore

$$\alpha_1 - \beta_1 = \mathbf{0} \Rightarrow \alpha_1 = \beta_1.$$

Also $\beta_2 - \alpha_2 = \mathbf{0} \Rightarrow \alpha_2 = \beta_2$.

Thus each vector $\alpha \in V$ is uniquely expressible as sum of an element of W_1 and an element of W_2 . Hence $V = W_1 \oplus W_2$.

6 Dimension of a Direct Sum

Theorem 1: If a finite dimensional vector space V (F) is a direct sum of two subspaces W_1 and W_2 , then

$$\dim V = \dim W_1 + \dim W_2. \quad (\text{Gorakhpur 2012, 14})$$

Proof: Let $\dim W_1 = m$ and $\dim W_2 = l$. Also let the sets of vectors

$$S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_m\} \quad \text{and} \quad S_2 = \{\beta_1, \beta_2, \dots, \beta_l\}$$

be the bases of W_1 and W_2 respectively.

We have $\dim W_1 + \dim W_2 = m + l$.

In order to prove that $\dim V = \dim W_1 + \dim W_2$, we should therefore prove that $\dim V = m + l$. We claim that the set

$$S = S_1 \cup S_2 = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l\}$$

is a basis of V .

First we show that the set S is linearly independent. Let

$$a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m + b_1\beta_1 + b_2\beta_2 + \dots + b_l\beta_l = \mathbf{0}.$$

$$\Rightarrow a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m = -(b_1\beta_1 + b_2\beta_2 + \dots + b_l\beta_l).$$

$$\text{Now } a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m \in W_1$$

$$\text{and } -(b_1\beta_1 + b_2\beta_2 + \dots + b_l\beta_l) \in W_2.$$

$$\text{Therefore } a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m \in W_1 \cap W_2$$

$$\text{and } -(b_1\beta_1 + b_2\beta_2 + \dots + b_l\beta_l) \in W_1 \cap W_2.$$

But V is the direct sum of W_1 and W_2 . Therefore $\mathbf{0}$ is the only vector belonging to $W_1 \cap W_2$. Then we have

$$a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m = \mathbf{0}, b_1\beta_1 + b_2\beta_2 + \dots + b_l\beta_l = \mathbf{0}.$$

Since both the sets $\{\alpha_1, \alpha_2, \dots, \alpha_m\}$ and $\{\beta_1, \beta_2, \dots, \beta_l\}$ are linearly independent, therefore we have

$$a_1 = 0, a_2 = 0, \dots, a_m = 0, b_1 = 0, b_2 = 0, \dots, b_l = 0.$$

Therefore S is linearly independent.

Now we shall show that $L(S) = V$. Let α be any element of V . Then

$$\alpha = \text{an element of } W_1 + \text{an element of } W_2.$$

$$= \text{a linear combination of } S_1 + \text{a linear combination of } S_2$$

$$= \text{a linear combination of elements of } S.$$

$$\therefore L(S) = V$$

$$\therefore S \text{ is a basis of } V. \text{ Therefore } \dim V = m + l.$$

Hence the theorem.

A sort of converse of this theorem is true. It has been proved in the following theorem.

Theorem 2: Let V be a finite dimensional vector space and let W_1, W_2 be subspaces of V such that $V = W_1 + W_2$ and $\dim V = \dim W_1 + \dim W_2$.

$$\text{Then } V = W_1 \oplus W_2.$$

Proof: Let $\dim W_1 = l$ and $\dim W_2 = m$. Then

$$\dim V = l + m.$$

$$\text{Let } S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_l\}$$

be a basis of W_1 and

$$S_2 = \{\beta_1, \beta_2, \dots, \beta_m\}$$

be a basis of W_2 . We shall show that $S_1 \cup S_2$ is a basis of V .

$$\text{Let } \alpha \in V. \text{ Since } V = W_1 + W_2, \text{ therefore we can write}$$

$$\alpha = \gamma + \delta \text{ where } \gamma \in W_1, \delta \in W_2.$$

Now $\gamma \in W_1$ can be expressed as a linear combination of the elements of S_1 and $\delta \in W_2$ can be expressed as a linear combination of the elements of S_2 . Therefore $\alpha \in V$ can be expressed as a linear combination of the elements of $S_1 \cup S_2$. Therefore $V = L(S_1 \cup S_2)$. Since $\dim V = l + m$ and $L(S_1 \cup S_2) = V$, therefore the number of distinct elements in $S_1 \cup S_2$ cannot be less than $l + m$. Thus $S_1 \cup S_2$ has $l + m$ distinct elements and therefore $S_1 \cup S_2$ is a basis of V . Therefore the set

$$\{\alpha_1, \alpha_2, \dots, \alpha_l, \beta_1, \beta_2, \dots, \beta_m\}$$

is linearly independent.

Now we shall show that

$$W_1 \cap W_2 = \{\mathbf{0}\}.$$

Let $\alpha \in W_1 \cap W_2$.

Then $\alpha \in W_1, \alpha \in W_2$.

Therefore $\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_l\alpha_l$

and $\alpha = b_1\beta_1 + b_2\beta_2 + \dots + b_m\beta_m$

for some a 's and b 's $\in F$.

$$\therefore a_1\alpha_1 + a_2\alpha_2 + \dots + a_l\alpha_l = b_1\beta_1 + b_2\beta_2 + \dots + b_m\beta_m$$

$$\Rightarrow a_1\alpha_1 + a_2\alpha_2 + \dots + a_l\alpha_l - b_1\beta_1 - b_2\beta_2 - \dots - b_m\beta_m = \mathbf{0}$$

$$\Rightarrow a_1 = 0, a_2 = 0, \dots, a_l = 0, b_1 = 0, b_2 = 0, \dots, b_m = 0$$

$$\Rightarrow \alpha = \mathbf{0}.$$

$$\therefore W_1 \cap W_2 = \{\mathbf{0}\}.$$

7 Complementary Subspaces

Definition: Let $V(F)$ be a vector space and W_1, W_2 be two subspaces of V . Then the subspace W_2 is called the complement of W_1 in V if V is the direct sum of W_1 and W_2 .

8 Existence of Complementary Subspaces

Theorem: Corresponding to each subspace W_1 of a finite dimensional vector space $V(F)$, there exists a subspace W_2 such that V is the direct sum of W_1 and W_2 .

Proof: Let $\dim W_1 = m$. Let the set

$$S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$$

be a basis of W_1 .

Since S_1 is a linearly independent subset of V , therefore S_1 can be extended to form a basis of V . Let the set

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l\}$$

be a basis of V .

Let W_2 be the subspace of V generated by the set

$$S_2 = \{\beta_1, \beta_2, \dots, \beta_l\}.$$

We shall prove that V is the direct sum of W_1 and W_2 . For this we shall prove that $V = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$.

Let α be any element of V . Then we can express

$$\begin{aligned}\alpha &= \text{a linear combination of elements of } S \\ &= \text{a linear combination of } S_1 + \text{a linear combination of } S_2 \\ &= \text{an element of } W_1 + \text{an element of } W_2.\end{aligned}$$

$$\therefore V = W_1 + W_2.$$

Again let $\beta \in W_1 \cap W_2$. Then β can be expressed as a linear combination of S_1 and also as a linear combination of S_2 . So we have

$$\begin{aligned}\beta &= a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m = b_1\beta_1 + b_2\beta_2 + \dots + b_l\beta_l. \\ \Rightarrow a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m - b_1\beta_1 - b_2\beta_2 - \dots - b_l\beta_l &= \mathbf{0} \\ \Rightarrow a_1 = 0, a_2 = 0, \dots, a_m = 0, b_1 = 0, b_2 = 0, \dots, b_l = 0 \\ &\quad \text{since } \alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l \text{ are linearly independent.} \\ \therefore \beta &= \mathbf{0} \text{ (zero vector).}\end{aligned}$$

$$\text{Thus } W_1 \cap W_2 = \{\mathbf{0}\}.$$

Hence V is the direct sum of W_1 and W_2 .

9 Direct Sum of Several Subspaces

Now we shall have some discussion on the direct sum of several subspaces. For this purpose we shall first define the concept of independence of subspaces analogous to the disjointness condition on two subspaces.

Definition: Suppose $W_1, W_2, \dots, W_k$ are subspaces of the vector space V . We shall say that $W_1, W_2, \dots, W_k$ are **independent**, if

$$\alpha_1 + \alpha_2 + \dots + \alpha_k = \mathbf{0}, \alpha_i \in W_i \text{ implies that each } \alpha_i = \mathbf{0}.$$

Theorem 1: Suppose V (F) is a vector space. Let $W_1, W_2, \dots, W_k$ be subspaces of V and let $W = W_1 + W_2 + \dots + W_k$. The following are equivalent :

- (i) $W_1, W_2, \dots, W_k$ are independent.
- (ii) Each vector α in W can be uniquely expressed in the form $\alpha = \alpha_1 + \alpha_2 + \dots + \alpha_k$ with α_i in $W_i, i = 1, \dots, k$.
- (iii) For each $j, 2 \leq j \leq k$, the subspace W_j is disjoint from the sum $(W_1 + \dots + W_{j-1})$.

Proof: In order to prove the equivalence of the three statements we shall prove that

$$(i) \Rightarrow (ii), (ii) \Rightarrow (iii) \text{ and } (iii) \Rightarrow (i).$$

$$(i) \Rightarrow (ii). \text{ Suppose } W_1, \dots, W_k \text{ are independent. Let } \alpha \in W.$$

Since $W = W_1 + \dots + W_k$, therefore we can write

$$\alpha = \alpha_1 + \dots + \alpha_k \text{ with } \alpha_i \text{ in } W_i.$$

Suppose that also $\alpha = \beta_1 + \dots + \beta_k$ with β_i in W_i .

$$\text{Then } \alpha_1 + \dots + \alpha_k = \beta_1 + \dots + \beta_k$$

$$\Rightarrow (\alpha_1 - \beta_1) + \dots + (\alpha_k - \beta_k) = \mathbf{0} \text{ with } \alpha_i - \beta_i \text{ in } W_i \text{ as } W_i \text{ is a subspace}$$

$$\Rightarrow \alpha_i - \beta_i = \mathbf{0} \quad [\because W_1, \dots, W_k \text{ are independent}]$$

$$\Rightarrow \alpha_i = \beta_i, i = 1, \dots, k.$$

Therefore the α_i 's are uniquely determined by α .

(ii) $\Rightarrow$ (iii). Let $\alpha \in W_j \cap (W_1 + \dots + W_{j-1})$.

Then $\alpha \in W_j$ and $\alpha \in W_1 + \dots + W_{j-1}$.

Now $\alpha \in W_1 + \dots + W_{j-1}$ implies that there exist vectors $\alpha_1, \dots, \alpha_{j-1}$ with α_i in W_i such that

$$\alpha = \alpha_1 + \dots + \alpha_{j-1}.$$

Also $\alpha \in W_j$.

Therefore we get two expressions for α as a sum of vectors, one in each W_i . These are

$$\alpha = \alpha_1 + \dots + \alpha_{j-1} + \mathbf{0} + \dots + \mathbf{0}$$

in which the vector belonging to W_j is $\mathbf{0}$

and $\alpha = \mathbf{0} + \dots + \mathbf{0} + \alpha + \dots + \mathbf{0}$

in which the vector belonging to W_j is α .

Since the expression for α is given to be unique, therefore we must have

$$\alpha_1 = \dots = \alpha_{j-1} = \mathbf{0} = \alpha.$$

Thus $W_j \cap (W_1 + \dots + W_{j-1}) = \{\mathbf{0}\}$.

(iii) $\Rightarrow$ (i).

Let $\alpha_1 + \dots + \alpha_k = \mathbf{0}$ where $\alpha_i \in W_i, i = 1, \dots, k$(1)

Then we are to prove that each $\alpha_i = \mathbf{0}$.

Suppose that for some i we have $\alpha_i \neq \mathbf{0}$.

Let j be the largest integer i between 1 and k such that $\alpha_i \neq \mathbf{0}$. Obviously j must be ≥ 2 and at the most j can be equal to k . Then (1) reduces to

$$\alpha_1 + \dots + \alpha_j = \mathbf{0}, \alpha_j \neq \mathbf{0}$$

$$\Rightarrow \alpha_j = -\alpha_1 - \dots - \alpha_{j-1}$$

$$\Rightarrow \alpha_j \in W_1 + \dots + W_{j-1} \quad [\because -\alpha_1 - \dots - \alpha_{j-1} \in W_1 + \dots + W_{j-1}]$$

$$\Rightarrow \alpha_j \in W_j \cap (W_1 + \dots + W_{j-1}) \quad [\Rightarrow \alpha_j \in W_j]$$

$$\Rightarrow \alpha_j = \mathbf{0}.$$

Thus we get a contradiction. Hence each $\alpha_i = \mathbf{0}$.

Note: If any (and hence all) of the three conditions of theorem 1 hold for $W_1, \dots, W_k$, then we shall say that W is the **direct sum** of $W_1, \dots, W_k$ and we write

$$W = W_1 \oplus \dots \oplus W_k.$$

Theorem 2: Let $V (F)$ be a vector space. Let $W_1, \dots, W_n$ be subspaces of V . Suppose that $V = W_1 + \dots + W_n$

and that $W_i \cap (W_1 + \dots + W_{i-1} + W_{i+1} + \dots + W_n) = \{\mathbf{0}\}$

for every $i = 1, 2, \dots, n$. Prove that V is the direct sum of $W_1, \dots, W_n$.

Proof: In order to prove that V is the direct sum of $W_1, \dots, W_n$, we should prove that each vector $\alpha \in V$ can be uniquely expressed as

$$\alpha = \alpha_1 + \dots + \alpha_n \text{ where } \alpha_i \in W_i, i = 1, \dots, n.$$

Since $V = W_1 + \dots + W_n$, therefore any vector α in V can be written as

$$\alpha = \alpha_1 + \dots + \alpha_n \text{ where } \alpha_i \in W_i. \quad \dots(1)$$

To show that $\alpha_1, \dots, \alpha_n$ are unique.

$$\text{Let } \alpha = \beta_1 + \dots + \beta_n \text{ where } \beta_i \in W_i. \quad \dots(2)$$

From (1) and (2), we get

$$\begin{aligned}\alpha_1 + \dots + \alpha_n &= \beta_1 + \dots + \beta_n \\ (\alpha_1 - \beta_1) + \dots + (\alpha_{i-1} - \beta_{i-1}) + (\alpha_i - \beta_i) \\ &\quad + (\alpha_{i+1} - \beta_{i+1}) + \dots + (\alpha_n - \beta_n) = \mathbf{0}. \quad \dots(3)\end{aligned}$$

Now each W_i is a subspace of V . Therefore $\alpha_i - \beta_i$ and also its additive inverse

$$\beta_i - \alpha_i \in W_i, i = 1, \dots, n.$$

From (3), we get

$$\begin{aligned}(\alpha_i - \beta_i) &= (\beta_1 - \alpha_1) + \dots + (\beta_{i-1} - \alpha_{i-1}) \\ &\quad + (\beta_{i+1} - \alpha_{i+1}) + \dots + (\beta_n - \alpha_n). \quad \dots(4)\end{aligned}$$

Now the vector on the right hand side of (4) and consequently the vector $\alpha_i - \beta_i$ is in $W_1 + \dots + W_{i-1} + W_{i+1} + \dots + W_n$.

Also $\alpha_i - \beta_i \in W_i$.

$$\therefore \alpha_i - \beta_i \in W_i \cap (W_1 + \dots + W_{i-1} + W_{i+1} + \dots + W_n).$$

But for every $i = 1, \dots, n$, it is given that

$$W_i \cap (W_1 + \dots + W_{i-1} + W_{i+1} + \dots + W_n) = \{\mathbf{0}\}.$$

Therefore $\alpha_i - \beta_i = \mathbf{0}, i = 1, \dots, n$

$$\Rightarrow \alpha_i = \beta_i, i = 1, \dots, n$$

$\Rightarrow$ the expression (1) for α is unique.

Hence V is the direct sum of $W_1, \dots, W_n$.

Theorem 3: Let $V (F)$ be a finite dimensional vector space and let $W_1, \dots, W_k$ be subspaces of V . Then the following two statements are equivalent.

(i) V is the direct sum of $W_1, \dots, W_k$.

(ii) If B_i is a basis of $W_i, i = 1, \dots, k$, then the union

$$B = \bigcup_{i=1}^k B_i \text{ is also a basis for } V.$$

Proof: Let $B_i = \{\alpha_1^i, \alpha_2^i, \dots, \alpha_{n_i}^i\}$ be a basis for W_i .

Here $n_i = \dim W_i =$ number of vectors in B_i . Also let B be the union of the bases B_i .

(i) $\Rightarrow$ (ii). It is given that V is the direct sum of $W_1, \dots, W_k$, therefore for any $\alpha \in V$, we can write

$$\alpha = \alpha_1 + \dots + \alpha_k \text{ for } \alpha_i \in W_i, i = 1, \dots, k.$$

Now α_i can be expressed as a linear combination of the vectors in B_i which is a basis of W_i . Therefore α can be expressed as a linear combination of the elements of $B = \bigcup_{i=1}^k B_i$.

Therefore $L(B) = V$ i.e., B spans V .

Now to show that B is linearly independent. Let

$$\sum_{i=1}^k (a_1^i \alpha_1^i + a_2^i \alpha_2^i + \dots + a_{n_i}^i \alpha_{n_i}^i) = \mathbf{0}. \quad \dots(1)$$

Since V is the direct sum of $W_1, \dots, W_k$, therefore $\mathbf{0} \in V$ can be uniquely expressed as a sum of vectors one in each W_i . This unique expression is

$$\mathbf{0} = \mathbf{0} + \dots + \mathbf{0} \text{ where } \mathbf{0} \in W_i, i = 1, \dots, k.$$

Now $a_1^i \alpha_1^i + \dots + a_{n_i}^i \alpha_{n_i}^i \in W_i$. Therefore from (1) which is an expression for $\mathbf{0} \in V$ as a sum of vectors one in each W_i , we get

$$a_1^i \alpha_1^i + \dots + a_{n_i}^i \alpha_{n_i}^i = \mathbf{0}, i = 1, \dots, k$$

$$a_1^i = \dots = a_{n_i}^i = 0$$

since $\{\alpha_1^i, \dots, \alpha_{n_i}^i\}$ is linearly independent being a basis for W_i .

Therefore $B = \bigcup_{i=1}^k B_i$ is linearly independent. Therefore B is a basis of V .

(ii) $\Rightarrow$ (i). It is given that $B = \bigcup_{i=1}^k B_i$ is a basis of V .

Therefore for any $\alpha \in V$, we can write

$$\begin{aligned} \alpha &= \sum_{i=1}^k (a_1^i \alpha_1^i + a_2^i \alpha_2^i + \dots + a_{n_i}^i \alpha_{n_i}^i) \\ &= \alpha_1 + \alpha_2 + \dots + \alpha_k \end{aligned} \quad \dots(2)$$

where $\alpha_i = a_1^i \alpha_1^i + \dots + a_{n_i}^i \alpha_{n_i}^i \in W_i$.

Thus each vector in V can be expressed as a sum of vectors one in each W_i .

Now V will be the direct sum of $W_1, \dots, W_k$ if the expression (2) for α is unique. Let

$$\alpha = \beta_1 + \beta_2 + \dots + \beta_k \quad \dots(3)$$

where $\beta_i = b_1^i \alpha_1^i + \dots + b_{n_i}^i \alpha_{n_i}^i \in W_i$.

From (2) and (3), we get

$$\alpha_1 + \dots + \alpha_k = \beta_1 + \dots + \beta_k$$

$$\Rightarrow (\alpha_1 - \beta_1) + \dots + (\alpha_i - \beta_i) + \dots + (\alpha_k - \beta_k) = \mathbf{0}$$

$$\Rightarrow \sum_{i=1}^k (\alpha_i - \beta_i) = \mathbf{0}$$

$$\Rightarrow \sum_{i=1}^k [(a_1^i - b_1^i) \alpha_1^i + \dots + (a_{n_i}^i - b_{n_i}^i) \alpha_{n_i}^i] = \mathbf{0}$$

$$\Rightarrow a_1^i - b_1^i = \dots = a_{n_i}^i - b_{n_i}^i = 0, i = 1, \dots, k$$

$$[\because \bigcup_{i=1}^k B_i \text{ is linearly independent being a basis of } V]$$

$$\Rightarrow a_1^i = b_1^i, \dots, a_{n_i}^i = b_{n_i}^i, i = 1, \dots, k$$

$$\Rightarrow \alpha_i = \beta_i, i = 1, \dots, k$$

$\Rightarrow$ the expression (2) for α is unique.

Hence V is the direct sum of $W_1, \dots, W_k$.

Note: While proving this theorem we have proved that if a finite dimensional vector space V (F) is the direct sum of its subspaces $W_1, \dots, W_k$, then

$$\dim V = \dim W_1 + \dots + \dim W_k.$$

Illustrative Examples

Example 2: Find dimension of V / W where $V = \mathbf{R}^3$, $W = \{(a, 0, 0) : a \in \mathbf{R}\}$.

Solution: Here $V = \mathbf{R}^3$. $\dim V = 3$.

$$W = \{(a, 0, 0) : a \in \mathbf{R}\}. \{(1, 0, 0)\} \text{ is a basis of } W \text{ so } \dim W = 1.$$

$$\text{Now } \dim V / W = \dim V - \dim W = 3 - 1 = 2.$$

Example 3: If $W_1 = \{(x, 0, z) : x, z \in \mathbf{R}\}$, $W_2 = \{(0, y, z) : y, z \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$ then show that $\mathbf{R}^3 = W_1 + W_2$ but $\mathbf{R}^3 \neq W_1 \oplus W_2$.

Solution: Any $(x, y, z) \in \mathbf{R}^3$ can be written as a sum of a vector of W_1 and a vector of W_2 as shown below.

$$\text{We have } (x, y, z) = \left(x, 0, \frac{1}{2}z\right) + \left(0, y, \frac{1}{2}z\right), \text{ where } \left(x, 0, \frac{1}{2}z\right) \in W_1$$

$$\text{and } \left(0, y, \frac{1}{2}z\right) \in W_2.$$

$$\text{Thus } \mathbf{R}^3 = W_1 + W_2.$$

$$\text{Also we have } (x, y, z) = \left(x, 0, \frac{1}{4}z\right) + \left(0, y, \frac{3}{4}z\right) \text{ where } \left(x, 0, \frac{1}{4}z\right) \in W_1 \text{ and}$$

$$\left(0, y, \frac{3}{4}z\right) \in W_2.$$

It follows that representation of (x, y, z) as a sum of a vector of W_1 and a vector of W_2 is not unique.

$$\text{Hence } \mathbf{R}^3 \neq W_1 \oplus W_2.$$

Example 4: If $W_1 = \{(a, b, c) : a = b = c\}$, $W_2 = \{(a, b, c) : a = 0\}$ are two subspaces of $\mathbf{R}^3$, show that $\mathbf{R}^3 = W_1 \oplus W_2$.

Solution: We claim that $\mathbf{R}^3 = W_1 + W_2$.

For if $\alpha = (a, b, c) \in \mathbf{R}^3$, then $\alpha = (a, a, a) + (0, b - a, c - a)$ where $(a, a, a) \in W_1$ and $(0, b - a, c - a) \in W_2$.

$$\text{Also we have } W_1 \cap W_2 = \{\mathbf{0}\}.$$

$$\begin{aligned} \text{For } \alpha = (a, b, c) \in W_1 \cap W_2 &\Rightarrow a = b = c \text{ and } a = 0 \\ \Rightarrow a = 0, b = 0, c = 0 &\text{ i.e., } \alpha = \mathbf{0}. \end{aligned}$$

$$\text{Thus } \mathbf{R}^3 = W_1 + W_2 \text{ and } W_1 \cap W_2 = \{\mathbf{0}\}.$$

$$\text{Hence } \mathbf{R}^3 = W_1 \oplus W_2.$$

Example 5: Construct three subspaces W_1, W_2, W_3 of a vector space V so that $V = W_1 \oplus W_2 = W_1 \oplus W_3$ but $W_2 \neq W_3$.

Solution: Take the vector space $V = \mathbf{R}^2$.

Obviously $W_1 = \{(a, 0) : a \in \mathbf{R}\}$,

$W_2 = \{(0, a) : a \in \mathbf{R}\}$, and $W_3 = \{(a, a) : a \in \mathbf{R}\}$,

are three subspaces of $\mathbf{R}^2$.

We have $V = W_1 + W_2$ and $W_1 \cap W_2 = \{(0, 0)\}$. $\therefore V = W_1 \oplus W_2$.

Also it can be easily shown that

$V = W_1 + W_3$ and $W_1 \cap W_3 = \{(0, 0)\}$. $\therefore V = W_1 \oplus W_3$.

Thus $V = W_1 \oplus W_2 = W_1 \oplus W_3$ but $W_2 \neq W_3$.

Comprehensive Exercise 2

- Let V be the vector space of square matrices of order n over the field $\mathbf{R}$. Let W_1 and W_2 be the subspaces of symmetric and antisymmetric matrices respectively. Show that $V = W_1 \oplus W_2$.
- Let V be the vector space of all functions from the real field $\mathbf{R}$ into $\mathbf{R}$. Let U be the subspace of even functions and W the subspace of odd functions. Show that $V = U \oplus W$.
- Let W_1, W_2 and W_3 be the following subspaces of $\mathbf{R}^3$:

$$W_1 = \{(a, b, c) : a + b + c = 0\}, W_2 = \{(a, b, c) : a = c\},$$

$$W_3 = \{(0, 0, c) : c \in \mathbf{R}\}.$$

Show that (i) $\mathbf{R}^3 = W_1 + W_2$; (ii) $\mathbf{R}^3 = W_1 + W_3$; (iii) $\mathbf{R}^3 = W_2 + W_3$.

When is the sum direct?

- Let W be a subspace of a vector space V over a field F . Show that $\alpha \in \beta + W$ iff $\alpha - \beta \in W$.
- Let W_1 and W_2 be two subspaces of a finite dimensional vector space V . If $\dim V = \dim W_1 + \dim W_2$ and $W_1 \cap W_2 = \{0\}$, prove that $V = W_1 \oplus W_2$.

Answers 2

- The sum is direct in (ii) and (iii)

10 Co-ordinates

Let $V(F)$ be a finite dimensional vector space. Let

$$B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$$

be an ordered basis for V . By an ordered basis we mean that the vectors of B have been enumerated in some well-defined way *i.e.*, the vectors occupying the first, second, ..., n th places in the set B are fixed.

Let $\alpha \in V$. Then there exists a unique n -tuple $(x_1, x_2, \dots, x_n)$ of scalars such that

$$\alpha = x_1 \alpha_1 + x_2 \alpha_2 + \dots + x_n \alpha_n = \sum_{i=1}^n x_i \alpha_i.$$

The n -tuple $(x_1, x_2, \dots, x_n)$ is called the n -tuple of **co-ordinates of α relative to the ordered basis B** . The scalar x_i is called the i th **coordinate of α relative to the ordered basis B** . The $n \times 1$ matrix

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$$

is called the **coordinate matrix of α relative to the ordered basis B** . We shall use the symbol $[\alpha]_B$ for the coordinate matrix of the vector α relative to the ordered basis B .

(Garhwal 2008)

It should be noted that for the same basis set B , the coordinates of the vector α are unique only with respect to a particular ordering of B . The basis set B can be ordered in several ways. The coordinates of α may change with a change in the ordering of B .

Note: If $V(F)$ is a vector space of dimension n , then coordinate vector of $\mathbf{0} \in V$ relative to any basis of V is always the zero n -tuple $(0, 0, \dots, 0)$.

11 Change of Basis

Suppose V is an n -dimensional vector space over the field F . Let B and B' be two ordered bases for V . If α is any vector in V , then we are now interested to know what is the relation between its coordinates with respect to B and its coordinates with respect to B' .

(Garhwal 2010)

Theorem 1: Let $V(F)$ be an n -dimensional vector space and let B and B' be two ordered bases for V . Then there is a unique necessarily invertible, $n \times n$ matrix A with entries in F such that

$$(1) \quad [\alpha]_B = A [\alpha]_{B'} \quad (\text{Garhwal 2008})$$

$$(2) \quad [\alpha]_{B'} = A^{-1} [\alpha]_B \text{ for every vector } \alpha \text{ in } V.$$

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_n\}$.

Then there exists a unique linear transformation T from V into V such that

$$T(\alpha_j) = \beta_j, \quad j = 1, 2, \dots, n. \quad \dots(1)$$

Since T maps a basis B onto a basis B' , therefore T is necessarily invertible. The matrix of T relative to B i.e., $[T]_B$ will be a unique $n \times n$ matrix with elements in F . Also this matrix will be invertible because T is invertible.

Let $[T]_B = A = [a_{ij}]_{n \times n}$. Then

$$T(\alpha_j) = \sum_{i=1}^n a_{ij} \alpha_i, \quad j = 1, 2, \dots, n. \quad \dots(2)$$

Let $x_1, x_2, \dots, x_n$ be the coordinates of α with respect to B and $y_1, y_2, \dots, y_n$ be the coordinates of α with respect to B' . Then

$$\alpha = y_1 \beta_1 + y_2 \beta_2 + \dots + y_n \beta_n = \sum_{j=1}^n y_j \beta_j$$

$$= \sum_{j=1}^n y_j T(\alpha_j) \quad [\text{From (1)}]$$

$$= \sum_{j=1}^n y_j \sum_{i=1}^n a_{ij} \alpha_i \quad [\text{From (2)}]$$

$$= \sum_{i=1}^n \left(\sum_{j=1}^n a_{ij} y_j \right) \alpha_i.$$

$$\text{Also} \quad \alpha = \sum_{i=1}^n x_i \alpha_i.$$

$$\therefore \quad x_i = \sum_{j=1}^n a_{ij} y_j$$

because the expression for α as a linear combination of elements of B is unique.

Now $[\alpha]_B$ is a column matrix of the type $n \times 1$. Also $[\alpha]_{B'}$ is a column matrix of the type $n \times 1$. The product matrix $A[\alpha]_{B'}$ will also be of the type $n \times 1$.

The i^{th} entry of $[\alpha]_B = x_i = \sum_{j=1}^n a_{ij} y_j = i^{\text{th}}$ entry of $A[\alpha]_{B'}$.

$$\therefore \quad [\alpha]_B = A[\alpha]_{B'} \quad \Rightarrow \quad A^{-1}[\alpha]_B = A^{-1}A[\alpha]_{B'}$$

$$\Rightarrow \quad A^{-1}[\alpha]_B = I[\alpha]_B \quad \Rightarrow \quad A^{-1}[\alpha]_B = [\alpha]_{B'}.$$

Note: The matrix $A = [T]_B$ is called the transition matrix from B to B' . It expresses the coordinates of each vector in V relative to B in terms of its coordinates relative to B' .

How to write the transition matrix from one basis to another ?

Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_n\}$ be two ordered bases for the n -dimensional vector space $V(F)$. Let A be the transition matrix from the basis B to the basis B' . Let T be the linear transformation from V into V which maps the basis B onto the basis B' . Then A is the matrix of T relative to B i.e., $A = [T]_B$. So in order to find the matrix A , we should first express each vector in the basis B' as a linear combination over F of the vectors in B . Thus we write the relations

$$\beta_1 = a_{11} \alpha_1 + a_{21} \alpha_2 + \dots + a_{n1} \alpha_n$$

$$\beta_2 = a_{12} \alpha_1 + a_{22} \alpha_2 + \dots + a_{n2} \alpha_n$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$\beta_n = a_{1n} \alpha_1 + a_{2n} \alpha_2 + \dots + a_{nn} \alpha_n.$$

Then the matrix $A = [a_{ij}]_{n \times n}$ i.e., A is the transpose of the matrix of coefficients in the above relations. Thus

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}.$$

Now suppose α is any vector in V . If $[\alpha]_B$ is the coordinate matrix of α relative to the basis B and $[\alpha]_{B'}$ its coordinate matrix relative to the basis B' then

$$[\alpha]_B = A [\alpha]_{B'}$$

$$\text{and} \quad [\alpha]_{B'} = A^{-1} [\alpha]_B.$$

Theorem 2: Let $B = \{\alpha_1, \dots, \alpha_n\}$ and $B' = \{\beta_1, \dots, \beta_n\}$ be two ordered bases for an n -dimensional vector space V (F). If $(x_1, \dots, x_n)$ is an ordered set of n scalars, let $\alpha = \sum_{i=1}^n x_i \alpha_i$

$$\text{and} \quad \beta = \sum_{i=1}^n x_i \beta_i.$$

Then show that $T(\alpha) = \beta$, where T is the linear operator on V defined by

$$T(\alpha_i) = \beta_i, \quad i = 1, 2, \dots, n.$$

Proof: We have $T(\alpha) = T\left(\sum_{i=1}^n x_i \alpha_i\right) = \sum_{i=1}^n x_i T(\alpha_i)$ [$\because T$ is linear]

$$= \sum_{i=1}^n x_i \beta_i = \beta.$$

Illustrative Examples

Example 6: Show that the set

$$S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$$

is a basis of $\mathbf{R}^3(\mathbf{R})$ where $\mathbf{R}$ is the field of real numbers. Hence find the coordinates of the vector (a, b, c) with respect to the above basis. (Kumaun 2010)

Solution: The dimension of the vector space $\mathbf{R}^3(\mathbf{R})$ is 3. If the set S is linearly independent, then S will form a basis of $\mathbf{R}^3(\mathbf{R})$ because S contains 3 vectors. Let x, y, z be scalars in $\mathbf{R}$ such that

$$x(1, 0, 0) + y(1, 1, 0) + z(1, 1, 1) = \mathbf{0} = (0, 0, 0)$$

$$\Rightarrow (x + y + z, y + z, z) = (0, 0, 0)$$

$$\Rightarrow x + y + z = 0, y + z = 0, z = 0$$

$$\Rightarrow x = 0, y = 0, z = 0$$

$$\Rightarrow \text{the set } S \text{ is linearly independent.}$$

$$\therefore S \text{ is a basis of } \mathbf{R}^3(\mathbf{R}).$$

Now to find the coordinates of (a, b, c) with respect to the ordered basis S . Let p, q, r be scalars in $\mathbf{R}$ such that

$$(a, b, c) = p(1, 0, 0) + q(1, 1, 0) + r(1, 1, 1)$$

$$\Rightarrow (a, b, c) = (p + q + r, q + r, r)$$

$$\Rightarrow p + q + r = a, q + r = b, r = c$$

$$\Rightarrow r = c, q = b - c, p = a - b.$$

Hence the coordinates of the vector (a, b, c) are (p, q, r) i.e., $(a - b, b - c, c)$.

Example 7: Find the coordinates of the vector $(2, 1, -6)$ of $\mathbf{R}^3$ relative to the basis

$$\alpha_1 = (1, 1, 2), \alpha_2 = (3, -1, 0), \alpha_3 = (2, 0, -1).$$

Solution: To find the coordinates of the vector $(2, 1, -6)$ relative to the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$, we shall express the vector $(2, 1, -6)$ as a linear combination of the vectors $\alpha_1, \alpha_2, \alpha_3$. Let p, q, r be scalars in $\mathbf{R}$ such that

$$(2, 1, -6) = p\alpha_1 + q\alpha_2 + r\alpha_3$$

$$\Rightarrow (2, 1, -6) = p(1, 1, 2) + q(3, -1, 0) + r(2, 0, -1)$$

$$\Rightarrow (2, 1, -6) = (p + 3q + 2r, p - q, 2p - r).$$

$$\therefore p + 3q + 2r = 2, \quad p - q = 1, \quad \dots(1)$$

$$\text{and } 2p - r = -6.$$

Solving the equations (1), we have

$$p = -7/8, q = -15/8$$

$$\text{and } r = 17/4.$$

Hence the required coordinates of the vector $(2, 1, -6)$ relative to the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$ are (p, q, r) i.e., $(-7/8, -15/8, 17/4)$.

Comprehensive Exercise 3

1. Show that the set $S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a basis of $\mathbf{C}^3$ ($\mathbf{C}$ is the field of complex numbers. Hence find the coordinates of the vector $(3 + 4i, 6i, 3 + 7i)$ in $\mathbf{C}^3$ with respect to the above basis.
2. Find the co-ordinate vector of $(3, 5, -2)$ relative to the basis $\{e_1, e_2, e_3\}$ where $e_1 = (1, 1, 1), e_2 = (0, 2, 3), e_3 = (0, 2, -1)$.
3. Let $B = \{\alpha_1, \alpha_2, \alpha_3\}$ be an ordered basis for $\mathbf{R}^3$, where

$$\alpha_1 = (1, 0, -1), \alpha_2 = (1, 1, 1), \alpha_3 = (1, 0, 0).$$

Obtain the coordinates of the vector (a, b, c) in the ordered basis B .

4. Let V be the vector space of all polynomial functions of degree less than or equal to two from the field of real numbers $\mathbf{R}$ into itself. For a fixed $t \in \mathbf{R}$, let $g_1(x) = 1, g_2(x) = x + t, g_3(x) = (x + t)^2$.

Prove that $\{g_1, g_2, g_3\}$ is a basis for V and obtain the coordinates of $c_0 + c_1x + c_2x^2$ in this ordered basis.

Answers 3

- | | |
|-------------------------------|--|
| 1. $(3 - 2i, -3 - i, 3 + 7i)$ | 2. $(3, -1, 2)$ |
| 3. $(b - c, b, a - 2b + c)$ | 4. $(c_0 - c_1t + c_2t^2, c_1 - 2c_2t, c_2)$ |

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- If W be a subspace of a finite dimensional vector space $V(F)$, then $\dim V/W =$
 - $\dim V - \dim W$
 - $\dim V + \dim W$
 - $(\dim V) / (\dim W)$
 - None of these.

(Garhwal 2010, 12)
- The set $S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a basis of $\mathbf{R}^3(\mathbf{R})$, then coordinates of the vector (a, b, c) with respect to the basis are
 - $(a + b, b - c, c)$
 - $(a - b, b - c, c)$
 - $(a - b, b + c, c)$
 - $(a - b, b - c, -c)$
- The dimension of $\frac{V}{W}$, where $V = \mathbf{R}^3$, $W = \{(a, 0, 0) : a \in \mathbf{R}\}$ is
 - 1
 - 2
 - 3
 - 4

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- Let V be the vector space of all functions from the real field $\mathbf{R}$ into $\mathbf{R}$. Let W_1 be the subspace of even functions and W_2 be the subspace of odd functions. Then $V = \dots\dots\dots$
- If W is an m -dimensional subspace of an n -dimensional vector space V , then the dimension of the quotient space $\frac{V}{W}$ is
(Garhwal 2006)
- Two subspaces W_1 and W_2 of the vector space $V(F)$ are said to be disjoint if $W_1 \cap W_2 = \dots\dots\dots$
- If a finite dimensional vector space $V(F)$ is a direct sum of two subspaces W_1 and W_2 , then $\dim V = \dots\dots\dots$

True or False

Write 'T' for true and 'F' for false statement.

- If a finite dimensional vector space $V(F)$ is a direct sum of two subspaces W_1 and W_2 , then $\dim V = \dim W_1 + \dim W_2$.
- The dimension of $\frac{V}{W}$, where $V = \mathbf{R}^3$, $W = \{(a, 0, 0) : a \in \mathbf{R}\}$ is 3.
- If $W_1 = \{(x, 0, z) : x, z \in \mathbf{R}\}$, $W_2 = \{(0, y, z) : y, z \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$ then $\mathbf{R}^3 \neq W_1 + W_2$.

Answers

Multiple Choice Questions

1. (a) 2. (b) 3. (c)

Fill in the Blank(s)

1. $W_1 \oplus W_2$ 2. $n - m$
3. $\{0\}$ 4. $\dim W_1 + \dim W_2$

True or False

1. T 2. F 3. F



Chapter

5



Matrices

1 Idempotent, Involutory, Nilpotent and Periodic Matrices

Idempotent Matrix. A matrix such that $\mathbf{A}^2 = \mathbf{A}$ is called **idempotent matrix**.

Example,

$$\mathbf{A} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

Involutory Matrix. A matrix $\mathbf{A}$ is said to be an **involutory matrix** if $\mathbf{A}^2 = \mathbf{I}$ (unit matrix). Example,

$$\mathbf{A} = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

Since $\mathbf{I}^2 = \mathbf{I}$ always, we conclude that a **Unit matrix** is always involutory.

Nilpotent Matrix. A matrix is said to be a **nilpotent matrix** if $\mathbf{A}^k = \mathbf{0}$ if (null matrix) where k is a +ve integer; if however k is the least +ve integer for which $\mathbf{A}^k = \mathbf{0}$, then k is called as the **index** of the *nilpotent matrix*. Examples,

$$\mathbf{A} = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} \text{ is nilpotent of index 2.}$$

$$\mathbf{A} = \begin{bmatrix} 1 & -3 & 4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix} \text{ is nilpotent of index 2.}$$

Periodic Matrix. A matrix $\mathbf{A}$ is said to be a **periodic matrix** if $\mathbf{A}^{k+1} = \mathbf{A}$, where k is a +ve integer. If k is the least +ive integer for which $\mathbf{A}^{k+1} = \mathbf{A}$ then k is said to be **period** of $\mathbf{A}$. If we choose $k = 1$, then $\mathbf{A}^2 = \mathbf{A}$ and we call it to be an **idempotent matrix**.

Illustrative Examples

Example 1: A matrix $\mathbf{A}$ such that $\mathbf{A}^2 = \mathbf{A}$ is called **Idempotent**. Determine all the idempotent diagonal matrices of order n .

Solution: Let $\mathbf{A} = \text{diag. } [d_1, d_2, d_3, \dots, d_n]$ be an idempotent matrix of order n .

Then $\mathbf{A}^2 = \mathbf{A}$.

$$\begin{aligned} \text{Now } \mathbf{A}^2 &= \begin{bmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & \dots & d_n \end{bmatrix} \begin{bmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & \dots & d_n \end{bmatrix} \\ &= \begin{bmatrix} d_1^2 & 0 & 0 & \dots & 0 \\ 0 & d_2^2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & \dots & d_n^2 \end{bmatrix}. \end{aligned}$$

$\therefore \mathbf{A}^2 = \mathbf{A}$ gives

$$d_1^2 = d_1, d_2^2 = d_2, \dots, d_n^2 = d_n$$

i.e., $d_1 = 0, 1; d_2 = 0, 1; \dots; d_n = 0, 1$.

Hence $\text{Diag. } [d_1, d_2, \dots, d_n]$, $d_1, d_2, \dots, d_n = 0, 1$ is the required idempotent diagonal matrix of order n .

Example 2: If $\mathbf{B}$ is an idempotent matrix, show that $\mathbf{A} = \mathbf{I} - \mathbf{B}$ is also idempotent and that $\mathbf{AB} = \mathbf{BA} = \mathbf{O}$.

Solution: Since $\mathbf{B}$ is idempotent, therefore $\mathbf{B}^2 = \mathbf{B}$.

$$\begin{aligned}
 \text{Now } \mathbf{A}^2 &= (\mathbf{I} - \mathbf{B})^2 = (\mathbf{I} - \mathbf{B})(\mathbf{I} - \mathbf{B}) \\
 &= \mathbf{I}^2 - \mathbf{IB} - \mathbf{BI} + \mathbf{B}^2 \\
 &= \mathbf{I} - \mathbf{B} - \mathbf{B} + \mathbf{B}^2 && [\because \mathbf{IB} = \mathbf{B} = \mathbf{BI}] \\
 &= \mathbf{I} - \mathbf{B} - \mathbf{B} + \mathbf{B} && [\because \mathbf{B}^2 = \mathbf{B}] \\
 &= \mathbf{I} - \mathbf{B} && [\because -\mathbf{B} + \mathbf{B} = \mathbf{O}] \\
 &= \mathbf{A}.
 \end{aligned}$$

Since $\mathbf{A}^2 = \mathbf{A}$, therefore $\mathbf{A}$ is idempotent.

$$\text{Again } \mathbf{AB} = (\mathbf{I} - \mathbf{B})\mathbf{B} = \mathbf{IB} - \mathbf{B}^2 = \mathbf{B} - \mathbf{B} = \mathbf{O}.$$

$$\text{Similarly } \mathbf{BA} = \mathbf{B}(\mathbf{I} - \mathbf{B}) = \mathbf{BI} - \mathbf{B}^2 = \mathbf{B} - \mathbf{B} = \mathbf{O}.$$

Example 3: Show that the matrix $\mathbf{A} = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is involutory.

Solution: The matrix $\mathbf{A}$ is involutory if $\mathbf{A}^2 = \mathbf{I}$.

$$\begin{aligned}
 \text{We have } \mathbf{A}^2 &= \mathbf{AA} = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \\
 &= \begin{bmatrix} 25 - 24 + 0 & 40 - 40 + 0 & 0 + 0 + 0 \\ -15 + 15 + 0 & -24 + 25 + 0 & 0 + 0 + 0 \\ -5 + 6 - 1 & -8 + 10 - 2 & 0 + 0 + 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}.
 \end{aligned}$$

$\therefore$ the matrix $\mathbf{A}$ is involutory.

Example 4: Show that the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \text{ is nilpotent and find its index.}$$

Solution: We have $\mathbf{A}^2 = \mathbf{AA}$

$$= \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}.$$

Again $A^3 = AA^2 = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{O}.$$

Thus 3 is the least positive integer such that $A^3 = \mathbf{O}$. Hence the matrix A is nilpotent of index 3.

2 Determinant of a Square Matrix

Let A be any square matrix. The determinant formed by the elements of A is said to be the determinant of matrix A . This is denoted by $|A|$ or $\det A$. Since in a determinant the number of rows is equal to the number of columns, therefore **only square matrices can have determinants**.

Hence, if $A = \begin{bmatrix} 12 & 0 & 13 \\ 15 & 12 & 11 \\ 13 & 11 & 14 \end{bmatrix}$, then $\det A = |A| = \begin{bmatrix} 12 & 0 & 13 \\ 15 & 12 & 11 \\ 13 & 11 & 14 \end{bmatrix}$.

Difference between a matrix and a determinant.

- (i) A matrix " A " cannot be reduced to a number whereas the determinant can be reduced to a number.
- (ii) The number of rows may or may not be equal to number of columns in a matrix while in a determinant the number of rows is equal to the number of columns.
- (iii) Interchanging the rows and columns, a different matrix is formed while in a determinant, an interchange of rows and columns does not change the value of the determinant.

3 Non-singular and Singular Matrices

Definition: A square matrix A is said to be non-singular or singular according as

$$|A| \neq 0 \quad \text{or} \quad |A| = 0.$$

4 Negative Integral Powers of a Non-singular Matrix

The negative integral powers are defined only for non-singular matrices.

Definition: Let $\mathbf{A}$ and $\mathbf{B}$ be non-singular matrices. If p is any positive integer, then we define

$$(\mathbf{A}^{-p}) = (\mathbf{A}^p)^{-1}.$$

$\mathbf{A}^{-1}$ is the unique inverse of $\mathbf{A}$.

If $\mathbf{A}$ is non-singular and its negative integral powers are considered, then for any integers r, s , we have

$$(i) \quad \mathbf{A}^r \mathbf{A}^s = \mathbf{A}^{r+s} \qquad (ii) \quad (\mathbf{A}^r)^s = \mathbf{A}^{rs}.$$

For two non-singular matrices $\mathbf{A}$ and $\mathbf{B}$ of the same order, the index law $(\mathbf{AB})^m = \mathbf{A}^m \mathbf{B}^m$ fails. However it becomes valid if $\mathbf{A}$ and $\mathbf{B}$ commute with respect to multiplication.

Theorem 1: If $\mathbf{A}$ is any n -rowed non-singular matrix, then

$$(\mathbf{A}^{-k}) = (\mathbf{A}^{-1})^k, \text{ for all } k \in \mathbb{N}.$$

Proof: If $\mathbf{A}$ be a non-singular matrix and $k \in \mathbb{N}$, by definition, we have

$$\begin{aligned} \mathbf{A}^{-k} &= (\mathbf{A}^k)^{-1} \\ &= (\mathbf{A} \cdot \mathbf{A} \cdot \dots k \text{ times})^{-1} \\ &= (\mathbf{A}^{-1} \cdot \mathbf{A}^{-1} \dots k \text{ times}) \\ &= (\mathbf{A}^{-1})^k. \end{aligned}$$

Hence, $\mathbf{A}^{-k} = (\mathbf{A}^k)^{-1} = (\mathbf{A}^{-1})^k$.

Theorem 2: Let $\mathbf{A}, \mathbf{B}$ be invertible matrices, then

- (i) $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
- (ii) $(k\mathbf{A})^{-1} = k^{-1}\mathbf{A}^{-1}, 0 \neq k \in \mathbb{C}$
- (iii) $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$ (Reversal Law for the Inverse of a Product)
- (iv) $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$
- (v) $(\mathbf{A}^\theta)^{-1} = (\mathbf{A}^{-1})^\theta.$

Proof: (i) Since $\mathbf{A}$ is an invertible matrix therefore there exists a matrix $\mathbf{X} = \mathbf{A}^{-1}$ such that $\mathbf{AX} = \mathbf{XA} = \mathbf{I}$.

By definition of inverse, we have

$$\mathbf{X}^{-1} = \mathbf{A}.$$

Hence, $(\mathbf{A}^{-1})^{-1} = \mathbf{A}.$

(ii) Let $\mathbf{X} = k^{-1}\mathbf{A}^{-1}$. Then, we have

$$\begin{aligned} (\mathbf{kA}) \mathbf{X} &= (\mathbf{kA}) (k^{-1}\mathbf{A}^{-1}) \\ &= k k^{-1} \mathbf{A} \mathbf{A}^{-1} \\ &= \mathbf{I}. \end{aligned}$$

Similarly, $\mathbf{X}(\mathbf{kA}) = \mathbf{I}.$

$$\therefore (\mathbf{kA}) \mathbf{X} = \mathbf{X}(\mathbf{kA}) = \mathbf{I}.$$

Hence, $(\mathbf{kA})^{-1} = \mathbf{X} = k^{-1}\mathbf{A}^{-1}.$

(iii) Let $\mathbf{X} = \mathbf{B}^{-1}\mathbf{A}^{-1}$. Then, we have

$$\begin{aligned} (\mathbf{AB}) \mathbf{X} &= (\mathbf{AB}) (\mathbf{B}^{-1}\mathbf{A}^{-1}) \\ &= \mathbf{A} [\mathbf{B} (\mathbf{B}^{-1}\mathbf{A}^{-1})] \\ &= \mathbf{A} (\mathbf{B}\mathbf{B}^{-1}\mathbf{A}^{-1}) \\ &= \mathbf{A} (\mathbf{I}\mathbf{A}^{-1}) \\ &= \mathbf{A}\mathbf{A}^{-1} \\ &= \mathbf{I}. \end{aligned}$$

Similarly, $\mathbf{X}(\mathbf{AB}) = \mathbf{I}.$

$$\therefore (\mathbf{AB}) \mathbf{X} = \mathbf{X}(\mathbf{AB}) = \mathbf{I}.$$

Hence, $(\mathbf{AB})^{-1} = \mathbf{X} = \mathbf{B}^{-1}\mathbf{A}^{-1}.$

(iv) By definition of inverse, we have

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

Taking transpose, we get

$$(\mathbf{A}\mathbf{A}^{-1})' = (\mathbf{A}^{-1}\mathbf{A})' = \mathbf{I}'$$

$$\text{or } (\mathbf{A}^{-1})' \mathbf{A}' = \mathbf{A}' (\mathbf{A}^{-1})' = \mathbf{I} \quad [\because (\mathbf{AB})' = \mathbf{B}' \mathbf{A}' \text{ and } \mathbf{I}' = \mathbf{I}]$$

This equality shows that $(\mathbf{A}^{-1})'$ is the inverse of the matrix $\mathbf{A}'$.

Hence, $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'.$

(v) The proof is similar to that of part (iv). Take transposed conjugate instead of transpose.

5 Trace of a Matrix

Definition: Let $\mathbf{A}$ be a square matrix of order n . The sum of the elements of $\mathbf{A}$ lying along the principal diagonal is called the trace of $\mathbf{A}$. We shall write the trace of $\mathbf{A}$ as $\text{tr } \mathbf{A}$. Thus if

$$\mathbf{A} = [a_{ij}]_{n \times n}, \text{ then } \text{tr } \mathbf{A} = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}.$$

In the following theorem we have given some fundamental properties of the trace function.

Theorem: Let $\mathbf{A}$ and $\mathbf{B}$ be two square matrices of order n and λ be a scalar. Then

- (1) $\text{tr } (\lambda \mathbf{A}) = \lambda \text{tr } \mathbf{A}$;
- (2) $\text{tr } (\mathbf{A} + \mathbf{B}) = \text{tr } \mathbf{A} + \text{tr } \mathbf{B}$;
- (3) $\text{tr } (\mathbf{AB}) = \text{tr } (\mathbf{BA})$.

Proof: Let $\mathbf{A} = [a_{ij}]_{n \times n}$ and $\mathbf{B} = [b_{ij}]_{n \times n}$.

(1) We have $\lambda \mathbf{A} = [\lambda a_{ij}]_{n \times n}$ by def. of multiplication of a matrix by a scalar.

$$\therefore \text{tr } (\lambda \mathbf{A}) = \sum_{i=1}^n \lambda a_{ii} = \lambda \sum_{i=1}^n a_{ii} = \lambda \text{tr } \mathbf{A}.$$

(2) We have $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]_{n \times n}$.

$$\begin{aligned} \therefore \text{tr } (\mathbf{A} + \mathbf{B}) &= \sum_{i=1}^n (a_{ii} + b_{ii}) \\ &= \sum_{i=1}^n a_{ii} + \sum_{i=1}^n b_{ii} \\ &= \text{tr } \mathbf{A} + \text{tr } \mathbf{B}. \end{aligned}$$

(3) We have $\mathbf{AB} = [c_{ij}]_{n \times n}$, where $c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$.

Also $\mathbf{BA} = [d_{ij}]_{n \times n}$, where $d_{ij} = \sum_{k=1}^n b_{ik} a_{kj}$.

$$\begin{aligned} \text{Now } \text{tr } (\mathbf{AB}) &= \sum_{i=1}^n c_{ii} = \sum_{i=1}^n \left(\sum_{k=1}^n a_{ik} b_{ki} \right) \\ &= \sum_{k=1}^n \sum_{i=1}^n a_{ik} b_{ki}, \quad \text{interchanging the order of summation in the last sum} \\ &= \sum_{k=1}^n \left(\sum_{i=1}^n b_{ki} a_{ik} \right) = \sum_{k=1}^n d_{kk} \\ &= d_{11} + d_{22} + \dots + d_{nn} = \text{tr } (\mathbf{BA}). \end{aligned}$$

6 Transpose of a Matrix

Definition: Let $\mathbf{A} = [a_{ij}]_{m \times n}$. Then the $n \times m$ matrix obtained from $\mathbf{A}$ by changing its rows into columns and its columns into rows is called the transpose of $\mathbf{A}$ and is denoted by the symbol $\mathbf{A}'$ or $\mathbf{A}^T$.

The operation of interchanging rows with columns is called transposition. Symbolically if

$$\mathbf{A} = [a_{ij}]_{m \times n},$$

then $\mathbf{A}' = [b_{ji}]_{n \times m}$, where $b_{ji} = a_{ij}$,

i.e., the $(j, i)^{th}$ element of $\mathbf{A}'$ is the $(i, j)^{th}$ element of $\mathbf{A}$.

For example, the transpose of the 3×4 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 2 & 1 \end{bmatrix}_{3 \times 4} \quad \text{is the } 4 \times 3 \text{ matrix } \mathbf{A}' = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 2 \\ 4 & 1 & 1 \end{bmatrix}_{4 \times 3}.$$

The first row of $\mathbf{A}$ is the first column of $\mathbf{A}'$. The second row of $\mathbf{A}$ is the second column of $\mathbf{A}'$. The third row of $\mathbf{A}$ is the third column of $\mathbf{A}'$.

Theorems: If $\mathbf{A}'$ and $\mathbf{B}'$ be the transposes of $\mathbf{A}$ and $\mathbf{B}$ respectively, then

- (i) $(\mathbf{A}')' = \mathbf{A}$;
- (ii) $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$, $\mathbf{A}$ and $\mathbf{B}$ being of the same size.
- (iii) $(k\mathbf{A})' = k\mathbf{A}'$, k being any complex number.
- (iv) $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$, $\mathbf{A}$ and $\mathbf{B}$ being conformable to multiplication.

The above law (iv) is called the reversal law for transposes i.e., the transpose of the product is the product of the transposes taken in the reverse order.

7 Orthogonal Matrix

Definition: A square matrix $\mathbf{A}$ is said to be orthogonal if $\mathbf{A}'\mathbf{A} = \mathbf{I}$. (Kanpur 2009)

If $\mathbf{A}$ is an orthogonal matrix, then $\mathbf{A}'\mathbf{A} = \mathbf{I}$

$$\Rightarrow |\mathbf{A}'\mathbf{A}| = |\mathbf{I}| \Rightarrow |\mathbf{A}'| \cdot |\mathbf{A}| = 1 \quad [\because \det(\mathbf{AB}) = (\det \mathbf{A}) \cdot (\det \mathbf{B})]$$

$$\Rightarrow |\mathbf{A}| \cdot |\mathbf{A}| = 1 \quad [\because |\mathbf{A}'| = |\mathbf{A}|]$$

$$\Rightarrow |\mathbf{A}|^2 = 1 \Rightarrow |\mathbf{A}| = \pm 1 \Rightarrow |\mathbf{A}| \neq 0$$

$$\Rightarrow \mathbf{A} \text{ is invertible.}$$

Also then $\mathbf{A}'\mathbf{A} = \mathbf{I} \Rightarrow \mathbf{A}' = \mathbf{A}^{-1}$ which in turn implies $\mathbf{A}\mathbf{A}' = \mathbf{I}$.

Thus $\mathbf{A}$ is an orthogonal matrix if and only if

$$\mathbf{A}'\mathbf{A} = \mathbf{I} = \mathbf{A}\mathbf{A}'.$$

Theorem: If $\mathbf{A}, \mathbf{B}$ be n -rowed orthogonal matrices, $\mathbf{AB}$ and $\mathbf{BA}$ are also orthogonal matrices.

Proof: Since $\mathbf{A}$ and $\mathbf{B}$ are both n -rowed square matrices, therefore $\mathbf{AB}$ is also an n -rowed square matrix.

Since $|\mathbf{AB}| = |\mathbf{A}| \cdot |\mathbf{B}|$ and $|\mathbf{A}| \neq 0$, also $|\mathbf{B}| \neq 0$, therefore $|\mathbf{AB}| \neq 0$. Hence, $\mathbf{AB}$ is a non-singular matrix.

$$\text{Now} \quad (\mathbf{AB})' = \mathbf{B}'\mathbf{A}'.$$

$$\begin{aligned} \therefore (\mathbf{AB})'(\mathbf{AB}) &= (\mathbf{B}'\mathbf{A}')(\mathbf{AB}) \\ &= \mathbf{B}'(\mathbf{A}'\mathbf{A})\mathbf{B} \\ &= \mathbf{B}'\mathbf{I}\mathbf{B} && [\because \mathbf{A}'\mathbf{A} = \mathbf{I}] \\ &= \mathbf{B}'\mathbf{B} \\ &= \mathbf{I} && [\because \mathbf{B}'\mathbf{B} = \mathbf{I}] \end{aligned}$$

$\therefore \mathbf{AB}$ is orthogonal. Similarly we can prove that $\mathbf{BA}$ is also orthogonal.

Illustrative Examples

Example 5: Determine the values of α, β, γ when $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal.

Solution : Let $\mathbf{A} = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}.$

Then $\mathbf{A}' = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix}.$

If $\mathbf{A}$ is orthogonal, then $\mathbf{A}\mathbf{A}' = \mathbf{I}$.

$$\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix} \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

or
$$\begin{bmatrix} 4\beta^2 + \gamma^2 & 2\beta^2 - \gamma^2 & -2\beta^2 + \gamma^2 \\ 2\beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 \\ -2\beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now equating the corresponding elements, we get

$$4\beta^2 + \gamma^2 = 1 \quad \dots(1)$$

$$2\beta^2 - \gamma^2 = 0 \quad \dots(2)$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1. \quad \dots(3)$$

From (1) and (2), we get $\beta = \pm \frac{1}{\sqrt{6}}, \gamma = \pm \frac{1}{\sqrt{3}}$.

From (3), we get $\alpha = \pm \frac{1}{\sqrt{2}}$.

Hence $\alpha = \pm \frac{1}{\sqrt{2}}, \beta = \pm \frac{1}{\sqrt{6}}, \gamma = \pm \frac{1}{\sqrt{3}}$.

Example 6: Show that the matrix

$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \text{ is orthogonal.}$$

Solution: Let $\mathbf{A} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$.

Then $\mathbf{A}' = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.

We have
$$\begin{aligned} \mathbf{A}\mathbf{A}' &= \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \mathbf{I}_2. \end{aligned}$$

Hence the matrix $\mathbf{A}$ is orthogonal.

Example 7: Verify that the matrix

$$\frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \text{ is orthogonal.}$$

Solution: Let

$$\mathbf{A} = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}.$$

Then

$$\mathbf{A}' = \frac{1}{3} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$$

We have

$$\begin{aligned} \mathbf{AA}' &= \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}_3. \end{aligned}$$

Hence, the matrix $\mathbf{A}$ is orthogonal.

8 Conjugate of a Matrix

If $i = \sqrt{-1}$, then $z = x + iy$ is called a complex number where x and y are any real numbers. If $z = x + iy$, then $\bar{z} = x - iy$ is called the **conjugate** of the complex number z .

We have $z\bar{z} = (x + iy)(x - iy) = x^2 + y^2$ i.e., is real.

Also if $z = \bar{z}$, then $x + iy = x - iy$ i.e., $2iy = 0$ i.e., $y = 0$ i.e., z is real.

Conversely, if z is real then $\bar{z} = z$.

If $z = x + iy$, then $\bar{z} = x - iy$. $\therefore \overline{(\bar{z})} = x + iy = z$.

If z_1 and z_2 are two complex numbers, then it can be easily seen that

$$(i) \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2 \quad \text{and} \quad (ii) \overline{z_1 z_2} = \overline{z_1} \overline{z_2}.$$

Conjugate of a Matrix

Definition: The matrix obtained from any given matrix $\mathbf{A}$ on replacing its elements by the corresponding conjugate complex numbers is called the conjugate of $\mathbf{A}$ and is denoted by $\overline{\mathbf{A}}$.

Thus if $\mathbf{A} = [a_{ij}]_{m \times n}$, then $\overline{\mathbf{A}} = [\overline{a_{ij}}]_{m \times n}$ where $\overline{a_{ij}}$ denotes the conjugate complex of a_{ij} .

If $\mathbf{A}$ be a matrix over the field of **real numbers**, then obviously $\overline{\mathbf{A}}$ coincides with $\mathbf{A}$.

Illustration: If $\mathbf{A} = \begin{bmatrix} 2+3i & 4-7i & 8 \\ -i & 6 & 9+i \end{bmatrix}$, then $\overline{\mathbf{A}} = \begin{bmatrix} 2-3i & 4+7i & 8 \\ i & 6 & 9-i \end{bmatrix}$.

Theorem: If $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ be the conjugates of $\mathbf{A}$ and $\mathbf{B}$ respectively, then

- (i) $\overline{(\overline{\mathbf{A}})} = \mathbf{A}$
- (ii) $\overline{(\mathbf{A} + \mathbf{B})} = \overline{\mathbf{A}} + \overline{\mathbf{B}}$, $\mathbf{A}$ and $\mathbf{B}$ being of the same size
- (iii) $\overline{(k \mathbf{A})} = \overline{k} \overline{\mathbf{A}}$, k being any complex number
- (iv) $\overline{(\mathbf{A}\mathbf{B})} = \overline{\mathbf{A}} \overline{\mathbf{B}}$, $\mathbf{A}$ and $\mathbf{B}$ being conformable to multiplication.

9 Transposed Conjugate of a Matrix

Definition: The transpose of the conjugate of a matrix $\mathbf{A}$ is called transposed conjugate of $\mathbf{A}$ and is denoted by $\mathbf{A}^\theta$ or by $\mathbf{A}^*$.

Obviously the conjugate of the transpose of $\mathbf{A}$ is the same as the transpose of the conjugate of $\mathbf{A}$ i.e.,

$$(\overline{\mathbf{A}'}) = (\overline{\mathbf{A}})' = \mathbf{A}^\theta.$$

If $\mathbf{A} = [a_{ij}]_{m \times n}$, then $\mathbf{A}^\theta = [b_{ji}]_{n \times m}$

where $b_{ji} = \overline{a_{ij}}$ i.e., the $(j, i)^{th}$ element of $\mathbf{A}^\theta$ = the conjugate complex of the $(i, j)^{th}$ element of $\mathbf{A}$.

Illustration: If $\mathbf{A} = \begin{bmatrix} 1+2i & 2-3i & 3+4i \\ 4-5i & 5+6i & 6-7i \\ 8 & 7+8i & 7 \end{bmatrix}$,

then $\mathbf{A}' = \begin{bmatrix} 1+2i & 4-5i & 8 \\ 2-3i & 5+6i & 7+8i \\ 3+4i & 6-7i & 7 \end{bmatrix}$

and
$$\overline{(\mathbf{A}')} = \mathbf{A}^\theta = \begin{bmatrix} 1-2i & 4+5i & 8 \\ 2+3i & 5-6i & 7-8i \\ 3-4i & 6+7i & 7 \end{bmatrix}.$$

Theorem: If $\mathbf{A}^\theta$ and $\mathbf{B}^\theta$ be the transposed conjugates of $\mathbf{A}$ and $\mathbf{B}$ respectively, then

- (i) $(\mathbf{A}^\theta)^\theta = \mathbf{A}$
- (ii) $(\mathbf{A} + \mathbf{B})^\theta = \mathbf{A}^\theta + \mathbf{B}^\theta$, $\mathbf{A}$ and $\mathbf{B}$ being of the same size
- (iii) $(k\mathbf{A})^\theta = \bar{k} \mathbf{A}^\theta$, k being any complex number
- (iv) $(\mathbf{AB})^\theta = \mathbf{B}^\theta \mathbf{A}^\theta$, $\mathbf{A}$ and $\mathbf{B}$ being conformable to multiplication.

10 Invertible Matrices

Inverse or reciprocal of a matrix.

Definition: Let $\mathbf{A}$ be any n -rowed square matrix. Then a matrix $\mathbf{B}$, if it exists, such that $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$ is called inverse of $\mathbf{A}$.

Note: For the products $\mathbf{AB}$, $\mathbf{BA}$ to be both defined and be equal, it is necessary that $\mathbf{A}$ and $\mathbf{B}$ are both square matrices of the same order. Thus **non-square matrices cannot possess inverse**.

Existence of the Inverse. Theorem. The necessary and sufficient condition for a square matrix $\mathbf{A}$ to possess the inverse is that $|\mathbf{A}| \neq 0$.

Important: If $\mathbf{A}$ be an invertible matrix, then the inverse of $\mathbf{A}$ is $\frac{1}{|\mathbf{A}|} \text{Adj. } \mathbf{A}$. It is usual to denote the inverse of $\mathbf{A}$ by $\mathbf{A}^{-1}$.

Thus,
$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{Adj. } \mathbf{A}, \text{ provided } |\mathbf{A}| \neq 0.$$

Thus the necessary and sufficient condition for a matrix to be invertible is that it is non-singular.

11 Reversal Law for the Inverse of a Product

Theorem 1: If $\mathbf{A}$, $\mathbf{B}$ be two n -rowed non-singular matrices, then $\mathbf{AB}$ is also non-singular and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$, i.e., the inverse of a product is the product of the inverses taken in the reverse order.

Theorem 2: If $\mathbf{A}$ be an $n \times n$ non-singular matrix, then $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$ i.e., the operations of transposing and inverting are commutative.

Theorem 3: If $\mathbf{A}$ be an $n \times n$ non-singular matrix, then $(\mathbf{A}^{-1})^0 = (\mathbf{A}^0)^{-1}$.

Illustrative Examples

Example 8: Find the :

(i) Transpose

(ii) Adjoint

(iii) Inverse of the matrix :

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}.$$

Solution: (i) We have the transpose of $\mathbf{A} = \mathbf{A}^T = \begin{bmatrix} 2 & -5 & -3 \\ -1 & 3 & 2 \\ 3 & 1 & 3 \end{bmatrix}$.

(ii) We have $|\mathbf{A}| = \begin{vmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{vmatrix}$

$$= 2(9 - 2) - (-1)(-15 + 3) + 3(-10 + 9)$$

$$= 14 - 12 - 3 = -1.$$

Let $\mathbf{A}_{ij}$ denote the cofactor of the element a_{ij} of $|\mathbf{A}|$.

Then $A_{11} = \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} = 7, A_{12} = -\begin{vmatrix} -5 & 1 \\ -3 & 3 \end{vmatrix} = 12,$

$$A_{13} = \begin{vmatrix} -5 & 3 \\ -3 & 2 \end{vmatrix} = -1,$$

$$A_{21} = -\begin{vmatrix} -1 & 3 \\ 2 & 3 \end{vmatrix} = 9, A_{22} = \begin{vmatrix} 2 & 3 \\ -3 & 3 \end{vmatrix} = 15,$$

$$A_{23} = -\begin{vmatrix} 2 & -1 \\ -3 & 2 \end{vmatrix} = -1,$$

$$A_{31} = \begin{vmatrix} -1 & 3 \\ 3 & 1 \end{vmatrix} = -10,$$

$$A_{32} = - \begin{vmatrix} 2 & 3 \\ -5 & 1 \end{vmatrix} = -17,$$

$$A_{33} = \begin{vmatrix} 2 & -1 \\ -5 & 3 \end{vmatrix} = 1.$$

∴ The matrix **B** formed of the cofactors of the elements of $|\mathbf{A}|$ is,

$$\mathbf{B} = \begin{bmatrix} 7 & 12 & -1 \\ 9 & 15 & -1 \\ -10 & -17 & 1 \end{bmatrix}.$$

Now adj. **A** (= the transpose of the matrix **B**) = $\begin{bmatrix} 7 & 9 & -10 \\ 12 & 15 & -17 \\ -1 & -1 & 1 \end{bmatrix}.$

(iii) We have $\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj. } \mathbf{A} = \frac{1}{(-1)} \begin{bmatrix} 7 & 9 & -10 \\ 12 & 15 & -17 \\ -1 & 1 & 1 \end{bmatrix}$

$$= (-1) \begin{bmatrix} 7 & 9 & -10 \\ 12 & 15 & -17 \\ -1 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -7 & -9 & 10 \\ -12 & -15 & 17 \\ 1 & 1 & -1 \end{bmatrix}.$$

Note: To check that the answer is correct the students should verify that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

12 Unitary Matrix

Definition: A square matrix **A** is said to be unitary if $\mathbf{A}^\theta \mathbf{A} = \mathbf{I}$.

Since $|\mathbf{A}^\theta| = \overline{|\mathbf{A}|}$ and $|\mathbf{A}^\theta \mathbf{A}| = |\mathbf{A}^\theta| |\mathbf{A}|$, therefore if $\mathbf{A}^\theta \mathbf{A} = \mathbf{I}$, we have $|\mathbf{A}| \overline{|\mathbf{A}|} = 1$.

Thus the determinant of a unitary matrix is of unit modulus.

If **A** is a unitary matrix, then $|\mathbf{A}| |\mathbf{A}^\theta| = 1$ and so $|\mathbf{A}| \neq 0$ i.e., **A** is non-singular and so invertible.

Hence $\mathbf{A}^\theta \mathbf{A} = \mathbf{I}$ implies $\mathbf{A}\mathbf{A}^\theta = \mathbf{I}$.

Thus $\mathbf{A}$ is a unitary matrix iff $\mathbf{A}^\theta \mathbf{A} = \mathbf{I} = \mathbf{A}\mathbf{A}^\theta$.

Theorem: If $\mathbf{A}, \mathbf{B}$ be n -rowed unitary matrices, $\mathbf{AB}$ and $\mathbf{BA}$ are also unitary matrices.

Proof: Since $\mathbf{A}$ and $\mathbf{B}$ are both n -rowed square matrices, therefore $\mathbf{AB}$ is also an n -rowed square matrix.

Since $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$ and $|\mathbf{A}| \neq 0$, also $|\mathbf{B}| \neq 0$, therefore $|\mathbf{AB}| \neq 0$.

Hence $\mathbf{AB}$ is a non-singular matrix.

Now $(\mathbf{AB})^\theta = \mathbf{B}^\theta \mathbf{A}^\theta$.

$$\begin{aligned} \therefore (\mathbf{AB})^\theta (\mathbf{AB}) &= (\mathbf{B}^\theta \mathbf{A}^\theta) (\mathbf{AB}) \\ &= \mathbf{B}^\theta (\mathbf{A}^\theta \mathbf{A}) \mathbf{B} \\ &= \mathbf{B}^\theta \mathbf{I} \mathbf{B} && [\because \mathbf{A}^\theta \mathbf{A} = \mathbf{I}] \\ &= \mathbf{B}^\theta \mathbf{B} \\ &= \mathbf{I} && [\because \mathbf{B}^\theta \mathbf{B} = \mathbf{I}] \end{aligned}$$

$\therefore \mathbf{AB}$ is unitary. Similarly we can prove that $\mathbf{BA}$ is also unitary.

Illustrative Examples

Example 9: Prove that

$$\mathbf{B} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \text{ is unitary.} \quad (\text{Bundelkhand 2006; Garhwal 11})$$

Solution: We have $\overline{\mathbf{B}} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1-i \\ 1+i & -1 \end{bmatrix}$.

$$\therefore \mathbf{B}^\theta = (\overline{\mathbf{B}})' = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}.$$

$$\begin{aligned} \text{Now } \mathbf{B}^\theta \mathbf{B} &= \frac{1}{3} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} 1.1 + (1+i)(1-i) & 1.(1+i) + (1+i)(-1) \\ (1-i).1 + (-1).(1-i) & (1-i)(1+i) + (-1)(-1) \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}.
 \end{aligned}$$

$\therefore$ $\mathbf{B}$ is unitary.

Example 10: Show that the matrix $\mathbf{A} = \begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix}$ is a unitary matrix, if

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 1.$$

Solution: We have $\mathbf{A} = \begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix}$

$$\therefore \mathbf{A}' = \begin{bmatrix} \alpha + i\gamma & \beta + i\delta \\ -\beta + i\delta & \alpha - i\gamma \end{bmatrix}$$

$$\mathbf{A}^\theta = \overline{\mathbf{A}'} = \begin{bmatrix} \alpha - i\gamma & \beta - i\delta \\ -\beta - i\delta & \alpha + i\gamma \end{bmatrix}.$$

A square matrix $\mathbf{A}$ is said to be unitary if $\mathbf{A}\mathbf{A}^\theta = \mathbf{I}$.

$$\therefore \begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix} \begin{bmatrix} \alpha - i\gamma & \beta - i\delta \\ -\beta - i\delta & \alpha + i\gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
 \text{or} \quad & \begin{bmatrix} \alpha^2 + \beta^2 + \gamma^2 + \delta^2 & \alpha\beta - i\beta\gamma + i\alpha\delta + \gamma\delta - \alpha\beta - i\alpha\delta + i\beta\gamma - \delta\gamma \\ \alpha\beta - i\alpha\delta + i\beta\gamma + \gamma\delta - \alpha\beta - i\beta\gamma + i\alpha\delta - \delta\gamma & \beta^2 + \delta^2 + \alpha^2 + \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}
 \end{aligned}$$

$$\text{or} \quad \begin{bmatrix} \alpha^2 + \beta^2 + \gamma^2 + \delta^2 & 0 \\ 0 & \alpha^2 + \beta^2 + \gamma^2 + \delta^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{if} \quad \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 1.$$

Example 11: If $\mathbf{A}$ is a unitary matrix, show that $\mathbf{A}^{-1}$ is also unitary.

Solution: We have $\mathbf{A}\mathbf{A}^\theta = \mathbf{A}^\theta\mathbf{A} = \mathbf{I}$, since $\mathbf{A}$ is a unitary matrix.

$$\Rightarrow (\mathbf{A}\mathbf{A}^\theta)^{-1} = (\mathbf{A}^\theta\mathbf{A})^{-1} = (\mathbf{I})^{-1} \quad [\text{Taking inverse}]$$

$$\Rightarrow (\mathbf{A}^\theta)^{-1} \cdot \mathbf{A}^{-1} = \mathbf{A}^{-1} (\mathbf{A}^\theta)^{-1} = \mathbf{I}$$

$$\Rightarrow (\mathbf{A}^{-1})^\theta \cdot \mathbf{A}^{-1} = \mathbf{A}^{-1} (\mathbf{A}^{-1})^\theta = \mathbf{I}.$$

Hence $\mathbf{A}^{-1}$ is a unitary matrix.

13 Matrices of Rotation and Reflection

A rotation about the origin O by an angle θ is denoted by $\text{Rot}(\theta)$ and a reflection about a line L through the origin which makes an angle θ with the x -axis is denoted by $\text{Ref}(\theta)$. If these rotations and reflections operate on all points on the plane and if these points be represented by position vectors, then a rotation is represented by a matrix.

$$\text{Rot}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

and a reflection is represented by a matrix

$$\text{Ref}(\theta) = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}.$$

$\text{Rot}(\theta)$ rotates column vectors by means of the following matrix multiplication

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Thus the coordinates (x', y') of the point (x, y) after rotation are

$$x' = x \cos \theta - y \sin \theta,$$

$$y' = x \sin \theta + y \cos \theta.$$

The direction of vector rotation is positive if angle θ is measured counter clockwise (e.g. 90°), and negative if angle θ is measured clockwise (e.g. -90°).

The clockwise rotation is represented by a matrix

$$\text{Rot}(-\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}.$$

Reflection transformation matrix is the matrix used to make reflection transformation of a figure.

The following relations hold true for rotation and reflection

$$(i) \quad \text{Rot}(\theta) \text{Rot}(\phi) = \text{Rot}(\theta + \phi)$$

$$(ii) \quad \text{Ref}(\theta) \text{Ref}(\phi) = \text{Rot}(2\theta - 2\phi)$$

$$(iii) \quad \text{Rot}(\theta) \text{Ref}(\phi) = \text{Ref}\left(\phi + \frac{1}{2}\theta\right)$$

$$(iv) \quad \text{Ref}(\phi) \text{Rot}(\theta) = \text{Ref}\left(\phi - \frac{1}{2}\theta\right).$$

These relations can be proved using matrix multiplication and application of trigonometric identities.

The set of all reflections in lines through the origin and rotations about the origin, together with the operation of composition of reflections and rotations, forms a group. $\text{Rot}(0)$ is the identity of the group. There exists an inverse $\text{Rot}(-\phi)$ for every rotation $(\text{Rot } \phi)$. Every reflection $\text{Ref. } (\theta)$ is its own inverse. Matrix multiplication being associative, the composition has closure and is associative. $\text{Ref } (\theta)$ and $\text{Rot } (\theta)$ both being represented by orthogonal matrices have a determinant whose absolute value is unity. Rotation matrices have a determinant of value $+1$ and reflection matrices have a determinant of value -1 .

Note : The set of all two dimensional matrices together with matrix multiplication form the orthogonal group $O(2)$.

Rotation Matrix Transformations

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	Identity matrix. Right remains right, up remains up.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} x \\ y \end{bmatrix}$
$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	Rotation by 90° clockwise. Right has become down, up has become right.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} y \\ -x \end{bmatrix}$
$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	Rotation by 90° anticlockwise. Right has become up, up has become left.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} -y \\ x \end{bmatrix}$
$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$	Rotation by 180° . Right has become left, up has become down.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} -x \\ -y \end{bmatrix}$
$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	Rotation by 270° anticlockwise. Right has become down, up has become right A positive 90° rotation (left) after a 180° rotation gives a negative 90° rotation (right).	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} y \\ -x \end{bmatrix}$

Reflection Matrix Transformations

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Reflection in the x -axis. Right remains right, up has become down.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} x \\ -y \end{bmatrix}$
$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	Reflection in the y -axis. Right has become left, up remains up.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} -x \\ y \end{bmatrix}$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Reflection in the line $y = x$. Right has become up, up has become right.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} y \\ x \end{bmatrix}$
$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$	Reflection in the line $y = -x$. Right has become down, up has become left.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} -y \\ -x \end{bmatrix}$
$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$	Reflection about the origin. Right has become left, up has become down.	$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} -x \\ -y \end{bmatrix}$

Note: Combinations of these matrices give multiple transformations. Two reflections generate a rotations.

Comprehensive Exercise 1

- Show that the matrix $\mathbf{A} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is idempotent.
- If $\mathbf{AB} = \mathbf{A}$ and $\mathbf{BA} = \mathbf{B}$, show that $\mathbf{A}$ and $\mathbf{B}$ are idempotent.
- A matrix $\mathbf{A}$ such that $\mathbf{A}^2 = \mathbf{I}$ is called **involutory**. Show that $\mathbf{A}$ is involutory if and only if $(\mathbf{I} + \mathbf{A})(\mathbf{I} - \mathbf{A}) = \mathbf{O}$.
- Show that the matrix $\mathbf{A} = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$ is nilpotent such that $\mathbf{A}^2 = \mathbf{O}$.
- Show that the matrix $\frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$ is orthogonal.
- Verify that the matrix $\begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ is orthogonal.
- Show that the matrix $= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ -i & -1 \end{bmatrix}$ is unitary.

8. Prove that the matrix $\begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$ is unitary. (Rohilkhand 2010)
9. If (l_r, m_r, n_r) , $r = 1, 2, 3$ be the direction cosines of three mutually perpendicular lines referred to an orthogonal cartesian co-ordinate system, then prove that $\begin{bmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{bmatrix}$ is an orthogonal matrix.
10. Show that if $\mathbf{A}$ is an orthogonal matrix, then $\mathbf{A}'$ and $\mathbf{A}^{-1}$ are also orthogonal.
11. If $\mathbf{N} = \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix}$ is a matrix, then show that $(\mathbf{I} - \mathbf{N})(\mathbf{I} + \mathbf{N})^{-1}$ is a unitary matrix, where $\mathbf{I}$ is an identity matrix.

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. A matrix $\mathbf{A} = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$ is nilpotent matrix of index
 (a) 1 (b) 2
 (c) 3 (d) 4
2. A matrix $\mathbf{A} = \begin{bmatrix} 6 & 9 \\ -4 & -6 \end{bmatrix}$ is
 (a) unitary (b) orthogonal
 (c) idempotent (d) nilpotent
3. A square matrix $\mathbf{A}$ is said to be orthogonal if
 (a) $\mathbf{A}'\mathbf{A} = \mathbf{A}$ (b) $\mathbf{A}'\mathbf{A} = \mathbf{I}$
 (c) $\mathbf{A}'\mathbf{A} = \mathbf{O}$ (d) $\mathbf{A}'\mathbf{A} = \mathbf{A}^2$
4. The necessary and sufficient condition for a matrix to be invertible is that it is
 (a) singular (b) non-singular
 (c) zero matrix (d) none of these (Rohilkhand 2007)

Fill in the Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

1. A matrix A is said to be involutory if
2. A square matrix A is said to be idempotent if
3. A square matrix A is said to be unitary if

True or False

Write ‘T’ for true and ‘F’ for false statement.

1. If A is a unitary matrix then A^{-1} is also unitary.
2. If A is an orthogonal matrix then A' is also orthogonal.
3. A matrix $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ is unitary matrix.

Answers

Multiple Choice Questions

1. (b)
2. (d)
3. (b)
4. (b)

Fill in the Blank(s)

1. $A^2 = I$
2. $A^2 = A$
3. $A^{\theta} A = I$

True or False

1. T
2. T
3. F



Chapter

6



Rank of a Matrix

1 Submatrix of a Matrix

Suppose $\mathbf{A}$ is any matrix of the type $m \times n$. Then a matrix obtained by leaving some rows and columns from $\mathbf{A}$ is called a **submatrix** of $\mathbf{A}$. In particular the matrix $\mathbf{A}$ itself is a sub-matrix of $\mathbf{A}$ because it is obtained from $\mathbf{A}$ by leaving no rows or columns.

Minors of a matrix. We know that every square matrix possesses a determinant. If $\mathbf{A}$ be an $m \times n$ matrix, then the determinant of every square sub-matrix of $\mathbf{A}$ is called a **minor** of the matrix $\mathbf{A}$. If we leave $m - p$ rows and $n - p$ columns from $\mathbf{A}$, we shall get a square submatrix of $\mathbf{A}$ of order p . The determinant of this square submatrix is called a p -rowed minor of $\mathbf{A}$.

For example, let

$$\mathbf{A} = \begin{bmatrix} 2 & 4 & 1 & 9 & 1 \\ 0 & 5 & 2 & 5 & 2 \\ 1 & 9 & 7 & 3 & 4 \\ 3 & -2 & 8 & 1 & 8 \end{bmatrix}_{4 \times 5}.$$

In a determinant the number of rows is equal to the number of columns. Therefore there can be no 5-rowed minor of $\mathbf{A}$.

If we leave any columns from $\mathbf{A}$, we shall get a square sub-matrix of $\mathbf{A}$ of order 4.

Thus

$$\begin{vmatrix} 2 & 4 & 1 & 9 \\ 0 & 5 & 2 & 5 \\ 1 & 9 & 7 & 3 \\ 3 & -2 & 8 & 1 \end{vmatrix}, \begin{vmatrix} 2 & 4 & 9 & 1 \\ 0 & 5 & 5 & 2 \\ 1 & 9 & 3 & 4 \\ 3 & -2 & 1 & 8 \end{vmatrix}, \text{etc.}$$

are 4-rowed minors of $\mathbf{A}$.

If we leave two columns and one row from $\mathbf{A}$, we shall get a square submatrix of $\mathbf{A}$ of order 3.

Thus

$$\begin{vmatrix} 2 & 4 & 1 \\ 0 & 5 & 2 \\ 1 & 9 & 7 \end{vmatrix}, \begin{vmatrix} 4 & 1 & 9 \\ 5 & 2 & 5 \\ 9 & 7 & 3 \end{vmatrix}, \begin{vmatrix} 5 & 2 & 5 \\ 9 & 7 & 3 \\ -2 & 8 & 1 \end{vmatrix}, \text{etc.}$$

are 3-rowed minors of $\mathbf{A}$.

If we leave three columns and two rows from $\mathbf{A}$, we shall get a square submatrix of $\mathbf{A}$ of order 2.

Thus

$$\begin{vmatrix} 2 & 4 \\ 0 & 5 \end{vmatrix}, \begin{vmatrix} 4 & 1 \\ 5 & 2 \end{vmatrix}, \begin{vmatrix} 5 & 2 \\ 9 & 7 \end{vmatrix}, \text{etc. are 2-rowed minors of } \mathbf{A}.$$

The numbers 2, 4, 1, 9, 1, 0, 5 etc. are all 1-rowed minors of $\mathbf{A}$.

2 Rank of a Matrix

(Lucknow 2006)

Definition: A number r is said to be the rank of a matrix $\mathbf{A}$ if it possesses the following two properties :

- (i) There is at least one square submatrix of $\mathbf{A}$ of order r whose determinant is not equal to zero.
- (ii) If the matrix $\mathbf{A}$ contains any square submatrix of order $r + 1$, then the determinant of every square submatrix of $\mathbf{A}$ of order $r + 1$ should be zero.

In short the rank of a matrix is the order of any highest order non-vanishing minor of the matrix.

Thus the rank of a matrix $\mathbf{A}$ is the order of any highest order square submatrix of $\mathbf{A}$ whose determinant is not equal to zero.

We shall denote the rank of a matrix $\mathbf{A}$ by the symbol $\rho(\mathbf{A})$.

It is obvious that the rank r of an $(m \times n)$ matrix can at most be equal to the smaller of the numbers m and n , but it may be less.

If there is a matrix $\mathbf{A}$ which has at least one non-zero minor of order n and there is no minor of $\mathbf{A}$ of order $n + 1$, then the rank of $\mathbf{A}$ is n . **Thus the rank of every non-singular matrix of order n is n .** The rank of a square matrix $\mathbf{A}$ of order n can be less than n if and only if $\mathbf{A}$ is singular i.e., $|\mathbf{A}| = 0$.

Note 1: Since the rank of every non-zero matrix is ≥ 1 , we agree to assign the rank, zero, to every null matrix :

Note 2: Every $(r + 1)$ -rowed minor of a matrix can be expressed as a linear combination of its r -rowed minors. Therefore if all the r -rowed minors of a matrix are equal to zero, then obviously all its $(r + 1)$ -rowed minors will also be equal to zero.

Important : The following two simple results will help us very much in finding the rank of a matrix:

- (i) *The rank of a matrix is $\leq r$, if all $(r + 1)$ - rowed minors of the matrix vanish.*
- (ii) *The rank of a matrix is $\geq r$, if there is at least one r -rowed minor of the matrix which is not equal to zero.*

Illustration:

$$(a) \quad \text{Let} \quad \mathbf{A} = \mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ be a unit matrix of order } 3.$$

We have $|\mathbf{A}| = 1$. Therefore $\mathbf{A}$ is a non-singular matrix. Hence rank $\mathbf{A} = 3$. In particular, the rank of a unit matrix of order n is n .

$$(b) \quad \text{Let} \quad \mathbf{A} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since $\mathbf{A}$ is a null matrix, therefore rank $\mathbf{A} = 0$.

$$(c) \quad \text{Let} \quad \mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 2 & 2 \end{bmatrix}.$$

We have $|\mathbf{A}| = 1(6 - 8) - 2(4 - 6) = 2 \neq 0$. Thus $\mathbf{A}$ is a non-singular matrix. Therefore rank $\mathbf{A} = 3$.

(d) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}$.

We have $|\mathbf{A}| = 1(24 - 25) - 2(18 - 20) + 3(15 - 16) = 0$

Therefore the rank of $\mathbf{A}$ is less than 3. Now there is at least one minor of $\mathbf{A}$ of order 2, namely $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$ which is not equal to zero. Hence rank $\mathbf{A} = 2$.

(e) Let $\mathbf{A} = \begin{bmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ 3 & 1 & 2 \end{bmatrix}$.

We have $|\mathbf{A}| = 0$, since the first two columns are identical.

Also each 2-rowed minor of $\mathbf{A}$ is equal to zero. But $\mathbf{A}$ is not a null matrix. Hence rank $\mathbf{A} = 1$.

(f) Let $\mathbf{A} = \begin{bmatrix} 2 & 4 & 3 & 2 \\ 3 & 5 & 1 & 4 \end{bmatrix}$.

Here we see that there is at least one minor of $\mathbf{A}$ of order 2 i.e., $\begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix}$ which is not equal to zero. Also there is no minor of $\mathbf{A}$ of order greater than 2. Hence the rank of $\mathbf{A} = 2$.

3 Echelon Form of a Matrix

Definition: A matrix $\mathbf{A}$ is said to be in Echelon form if :

- (i) Every row of $\mathbf{A}$ which has all its entries 0 occurs below every row which has a non-zero entry.
- (ii) The number of zeros before the first non-zero element in a row is less than the number of such zeros in the next row.

Important result: The rank of a matrix in Echelon form is equal to the number of non-zero rows of the matrix.

Illustration: Find the rank of the matrix $\begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

The matrix $\mathbf{A}$ has one zero row. We see that it occurs below every non-zero row.

Further the number of zeros before the first non-zero element in the first row is one. The number of zeros before the first non-zero element in the second row is two. Thus the number of zeros before the first non-zero element in any row is less than the number of such zeros in the next row.

Thus the matrix $\mathbf{A}$ is in Echelon form.

$\therefore$ rank $\mathbf{A}$ = the number of non-zero rows of $\mathbf{A}$ = 2.

4 Theorem

The rank of the transpose of a matrix is the same as that of the original matrix.

Proof: Let $\mathbf{A}$ be any matrix and $\mathbf{A}'$ be the transpose of the matrix $\mathbf{A}$. Let rank $\mathbf{A} = r$ and rank $\mathbf{A}' = s$. Then to prove that $r = s$.

We have rank $\mathbf{A} = r$

$\Rightarrow$ there exists at least one r -rowed square submatrix, say $\mathbf{R}$, of $\mathbf{A}$ such that

$$|\mathbf{R}'| = |\mathbf{R}| \neq 0$$

$\Rightarrow$ there exists at least one r -rowed square submatrix $\mathbf{R}'$ of $\mathbf{A}'$ such that

$$|\mathbf{R}'| = |\mathbf{R}| \neq 0$$

$\Rightarrow$ rank $\mathbf{A}' \geq r \Rightarrow s \geq r$(1)

Since $(\mathbf{A}')' = \mathbf{A}$, therefore interchanging the roles of $\mathbf{A}$ and $\mathbf{A}'$ in the above result (1), we have

$$r \geq s. \quad \text{...(2)}$$

Illustrative Examples

From (1) and (2), we conclude that $r = s$. Hence the result.

Example 1: Find the ranks of the following matrices :

$$(i) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix},$$

$$(ii) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}$$

(Bundelkhand 2009)

Solution : (i) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$.

We have $|\mathbf{A}| = 1(2 - 0) - 2(4 - 0) + 3(2 - 0)$, expanding along the first row
 $= 2 - 8 + 6 = 0$.

But there is at least one minor of order 2 of the matrix $\mathbf{A}$, namely $\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$ which is not equal to zero. Hence rank $\mathbf{A} = 2$.

(ii) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}$.

Here there is at least one minor of order 2 of the matrix $\mathbf{A}$, namely $\begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix}$ which is not equal to 0. Also there is no minor of the matrix $\mathbf{A}$ of order greater than 2. Hence rank $\mathbf{A} = 2$.

Example 2: Under what conditions the rank of the following matrix is 3

$$\mathbf{A} = \begin{bmatrix} 2 & 4 & 2 \\ 2 & 1 & 2 \\ 1 & 0 & x \end{bmatrix} ?$$

Solution: The rank of the given matrix $\mathbf{A} = 3$.

$\therefore$ The minor of order 3 of matrix $\mathbf{A} \neq 0$

i.e., $\begin{vmatrix} 2 & 4 & 2 \\ 2 & 1 & 2 \\ 1 & 0 & x \end{vmatrix} \neq 0$

or $2(x - 0) - 4(2x - 2) + 2(0 - 1) \neq 0$

or $2x - 8x + 8 - 2 \neq 0$

or $-6x + 6 \neq 0$

or $x \neq 1$.

Example 3: For which value of 'b' the rank of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ b & 13 & 10 \end{bmatrix}$ is 2?

Solution: The rank of the given matrix $\mathbf{A} = 2$.

$\therefore$ The minor of order 3 of matrix $\mathbf{A} = 0$

$$\text{i.e.,} \quad \begin{vmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ b & 13 & 10 \end{vmatrix} = 0$$

$$\text{or} \quad 1(30 - 26) - 5(0 - 2b) + 4(0 - 3b) = 0$$

$$\text{or} \quad 4 + 10b - 12b = 0$$

$$\text{or} \quad 4 - 2b = 0$$

$$\text{or} \quad b = 2.$$

Example 4: Find the values of a so that $\text{rank}(\mathbf{A}) < 3$, where $\mathbf{A}$ is the matrix

$$\mathbf{A} = \begin{bmatrix} 3a - 8 & 3 & 3 \\ 3 & 3a - 8 & 3 \\ 3 & 3 & 3a - 8 \end{bmatrix}.$$

Solution: The rank of the given matrix $\mathbf{A} < 3$

$\therefore$ The minor of order 3 of matrix $\mathbf{A} = 0$

$$\text{i.e.,} \quad \begin{vmatrix} 3a - 8 & 3 & 3 \\ 3 & 3a - 8 & 3 \\ 3 & 3 & 3a - 8 \end{vmatrix} = 0$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$, we get

$$\begin{vmatrix} 3a - 2 & 3a - 2 & 3a - 2 \\ 3 & 3a - 8 & 3 \\ 3 & 3 & 3a - 8 \end{vmatrix} = 0$$

$$\text{or} \quad (3a - 2) \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3a - 8 & 3 \\ 3 & 3 & 3a - 8 \end{vmatrix} = 0$$

Applying $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$, we get

$$(3a-2) \begin{vmatrix} 1 & 0 & 0 \\ 3 & 3a-11 & 0 \\ 3 & 0 & 3a-11 \end{vmatrix} = 0$$

or $(3a-2)(3a-11)(3a-11) = 0$

or $a = \frac{2}{3}, \frac{11}{3}, \frac{11}{3}.$

Example 5: Prove that the points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) are collinear if and only if the rank of the matrix.

$$A = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \text{ is less than three.}$$

(Kanpur 2001; Bundelkhand 07)

Solution: The condition is necessary:

Given that the points $(x_i, y_i); i = 1, 2, 3$ are collinear.

We are to prove that the rank of matrix $A = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$ is less than three.

The given points are collinear

$\Rightarrow$ The area of the triangle formed by these points is zero.

$$\Rightarrow \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \text{The rank of matrix } \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \text{ is less than } 3.$$

Hence the condition is necessary.

The condition is sufficient:

The rank of the given matrix $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ is less than 3.

We are to prove that the points $(x_i, y_i); i = 1, 2, 3$ are collinear.

$$\text{rank } \mathbf{A} < 3$$

$$\Rightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

The given points are collinear.

Hence the condition is sufficient.

Example 6: $\mathbf{A}$ is a non-zero column and $\mathbf{B}$ a non-zero row matrix, show that $\text{rank}(\mathbf{AB}) = 1$.

Solution : Let $\mathbf{A} = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \dots \\ a_{m1} \end{bmatrix}$ and $\mathbf{B} = [b_{11} \quad b_{12} \quad b_{13} \quad \dots \quad b_{1n}]$

be two non-zero column and row matrices respectively.

We have $\mathbf{AB} = \begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{11}b_{13} & \dots & a_{11}b_{1n} \\ a_{21}b_{11} & a_{21}b_{12} & a_{21}b_{13} & \dots & a_{21}b_{1n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1}b_{11} & a_{m1}b_{12} & a_{m1}b_{13} & \dots & a_{m1}b_{1n} \end{bmatrix}.$

Since $\mathbf{A}$ and $\mathbf{B}$ are non-zero matrices, therefore the matrix $\mathbf{AB}$ will also be non-zero. The matrix $\mathbf{AB}$ will have at least one non-zero element obtained by multiplying corresponding non-zero elements of $\mathbf{A}$ and $\mathbf{B}$.

All the two-rowed minors of $\mathbf{A}$ obviously vanish. But $\mathbf{A}$ is a non-zero matrix. Hence $\text{rank } \mathbf{A} = 1$.

Comprehensive Exercise 1

1. Determine the rank of each of the following matrices :

$$(i) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(iv) \begin{bmatrix} 5 & 10 \\ 3 & 6 \end{bmatrix}$$

(Meerut 2006B)

$$(v) \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

$$(vi) \begin{bmatrix} 1 & 2 & -7 & 5 \\ 0 & 5 & 0 & 8 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

$$(vii) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}$$

2. Show that the rank of a matrix is $\geq$ the rank of every sub-matrix thereof.

3. Show that the rank of a matrix does not alter on affixing any number of additional rows or columns of zeros.

4. If $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, find the rank of A and A^2 .

5. Under what conditions the rank of the following matrix is 3? Is it possible for the

rank to be 1? Why? $A = \begin{bmatrix} 2 & 4 & 2 \\ 3 & 1 & 4 \\ 1 & 0 & x \end{bmatrix}$.

(Kanpur 2010)

6. A is an n -rowed square matrix of rank $(n - 1)$, show that $\text{Adj. } A$ is not a null matrix.

Answers 1

1. (i) 4 (ii) 1 (iii) 0 (iv) 1 (v) 1
 (vi) 3 (vii) 2

4. Rank $\mathbf{A} = 3$, rank $\mathbf{A}^2 = 2$
5. $x \neq \frac{7}{5}$; No, because one minor of order 2 of $\mathbf{A}$ is non-zero

5 Elementary Operations or Elementary Transformations of a Matrix

An elementary transformation (or, an E -transformation) is an operation of any one of the following types :

1. *The interchange of any two rows(or columns).*
2. *The multiplication of the elements of any row (or column) by any non-zero number.*
3. *The addition to the elements of any other row (or column) the corresponding elements of any other row (or column) multiplied by any number.*

An elementary transformation is called a **row transformation** or a **column transformation** according as it applies to rows or columns.

6 Symbols to be Employed for the Elementary Transformations

The following notation will be used to denote the six elementary transformations :

1. The interchange of i^{th} and j^{th} rows will be denoted by $R_i \leftrightarrow R_j$.
2. The multiplication of the i^{th} row by a non-zero number k will be denoted by $R_i \rightarrow k R_i$.
3. The addition of k times the j^{th} row to the i^{th} row will be denoted by $R_i \rightarrow R_i + kR_j$.

The corresponding column transformations will be denoted by writing C , in place of R i.e., by $C_i \leftrightarrow C_j$, $C_i \rightarrow k C_i$, $C_i \rightarrow C_i + kC_j$ respectively.

Important: It is quite obvious that if a matrix $\mathbf{B}$ is obtained from $\mathbf{A}$ by an elementary transformation, $\mathbf{A}$ can also be obtained from $\mathbf{B}$ by an elementary transformation of the same type.

For example, let

$$\mathbf{A} = \begin{bmatrix} 1 & 4 & 2 & 9 \\ 2 & 5 & 1 & 3 \\ 3 & 7 & 8 & 4 \end{bmatrix}_{3 \times 4}.$$

The elementary transformation $R_2 \rightarrow R_2 + 2R_3$ transforms **A** into a matrix **B**, where

$$\mathbf{B} = \begin{bmatrix} 1 & 4 & 2 & 9 \\ 8 & 19 & 17 & 11 \\ 3 & 7 & 8 & 4 \end{bmatrix}_{3 \times 4}.$$

Now if we apply the elementary transformation $R_2 \rightarrow R_2 - 2R_3$ to the matrix **B**, we see that the matrix **B** transforms to the matrix **A**.

Again suppose we apply the elementary transformation $R_2 \rightarrow 3R_2$ to the matrix **A**.

Then **A** transforms into a matrix **C**, where

$$\mathbf{C} = \begin{bmatrix} 1 & 4 & 2 & 9 \\ 6 & 15 & 3 & 9 \\ 3 & 7 & 8 & 4 \end{bmatrix}.$$

Now if we apply the elementary transformation $R_2 \rightarrow \frac{1}{3} R_2$ to the matrix **C**, we see that the matrix **C** transforms back to the matrix **A**.

7 Elementary Matrices

Definition: A matrix obtained from a unit matrix by a single elementary transformation is called an elementary matrix (or *E*-matrix). For example,

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

are the elementary matrices obtained from $\mathbf{I}_3$ by subjecting it to the elementary operations $C_1 \leftrightarrow C_3$, $R_2 \rightarrow 4R_2$, $R_1 \rightarrow R_1 + 2R_2$ respectively.

It may be worthwhile to note that an *E*-matrix can be obtained from **I** by subjecting it to a row transformation or a column transformation. We shall use the following symbols to denote elementary matrices of different types :

- (i) E_{ij} will denote the *E*-matrix obtained by interchanging the i^{th} and j^{th} rows of a unit matrix. The students can easily see that the matrices obtained by interchanging the i^{th} and j^{th} rows or the i^{th} and j^{th} columns of a unit matrix are the same. Therefore E_{ij} will also denote the elementary matrix obtained by interchanging the i^{th} and j^{th} columns of a unit matrix.

- (ii) $\mathbf{E}_i(k)$ will denote the E -matrix obtained by multiplying the i^{th} row of a unit matrix by a non-zero number k . It can be easily seen that the matrices obtained by multiplying the i^{th} row or the i^{th} column of a unit matrix by k are the same. Therefore $\mathbf{E}_i(k)$ will also denote the elementary matrix obtained by multiplying the i^{th} column of a unit matrix by a non-zero number k .
- (iii) $\mathbf{E}_{ij}(m)$ will denote the elementary matrix, obtained by adding to the elements of the i^{th} row of a unit matrix, the products by any number m of the corresponding elements of the j^{th} row. It may be easily seen that the E -matrix, $\mathbf{E}_{ij}(m)$ can also be obtained by adding to the elements of the j^{th} column of a unit matrix, the products by m of the corresponding elements of the i^{th} column.

It can be easily seen that

$$|\mathbf{E}_{ij}| = -1, |\mathbf{E}_i(k)| = k \neq 0, |\mathbf{E}_{ij}(m)| = 1.$$

Thus all the elementary matrices are non-singular.

Therefore each elementary matrix possesses inverse.

Now it is very interesting to note that the elementary transformations of a matrix can also be obtained by algebraic operations on the same by the corresponding elementary matrices. In this connection we have the following theorem.

8 Theorem

Every elementary row (column) transformation of a matrix can be obtained by pre-multiplication (post-multiplication) with the corresponding elementary matrix.

We shall first prove that every elementary row transformation of a product $\mathbf{AB}$ of two matrices $\mathbf{A}$ and $\mathbf{B}$ can be obtained by subjecting the pre-factor $\mathbf{A}$ to the same elementary row transformation. Similarly every elementary column transformation of a product $\mathbf{AB}$ of two matrices $\mathbf{A}$ and $\mathbf{B}$ can be obtained by subjecting the post-factor $\mathbf{B}$ to the same elementary column transformation.

Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{jk}]$ be two $m \times n$ and $n \times p$ matrices respectively so that the product $\mathbf{AB}$ is defined.

Let $\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \dots, \mathbf{R}_m$ denote the row vectors of the matrix $\mathbf{A}$ and $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_p$ denote the column vectors of the matrix $\mathbf{B}$.

We can then write, $\mathbf{A} = \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \\ \mathbf{R}_3 \\ \dots \\ \mathbf{R}_m \end{bmatrix}, \mathbf{B} = [\mathbf{C}_1 \ \mathbf{C}_2 \ \mathbf{C}_3 \ \dots \ \mathbf{C}_p].$

$$\therefore \mathbf{AB} = \begin{bmatrix} \mathbf{R}_1\mathbf{C}_1 & \mathbf{R}_1\mathbf{C}_2 & \mathbf{R}_1\mathbf{C}_3 & \dots & \mathbf{R}_1\mathbf{C}_p \\ \mathbf{R}_2\mathbf{C}_1 & \mathbf{R}_2\mathbf{C}_2 & \mathbf{R}_2\mathbf{C}_3 & \dots & \mathbf{R}_2\mathbf{C}_p \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \mathbf{R}_m\mathbf{C}_1 & \mathbf{R}_m\mathbf{C}_2 & \mathbf{R}_m\mathbf{C}_3 & \dots & \mathbf{R}_m\mathbf{C}_p \end{bmatrix}$$

Now if σ denotes any elementary row transformation, it is quite obvious from the above representation that $(\sigma \mathbf{A}) \mathbf{B} = \sigma (\mathbf{AB})$. For example, if σ denotes the elementary row transformation $R_1 \leftrightarrow R_2$, it is quite obvious that $(\sigma \mathbf{A}) \mathbf{B} = \sigma (\mathbf{AB})$.

Similarly it is quite obvious that if the columns $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_p$ of $\mathbf{B}$ be subjected to any elementary column transformation, the columns of $\mathbf{AB}$ are also subjected to the same elementary column transformation. Hence the result.

Now to prove our main theorem, if $\mathbf{A}$ be an $m \times n$ matrix, we can write

$$\mathbf{A} = \mathbf{I}_m \mathbf{A}.$$

If σ denotes any elementary row transformation, we have

$$\sigma \mathbf{A} = \sigma (\mathbf{I}_m \mathbf{A}) = (\sigma \mathbf{I}_m) \mathbf{A} = \mathbf{EA},$$

where $\mathbf{E}$ is the E -matrix corresponding to the same row transformation σ .

Similarly, we can write $\mathbf{A} = \mathbf{A} \mathbf{I}_n$.

If σ denotes any elementary column transformation, we have

$$\sigma (\mathbf{A}) = \sigma (\mathbf{A} \mathbf{I}_n) = \mathbf{A} \sigma (\mathbf{I}_n) = \mathbf{AE}_1,$$

where $\mathbf{E}_1$ is the E -matrix corresponding to the same column transformation σ .

Illustration : Let $\mathbf{A} = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 7 & 1 \\ 3 & 8 & 4 \end{bmatrix}.$

The E -transformation $R_1 \rightarrow R_1 + 2R_3$ transforms $\mathbf{A}$ into $\mathbf{B}$, where

$$\mathbf{B} = \begin{bmatrix} 7 & 20 & 10 \\ 2 & 7 & 1 \\ 3 & 8 & 4 \end{bmatrix}.$$

Also if we apply the row transformation $R_1 \rightarrow R_1 + 2R_3$ to the unit matrix

$$\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

then the E -matrix $\mathbf{E}$ thus obtained is $\mathbf{E} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Now

$$\begin{aligned} \mathbf{E}\mathbf{A} &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 2 \\ 2 & 7 & 1 \\ 3 & 8 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 \cdot 1 + 0 \cdot 2 + 2 \cdot 3 & 1 \cdot 4 + 0 \cdot 7 + 2 \cdot 8 & 1 \cdot 2 + 0 \cdot 1 + 2 \cdot 4 \\ 0 \cdot 1 + 1 \cdot 2 + 0 \cdot 3 & 0 \cdot 4 + 1 \cdot 7 + 0 \cdot 8 & 0 \cdot 2 + 1 \cdot 1 + 0 \cdot 4 \\ 0 \cdot 1 + 0 \cdot 2 + 1 \cdot 3 & 0 \cdot 4 + 0 \cdot 7 + 1 \cdot 8 & 0 \cdot 2 + 0 \cdot 1 + 1 \cdot 4 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 20 & 10 \\ 2 & 7 & 1 \\ 3 & 8 & 4 \end{bmatrix} = \mathbf{B}. \end{aligned}$$

Similarly we can see that a column transformation of $\mathbf{A}$ can be effected by post-multiplying $\mathbf{A}$ with the corresponding elementary matrix.

Non-singularity and Inverses of the Elementary Matrices

(i) The elementary matrix corresponding to the E -operation $R_i \leftrightarrow R_j$ is its own inverse.

Let $\mathbf{E}_{ij}$ denote the elementary matrix obtained by interchanging the i^{th} and j^{th} rows of a unit matrix.

The interchange of the i^{th} and j^{th} rows of $\mathbf{E}_{ij}$ will transform $\mathbf{E}_{ij}$ to the unit matrix. But every elementary row transformation of a matrix can be brought about by pre-multiplication with the corresponding elementary matrix. Therefore the row transformation which changes $\mathbf{E}_{ij}$ to $\mathbf{I}$ can be effected on pre-multiplication by $\mathbf{E}_{ij}$.

Thus $\mathbf{E}_{ij} \mathbf{E}_{ij} = \mathbf{I}$ or $(\mathbf{E}_{ij})^{-1} = \mathbf{E}_{ij}$.

Hence $\mathbf{E}_{ij}$ is its own inverse.

Similarly, we can show that the elementary matrix corresponding to the E-operation $C_i \leftrightarrow C_j$ is its own inverse.

(ii) The inverse of the E-matrix corresponding to the E-operation $R_i \rightarrow kR_i$, ($k \neq 0$), is the E-matrix corresponding to the E-operation $R_i \rightarrow k^{-1} R_i$.

Let $\mathbf{E}_i(k)$ denote the elementary matrix obtained by multiplying the elements of the i^{th} row of a unit matrix $\mathbf{I}$ by a non-zero number k .

The operation of the multiplication of the i^{th} row of $\mathbf{E}_i(k)$, by k^{-1} will transform $\mathbf{E}_i(k)$ to the unit matrix $\mathbf{I}$. This row transformation of $\mathbf{E}_i(k)$ can be effected on pre-multiplication by the corresponding elementary matrix $\mathbf{E}_i(k^{-1})$.

Thus $\mathbf{E}_i(k^{-1}) \mathbf{E}_i(k) = \mathbf{I}$ or $\{\mathbf{E}_i(k)\}^{-1} = \mathbf{E}_i(k^{-1})$.

Similarly, we can show that the inverse of the E-matrix corresponding to the E-operation $C_i \rightarrow kC_i$, $k \neq 0$, is the E-matrix corresponding to the E-operation $C_i \rightarrow k^{-1} C_i$.

(iii) The inverse of the E-matrix corresponding to the E-operation $R_i \rightarrow R_i + kR_j$ is the E-matrix corresponding to the E-operation $R_i \rightarrow R_i - kR_j$.

Let $\mathbf{E}_{ij}(k)$ denote the elementary matrix obtained by adding to the elements of the i^{th} row of a unit matrix $\mathbf{I}$, the products by any number k of the corresponding elements of the j^{th} row of $\mathbf{I}$.

If we add to elements of the i^{th} row of $\mathbf{E}_{ij}(k)$, the products by $-k$ of the corresponding elements of its j^{th} row, then this row operation will transform $\mathbf{E}_{ij}(k)$ to the unit matrix $\mathbf{I}$. Now this row transformation of $\mathbf{E}_{ij}(k)$ can be effected on pre-multiplication by the corresponding elementary matrix $\mathbf{E}_{ij}(-k)$.

Therefore $\mathbf{E}_{ij}(-k) \mathbf{E}_{ij}(k) = \mathbf{I}$ or $\{\mathbf{E}_{ij}(k)\}^{-1} = \mathbf{E}_{ij}(-k)$.

Similarly, we can show that the inverse of the E-matrix corresponding to the E-operation $C_i \rightarrow C_i + kC_j$ is the E-matrix corresponding to the E-operation $C_i \rightarrow C_i - kC_j$.

From the above theorem, we thus conclude that the **inverse of an elementary matrix is also an elementary matrix of the same type.**

9 Invariance of Rank Under Elementary Transformations

Theorem: *Elementary transformations do not change the rank of a matrix.*

(Lucknow 2005, 06)

Proof: Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix of rank r . We shall prove the theorem in three stages.

Case I. *Interchange of a pair of rows does not change the rank.*

Let $\mathbf{B}$ be the matrix obtained from the matrix $\mathbf{A}$ by the E -transformation $R_p \leftrightarrow R_q$. Let r be the rank of $\mathbf{A}$ and s be the rank of $\mathbf{B}$. Then to prove that $r = s$.

We have $\text{rank } \mathbf{A} = r \Rightarrow$ there exists at least one r -rowed square submatrix, say $\mathbf{R}$, of $\mathbf{A}$ such that $|\mathbf{R}| \neq 0$.

Let $\mathbf{S}$ be the r -rowed square submatrix of $\mathbf{B}$ which has the same rows as are in $\mathbf{R}$ though they may be in different relative positions.

Then either $|\mathbf{S}| = |\mathbf{R}|$ or $|\mathbf{S}| = -|\mathbf{R}|$.

$$\therefore |\mathbf{R}| \neq 0 \Rightarrow |\mathbf{S}| \neq 0.$$

$$\therefore \text{rank } \mathbf{B} \geq r \Rightarrow s \geq r.$$

Again, as $\mathbf{A}$ can also be obtained from $\mathbf{B}$ by an interchange of rows, we have $r \geq s$.

Hence $r = s$.

Case II. *Multiplication of the elements of a row by a non-zero number does not change the rank.*

Let $\mathbf{B}$ be the matrix obtained from the matrix $\mathbf{A}$ by the E -transformation $R_p \rightarrow k R_p, (k \neq 0)$, and let s be the rank of the matrix $\mathbf{B}$.

Now if $|\mathbf{B}_0|$ be any $(r+1)$ -rowed minor of $\mathbf{B}$ there exists a uniquely determined minor $|\mathbf{A}_0|$ of $\mathbf{A}$ such that

$$|\mathbf{B}_0| = |\mathbf{A}_0| \quad (\text{this happens if the } p^{\text{th}} \text{ row of } \mathbf{B} \text{ is one of those rows which}$$

are struck off to obtain $\mathbf{B}_0$ from $\mathbf{B}$),

$$\text{or} \quad |\mathbf{B}_0| = k |\mathbf{A}_0|$$

(this happens when p^{th} row is retained while obtaining $\mathbf{B}_0$ from $\mathbf{B}$).

Since the matrix $\mathbf{A}$ is of rank r , therefore every $(r + 1)$ -rowed minor of $\mathbf{A}$ vanishes *i.e.*, $|\mathbf{A}_0| = 0$. Hence $|\mathbf{B}_0| = 0$. Thus we see that every $(r + 1)$ -rowed minor of $\mathbf{B}$ also vanishes. Therefore, s (the rank of $\mathbf{B}$) cannot exceed r (the rank of $\mathbf{A}$).

$$\therefore \quad s \leq r.$$

Also, since $\mathbf{A}$ can be obtained from $\mathbf{B}$ by E -transformation of the same type *i.e.*, $R_p \rightarrow (1/k) R_p$, therefore, by interchanging the roles of $\mathbf{A}$ and $\mathbf{B}$ we find that

$$r \leq s.$$

Thus $r = s.$

Case III. *Addition to the elements of a row the products by any number k of the corresponding elements of any other row does not change the rank.*

Let $\mathbf{B}$ be the matrix obtained from the matrix $\mathbf{A}$ by the E -transformation $R_p \rightarrow R_p + kR_q$ and let s be the rank of the matrix $\mathbf{B}$.

Let $\mathbf{B}_0$ be any $(r + 1)$ -rowed square sub-matrix of $\mathbf{B}$ and $\mathbf{A}_0$ be the correspondingly placed sub-matrix of $\mathbf{A}$.

The transformation $R_p \rightarrow R_p + kR_q$ has changed only the p^{th} row of the matrix $\mathbf{A}$. Also the value of a determinant does not change if we add to the elements of any row the corresponding elements of any other row multiplied by some number k .

Therefore, if no row of the sub-matrix $\mathbf{A}_0$ is part of the p^{th} row of $\mathbf{A}$, or if two rows of $\mathbf{A}_0$ are parts of the p^{th} and q^{th} rows of $\mathbf{A}$, then $|\mathbf{B}_0| = |\mathbf{A}_0|$.

Since the rank of $\mathbf{A}$ is r , therefore $|\mathbf{A}_0| = 0$, and consequently $|\mathbf{B}_0| = 0$.

Again, if a row of $\mathbf{A}_0$ is a part of the p^{th} row of $\mathbf{A}$, but no row is a part of the q^{th} row, then $|\mathbf{B}_0| = |\mathbf{A}_0| + k |\mathbf{C}_0|$, where $\mathbf{C}_0$ is an $(r + 1)$ -rowed square matrix which can be obtained from $\mathbf{A}_0$ by replacing the elements of $\mathbf{A}_0$ in the row which corresponds to the p^{th} row of $\mathbf{A}$ by the corresponding elements in the q^{th} row of $\mathbf{A}$. Obviously all the $r + 1$ rows of the matrix $\mathbf{C}_0$ are exactly the same as the rows of some $(r + 1)$ -rowed square sub-matrix of $\mathbf{A}$, though arranged in some different order. Therefore $|\mathbf{C}_0|$ is ± 1 times some $(r + 1)$ -rowed minor of $\mathbf{A}$. Since the rank of $\mathbf{A}$ is r , therefore, every $(r + 1)$ -rowed minor of $\mathbf{A}$ is zero, so that $|\mathbf{A}_0| = 0, |\mathbf{C}_0| = 0$, and consequently $|\mathbf{B}_0| = 0$.

Thus we see that every $(r + 1)$ -rowed minor of $\mathbf{B}$ also vanishes. Hence, s (the rank of $\mathbf{B}$) cannot exceed r (the rank of $\mathbf{A}$).

$$\therefore \quad s \leq r.$$

Also, since $\mathbf{A}$ can be obtained from $\mathbf{B}$ by an E -transformation of the same type *i.e.*, $R_p \rightarrow R_p - kR_q$, therefore, interchanging the roles of $\mathbf{A}$ and $\mathbf{B}$, we have $r \leq s$.

Thus $r = s$.

We have thus shown that rank of a matrix remains unaltered by any E -row transformation. Therefore we can also say that the rank of a matrix remains unaltered by a series of elementary row transformations.

Similarly we can show that the rank of a matrix remains unaltered by a series of elementary column transformations.

Finally, we conclude that the rank of a matrix remains unaltered by a finite chain of elementary operations.

Corollary: We have already proved that every elementary row (column) transformation of a matrix can be effected by pre-multiplication (post-multiplication) with the corresponding elementary matrix. Combining this theorem with the theorem just established, we conclude the following important results :

The pre-multiplication or post-multiplication by any elementary matrix, and as such by any series of elementary matrices, does not alter the rank of a matrix.

10 Reduction to Normal Form

Theorem: Every $m \times n$ matrix of rank r can be reduced to the form $\begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$ by a finite chain of E -operations where $\mathbf{I}_r$, is the r -rowed unit matrix.

Proof: Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix of rank r . If $\mathbf{A}$ is a zero matrix, then r is equal to zero and we have nothing to prove. So let us take $\mathbf{A}$ as a non-zero matrix.

Since $\mathbf{A}$ is a non-zero matrix, therefore $\mathbf{A}$ has at least one element different from zero, say $a_{pq} = k \neq 0$.

By interchanging the p^{th} row with the first row and the q^{th} column with the first column respectively, we obtain a matrix $\mathbf{B}$ whose leading element is equal to k which is not equal to zero.

Multiplying the elements of the first row of the matrix $\mathbf{B}$ by $1/k$, we obtain a matrix $\mathbf{C}$ whose leading element is equal to unity.

Let
$$\mathbf{C} = \begin{bmatrix} 1 & c_{12} & c_{13} & \cdots & c_{1n} \\ c_{21} & c_{22} & c_{23} & \cdots & c_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ c_{m1} & c_{m2} & c_{m3} & \cdots & c_{mn} \end{bmatrix}.$$

Subtracting suitable multiples of the first column of $\mathbf{C}$ from the remaining columns, and suitable multiples of the first row from the remaining rows, we obtain a matrix $\mathbf{D}$ in which all elements of the first row and first column except the leading element are equal to zero.

Let
$$\mathbf{D} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & & & & & \\ 0 & & \mathbf{A}_1 & & & \\ 0 & & & & & \\ \cdots & & & & & \\ 0 & & & & & \end{bmatrix},$$

where $\mathbf{A}_1$ is an $(m-1) \times (n-1)$ matrix.

If now, $\mathbf{A}_1$ be a non-zero matrix, we can deal with it as we did with $\mathbf{A}$. If the elementary operations applied to $\mathbf{A}_1$ for this purpose be applied to $\mathbf{D}$, they will not affect the first row and the first column of $\mathbf{D}$. Continuing this process, we shall finally obtain a matrix $\mathbf{M}$, such that

$$\mathbf{M} = \begin{bmatrix} \mathbf{I}_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}.$$

The matrix $\mathbf{M}$ has the rank k . Since the matrix $\mathbf{M}$ has been obtained from the matrix $\mathbf{A}$ by elementary transformations and elementary transformations do not alter the rank, therefore we must have $k = r$.

Hence every $m \times n$ matrix of rank r can be reduced to the form $\begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$ by a finite chain of elementary transformations.

Note: The above form is usually called the **first canonical form** or **normal form** of a matrix.

Corollary 1: The rank of an $m \times n$ matrix $\mathbf{A}$ is r if and only if (iff) it can be reduced to the form

$$\begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \text{ by a finite chain of E-operations.}$$

The condition is necessary. The proof has been given in the above theorem.

The condition is also sufficient. The matrix $\mathbf{A}$ has been transformed into the form $\begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$ by elementary transformations which do not alter the rank of the matrix.

Since the rank of the matrix $\begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$ is r , therefore the rank of the matrix $\mathbf{A}$ must also be r .

Corollary 2: If $\mathbf{A}$ be an $m \times n$ matrix of rank r , there exist non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$ such that

$$\mathbf{PAQ} = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}.$$

Proof: If $\mathbf{A}$ be an $m \times n$ matrix of rank r , it can be transformed into the form $\begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$ by elementary operations. Since E -row (column) operations are equivalent to pre-(post)-multiplication by the corresponding elementary matrices, we have the following result :

If $\mathbf{A}$ be an $m \times n$ matrix of rank r , there exist E -matrices $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_s, \mathbf{Q}_1, \mathbf{Q}_2, \dots, \mathbf{Q}_t$ such that

$$\mathbf{P}_s \mathbf{P}_{s-1} \dots \mathbf{P}_1 \mathbf{A} \mathbf{Q}_1 \mathbf{Q}_2 \dots \mathbf{Q}_t = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}.$$

Now each elementary matrix is non-singular and the product of non-singular matrices is also non-singular. Therefore if $\mathbf{P} = \mathbf{P}_s \mathbf{P}_{s-1} \dots \mathbf{P}_1$ and $\mathbf{Q} = \mathbf{Q}_1 \mathbf{Q}_2 \dots \mathbf{Q}_t$, then $\mathbf{P}$ and $\mathbf{Q}$ are non-singular matrices. Hence

$$\mathbf{PAQ} = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}.$$

11 Equivalence of Matrices

Definition: If $\mathbf{B}$ be an $m \times n$ matrix obtained from an $m \times n$ matrix $\mathbf{A}$ by finite number of elementary transformations of $\mathbf{A}$ then $\mathbf{A}$ is called equivalent to $\mathbf{B}$. Symbolically, we write $\mathbf{A} \sim \mathbf{B}$, which is read as ' $\mathbf{A}$ is equivalent to $\mathbf{B}$.'

The following three properties of the relation ' $\sim$ ' in the set of all $m \times n$ matrices are quite obvious :

(i) **Reflexivity:** If $\mathbf{A}$ is any $m \times n$ matrix, then $\mathbf{A} \sim \mathbf{A}$. Obviously $\mathbf{A}$ can be obtained from $\mathbf{A}$ by the elementary transformation $R_i \rightarrow kR_i$, where $k = 1$.

(ii) **Symmetry:** If $\mathbf{A} \sim \mathbf{B}$, then $\mathbf{B} \sim \mathbf{A}$. If $\mathbf{B}$ can be obtained from $\mathbf{A}$ by a finite number of elementary transformations of $\mathbf{A}$, then $\mathbf{A}$ can also be obtained from $\mathbf{B}$ by a finite number of elementary transformations of $\mathbf{B}$.

(iii) **Transitivity:** If $\mathbf{A} \sim \mathbf{B}$, $\mathbf{B} \sim \mathbf{C}$, then $\mathbf{A} \sim \mathbf{C}$.

If $\mathbf{B}$ can be obtained from $\mathbf{A}$ by a series of elementary transformations of $\mathbf{A}$ and $\mathbf{C}$ can be obtained from $\mathbf{B}$ by a series of elementary transformations of $\mathbf{B}$, then $\mathbf{C}$ can also be obtained from $\mathbf{A}$ by a series of elementary transformations of $\mathbf{A}$.

Therefore the relation ' $\sim$ ' in the set of all $m \times n$ matrices is an equivalence relation.

Illustrative Examples

Example 7: If $\mathbf{A}$ and $\mathbf{B}$ be two equivalent matrices, then show that $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$.

Solution : If $\mathbf{A} \sim \mathbf{B}$, then $\mathbf{B}$ can be obtained from $\mathbf{A}$ by a finite number of elementary transformations of $\mathbf{A}$. Now the elementary transformations do not change the rank of a matrix.

$\therefore$ If $\mathbf{A} \sim \mathbf{B}$, then $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$.

Example 8: Show that if two matrices $\mathbf{A}$ and $\mathbf{B}$ have the same size and the same rank, they are equivalent.

Solution: Let $\mathbf{A}$ and $\mathbf{B}$ be two $m \times n$ matrices of the same rank r . Then by article 10, we have

$$\mathbf{A} \sim \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \text{ and also } \mathbf{B} \sim \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}.$$

By the symmetry of the equivalence relation,

$$\mathbf{B} \sim \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \text{ implies } \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \sim \mathbf{B}.$$

Now by the transitivity of the equivalence relation

$$\mathbf{A} \sim \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \text{ and } \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \sim \mathbf{B} \text{ implies } \mathbf{A} \sim \mathbf{B}.$$

Example 9: (i) Use elementary transformations to reduce the following matrix **A** to triangular form and hence find rank **A** :

$$\mathbf{A} = \begin{bmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix}.$$

(Bundelkhand 2010)

(ii) Find the rank of the matrix $\begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}.$

Solution : (i) We have the matrix

$$\mathbf{A} \sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 5 & 3 & 14 & 4 \end{bmatrix} \text{ by } R_1 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 8 & 4 & 4 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - 5R_1$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -12 & -4 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - 8R_2.$$

The last equivalent matrix is in Echelon form (or in triangular form). The number of non-zero rows in this matrix is 3. Therefore its rank is 3. Hence rank **A** = 3.

(ii) Let us denote the given matrix by **A**. To find the rank of **A**, we shall reduce it to Echelon form. Performing the column operation $C_1 \rightarrow \frac{1}{8}C_1$, we get

$$\mathbf{A} \sim \begin{bmatrix} 1 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -1 & -1 & -3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ 0 & 0 & 0 & 10 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 + R_1.$$

The last equivalent matrix is in Echelon form. The number of non-zero rows in this matrix is 3. Therefore its rank is 3. Hence rank **A** = 3.

Example 10: Is the matrix $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix}$ equivalent to **I**₃?

(Meerut 2008)

Solution: Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix}$.

We have $|\mathbf{A}| = \begin{vmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & -3 & -5 \end{vmatrix} R_2 + R_1, R_3 - 2R_1$

$$= -10 + 9 = -1 \text{ i.e., } \neq 0.$$

Thus the matrix $\mathbf{A}$ is non-singular. Hence it is of rank 3. The rank of $\mathbf{I}_3$ is also 3. Since $\mathbf{A}$ and $\mathbf{I}_3$ are matrices of the same size and the same rank, therefore $\mathbf{A} \sim \mathbf{I}_3$.

Example 11: Reduce the matrix, $\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$ to the normal form $\begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$ and

hence determine its rank.

(Meerut 2001, 09B, 10)

Solution: We have the matrix $\mathbf{A}$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 5 & -8 & 14 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix} \text{ by } C_2 \rightarrow C_2 + C_1, C_3 \rightarrow C_3 - 2C_1, C_4 \rightarrow C_4 + 3C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & -8 & 14 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix} \text{ by } R_2 \rightarrow R_2 - 4R_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 5 & -8 & 14 \end{bmatrix} \text{ by } R_2 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 3 & 0 & -2 \\ 0 & 5 & -8 & 4 \end{bmatrix} \text{ by } C_4 \rightarrow C_4 - 2C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -8 & 4 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - 3R_2, R_4 \rightarrow R_4 - 5R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 4 & -8 \end{bmatrix} \text{ by } C_3 \leftrightarrow C_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{bmatrix} \text{ by } C_3 \rightarrow -\frac{1}{2}C_3, C_4 \rightarrow -\frac{1}{8}C_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ by } R_4 \rightarrow R_4 + 2R_3$$

$\sim I_4$. Hence the matrix **A** is of rank 4.

Example 12: Determine the rank of the following matrices :

$$(i) \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$$

(Kanpur 2010)

$$(ii) \begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

(Meerut, 2003, 09 B, 10B; Garhwal 14)

Solution : (i) Let us denote the given matrix by **A**. Performing the elementary operations $R_3 \rightarrow R_3 - R_1, R_4 \rightarrow R_4 - R_1$, we see that

$$\begin{aligned}
 \mathbf{A} &\sim \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 4 & 1 \\ 0 & 6 & 8 & 2 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_4 \rightarrow R_4 - 2R_2 \\
 &\sim \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 12 & 16 & 4 \\ 0 & 12 & 12 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_2 \rightarrow 4R_2, R_3 \rightarrow 3R_3 \\
 &\sim \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 12 & 16 & 4 \\ 0 & 0 & -4 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - R_2.
 \end{aligned}$$

The last equivalent matrix is in Echelon form. The number of non-zero rows in this matrix is 3. Therefore its rank is 3. Hence $\text{rank } \mathbf{A} = 3$.

(ii) Let us denote the given matrix by $\mathbf{A}$. Performing the elementary operation $R_1 \leftrightarrow R_2$, we see that

$$\begin{aligned}
 \mathbf{A} &\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ -2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} \\
 &\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 3 & 3 & -3 \\ 0 & -2 & -2 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix} \text{ by } R_2 \rightarrow R_2 + 2R_1, R_3 \rightarrow R_3 - R_1 \\
 &\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \end{bmatrix} \text{ by } R_2 \rightarrow \frac{1}{3}R_2, R_3 \rightarrow -\frac{1}{2}R_3 \\
 &\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - R_2.
 \end{aligned}$$

The last equivalent matrix is in Echelon form. The number of non-zero rows in this matrix is 2. Therefore rank $\mathbf{A} = 2$.

Example 13: Find the rank of the matrix $\mathbf{A} = \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$ by reducing it to normal form.

(Meerut 2008; Avadh 08; Rohilkhand 09)

Solution : Performing the operation $R_1 \rightarrow \frac{1}{2}R_1, R_2 \rightarrow \frac{1}{2}R_2$, we see that

$$\mathbf{A} \sim \begin{bmatrix} 1 & -1 & 0 & 3 \\ 2 & 1 & 0 & 1 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 3 & 0 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix}$$

by $R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - R_1, R_4 \rightarrow R_4 - R_1$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix} \text{ by } C_2 \rightarrow C_2 + C_1, C_4 \rightarrow C_4 - 3C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & -5 \end{bmatrix} \text{ by } R_2 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & -5 \end{bmatrix} \text{ by } R_2 \rightarrow (-1)R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & -8 \end{bmatrix} \text{ by } R_4 \rightarrow R_4 - 3R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & -8 \end{bmatrix} \text{ by } C_3 \rightarrow C_3 + C_2, C_4 \rightarrow C_4 - C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_4 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } C_3 \rightarrow \frac{1}{3}C_3, C_4 \rightarrow -\frac{1}{8}C_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } C_4 \rightarrow C_4 - C_3,$$

which is the normal form $\begin{bmatrix} \mathbf{I}_3 & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$. Hence rank $\mathbf{A} = 3$.

Example 14: Find two non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$ such that $\mathbf{PAQ}$ is in the normal form where

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix}.$$

Also find the rank of the matrix $\mathbf{A}$.

(Garhwal 2006, 10; Kumaun 10)

Solution: We write $\mathbf{A} = \mathbf{I}_3 \mathbf{A} \mathbf{I}_3$ i.e.,

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now we go on applying E -operations on the matrix $\mathbf{A}$ (the left hand member of the above equation) until it is reduced to the normal form. Every E -row operation will also be applied to the pre-factor $\mathbf{I}_3$ (or its transform) of the product on the right hand member of the above equation and every E -column operation to the post-factor $\mathbf{I}_3$ (or its transform).

Performing $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 3R_1$, we get

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -2 \\ 0 & -2 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & -2 \\ 0 & -2 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_2 \rightarrow -\frac{1}{2}R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -3 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_3 \rightarrow R_3 + 2R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -2 & -1 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $C_3 \rightarrow C_3 - C_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -2 & -1 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\therefore \mathbf{PAQ} = \begin{bmatrix} \mathbf{I}_2 & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix},$$

where $\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -2 & -1 & 1 \end{bmatrix}, \mathbf{Q} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$

Since $\mathbf{A} \sim \begin{bmatrix} \mathbf{I}_2 & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$, therefore $\text{rank } \mathbf{A} = 2.$

Comprehensive Exercise 2

Determine the rank of the following matrices :

1.
$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$$

2.
$$\begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$$

(Bundelkhand 2008;
Rohilkhand 08; Avadh 05; Lucknow 11)

3.
$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

(Avadh 2006, 09; Purvanchal 06)

4.
$$\begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 1 & a & b & 0 \\ 0 & c & d & 1 \end{bmatrix}$$

(Lucknow 2007)

5.
$$\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ 1 & 3 & 4 & 1 \end{bmatrix}$$

(Rohilkhand 2007)

6.
$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

7.
$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$$

(Lucknow 08)

8.
$$\begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

9.
$$\begin{bmatrix} 1 & 2 & -4 & 5 \\ 2 & -1 & 3 & 6 \\ 8 & 1 & 9 & 7 \end{bmatrix}$$

10.
$$\begin{bmatrix} 1 & -1 & 3 & 6 \\ 1 & 3 & -3 & -4 \\ 5 & 3 & 3 & 11 \end{bmatrix}$$

11.
$$\begin{bmatrix} 8 & 0 & 0 & 1 \\ 1 & 0 & 8 & 1 \\ 0 & 0 & 1 & 8 \\ 0 & 8 & 1 & 8 \end{bmatrix}$$

(Kumaun 2008)

12.
$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

(Garhwal 2006, 13; Avadh 10)

13. $\begin{bmatrix} 1 & -3 & 4 & 7 \\ 9 & 1 & 2 & 0 \end{bmatrix}$ (Kumaun 2008)

14. $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$

15. $\begin{bmatrix} 2 & 1 & 3 \\ 4 & 7 & 13 \\ 4 & -3 & -1 \end{bmatrix}$ (Kumaun 2012)

16. $\begin{bmatrix} 4 & 5 & 6 \\ 5 & 6 & 7 \\ 7 & 8 & 9 \end{bmatrix}$

17. $\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 1 & -1 & 4 & 0 \\ -2 & 2 & 8 & 0 \end{bmatrix}$

18. $\begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{bmatrix}$

(Meerut 2011; Kumaun 12)

19. With the help of elementary transformations find the rank of the following matrix:

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 3 & 0 & 3 \\ 1 & -2 & -3 & -3 \\ 1 & 1 & 2 & 3 \end{bmatrix}$$

(Rohilkhand 2010)

20. Reduce the following matrix to its Echelon form and find its rank :

$$\begin{bmatrix} 1 & 3 & 4 & 5 \\ 3 & 9 & 12 & 9 \\ -1 & -3 & -4 & -3 \end{bmatrix}$$

(Meerut 2004B; Rohilkhand 06, 10)

21. Find the rank of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$ after reducing it to normal form.

(Garhwal 2015)

22. Reduce the matrix $\begin{bmatrix} 0 & 1 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ to normal form and find its rank.

(Meerut 2009; Bundelkhand 11; Avadh 11)

23. (i) Reduce the matrix $\begin{bmatrix} 9 & 7 & 3 & 6 \\ 5 & -1 & 4 & 1 \\ 6 & 8 & 2 & 4 \end{bmatrix}$ to normal form and find its rank.

(Agra 2007)

- (ii) Reduce the following matrix $\mathbf{A}$ into normal form and hence find its rank:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}.$$

(Kanpur 2011)

24. Are the following pairs of matrices equivalent?

$$(i) \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 5 \end{bmatrix}, \begin{bmatrix} 1 & 0 & -5 & 6 \\ 3 & -2 & 1 & 2 \\ 5 & -2 & -9 & 14 \\ 4 & -2 & -4 & 8 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 4 & 0 & 2 \\ 3 & 1 & 0 \\ 5 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 3 & 9 & 0 & 2 \\ 7 & -2 & 0 & 1 \\ 8 & 1 & 1 & 5 \end{bmatrix}$$

25. Find the ranks of **A**, **B**, **A + B**, **AB** and **BA** where

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}.$$

26. Show that if **A** and **B** are equivalent matrices, then there exist non-singular matrices **P** and **Q** such that **B = PAQ**.

27. Show that the rank of a matrix is not altered if a column of it is multiplied by a non-zero scalar.

28. (i) What is the rank of a non-singular matrix of order n ?

(ii) What is the rank of an elementary matrix?

Answers 2

1. 3 2. 2 3. 3 4. 2 5. 2 6. 2 7. 3 8. 4 9. 3
 10. 3 11. 4 12. 2 13. 2 14. 1 15. 2 16. 2 17. 3
 18. $\text{rank}(\mathbf{A}) = 3$ if $a \neq b \neq c$ and $a + b + c \neq 0$; $\text{rank}(\mathbf{A}) = 2$ if $a \neq b \neq c$ and $a + b + c = 0$; Also $\text{rank}(\mathbf{A}) = 2$ if $a = b \neq c$; and $\text{rank}(\mathbf{A}) = 1$ if $a = b = c$
 19. 3 20. 2 21. 2 22. 3 23. (i) 3 (ii) 2
 24. (i) No, since the rank of the first matrix is 4 and that of the second matrix is 2.
 (ii) No, since the matrices are not of the same type
 25. Rank **A** = 2; Rank **B** = 1; Rank **A + B** = 2;
 Rank **AB** = 0; Rank **BA** = 1
 28. (i) n . (ii) Equal to the order of the matrix

12 Row and Column Equivalence of Matrices

Definition: A matrix $\mathbf{A}$ is said to be row equivalent to $\mathbf{B}$ if $\mathbf{B}$ is obtainable from $\mathbf{A}$ by a finite number of E -row transformations of $\mathbf{A}$. Symbolically, we then write $\mathbf{A} \stackrel{R}{\sim} \mathbf{B}$. Similarly a matrix $\mathbf{A}$ is said to be column equivalent to $\mathbf{B}$ if $\mathbf{B}$ is obtainable from $\mathbf{A}$ by a finite number of E -column transformations of $\mathbf{A}$. Symbolically, we then write $\mathbf{A} \stackrel{C}{\sim} \mathbf{B}$.

13 Employment of Only Row Transformations

Theorem: If $\mathbf{A}$ be an $m \times n$ matrix of rank r , then there exists a non-singular matrix $\mathbf{P}$ such that $\mathbf{PA} = \begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix}$, where $\mathbf{G}$ is an $r \times n$ matrix of rank r and $\mathbf{O}$ is $(m - r) \times n$.

Proof: Since $\mathbf{A}$ is an $m \times n$ matrix of rank r , therefore there exist non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$, such that

$$\mathbf{PAQ} = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}. \quad \dots(1)$$

Now every non-singular matrix can be expressed as the product of elementary matrices. So let

$$\mathbf{Q} = \mathbf{Q}_1 \mathbf{Q}_2 \dots \mathbf{Q}_t \text{ where } \mathbf{Q}_1, \mathbf{Q}_2, \dots, \mathbf{Q}_t \text{ are elementary matrices.}$$

Thus the relation (1) can be written as

$$\mathbf{PAQ}_1 \mathbf{Q}_2 \dots \mathbf{Q}_t = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}. \quad \dots(2)$$

Now every E -column transformation of a matrix, is equivalent to post-multiplication with the corresponding elementary matrix. Since no column transformation can affect the last $(m - r)$ rows of the right hand side of (2), therefore post-multiplying the L. H. S. of (2) by the elementary matrices $\mathbf{Q}_t^{-1}, \mathbf{Q}_{t-1}^{-1}, \dots, \mathbf{Q}_2^{-1}, \mathbf{Q}_1^{-1}$ successively and affecting the corresponding column transformations in the right hand side of (2), we get a relation of the form

$$\mathbf{PA} = \begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix}.$$

Since elementary transformations do not alter the rank, therefore the rank of the matrix $\mathbf{PA}$ is the same as that of the matrix $\mathbf{A}$ which is r . Thus the rank of the matrix $\begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix}$ is r and therefore the rank of the matrix $\mathbf{G}$ is also r as the matrix $\mathbf{G}$ has r rows and last $m - r$ rows of the matrix $\begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix}$ consist of zeros only.

14 Employment of Only Column Transformations

Theorem: If $\mathbf{A}$ be an $m \times n$ matrix of rank r , then there exists a non-singular matrix $\mathbf{Q}$ such that $\mathbf{AQ} = [\mathbf{H} \ \mathbf{O}]$, where $\mathbf{H}$ is an $m \times n$ matrix of rank r and $\mathbf{O}$ is $m \times (n - r)$.

Proof: Since $\mathbf{A}$ is an $m \times n$ matrix of rank r , therefore there exist non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$ such that

$$\mathbf{PAQ} = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}. \quad \dots(1)$$

Now every non-singular matrix can be expressed as the product of elementary matrices. So let

$$\mathbf{P} = \mathbf{P}_1 \mathbf{P}_2 \dots \mathbf{P}_s \text{ where } \mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_s \text{ are elementary matrices.}$$

Thus the relation (1) can be written as

$$\mathbf{P}_1 \mathbf{P}_2 \dots \mathbf{P}_s \mathbf{AQ} = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}. \quad \dots(2)$$

Now every E -row transformation of a matrix is equivalent to pre-multiplication with the corresponding elementary matrix. Again no row transformation can affect the last $n - r$ columns of $\begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$.

Therefore pre-multiplying the L.H.S. of (2) by the elementary matrices.

$$\mathbf{P}_1^{-1}, \mathbf{P}_2^{-1}, \dots, \mathbf{P}_s^{-1}$$

successively and affecting the corresponding row transformations in the R.H.S. of (2), we get a relation of the form $\mathbf{AQ} = [\mathbf{H} \ \mathbf{O}]$.

Now elementary transformations do not alter the rank. Therefore the rank of the matrix $\mathbf{AQ}$ is the same as that of $\mathbf{A}$ which is r . Thus the rank of the matrix $[\mathbf{H} \ \mathbf{O}]$ is r and therefore the rank of the matrix $\mathbf{H}$ is also r as the matrix $\mathbf{H}$ has r columns and the last $n - r$ columns of the matrix $[\mathbf{H} \ \mathbf{O}]$ consists of zero only.

15 The Rank of a Product

Theorem: *The rank of a product of two matrices cannot exceed the rank of either matrix.*
(Lucknow 2009; Kumaun 11)

Proof: Let $\mathbf{A}$ and $\mathbf{B}$ be two $m \times n$ and $n \times p$ matrices respectively. Let r_1, r_2 be the ranks of $\mathbf{A}$ and $\mathbf{B}$ respectively and let r be the rank of the product $\mathbf{AB}$.

To prove $r \leq r_1$ and $r \leq r_2$.

Since $\mathbf{A}$ is an $m \times n$ matrix of rank r_1 , therefore there exists a non-singular matrix $\mathbf{P}$ such that $\mathbf{PA} = \begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix}$, where $\mathbf{G}$ is an $r_1 \times n$ matrix of rank r_1 and $\mathbf{O}$ is $(m - r_1) \times n$.

$$\therefore \quad \mathbf{PAB} = \begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix} \mathbf{B}.$$

Since the rank of a matrix does not alter by multiplying it with a non-singular matrix, therefore

$$\text{Rank}(\mathbf{PAB}) = \text{Rank}(\mathbf{AB}) = r.$$

$$\therefore \quad \text{Rank of the matrix} \begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix} \mathbf{B} = r.$$

Since the matrix $\mathbf{G}$ has only r_1 non-zero rows, therefore the matrix $\begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix} \mathbf{B}$ cannot have more than r_1 non-zero rows which arise by multiplying the r_1 non-zero rows of $\mathbf{G}$ with the columns of $\mathbf{B}$.

$$\therefore \quad \text{Rank of the matrix} \begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix} \mathbf{B} \text{ is } \leq r_1 \text{ i.e., } t \leq r_1$$

$$\text{i.e., } \text{Rank}(\mathbf{AB}) \leq \text{Rank of the prefactor } \mathbf{A}.$$

$$\text{Again } r = \text{Rank}(\mathbf{AB}) = \text{Rank}(\mathbf{AB})' = \text{Rank}(\mathbf{B}'\mathbf{A}') \leq \text{Rank of the prefactor } \mathbf{B}'$$

$$= \text{Rank } \mathbf{B} \quad [\because \text{Rank } \mathbf{B} = \text{Rank } \mathbf{B}']$$

$$= r_2.$$

$$\therefore \quad r \leq r_2 \text{ i.e., } \text{Rank}(\mathbf{AB}) \leq \text{Rank of the post-factor } \mathbf{B}.$$

16 Theorem

Every non-singular matrix is row equivalent to a unit matrix.

Proof: We shall prove the theorem by induction on n , the order of the matrix. If the matrix be of order 1 i.e., if $\mathbf{A} = [a_{11}]$, the theorem obviously holds.

Let us assume that the theorem holds for all non-singular matrices of order $n - 1$.

Let $\mathbf{A} = [a_{ij}]$ be an $n \times n$ non-singular matrix. The first column of the matrix $\mathbf{A}$ has at least one element different from zero, for otherwise we shall have $|\mathbf{A}| = 0$ and the matrix $\mathbf{A}$ will not be non-singular.

Let $a_{p1} = k \neq 0$.

By interchanging the p^{th} row with the first row (if necessary), we obtain a matrix $\mathbf{B}$ whose leading element is equal to k which is not equal to zero.

Multiplying the elements of the first row of the matrix $\mathbf{B}$ by $1/k$, we obtain a matrix $\mathbf{C}$ whose leading element is equal to unity.

Let
$$\mathbf{C} = \begin{bmatrix} 1 & c_{12} & c_{13} & \dots & c_{1n} \\ c_{21} & c_{22} & c_{23} & \dots & c_{2n} \\ c_{31} & c_{32} & c_{33} & \dots & c_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ c_{n1} & c_{n2} & c_{n3} & \dots & c_{nn} \end{bmatrix}.$$

Subtracting suitable multiples of the first row of $\mathbf{C}$ from the remaining rows, we obtain a matrix $\mathbf{D}$ in which all elements of the first column except the leading element are equal to zero.

Let
$$\mathbf{D} = \begin{bmatrix} 1 & d_{12} & d_{13} & \dots & d_{1n} \\ 0 & & & & \\ 0 & & \mathbf{A}_1 & & \\ \dots & & & & \\ 0 & & & & \end{bmatrix}.$$

where $\mathbf{A}_1$ is an $(n - 1) \times (n - 1)$ matrix. The matrix $\mathbf{A}_1$ is non-singular, for otherwise $|\mathbf{A}_1| = 0$ and so $|\mathbf{D}|$ is also equal to zero. Thus the matrix $\mathbf{D}$ will be non-singular, and therefore $\mathbf{A}$, which is row equivalent to $\mathbf{D}$, will also not be non-singular.

By the inductive hypothesis, $\mathbf{A}_1$ can be transformed to $\mathbf{I}_{n-1}$ by E -row operations. If these elementary row operations be applied to $\mathbf{D}$, they will not affect the first row and the first column of $\mathbf{D}$ and we shall obtain a matrix $\mathbf{M}$ such that

$$\mathbf{M} = \begin{bmatrix} 1 & d_{12} & d_{13} & \dots & d_{1n} \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}.$$

By adding suitable multiples of the second, third, ..., n^{th} rows to the first row of $\mathbf{M}$, we obtain the matrix $\mathbf{I}_n$.

Thus the matrix $\mathbf{A}$ has been reduced to $\mathbf{I}_n$ by E -row operations only.

The proof is now complete by induction.

Corollary 1: *If $\mathbf{A}$ be an n -rowed non-singular matrix, there exist E -matrices $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_t$ such that*

$$\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} = \mathbf{I}_n.$$

If $\mathbf{A}$ be an n -rowed non-singular matrix, it can be reduced to $\mathbf{I}_n$ by E -row operations only. Since every E -row operation is equivalent to pre-multiplication by the corresponding E -matrix, therefore we can say that if $\mathbf{A}$ be an n -rowed non-singular matrix, there exist E -matrices $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_t$ such that

$$\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} = \mathbf{I}_n.$$

Corollary 2: *Every non-singular matrix $\mathbf{A}$ is expressible as the product of elementary matrices.*

If $\mathbf{A}$ be an n -rowed non-singular matrix, there exist E -matrices $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_t$ such that

$$\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} = \mathbf{I}_n. \quad \dots (1)$$

Pre-multiplying both sides of the relation (1) by $(\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1)^{-1}$, we get

$$(\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1)^{-1} (\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1) \mathbf{A} = (\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1)^{-1} \mathbf{I}_n$$

or

$$\mathbf{I}_n \mathbf{A} = \mathbf{E}_1^{-1} \mathbf{E}_2^{-1} \mathbf{E}_3^{-1} \dots \mathbf{E}_{t-1}^{-1} \mathbf{E}_t^{-1} \mathbf{I}_n$$

or

$$\mathbf{A} = \mathbf{E}_1^{-1} \mathbf{E}_2^{-1} \dots \mathbf{E}_{t-1}^{-1} \mathbf{E}_t^{-1}.$$

Since the inverse of an elementary matrix is also an elementary matrix of the same type hence we get the result.

Corollary 3: *The rank of a matrix does not alter by pre-multiplication or post-multiplication with a non-singular matrix.*

Every non-singular matrix can be expressed as the product of elementary matrices. Also E -row (column) transformations are equivalent to pre-(post)-multiplication with the corresponding elementary matrices and elementary transformations do not alter the rank of a matrix. Hence we get the result.

17 Use of Elementary Transformations to Find the Inverse of a Non-Singular Matrix

Let $\mathbf{A}$ be a non-singular matrix of order n . It can be easily shown that $\mathbf{A}$ can be reduced to the unit matrix $\mathbf{I}_n$ by a finite number of E -row transformations only. Now each E -row transformation of a matrix is equivalent to pre-multiplication by the corresponding E -matrix. Therefore there exist elementary matrices, say, $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_t$ such that $(\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1) \mathbf{A} = \mathbf{I}_n$.

Post-multiplying both sides by $\mathbf{A}^{-1}$, we get

$$(\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1) \mathbf{A} \mathbf{A}^{-1} = \mathbf{I}_n \mathbf{A}^{-1}$$

$$\text{or} \quad (\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1) \mathbf{I}_n = \mathbf{A}^{-1} \quad [\because \mathbf{A} \mathbf{A}^{-1} = \mathbf{I}_n, \mathbf{I}_n \mathbf{A}^{-1} = \mathbf{A}^{-1}]$$

$$\text{or} \quad \mathbf{A}^{-1} = (\mathbf{E}_t \mathbf{E}_{t-1} \dots \mathbf{E}_2 \mathbf{E}_1) \mathbf{I}_n.$$

Hence we get the following result :

If a non-singular matrix $\mathbf{A}$ of order n is reduced to the unit matrix $\mathbf{I}_n$ by a sequence of E -row transformations only, then the same sequence of E -row transformations applied to the unit matrix $\mathbf{I}_n$ gives the inverse of $\mathbf{A}$ (i.e., $\mathbf{A}^{-1}$).

18 Working Rule for Finding the Inverse of a Non-Singular Matrix by E -Row Transformations

Suppose $\mathbf{A}$ is a non-singular matrix of order n . Then we write $\mathbf{A} = \mathbf{I}_n \mathbf{A}$.

Now we go on applying E -row transformations only to the matrix $\mathbf{A}$ and the pre-factor $\mathbf{I}_n$ of the product $\mathbf{I}_n \mathbf{A}$ till we reach the result $\mathbf{I}_n = \mathbf{B} \mathbf{A}$.

Then obviously $\mathbf{B}$ is the inverse of $\mathbf{A}$.

Illustrative Examples

Example 15: Find the inverse of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ by using E -transformations.

(Avadh 2006, 08; Purvanchal 09; Rohilkhand 09; Lucknow 09; Garhwal 10, 12)

Solution : We write $\mathbf{A} = \mathbf{I}_3 \mathbf{A}$, i.e., $\begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A}$.

Now we go on applying E -row transformations to the matrix $\mathbf{A}$ (the left hand member of the above equation) until it is reduced to the form $\mathbf{I}_3$. Every E -row transformation will also be applied to the prefactor $\mathbf{I}_3$ (or its transform) of the product on the right hand side of the above equation.

Applying $R_2 \rightarrow R_2 - 3R_1$, $R_3 \rightarrow R_3 - R_1$, we get

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \mathbf{A}.$$

Now we should try to make 1 in the place of the second element of the second row of the matrix on the left hand side. So applying $R_2 \rightarrow -\frac{1}{4}R_2$, we get

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{4} & -\frac{1}{4} & 0 \\ -1 & 0 & 1 \end{bmatrix} \mathbf{A}.$$

Now we shall make zeros in the place of the second elements of the first and third rows with the help of the second row. So applying $R_1 \rightarrow R_1 - 2R_2$, $R_3 \rightarrow R_3 + R_2$, we get

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix} \mathbf{A}.$$

Now the third element of the third row is already 1. So to make the third element of the first row zero, we apply $R_1 \rightarrow R_1 - R_3$, and we get

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} & \frac{3}{4} & -1 \\ \frac{3}{4} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix} \mathbf{A}.$$

Thus $\mathbf{I}_3 = \mathbf{BA}$, where $\mathbf{B} = \begin{bmatrix} -\frac{1}{4} & \frac{3}{4} & -1 \\ \frac{3}{4} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix}.$

$\therefore \mathbf{A}^{-1} = \mathbf{B} = \begin{bmatrix} -\frac{1}{4} & \frac{3}{4} & -1 \\ \frac{3}{4} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix}.$

Example 16: Find the inverse of the matrix $\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 2 & 3 \\ 2 & 3 & 3 & 3 \end{bmatrix}$ by using E-transformations.

Solution : We write $\begin{bmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 2 & 3 \\ 2 & 3 & 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{A}.$

Performing $R_1 \leftrightarrow R_2$, we get

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 2 & 2 \\ 2 & 2 & 2 & 3 \\ 2 & 3 & 3 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{A}.$$

Performing $R_3 \rightarrow R_3 - 2R_1$, $R_4 \rightarrow R_4 - 2R_1$, we get

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & -2 & -3 \\ 0 & 1 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & -2 & 0 & 1 \end{bmatrix} \mathbf{A}.$$

Performing $R_4 \rightarrow R_4 - R_2$, $R_1 \rightarrow R_1 - R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & -2 & -3 \\ 0 & 1 & -3 & -5 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ -1 & -2 & 0 & 1 \end{bmatrix} \mathbf{A}.$$

Performing $R_4 \rightarrow R_4 - R_2$, $R_1 \rightarrow R_1 - R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & -2 & -3 \\ 0 & 0 & -3 & -5 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ -1 & -2 & 0 & 1 \end{bmatrix} \mathbf{A}.$$

Performing $R_3 \rightarrow -\frac{1}{2}R_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & -3 & -5 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ -1 & -2 & 0 & 1 \end{bmatrix} \mathbf{A}.$$

Performing $R_2 \rightarrow R_2 - 2R_3$, $R_4 \rightarrow R_4 + 3R_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & 0 & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ -1 & 1 & -\frac{3}{2} & 1 \end{bmatrix} \mathbf{A}.$$

Performing $R_4 \rightarrow -2R_4$, we get

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 2 & -2 & 3 & -2 \end{bmatrix} \mathbf{A}.$$

Performing $R_3 \rightarrow R_3 - \frac{3}{2}R_4, R_2 \rightarrow R_2 + R_4, R_1 \rightarrow R_1 - R_4$ we get

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 3 & -3 & 2 \\ 3 & -4 & 4 & -2 \\ -3 & 4 & -5 & 3 \\ 2 & -2 & 3 & -2 \end{bmatrix} \mathbf{A}.$$

$\therefore$

$$\mathbf{A}^{-1} = \begin{bmatrix} -3 & 3 & -3 & 2 \\ 3 & -4 & 4 & -2 \\ -3 & 4 & -5 & 3 \\ 2 & -2 & 3 & -2 \end{bmatrix}.$$

Comprehensive Exercise 3

1. Reduce the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & 3 \end{bmatrix}$ to $\mathbf{I}_3$ by E -row transformations only.
2. Compute the inverse of the following matrices by using elementary row-transformations :

(i) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

(ii) $\begin{bmatrix} i & -1 & 2i \\ 2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix}$

(iv) $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

(Avadh 2007; Garhwal 15)

(v) $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

(Rohilkhand 2011; Bundelkhand 08)

Answers 3

$$2. \quad (i) \frac{1}{157} \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ -2 & -1 & 0 \end{bmatrix} \quad (ii) \begin{bmatrix} 0 & \frac{1}{4} & -\frac{1}{2} \\ -1 & \frac{3i}{4} & \frac{i}{2} \\ 0 & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

$$(iii) \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix} \quad (iv) \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$(v) \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- I. If $\mathbf{A}$ is a matrix such that there exists a square submatrix of order r which is non-singular and every square submatrix of order $r + 1$ or more is singular, then
 - (a) $\text{rank } \mathbf{A} = r + 1$
 - (b) $\text{rank } \mathbf{A} = r$
 - (c) $\text{rank } \mathbf{A} > r$
 - (d) $\text{rank } \mathbf{A} \geq r + 1$

2. Let $\mathbf{A} = [a_{ij}]_{m \times n}$ be a matrix such that $a_{ij} = 1$ for all i, j . Then

- (a) $\text{rank } \mathbf{A} > 1$
- (b) $\text{rank } \mathbf{A} = 1$
- (c) $\text{rank } \mathbf{A} = m$
- (d) $\text{rank } \mathbf{A} = n$

3. The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$ is

- (a) 1
- (b) 2
- (c) 3
- (d) 4

4. If $\mathbf{A}$ and $\mathbf{B}$ are two matrices such that $\text{rank of } \mathbf{A} = m$ and $\text{rank of } \mathbf{B} = n$, then

- (a) $\text{rank } (\mathbf{A} \mathbf{B}) = m n$
- (b) $\text{rank } (\mathbf{A} \mathbf{B}) \geq \text{rank } \mathbf{A}$
- (c) $\text{rank } (\mathbf{A} \mathbf{B}) \geq \text{rank } \mathbf{B}$
- (d) $\text{rank } (\mathbf{A} \mathbf{B}) \leq \min (\text{rank } \mathbf{A}, \text{rank } \mathbf{B})$

(Agra 2007)

5. If $\mathbf{A}$ is an invertible matrix and $\mathbf{B}$ is a matrix, then

- (a) $\text{rank } (\mathbf{A} \mathbf{B}) = \text{rank } \mathbf{A}$
- (b) $\text{rank } (\mathbf{A} \mathbf{B}) = \text{rank } \mathbf{B}$
- (c) $\text{rank } (\mathbf{A} \mathbf{B}) > \text{rank } \mathbf{A}$
- (d) $\text{rank } (\mathbf{A} \mathbf{B}) > \text{rank } \mathbf{B}$

6. The rank of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is

- (a) 0
- (b) 1
- (c) 2
- (d) 3

(Bundelkhand 2001; Meerut 03)

Fill in the Blank(s)

Fill in the blanks “.....”, so that the following statements are complete and correct.

- The rank of a matrix is the of any highest order non-vanishing minor of the matrix.
- The rank of every non-singular matrix of order n is
(Bundelkhand 2005)
- The rank of a matrix in Echelon form is equal to the number of non-zero of the matrix.
(Bundelkhand 2008)
- $\mathbf{A}$ is a non-zero column matrix and $\mathbf{B}$ is a non-zero row matrix, then rank $(\mathbf{AB}) = \dots\dots$.
- If a matrix $\mathbf{B}$ is obtained from $\mathbf{A}$ by an elementary transformation, $\mathbf{A}$ can also be obtained from $\mathbf{B}$ by an elementary transformation of the type.
- A matrix obtained from a by a single elementary transformation is called an elementary matrix.
- If a matrix $\mathbf{A}$ of order $m \times n$ can be expressed as :

$$\mathbf{A} \sim \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}, \text{ then rank of } \mathbf{A} \text{ is } \dots\dots\dots$$

(Meerut 2001)
- If $\mathbf{A}$ be an $m \times n$ matrix of rank r , there exist non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$ such that $\mathbf{PAQ} = \dots\dots$.
- If $\mathbf{B}$ be an $m \times n$ matrix obtained from an $m \times n$ matrix $\mathbf{A}$ by finite number of elementary transformations of $\mathbf{A}$, then $\mathbf{A}$ is called to be $\mathbf{B}$.
- If two matrices $\mathbf{A}$ and $\mathbf{B}$ have the same size and the same rank, they are
- The rank of a matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is
(Agra 2008)

True or False

Write ‘T’ for true and ‘F’ for false statement.

- The rank of the transpose of a matrix is the same as that of the original matrix.
 - Every elementary row transformation of a matrix can be obtained by post-multiplication with the corresponding elementary matrix.
 - Elementary transformations change the rank of a matrix.
-

4. If $\mathbf{A}$ and $\mathbf{B}$ are two equivalent matrices, then $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$.
5. The rank of a matrix does not alter by pre-multiplication or post-multiplication with a non-singular matrix.

Answers

Multiple Choice Questions

- | | | |
|--------|--------|--------|
| 1. (b) | 2. (b) | 3. (c) |
| 4. (d) | 5. (b) | 6. (c) |

Fill in the Blank(s)

- | | | |
|----------------|---|----------------|
| 1. order | 2. n | 3. rows |
| 4. 1 | 5. same | 6. unit matrix |
| 7. r | 8. $\begin{bmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$ | 9. equivalent |
| 10. equivalent | 11. 2 | |

True or False

- | | | |
|--------|--------|--------|
| 1. T | 2. F | 3. F |
| 4. T | 5. T | |



Chapter

7

Applications of Matrices

1 Vectors

Definition: Any ordered n -tuple of numbers is called an n -vector. By an ordered n -tuple we mean a set consisting of n numbers in which the place of each number is fixed. If $x_1, x_2, \dots, x_n$ be any n numbers, then the ordered n -tuple $\mathbf{X} = (x_1, x_2, \dots, x_n)$ is called an n -vector. The ordered triad (x_1, x_2, x_3) is called a 3-vector. Similarly $(1, 0, 1, -1)$ and $(1, 8, -5, 7)$ are 4-vectors. The n numbers $x_1, x_2, \dots, x_n$ are called components of the n -vector $\mathbf{X} = (x_1, x_2, \dots, x_n)$. A vector may be written either as a *row vector* or as a *column vector*. If $\mathbf{A}$ be a matrix of the type $m \times n$, then each row of $\mathbf{A}$ will be an n -vector and each column of $\mathbf{A}$ will be an m -vector. A vector whose components are all zero is called a *zero vector* and will be denoted by $\mathbf{O}$.

If k be any number and $\mathbf{X}$ be any vector, then relative to the vector $\mathbf{X}$, k is called a scalar.

Algebra of vectors. Since an n -vector is nothing but a row matrix or a column matrix, therefore we can develop an algebra of vectors in the same manner as the algebra of matrices.

Equality of two vectors. Two n -vectors $\mathbf{X}$ and $\mathbf{Y}$ where $\mathbf{X} = (x_1, x_2, \dots, x_n)$ and $\mathbf{Y} = (y_1, y_2, \dots, y_n)$ are said to be **equal** if and only if their corresponding components are equal *i.e.*, if $x_i = y_i$, for all $i = 1, 2, \dots, n$.

For example if

$$\mathbf{X} = (1, 4, 7) \quad \text{and} \quad \mathbf{Y} = (1, 4, 7),$$

then $\mathbf{X} = \mathbf{Y}$.

But if $\mathbf{X} = (1, 4, 7)$ and $\mathbf{Y} = (4, 1, 7)$, then $\mathbf{X} \neq \mathbf{Y}$.

Addition of two vectors. If $\mathbf{X} = (x_1, x_2, \dots, x_n)$ and $\mathbf{Y} = (y_1, y_2, \dots, y_n)$ then by definition $\mathbf{X} + \mathbf{Y} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$.

Thus $\mathbf{X} + \mathbf{Y}$ is an n -vector whose components are the sums of corresponding components of $\mathbf{X}$ and $\mathbf{Y}$.

If $\mathbf{X} = (2, 4, -7)$ and $\mathbf{Y} = (1, -3, 5)$,

then $\mathbf{X} + \mathbf{Y} = (2 + 1, 4 - 3, -7 + 5) = (3, 1, -2)$.

Multiplication of a vector by a scalar (number).

If k be any number and

$$\mathbf{X} = (x_1, x_2, \dots, x_n),$$

then by definition

$$k\mathbf{X} = (kx_1, kx_2, \dots, kx_n).$$

The vector $k\mathbf{X}$ is called the scalar multiple of the vector $\mathbf{X}$ by the scalar k .

If $\mathbf{X} = (1, 3, 8)$, then $4\mathbf{X} = (4, 12, 32)$

and $0\mathbf{X} = (0, 0, 0)$.

Properties of addition and scalar multiplication of vectors. If $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ be any three n -vectors and p, q be any two numbers, then obviously

$$(i) \quad \mathbf{X} + \mathbf{Y} = \mathbf{Y} + \mathbf{X}.$$

$$(ii) \quad \mathbf{X} + (\mathbf{Y} + \mathbf{Z}) = (\mathbf{X} + \mathbf{Y}) + \mathbf{Z}.$$

$$(iii) \quad p(\mathbf{X} + \mathbf{Y}) = p\mathbf{X} + p\mathbf{Y}$$

$$(iv) \quad (p + q)\mathbf{X} = p\mathbf{X} + q\mathbf{X}.$$

$$(v) \quad p(q\mathbf{X}) = (pq)\mathbf{X}.$$

2 Linear Dependence and Linear Independence of Vectors

(Agra 2005)

Linearly dependent set of vectors. Definition:

A set of r n -vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_r$ is said to be linearly dependent if there exist r scalars (numbers) $k_1, k_2, \dots, k_r$, not all zero, such that

$$k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 + \dots + k_r \mathbf{X}_r = \mathbf{O},$$

where, $\mathbf{O}$ denotes the n -vector whose components are all zero.

Linearly independent set of vectors. Definition:

A set of r n -vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_r$ is said to be linearly independent if every relation of the type

$$k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 + \dots + k_r \mathbf{X}_r = \mathbf{O}$$

implies $k_1 = k_2 = k_3 = \dots = k_r = 0$.

Illustrative Examples

Example 1: Show that the vectors $\mathbf{X}_1 = (1, 2, 4)$, $\mathbf{X}_2 = (3, 6, 12)$ are linearly dependent.

Solution: By a little inspection, we see that

$$3\mathbf{X}_1 + (-1)\mathbf{X}_2 = (3, 6, 12) + (-3, -6, -12) = (0, 0, 0) = \mathbf{O}.$$

Thus there exist numbers $k_1 = 3, k_2 = -1$ which are not all zero such that

$$k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 = \mathbf{O}.$$

Hence the vectors $\mathbf{X}_1$ and $\mathbf{X}_2$ are linearly dependent.

Example 2: Show that the set consisting only of the zero vector, $\mathbf{O}$, is linearly dependent.

Solution : Let $\mathbf{X} = (0, 0, 0, \dots, 0)$ be an n -vector whose components are all zero. Then the relation $k\mathbf{X} = \mathbf{O}$ is true for some non-zero value of the number k . For example, $1\mathbf{X} = \mathbf{O}$ and $1 \neq 0$.

Hence the vector $\mathbf{O}$ is linearly dependent.

Example 3: Show that the vectors $\mathbf{X}_1 = (1, 2, 3)$ and $\mathbf{X}_2 = (4, -2, 7)$ are linearly independent.

Solution: Let k_1 and k_2 be two numbers such that $k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 = \mathbf{O}$,

i.e.,
$$k_1 (1, 2, 3) + k_2 (4, -2, 7) = (0, 0, 0)$$

$$\text{i.e.,} \quad (k_1 + 4k_2, 2k_1 - 2k_2, 3k_1 + 7k_2) = (0, 0, 0).$$

Equating the corresponding components, we get

$$k_1 + 4k_2 = 0, 2k_1 - 2k_2 = 0, 3k_1 + 7k_2 = 0.$$

The only common values of k_1 and k_2 which satisfy these equations are $k_1 = 0, k_2 = 0$.

$$\text{Thus} \quad k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 = \mathbf{O}, \quad \text{iff } k_1 = k_2 = 0.$$

Hence the vectors $\mathbf{X}_1$ and $\mathbf{X}_2$ are linearly independent.

Example 4: Show that the set of three 3-vectors

$$\mathbf{X}_1 = (1, 0, 0), \mathbf{X}_2 = (0, 1, 0), \mathbf{X}_3 = (0, 0, 1)$$

is linearly independent.

Solution : Let k_1, k_2, k_3 be three numbers such that

$$k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 + k_3 \mathbf{X}_3 = \mathbf{O},$$

$$\text{i.e.,} \quad k_1(1, 0, 0) + k_2(0, 1, 0) + k_3(0, 0, 1) = (0, 0, 0),$$

$$\text{i.e.,} \quad (k_1, 0, 0) + (0, k_2, 0) + (0, 0, k_3) = (0, 0, 0),$$

$$\text{i.e.,} \quad (k_1, k_2, k_3) = (0, 0, 0).$$

Obviously this relation is true if and only if $k_1 = 0, k_2 = 0, k_3 = 0$.

Hence the vectors $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ are linearly independent.

3 A Vector as a Linear Combination of Vectors

Definition: A vector $\mathbf{X}$ which can be expressed in the form

$$\mathbf{X} = k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 + \dots + k_r \mathbf{X}_r,$$

is said to be a linear combination of the set of vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_r$.

Here $k_1, k_2, \dots, k_r$ are any numbers.

The following two results are quite obvious :

- (i) If a set of vectors is linearly dependent, then at least one member of the set can be expressed as a linear combination of the remaining members.
- (ii) If a set of vectors is linearly independent then no member of the set can be expressed as a linear combination of the remaining members.

4 The n -Vector Space

The set of all n vectors of a field F is called the n -vector space over F . It is usually denoted by $V_n(F)$ or simply by V_n if the field is understood. Similarly the set of all 3-vectors is a vector space which is usually denoted by V_3 . The elements of the field F are known as *scalars* relatively to the vectors.

5 Sub-space of an n -Vector Space V_n

Definition: A non-empty set, S , of vectors of V_n is called a vector sub-space of V_n , if $\mathbf{a} + \mathbf{b}$ belongs to S whenever $\mathbf{a}, \mathbf{b}$ belong to S and $k\mathbf{a}$ belongs to S , where k is any scalar.

It is important to note that every sub-space of V_n contains the zero vector, being the scalar product of any vector with the scalar zero.

Illustration: If $\mathbf{a} = (a_1, a_2, a_3)$ is any non-zero vector of V_3 , then the set S of vectors $k\mathbf{a}$ is a subspace of V_3 , where k is a variable scalar which can take any value. The sum of any two members $k_1\mathbf{a}, k_2\mathbf{a}$ of S , is the vector $k_1\mathbf{a} + k_2\mathbf{a}$

$$\text{i.e.,} \quad (k_1a_1, k_1a_2, k_1a_3) + (k_2a_1, k_2a_2, k_2a_3)$$

$$\text{i.e.,} \quad (k_1 + k_2)\mathbf{a}$$

which is also a member of S . Also the scalar multiple by any scalar x of any vector $k_1\mathbf{a}$ of S is the vector $(xk_1)\mathbf{a}$ which is again a member of S .

Hence the set S of vectors $k\mathbf{a}$ is a subspace of V_3 .

6 Vector Subspace Spanned by a given System of Vectors

Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be any three vectors of V_3 . The set S of all vectors of the form $x\mathbf{a} + y\mathbf{b} + z\mathbf{c}$, where x, y, z are any scalars, is a subspace of V_3 . For, if $x_1\mathbf{a} + y_1\mathbf{b} + z_1\mathbf{c}, x_2\mathbf{a} + y_2\mathbf{b} + z_2\mathbf{c}$ be any two members of S , then

$$(x_1\mathbf{a} + y_1\mathbf{b} + z_1\mathbf{c}) + (x_2\mathbf{a} + y_2\mathbf{b} + z_2\mathbf{c}) = (x_1 + x_2)\mathbf{a} + (y_1 + y_2)\mathbf{b} + (z_1 + z_2)\mathbf{c},$$

which is also a member of S . Also if k be any scalar, then

$$k(x_1\mathbf{a} + y_1\mathbf{b} + z_1\mathbf{c}) = (kx_1)\mathbf{a} + (ky_1)\mathbf{b} + (kz_1)\mathbf{c},$$

which is again a member of S . Thus S is a vector subspace and we say that S is spanned by the vectors $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$. More generally, if $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r$ be a set of r fixed vectors of V_n , then the set S of all n -vectors of the form $p_1\mathbf{a}_1 + p_2\mathbf{a}_2 + \dots + p_r\mathbf{a}_r$ where $p_1, p_2, \dots, p_r$ are any scalars is a vector subspace of V_n .

This vector space is said to be spanned by the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r$. Thus a vector space which arises as a set of all linear combinations of any given set of vectors, is said to be spanned by the given set of vectors.

7 Basis and Dimension of a Subspace

A set of vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \dots, \mathbf{a}_k$ belonging to the subspace S is said to be a basis of S , if

- (i) the subspace S is spanned by the set $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ and
- (ii) the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ are linearly independent.

It can be easily shown that every subspace, S , of V_n possesses a basis.

It can be easily shown that the vectors

$$\begin{aligned}\mathbf{e}_1 &= (1, 0, 0, \dots, 0), \mathbf{e}_2 = (0, 1, 0, \dots, 0), \mathbf{e}_3 = (0, 0, 1, 0, \dots, 0), \dots, \\ \mathbf{e}_n &= (0, 0, 0, \dots, 1)\end{aligned}$$

constitute a basis of V_n .

We have already shown that these vectors are linearly independent. Moreover any vector $\mathbf{a} = (a_1, a_2, \dots, a_n)$ of V_n is expressible as $\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + \dots + a_n\mathbf{e}_n$. Hence the vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n$ constitute a basis of V_n .

Theorem: *A basis of a subspace, S , can always be selected out of a set of vectors which span S .*

Proof: Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r$ be a set of vectors which spans a subspace S . If these vectors are linearly independent, they already constitute a basis of S as they span S . In case they are linearly dependent, some member of the set is a linear combination of the members. Deleting this member we obtain another set which also spans.

Continuing in this manner we shall ultimately, in a finite number of steps arrive at a basis of S .

A vector subspace may (and in fact does) possess several bases

For example, if

$$\mathbf{a}_1 = (1, 0, 0), \mathbf{a}_2 = (0, 1, 0), \mathbf{a}_3 = (0, 0, 1), \mathbf{b}_1 = (1, 1, 1), \mathbf{b}_2 = (1, 1, 0), \mathbf{b}_3 = (1, 0, 0),$$

then $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ constitute a basis of V_3 . Also $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ constitute a basis of V_3 .

But it is important to note that *the number of members in any one basis of a subspace is the same as in any other basis. This number is called the dimension of the subspace.*

We have already shown that one basis of V_n possesses n members. Therefore every basis of V_n must possess n members. Thus V_n is of dimension n . In a particular the dimension of V_3 is 3.

Also it can be easily shown that if r be the dimension of a subspace S and if $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ be a linearly independent set of vectors belonging to S , then we can always find vectors $\mathbf{a}_{k+1}, \mathbf{a}_{k+2}, \dots, \mathbf{a}_r$ such that the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k, \mathbf{a}_{k+1}, \dots, \mathbf{a}_r$ constitute a basis of S . In other words we can say that every linearly independent set of vectors belonging to a subspace S can always be extended so as to constitute a basis of S .

Moreover if r be the dimension of a subspace S , then every set of more than r members of S will be linearly dependent.

Intersection of subspaces. If S and T be two subspaces of V_n , then the vectors common to both S and T also constitute a subspace. This subspace is called the intersection of the subspaces S and T .

8 Row Rank and Column Rank of a Matrix

Row rank of a matrix. Let $\mathbf{A} = [a_{ij}]$ be any $m \times n$ matrix. Each of the m rows of $\mathbf{A}$ consists of n elements. Therefore the row vectors of $\mathbf{A}$ are n -vectors. These row vectors of $\mathbf{A}$ will span a subspace R of V_n . This subspace R is called the *row space* of the matrix $\mathbf{A}$. The dimension r of R is called the *row rank* of $\mathbf{A}$. In other words the row rank of a matrix $\mathbf{A}$ is equal to the maximum number of linearly independent rows of $\mathbf{A}$.

Row rank of a matrix. Definition: *The maximum number of linearly independent rows of a matrix $\mathbf{A}$ is said to be the row rank of the matrix $\mathbf{A}$.*

Left nullity of a matrix. Suppose $\mathbf{X}$ is an m -vector written in the form of a row vector. Then the matrix product $\mathbf{XA}$ is defined. The subspace S of V_m generated by the row vectors $\mathbf{X}$ belonging to V_m such that $\mathbf{XA} = \mathbf{O}$ is called the *row null* space of the matrix $\mathbf{A}$. The dimension s of S is called the left nullity or *row nullity* of the matrix $\mathbf{A}$.

We shall now prove that the *sum of the row rank and the row nullity of a matrix is equal to the number of rows, i.e.,*

$$r + s = m.$$

Proof: Since the row space R of $\mathbf{A}$ is spanned by the row vectors of $\mathbf{A}$, therefore it will be a set of all vectors of the form

$$\begin{aligned} x_1(a_{11}, a_{12}, a_{13}, \dots, a_{1n}) + x_2(a_{21}, a_{22}, a_{23}, \dots, a_{2n}) + \dots \\ \dots + x_m(a_{m1}, a_{m2}, a_{m3}, \dots, a_{mn}) \end{aligned}$$

i.e., of the form $(x_1a_{11} + x_2a_{21} + \dots + x_ma_{m1}, x_1a_{12} + x_2a_{22} + \dots + x_ma_{m2},$

$$x_1a_{13} + x_2a_{23} + \dots + x_ma_{m3}, \dots, x_1a_{1n} + x_2a_{2n} + \dots + x_ma_{mn})$$

i.e., of the form $\mathbf{XA}$, where $\mathbf{X} = (x_1, x_2, \dots, x_m)$ is an m -vector.

Let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_s$ be a basis of the subspace S of V_m generated by all vectors $\mathbf{X}$ such that $\mathbf{XA} = \mathbf{O}$. Then, we have

$$\mathbf{u}_1\mathbf{A} = \mathbf{u}_2\mathbf{A} = \dots = \mathbf{u}_s\mathbf{A} = \mathbf{O}.$$

Since the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_s$ belong to V_m and form a linearly independent set, therefore we can find vectors $\mathbf{u}_{s+1}, \mathbf{u}_{s+2}, \dots, \mathbf{u}_m$ in V_m such that the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_s, \mathbf{u}_{s+1}, \dots, \mathbf{u}_m$ constitute a basis of V_m . Then every vector $\mathbf{X}$ belonging to V_m can be expressed in the form

$$\mathbf{X} = h_1\mathbf{u}_1 + h_2\mathbf{u}_2 + \dots + h_m\mathbf{u}_m.$$

Now every member of the subspace R is expressible as $\mathbf{XA}$

$$\text{i.e., as } (h_1\mathbf{u}_1 + h_2\mathbf{u}_2 + \dots + h_m\mathbf{u}_m)\mathbf{A}$$

$$\text{i.e., as } h_1\mathbf{u}_1\mathbf{A} + h_2\mathbf{u}_2\mathbf{A} + \dots + h_s\mathbf{u}_s\mathbf{A} + h_{s+1}\mathbf{u}_{s+1}\mathbf{A} + h_{s+2}\mathbf{u}_{s+2}\mathbf{A} + \dots + h_m\mathbf{u}_m\mathbf{A}$$

$$\text{i.e., as } h_{s+1}\mathbf{u}_{s+1}\mathbf{A} + h_{s+2}\mathbf{u}_{s+2}\mathbf{A} + \dots + h_m\mathbf{u}_m\mathbf{A}.$$

Therefore the $m-s$ n -vectors $\mathbf{u}_{s+1}\mathbf{A}, \mathbf{u}_{s+2}\mathbf{A}, \dots, \mathbf{u}_m\mathbf{A}$ span R . In fact, these vectors form a basis of R . For any relation of the form

$$k_{s+1}\mathbf{u}_{s+1}\mathbf{A} + k_{s+2}\mathbf{u}_{s+2}\mathbf{A} + \dots + k_m\mathbf{u}_m\mathbf{A} = \mathbf{O}$$

$$\text{implies } (k_{s+1}\mathbf{u}_{s+1} + k_{s+2}\mathbf{u}_{s+2} + \dots + k_m\mathbf{u}_m)\mathbf{A} = \mathbf{O},$$

which shows that $k_{s+1}\mathbf{u}_{s+1} + k_{s+2}\mathbf{u}_{s+2} + \dots + k_m\mathbf{u}_m$ is a member of the subspace S and as such it can be linearly expressed in terms of the basis $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_s$ of S .

But the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m$ are linearly independent. Therefore a relation of the form $k_{s+1}\mathbf{u}_{s+1} + k_{s+2}\mathbf{u}_{s+2} + \dots + k_m\mathbf{u}_m = p_1\mathbf{u}_1 + p_2\mathbf{u}_2 + \dots + p_s\mathbf{u}_s$ will exist if and only if $k_{s+1} = k_{s+2} = \dots = k_m = 0$. Hence the vectors $\mathbf{u}_{s+1}\mathbf{A}, \mathbf{u}_{s+2}\mathbf{A}, \dots, \mathbf{u}_m\mathbf{A}$ are linearly independent and form a basis of R . Thus the dimension of R is $m-s$.

$$\text{Hence } r = m - s \quad \text{or} \quad r + s = m.$$

9 Column Rank of a Matrix

Let $A = [a_{ij}]$ be any $m \times n$ matrix. Each of the n columns of $\mathbf{A}$ consists of m elements. Therefore the column vectors of $\mathbf{A}$ are m -vectors. These column vectors of $\mathbf{A}$ will span a subspace C of V_m . This subspace C is called the *column space* of the matrix $\mathbf{A}$. The dimension c of C is called the *column rank* of $\mathbf{A}$. In other words the column rank of a matrix $\mathbf{A}$ is equal to the maximum number of linearly independent columns of $\mathbf{A}$.

Column rank of a matrix. Definition. *The maximum number of linearly independent columns of a matrix $\mathbf{A}$ is said to be the column rank of the matrix $\mathbf{A}$.*

Right nullity of a matrix. Suppose $\mathbf{Y}$ is an n -vector written in the form of a column vector. Then the matrix product $\mathbf{A}\mathbf{Y}$ is defined. The subspace T of V_n generated by the column vectors $\mathbf{Y}$ belonging to V_n such that $\mathbf{A}\mathbf{Y} = \mathbf{O}$ is called the *column null space* of the matrix $\mathbf{A}$. The dimension t of T is called the *right nullity* or *column nullity* of the matrix $\mathbf{A}$.

As in article 8, we can show that $c + t = n$.

10 Invariance of Row Rank under E -row Operations

Theorem: *Row equivalent matrices have the same row rank.*

Proof: Let $\mathbf{A}$ be any given $m \times n$ matrix. Let $\mathbf{B}$ be a matrix row equivalent to $\mathbf{A}$. Since $\mathbf{B}$ is obtainable from $\mathbf{A}$ by a finite chain of E -row operations and every E -row operation is equivalent to pre-multiplication by the corresponding E -matrix, there exist E -matrices $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_k$ each of the type $m \times m$ such that

$$\mathbf{B} = \mathbf{E}_k \mathbf{E}_{k-1} \dots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A}, \text{ i.e., } \mathbf{B} = \mathbf{P} \mathbf{A},$$

where $\mathbf{P} = \mathbf{E}_k \mathbf{E}_{k-1} \dots \mathbf{E}_2 \mathbf{E}_1$ is a non-singular matrix of the type $m \times m$.

Let us write

$$\mathbf{B} = \mathbf{P} \mathbf{A} = \begin{bmatrix} p_{11} & p_{12} & \dots & p_{1m} \\ p_{21} & p_{22} & \dots & p_{2m} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ p_{m1} & p_{m2} & \dots & p_{mm} \end{bmatrix} \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \\ \dots \\ \mathbf{R}_m \end{bmatrix} \quad \dots (1)$$

where the matrix $\mathbf{A}$ has been expressed as a matrix of its row sub-matrices $\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_m$.

From the product of the matrices on the R.H.S. of (1), we observe that the rows of the matrix **B** are

$$\begin{aligned}
 & p_{11}\mathbf{R}_1 + p_{12}\mathbf{R}_2 + \dots + p_{1m}\mathbf{R}_m, \\
 & p_{21}\mathbf{R}_1 + p_{22}\mathbf{R}_2 + \dots + p_{2m}\mathbf{R}_m, \\
 & \dots \quad \dots \quad \dots \quad \dots \quad \dots \\
 & \dots \quad \dots \quad \dots \quad \dots \quad \dots \\
 & p_{m1}\mathbf{R}_1 + p_{m2}\mathbf{R}_2 + \dots + p_{mm}\mathbf{R}_m.
 \end{aligned}$$

Thus we see that the rows of **B** are all linear combinations of the rows $\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_m$ of **A**. Therefore every member of the row space of **B** is also a member of the row space of **A**.

Similarly by writing $\mathbf{A} = \mathbf{P}^{-1}\mathbf{B}$ and giving the same reasoning we can prove that every member of the row space of **A** is also a member of the row space of **B**. Therefore the row spaces of **A** and **B** are identical.

Thus we see that elementary row operations do not alter the row space of a matrix. Hence the row rank of a matrix remains invariant under *E*-row transformations.

Note: From the above theorem we also conclude that *pre-multiplication by a non-singular matrix does not alter the row rank of a matrix.*

11 Invariance of Column Rank under *E*-column Operations

Theorem: *Column equivalent matrices have the same column rank.*

Or

Post-multiplication by a non-singular matrix does not alter the column rank of a matrix.

Proof: Proceeding in the same way as in article 10, we can show that post-multiplication with a non-singular matrix does not alter the column space and therefore the column rank of a matrix.

Note: Since every *n*-rowed *E*-matrix is obtained from $\mathbf{I}_n$ by single *E*-operation (row or column operation as may be desired), therefore the row rank and column rank of an *E*-matrix are equal to *n*.

12 Invariance of Column Rank under E -row Operations

Theorem: *Row equivalent matrices have the same column rank.*

Proof: Let $\mathbf{A}$ be any given $m \times n$ matrix and let $\mathbf{B}$ be a matrix row equivalent to $\mathbf{A}$. Then there exists a non-singular matrix $\mathbf{P}$ such that $\mathbf{B} = \mathbf{PA}$.

For every column vector $\mathbf{X}$ such that $\mathbf{AX} = \mathbf{O}$, we have

$$\mathbf{BX} = (\mathbf{PA})\mathbf{X} = \mathbf{P}(\mathbf{AX}) = \mathbf{PO} = \mathbf{O}.$$

Since $\mathbf{B} = \mathbf{PA}$, therefore $\mathbf{A} = \mathbf{P}^{-1}\mathbf{B}$.

Therefore for every vector $\mathbf{X}$ such that $\mathbf{BX} = \mathbf{O}$, we have

$$\mathbf{AX} = (\mathbf{P}^{-1}\mathbf{B})\mathbf{X} = \mathbf{P}^{-1}(\mathbf{BX}) = \mathbf{P}^{-1}\mathbf{O} = \mathbf{O}.$$

Thus we see that the matrices $\mathbf{A}$ and $\mathbf{B}$ have the same right nullities and consequently their column ranks are equal.

Similarly we can prove that column equivalent matrices have the same row rank.

13 Theorem

If r be the row rank of an $m \times n$ matrix $\mathbf{A}$ then there exists a non-singular matrix, $\mathbf{P}$ such that

$$\mathbf{PA} = \begin{bmatrix} \mathbf{K} \\ \mathbf{O} \end{bmatrix},$$

where $\mathbf{K}$ is an $r \times n$ matrix consisting of a set of r linearly independent rows of $\mathbf{A}$.

Proof: If the row rank r of $\mathbf{A}$ is zero, we have nothing to prove. Therefore let us assume that $r > 0$. The matrix $\mathbf{A}$ has then r linearly independent rows. By elementary row operations on $\mathbf{A}$ we can bring these linearly independent rows in the first r places. Since the last $m - r$ rows are now linear combinations of the first r rows, they can be made zero by E -row operations without altering the first r rows.

Thus we see that the matrix $\mathbf{A}$ is row equivalent to a matrix $\mathbf{B}$ such that

$$\mathbf{B} = \begin{bmatrix} \mathbf{K} \\ \mathbf{O} \end{bmatrix},$$

where $\mathbf{K}$ is an $r \times n$ matrix consisting of a set of r linearly independent rows of $\mathbf{A}$.

Since every elementary row operation is equivalent to pre-multiplication by the corresponding E -matrix and the product of E -matrices is a non-singular matrix, therefore there exists a non-singular matrix $\mathbf{P}$ such that

$$\mathbf{PA} = \begin{bmatrix} \mathbf{K} \\ \mathbf{O} \end{bmatrix}.$$

Similarly considering column transformations instead of row transformations, we can show that if c be the column rank of a matrix $\mathbf{A}$, then there exists a non-singular matrix $\mathbf{R}$ such that

$$\mathbf{AR} = [\mathbf{L} \quad \mathbf{O}],$$

where $\mathbf{L}$ is an $m \times c$ matrix consisting of, c , linearly independent columns of $\mathbf{A}$.

14 Equality of Row Rank, Column Rank and Rank

Theorem 1: *The row rank of a matrix is the same as its rank.*

Proof: Let, s be the row rank and r , the rank of an $m \times n$ matrix $\mathbf{A}$. Since the matrix $\mathbf{A}$ is of row rank s , therefore by article 13, there exists a non-singular matrix $\mathbf{P}$ such that

$$\mathbf{PA} = \begin{bmatrix} \mathbf{K} \\ \mathbf{O} \end{bmatrix}, \text{ where } \mathbf{K} \text{ is an } s \times n \text{ matrix.}$$

Now we know that pre-multiplication by a non-singular matrix does not alter the rank of a matrix.

$$\therefore \text{Rank } (\mathbf{PA}) = \text{Rank } \mathbf{A} = r.$$

But each minor of order $(s + 1)$ of the matrix $\mathbf{PA}$ involves at least one row of zeros.

$$\therefore \text{Rank } (\mathbf{PA}) \leq s.$$

$$\therefore r \leq s$$

Again, since the rank of the matrix $\mathbf{A}$ is r , therefore by article 13 of chapter 6 there exists a non-singular matrix $\mathbf{R}$ such that

$$\mathbf{RA} = \begin{bmatrix} \mathbf{G} \\ \mathbf{O} \end{bmatrix},$$

where $\mathbf{G}$ is an $r \times n$ matrix.

Now we know that pre-multiplication by a non-singular matrix does not alter the row rank of a matrix.

$\therefore$ Row rank ($\mathbf{RA}$) = Row rank $\mathbf{A} = s$.

But the matrix $\mathbf{RA}$ has only r non-zero rows. Therefore the row rank of $\mathbf{RA}$ can, at the most, be equal to r .

$\therefore s \leq r$

Hence $r = s$.

Theorem 2: *The column rank of a matrix is the same as its rank.*

Proof: Let the matrix $\mathbf{A}'$ be the transpose of the matrix $\mathbf{A}$. Then the columns of $\mathbf{A}$ are the rows of $\mathbf{A}'$.

$\therefore$ the column rank of $\mathbf{A}$ = the row rank of $\mathbf{A}'$ = the rank of $\mathbf{A}'$ = the rank of $\mathbf{A}$.

Thus from theorems 1 and 2, we conclude that the rank, row rank and column rank of a matrix are all equal. In other words we can say that *the maximum number of linearly independent rows of a matrix is equal to the maximum number of its linearly independent columns and is equal to the rank of the matrix.*

Thus we have proved that the row rank, the column rank and the rank of a matrix are all equal. Therefore sometimes *the rank of a matrix is also defined as the maximum number of linearly independent row vectors or column vectors.*

We shall now first consider systems of linear homogeneous equations and then proceed to discuss systems of non-homogeneous linear equations.

15 Homogeneous Linear Equations

$$\text{Suppose } \left. \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0, \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0, \\ \dots \dots \dots \dots \dots \dots \dots \dots \\ \dots \dots \dots \dots \dots \dots \dots \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = 0 \end{array} \right\} \dots (1)$$

is a system of m homogeneous equations in n unknowns $x_1, x_2, \dots, x_n$. Let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \dots \\ x_n \end{bmatrix}_{n \times 1}, \quad \mathbf{O} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \dots \\ \dots \\ 0 \end{bmatrix}_{m \times 1},$$

where $\mathbf{A}, \mathbf{X}, \mathbf{O}$ are $m \times n, n \times 1, m \times 1$ matrices respectively. Then obviously we can write the system of equations (1) in the form of a single matrix equation

$$\mathbf{AX} = \mathbf{O}. \quad \dots(2)$$

The matrix $\mathbf{A}$ is called the **coefficient matrix** of the system of equations (1).

Obviously $x_1 = 0, x_2 = 0, \dots, x_n = 0$ i.e., $\mathbf{X} = \mathbf{O}$ is a solution of (1). It is a trivial (self-obvious) solution of (1).

Again suppose $\mathbf{X}_1$ and $\mathbf{X}_2$ are two solutions of (2). Then their linear combination $k_1\mathbf{X}_1 + k_2\mathbf{X}_2$, where k_1 and k_2 are any arbitrary numbers, is also a solution of (2).

We have

$$\begin{aligned} \mathbf{A}(k_1\mathbf{X}_1 + k_2\mathbf{X}_2) &= k_1(\mathbf{AX}_1) + k_2(\mathbf{AX}_2) \\ &= k_1\mathbf{O} + k_2\mathbf{O} = \mathbf{O}. \end{aligned} \quad [\because \mathbf{AX}_1 = \mathbf{O} \text{ and } \mathbf{AX}_2 = \mathbf{O}]$$

Hence $k_1\mathbf{X}_1 + k_2\mathbf{X}_2$ is also a solution of (2).

Therefore the collection of all the solutions of the system of equations $\mathbf{AX} = \mathbf{O}$ forms a sub-space of the n -vector space V_n .

Theorem: The number of linearly independent solutions of the system of m homogeneous linear equations in n variables, $\mathbf{AX} = \mathbf{O}$, is $(n - r)$, where r is the rank of the matrix $\mathbf{A}$.

Proof:

Let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}, \quad \text{and } \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \dots \\ x_n \end{bmatrix}_{n \times 1}.$$

Since the rank of the coefficient matrix $\mathbf{A}$ is r , therefore it has r linearly independent columns. Without loss of generality we can suppose that the first r columns from the left of the matrix $\mathbf{A}$ are linearly independent, because it amounts only to renaming the components of $\mathbf{X}$.

The matrix $\mathbf{A}$ can be written as $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_r, \dots, \mathbf{C}_n]_{1 \times n}$, where $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_n$ are the column vectors of the matrix $\mathbf{A}$ each of them being an m -vector.

The equation $\mathbf{AX} = \mathbf{O}$ can now be written as the vector equation

$$x_1\mathbf{C}_1 + x_2\mathbf{C}_2 + \dots + x_r\mathbf{C}_r + x_{r+1}\mathbf{C}_{r+1} + \dots + x_n\mathbf{C}_n = \mathbf{O}. \quad \dots(1)$$

It can now be easily seen that every solution of the equation $\mathbf{AX} = \mathbf{O}$ is some suitable linear combination of these $n - r$ solutions $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_{n-r}$.

Suppose the vector $\mathbf{X}$, with components $x_1, x_2, \dots, x_n$ is any solution of the equation $\mathbf{AX} = \mathbf{O}$. The vector

$$\mathbf{X} + x_{r+1}\mathbf{X}_1 + x_{r+2}\mathbf{X}_2 + \dots + x_n\mathbf{X}_{n-r}, \quad \dots(5)$$

which, being a linear combination of solutions, is also a solution. It is quite obvious that the last $n - r$ components of the vector (5) are all equal to zero. Let $z_1, z_2, \dots, z_r$ be the first r components of the vector (5). Then the vector whose components are $(z_1, z_2, \dots, z_r, 0, 0, \dots, 0)$ is a solution of the equation

$$\mathbf{AX} = \mathbf{O}.$$

Therefore from (1), we have

$$z_1\mathbf{C}_1 + z_2\mathbf{C}_2 + \dots + z_r\mathbf{C}_r = \mathbf{O}.$$

But the vectors $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_r$

are linearly independent. Therefore, we have $z_1 = 0, z_2 = 0, \dots, z_r = 0$. Hence (5) is a zero vector.

Therefore

$$\mathbf{X} = -x_{r+1}\mathbf{X}_1 - x_{r+2}\mathbf{X}_2 - \dots - x_n\mathbf{X}_{n-r}.$$

Thus every solution $\mathbf{X}$ is a linear combination of the $n - r$ linearly independent solutions $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_{n-r}$.

Therefore the set of solutions

$$\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_{n-r}\}$$

forms a basis of the vector space of all the solutions of the system of equations $\mathbf{AX} = \mathbf{O}$.

16 Some Important Conclusions About the Nature of Solutions of the Equation $\mathbf{AX} = \mathbf{O}$

Suppose we have m equations in n unknowns. Then the coefficient matrix $\mathbf{A}$ will be of the type $m \times n$. Let r be the rank of the matrix $\mathbf{A}$. Obviously r cannot be greater than n (the number of columns of the matrix $\mathbf{A}$). Therefore we have either $r = n$ or $r < n$.

Case I. If $r = n$, the equation $\mathbf{AX} = \mathbf{O}$ will have $n - n$ i.e., no linearly independent solutions. In this case the zero solution will be the only solution. We know that zero vector forms a linearly dependent set.

Case II. If $r < n$, we shall have $n - r$ linearly independent solutions. Any linear combination of these $n - r$ solutions will also be a solution of $\mathbf{AX} = \mathbf{O}$. Thus in this case the equation $\mathbf{AX} = \mathbf{O}$ will have an infinite number of solutions.

Case III. Suppose $m < n$ i.e., the number of equations is less than the number of unknowns. Since $r \leq m$, therefore r is definitely less than n . Hence in this case the given system of equations must possess a non-zero solution. The number of solutions of the equation $\mathbf{AX} = \mathbf{O}$ will be infinite.

17 Fundamental Set of Solutions of the Equation $\mathbf{AX} = \mathbf{O}$

Suppose the rank r of the coefficient matrix $\mathbf{A}$ is less than the number of the unknowns n . In this case the given equations have a set of $n - r$ linearly independent solutions and every possible solution is a linear combination of these $n - r$ solutions. This set of $n - r$ solutions is called a fundamental set of solutions of the equation $\mathbf{AX} = \mathbf{O}$.

Definition: A set of linearly independent solutions $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_k$ of the system of homogeneous equations $\mathbf{AX} = \mathbf{O}$ is called the fundamental system of solutions of $\mathbf{AX} = \mathbf{O}$, if every solution $\mathbf{X}$ of $\mathbf{AX} = \mathbf{O}$ can be written as a linear combination of these vectors i.e., in the form

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \dots + c_k\mathbf{X}_k,$$

where $c_1, c_2, \dots, c_k$ are suitable numbers.

18 Working Rule for Finding the Solutions of the Equation $\mathbf{AX} = \mathbf{O}$

Reduce the coefficient matrix $\mathbf{A}$ to **Echelon form** by applying **elementary row transformations only**. This Echelon form will help us to know the rank of the matrix $\mathbf{A}$. Suppose the matrix $\mathbf{A}$ is of the type $m \times n$ and its rank comes out to be r . If $r < m$, then in the process of reducing the matrix $\mathbf{A}$ to Echelon form, $(m - r)$ equations will be eliminated. The given system of m equations will thus be replaced by an equivalent system of r equations. Solving these r equations (by Cramer's rule or otherwise), we can express the values of some r unknowns in terms of the remaining $n - r$ unknowns. These $n - r$ unknowns can be given any arbitrarily chosen values.

If $r = n$, the zero solution (trivial solution) will be the only solution. If $r < n$, there will be an infinity of solutions.

Illustrative Examples

Example 5: Does the following system of equations possess a common non-zero solution?

$$x + 2y + 3z = 0, 3x + 4y + 4z = 0, 7x + 10y + 12z = 0. \quad (\text{Lucknow 2005})$$

Solution : The given system of equations can be written in the form of the single matrix equation

$$\mathbf{AX} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 4 \\ 7 & 10 & 12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{O}.$$

We shall start reducing the coefficient matrix $\mathbf{A}$ to triangular form by applying only E -row transformations on it. Applying $R_2 \rightarrow R_2 - 3R_1$, $R_3 \rightarrow R_3 - 7R_1$, the given system of equations is equivalent to

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -5 \\ 0 & -4 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{O}.$$

Here we find that the determinant of the matrix on the left hand side of this equation is not equal to zero. Therefore the rank of this matrix is 3. So there is no need of further applying E -row transformations on the coefficient matrix. The rank of the coefficient matrix $\mathbf{A}$ is 3, i.e., equal to the number of unknowns. Therefore the given system of equations does not possess any linearly independent solution. The zero solution, i.e., $x = y = z = 0$ is the only solution of the given system of equations.

Example 6: Solve completely the system of equations

$$x + 3y - 2z = 0, 2x - y + 4z = 0, x - 11y + 14z = 0. \quad (\text{Meerut 2005B})$$

Solution : The given system of equations is equivalent to the single matrix equation

$$\mathbf{AX} = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{O}.$$

We shall reduce the coefficient matrix $\mathbf{A}$ to Echelon form by applying only E -row operations on it. Performing $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - R_1$, we have

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

Performing $R_3 \rightarrow R_3 - 2R_2$, we have

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

The coefficient matrix is now triangular. The coefficient matrix being of rank 2, the given system of equations possess $3 - 2 = 1$ linearly independent solution. We shall assign arbitrary values to $n - r = 3 - 2 = 1$ variable and the remaining $r = 2$ variables shall be found in terms of these. The given system of equations is equivalent to

$$x + 3y - 2z = 0, -7y + 8z = 0.$$

Thus $y = \frac{8}{7}z, x = -\frac{10}{7}z.$

Choose $z = c.$

Then $y = \frac{8}{7}c, x = -\frac{10}{7}c.$

Hence $x = -\frac{10}{7}c, y = \frac{8}{7}c, z = c$

constitute the general solution of the given system, where c is an arbitrary parameter.

Remark: In matrix form, the general solution can be expressed as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -\frac{10}{7}c \\ \frac{8}{7}c \\ c \end{bmatrix} = c \begin{bmatrix} -\frac{10}{7} \\ \frac{8}{7} \\ 1 \end{bmatrix}, \text{ where } c \text{ is an arbitrary number.}$$

Example 7: Does the following system of equations possess a common non-zero solution?

$$x + y + z = 0, 2x - y - 3z = 0, 3x - 5y + 4z = 0, x + 17y + 4z = 0.$$

Solution: The given system of equations is equivalent to the single matrix equation

$$\mathbf{A}\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & -3 \\ 3 & -5 & 4 \\ 1 & 17 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

We shall first find the rank of the coefficient matrix $\mathbf{A}$ by reducing it to Echelon form by applying elementary row transformations only.

Applying $R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1, R_4 \rightarrow R_4 - R_1$, we get

$$\begin{aligned} \mathbf{A} &\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -5 \\ 0 & -8 & 1 \\ 0 & 16 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -5 \\ 0 & -24 & 3 \\ 0 & 48 & 9 \end{bmatrix} \text{ by } R_3 \rightarrow 3R_3, R_4 \rightarrow 3R_4 \\ &\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -5 \\ 0 & 0 & 43 \\ 0 & 0 & -71 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - 8R_2, R_4 \rightarrow R_4 + 16R_2 \\ &\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -5 \\ 0 & 0 & 43 \\ 0 & 0 & 0 \end{bmatrix} \text{ by } R_4 \rightarrow R_4 + \frac{71}{43}R_3. \end{aligned}$$

Above is the Echelon form of the coefficient matrix $\mathbf{A}$. We have $\text{rank } \mathbf{A} =$ the number of non-zero rows in this Echelon form $= 3$. The number of unknowns is also 3. Since $\text{rank } \mathbf{A}$ is equal to the number of unknowns, therefore the given system of equations does not possess any linearly independent solution. Thus the given system of equations possesses no non-zero solution. Hence the zero solution *i.e.*, $x = y = z = 0$ is the only solution of the given system of equations.

Example 8: Find all the solutions of the following system of equations :

$$3x + 4y - z - 6w = 0, \quad 2x + 3y + 2z - 3w = 0,$$

$$2x + y - 14z - 9w = 0, \quad x + 3y + 13z + 3w = 0.$$

Solution: The given system of equations is equivalent to the single matrix equation

$$\mathbf{A}\mathbf{X} = \begin{bmatrix} 3 & 4 & -1 & -6 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 1 & 3 & 13 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \mathbf{O}.$$

We shall first find the rank of the coefficient matrix $\mathbf{A}$ by reducing it to Echelon form by applying E -row transformations only.

Applying $R_4 \leftrightarrow R_1$, we get

$$\mathbf{A} \sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 3 & 4 & -1 & -6 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & -3 & -20 & -9 \\ 0 & -5 & -40 & -15 \\ 0 & -5 & -40 & -15 \end{bmatrix}$$

By $R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - 3R_1$

$$\sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & 1 & 8 & 3 \end{bmatrix}$$

by $R_2 \rightarrow -\frac{1}{3}R_2, R_3 \rightarrow -\frac{1}{5}R_3, R_4 \rightarrow -\frac{1}{5}R_4$

$$\sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - R_2.$$

The rank of $\mathbf{A}$ is obviously 2 which is less than the number of unknowns 4. Therefore the given system of equations possesses $4 - 2$, i.e., 2 linearly independent solutions. The given system of equations is equivalent to the equation

$$\begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Thus the given system of four equations is equivalent to the system of two equations, i.e.,

$$\left. \begin{aligned} x + 3y + 13z + 4w &= 0 \\ y + 8z + 3w &= 0 \end{aligned} \right\}.$$

From these equations, we get

$$y = -8z - 3w, x = -3(-8z - 3w) - 13z - 3w$$

$$\text{i.e., } y = -8z - 3w, x = 11z + 6w.$$

Hence $x = 11c_1 + 6c_2$, $y = -8c_1 - 3c_2$, $z = c_1$, $w = c_2$ constitute the general solution of the given system of equations, where c_1 and c_2 are arbitrary numbers. Since we can give any arbitrary values to c_1 and c_2 , therefore the given system of equations have an infinite number of solutions.

Remark: In matrix form, the general solution of the given system of equations can be expressed as

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 11c_1 + 6c_2 \\ -8c_1 - 3c_2 \\ c_1 + 0c_2 \\ 0c_1 + 1c_2 \end{bmatrix} = c_1 \begin{bmatrix} 11 \\ -8 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 6 \\ -3 \\ 0 \\ 1 \end{bmatrix}.$$

Here

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 11 \\ -8 \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -6 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

are two linearly independent solutions of the given system of equations and all their linear combinations will also be the solutions of the given system of equations.

Example 9: Show that the only real value of λ for which the following equations have non-zero solutions is 6 :

$$x + 2y + 3z = \lambda x, 3x + y + 2z = \lambda y, 2x + 3y + z = \lambda z.$$

Solution: The given system of equations is equivalent to the single matrix equation

$$\mathbf{AX} = \begin{bmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{O}.$$

If the given system of equations is to possess a non-zero solution, the coefficient matrix $\mathbf{A}$ must be of rank less than 3. If the matrix $\mathbf{A}$ is to be of rank less than 3 its determinant must be equal to zero. Thus we have

$$\begin{vmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0$$

$$\text{or} \quad \begin{vmatrix} 6-\lambda & 6-\lambda & 6-\lambda \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0,$$

adding the second and the third rows to the first

$$\text{or} \quad (6-\lambda) \begin{vmatrix} 1 & 1 & 1 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0$$

$$\text{or} \quad (6-\lambda) \begin{vmatrix} 1 & 1 & 0 \\ 3 & -\lambda-2 & -1 \\ 1 & 1 & -\lambda-1 \end{vmatrix} = 0, C_2 - C_1, C_3 - C_1$$

$$\text{or} \quad (6-\lambda) [(\lambda+2)(\lambda+1)+1] = 0$$

$$\text{or} \quad (6-\lambda) [\lambda^2 + 3\lambda + 3] = 0.$$

The roots of the equation $\lambda^2 + 3\lambda + 3 = 0$ are $\lambda = \frac{-3 \pm \sqrt{9-12}}{2}$ i.e., are imaginary.

Hence the only real value of λ for which the system of equations is to have a non-zero solution is 6.

Comprehensive Exercise 1

Find all the solutions of the following system of linear homogeneous equations :

- $2x - 3y + z = 0, x + 2y - 3z = 0, 4x - y - 2z = 0.$
- $x + y - 3z + 2w = 0, 2x - y + 2z - 3w = 0, 3x - 2y + z - 4w = 0,$
 $-4x + y - 3z + w = 0.$
- $x + y + z = 0, 2x + 5y + 7z = 0, 2x - 5y + 3z = 0.$
- $x + 2y + 3z = 0, 2x + 3y + 4z = 0, 7x + 13y + 19z = 0.$
- $4x + 2y + z + 3u = 0, 6x + 3y + 4z + 7u = 0, 2x + y + u = 0.$
- $2x - 2y + 5z + 3w = 0, 4x - y + z + w = 0, 3x - 2y + 3z + 4w = 0,$
 $x - 3y + 7z + 6w = 0.$
- $x - 2y + z - w = 0, x + y - 2z + 3w = 0, 4x + y - 5z + 8w = 0,$
 $5x - 7y + 2z - w = 0$

Answers 1

- $x = 0, y = 0, z = 0$
- $x = 0, y = 0, z = 0, w = 0$
- $x = 0, y = 0, z = 0$
- $x = 2c, y = -2c, z = c$
- $x = c_1, u = c_2, y = -2c_1 - c_2, z = -c_2$
- $x = \frac{5}{9}c, y = 4c, z = \frac{7}{9}c, w = c$
- $x = c_1 - \frac{5}{3}c_2, y = c_1 - \frac{4}{3}c_2, z = c_1, w = c_2$

19 System of Linear Non-homogeneous Equations

Sometimes we think that we can solve every two simultaneous equations of the type

$$\left. \begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned} \right\}.$$

But it is not so. For example, consider the simultaneous equations

$$\left. \begin{aligned} 3x + 4y &= 5 \\ 6x + 8y &= 13 \end{aligned} \right\}.$$

There is no set of values of x and y which satisfies both these equations. Such equations are said to be **inconsistent**.

Let us take another example. Consider the simultaneous equations

$$\left. \begin{array}{l} 3x + 4y = 5 \\ 6x + 8y = 10 \end{array} \right\}.$$

These equations are consistent since there exist values of x and y which satisfy both of these equations. We see that $x = -\frac{4}{3}c + \frac{5}{3}$, $y = c$ constitute a solution of these equations, where c is arbitrary. Thus these equations possess an infinite number of solutions.

Now we shall discuss the nature of solutions of a system of non-homogeneous linear equations :

$$\text{Let } \left. \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{array} \right\} \dots(1)$$

be a system of m non-homogeneous equations in n unknown $x_1, x_2, \dots, x_n$.

If we write

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}, \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}_{n \times 1}, \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix}_{m \times 1}$$

where $\mathbf{A}$, $\mathbf{X}$, $\mathbf{B}$ are $m \times n$, $n \times 1$ and $m \times 1$ matrices respectively, the above equations can be written in the form of a single matrix equation $\mathbf{AX} = \mathbf{B}$.

Any set of values of $x_1, x_2, \dots, x_n$ which simultaneously satisfy all these equations is called a solution of the system (1). When the system of equations has one or more solutions, the equations are said to be **consistent** otherwise they are said to be **inconsistent**.

The matrix

$$[\mathbf{A} \quad \mathbf{B}] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

is called the **augmented matrix** of the given system of equations.

20 Condition for Consistency

Theorem: The system of equations $\mathbf{AX} = \mathbf{B}$ is consistent i.e., possesses a solution, if and only if the coefficient matrix $\mathbf{A}$ and the augmented matrix $[\mathbf{A} \ \mathbf{B}]$ are of the same rank.

(Lucknow 2005; Meerut 07B; Kumaun 10, 12, 13; Kanpur 11)

Proof: Let $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_n$ denote the column vectors of the matrix $\mathbf{A}$. The equation $\mathbf{AX} = \mathbf{B}$ is then equivalent to

$$[\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_n] \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = \mathbf{B} \quad \text{i.e.,} \quad x_1\mathbf{C}_1 + x_2\mathbf{C}_2 + \dots + x_n\mathbf{C}_n = \mathbf{B}. \quad \dots(1)$$

Let now r be the rank of the matrix $\mathbf{A}$. The matrix $\mathbf{A}$ has then r linearly independent columns and without loss of generality, we can suppose that the first r columns $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_r$ form a linearly independent set so that each of the remaining $n - r$ columns is a linear combination of these r columns.

The condition is necessary. If the given system of equations is consistent, there must exist n scalars (numbers) $k_1, k_2, \dots, k_n$ such that

$$k_1\mathbf{C}_1 + k_2\mathbf{C}_2 + \dots + k_n\mathbf{C}_n = \mathbf{B}. \quad \dots(2)$$

Since each of the $n - r$ columns $\mathbf{C}_{r+1}, \mathbf{C}_{r+2}, \dots, \mathbf{C}_n$ is a linear combination of first r columns $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_r$, it is obvious from (2) that $\mathbf{B}$ is also a linear combination of $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_r$. Thus the maximum number of linearly independent columns of the matrix $[\mathbf{A} \ \mathbf{B}]$ is also r . Therefore the matrix $[\mathbf{A} \ \mathbf{B}]$ is also of rank r . Hence the matrices $\mathbf{A}$ and $[\mathbf{A} \ \mathbf{B}]$ are of the same rank.

The condition is sufficient. Now suppose that the matrices $\mathbf{A}$ and $[\mathbf{A} \ \mathbf{B}]$ are of the same rank r . The maximum number of linearly independent columns of the matrix $[\mathbf{A} \ \mathbf{B}]$ is then r . But the first r columns $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_r$ of the matrix $[\mathbf{A} \ \mathbf{B}]$ already form a linearly independent set. Therefore the column $\mathbf{B}$ should be expressed as a linear combination of the columns $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_r$.

Thus there exist r scalars $k_1, k_2, \dots, k_r$ such that

$$k_1\mathbf{C}_1 + k_2\mathbf{C}_2 + \dots + k_r\mathbf{C}_r = \mathbf{B}. \quad \dots(3)$$

Now (3) may be written as

$$k_1\mathbf{C}_1 + k_2\mathbf{C}_2 + \dots + k_r\mathbf{C}_r + 0.\mathbf{C}_{r+1} + 0.\mathbf{C}_{r+2} + \dots + 0.\mathbf{C}_n = \mathbf{B}. \quad \dots(4)$$

Comparing (1) and (4), we see that

$$x_1 = k_1, x_2 = k_2, \dots, x_r = k_r, x_{r+1} = 0, x_{r+2} = 0, \dots, x_n = 0$$

constitute a solution of the equation $\mathbf{AX} = \mathbf{B}$.

Therefore the given system of equations is consistent.

21 Condition for a System of n Equations in n Unknowns to have a Unique Solution

Theorem: If $\mathbf{A}$ be an n -rowed non-singular matrix, $\mathbf{X}$ be an $n \times 1$ matrix, $\mathbf{B}$ be an $n \times 1$ matrix, the system of equations $\mathbf{AX} = \mathbf{B}$ has a unique solution.

Proof: If $\mathbf{A}$ be an n -rowed non-singular matrix, the ranks of the matrices $\mathbf{A}$ and $[\mathbf{A} \ \mathbf{B}]$ are both n . Therefore the system of equations $\mathbf{AX} = \mathbf{B}$ is consistent i.e., possesses a solution.

Pre-multiplying both sides of $\mathbf{AX} = \mathbf{B}$ by $\mathbf{A}^{-1}$, we have

$$\mathbf{A}^{-1}\mathbf{AX} = \mathbf{A}^{-1}\mathbf{B} \quad \text{i.e.,} \quad \mathbf{IX} = \mathbf{A}^{-1}\mathbf{B} \quad \text{i.e.,} \quad \mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$$

is a solution of the equation $\mathbf{AX} = \mathbf{B}$.

To show that the solution is unique, let us suppose that $\mathbf{X}_1$ and $\mathbf{X}_2$ be two solutions of $\mathbf{AX} = \mathbf{B}$.

$$\text{Then} \quad \mathbf{AX}_1 = \mathbf{B}, \mathbf{AX}_2 = \mathbf{B} \Rightarrow \mathbf{AX}_1 = \mathbf{AX}_2 \Rightarrow \mathbf{A}^{-1}\mathbf{AX}_1 = \mathbf{A}^{-1}\mathbf{AX}_2$$

$$\Rightarrow \mathbf{IX}_1 = \mathbf{IX}_2 \Rightarrow \mathbf{X}_1 = \mathbf{X}_2.$$

Hence the solution is unique.

22 Working Rule for Finding the Solution of the Equation $\mathbf{AX} = \mathbf{B}$

Suppose the coefficient matrix $\mathbf{A}$ is of the type $m \times n$, i.e., we have m equations in n unknowns. Write the augmented matrix $[\mathbf{A} \ \mathbf{B}]$ and reduce it to Echelon form by applying only E -row transformations on it. This Echelon form will enable us to know the ranks of the augmented matrix $[\mathbf{A} \ \mathbf{B}]$ and the coefficient matrix $\mathbf{A}$. Then the following different cases arise :

Case I. Rank $\mathbf{A} <$ Rank $[\mathbf{A} \ \mathbf{B}]$.

In this case the equations $\mathbf{AX} = \mathbf{B}$ are inconsistent i.e., they have no solution.

Case II. Rank $\mathbf{A} = \text{Rank} [\mathbf{A} \ \mathbf{B}] = r$ (say).

In this case the equations $\mathbf{AX} = \mathbf{B}$ are consistent *i.e.*, they possess a solution. If $r < m$, then in the process of reducing the matrix $[\mathbf{A} \ \mathbf{B}]$ to Echelon form, $(m - r)$ equations will be eliminated. The given system of m equations will then be replaced by an equivalent system of r equations. From these r equations we shall be able to express the values of some r unknowns in terms of the remaining $n - r$ unknowns which can be given any arbitrarily chosen values.

If $r = n$, then $n - r = 0$, so that no variable is to be assigned arbitrary values and therefore in this case there will be a unique solution.

If $r < n$, then $n - r$ variables can be assigned arbitrary values. So in this case there will be an infinite number of solutions. Only $n - r + 1$ solutions will be linearly independent and the rest of the solutions will be linear combinations of them.

If $m < n$, then $r \leq m < n$. Thus in this case $n - r > 0$. Therefore when the number of equations is less than the number of unknowns, the equations will always have an infinite number of solutions, provided they are consistent.

Illustrative Examples

Example 10: Show that the equations

$$x + y + z = -3, \quad 3x + y - 2z = -2, \quad 2x + 4y + 7z = 7$$

are not consistent.

(Avadh 2008; Purvanchal 07)

Solution: The given system of equations is equivalent to the single matrix equation

$$\mathbf{AX} = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & -2 \\ 2 & 4 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ -2 \\ 7 \end{bmatrix} = \mathbf{B}.$$

The augmented matrix $[\mathbf{A} \ \mathbf{B}] = \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 3 & 1 & -2 & : & -2 \\ 2 & 4 & 7 & : & 7 \end{bmatrix}.$

We shall reduce the augmented matrix $[\mathbf{A} \ \mathbf{B}]$ to Echelon form by applying *E*-row transformations only. Applying

$$R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 2R_1, \text{ we get}$$

$$\begin{aligned}
 [\mathbf{A} \quad \mathbf{B}] &\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 2 & 5 & : & 13 \end{bmatrix} \\
 &\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 0 & 0 & : & 20 \end{bmatrix}, \text{ applying } R_3 \rightarrow R_3 + R_2.
 \end{aligned}$$

Above is the Echelon form of the matrix $[\mathbf{A} \quad \mathbf{B}]$. We have $\text{rank} [\mathbf{A} \quad \mathbf{B}] =$ the number of non-zero rows in this Echelon form $= 3$

Also by the same E -row transformations, we get

$$\mathbf{A} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix}.$$

Obviously $\text{rank } \mathbf{A} = 2$.

Since $\text{Rank } \mathbf{A} \neq \text{Rank} [\mathbf{A} \quad \mathbf{B}]$, therefore the given equations are inconsistent *i.e.*, they have no solution.

Remark: The inconsistency of the given equations can also be shown as below :

The given system of equations is equivalent to the matrix equation

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ 7 \\ 20 \end{bmatrix}.$$

This matrix equation is equivalent to the system of equations

$$\left. \begin{aligned} x + y + z &= -3 \\ 0x - 2y - 5z &= 7 \\ 0x + 0y + 0z &= 20 \end{aligned} \right\}.$$

The last equation shows that $0 = 20$, which is not possible. Hence the given equations are inconsistent.

Example 11: Show that the equations

$$x + y + z = 6, x + 2y + 3z = 14, x + 4y + 7z = 30$$

are consistent and solve them.

(Meerut 2010 B; Avadh 09)

Solution : The given system of equations is equivalent to the single matrix equation

$$\mathbf{A}\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \\ 30 \end{bmatrix} = \mathbf{B}.$$

The augmented matrix $[\mathbf{A} \ \mathbf{B}] = \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 1 & 2 & 3 & \vdots & 14 \\ 1 & 4 & 7 & \vdots & 30 \end{bmatrix}.$

We shall reduce the augmented matrix $[\mathbf{A} \ \mathbf{B}]$ to Echelon form by applying elementary row transformations only. Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$, we get

$$[\mathbf{A} \ \mathbf{B}] \sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 0 & 1 & 2 & \vdots & 8 \\ 0 & 3 & 6 & \vdots & 24 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 0 & 1 & 2 & \vdots & 8 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}, \text{ by } R_3 \rightarrow R_3 - 3R_2.$$

Above is the Echelon form of the matrix $[\mathbf{A} \ \mathbf{B}]$. We have $\text{rank}[\mathbf{A} \ \mathbf{B}] =$ the number of non-zero rows in this Echelon form $= 2$.

By the same elementary transformations, we get $\mathbf{A} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$

Obviously $\text{rank } \mathbf{A} = 2$. Since $\text{rank } \mathbf{A} = \text{rank} [\mathbf{A} \ \mathbf{B}]$, therefore the given equations are consistent. Here the number of unknowns is 3. Since $\text{rank } \mathbf{A}$ is less than the number of unknowns therefore the given system will have an infinite number of solutions. We see that the given system of equations is equivalent to the matrix equation

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 0 \end{bmatrix}.$$

This matrix equation is equivalent to the system of equations

$$\left. \begin{aligned} x + y + z &= 6 \\ y + 2z &= 8 \end{aligned} \right\}.$$

$$\therefore y = 8 - 2z, x = 6 - y - z = 6 - (8 - 2z) - z = z - 2.$$

Taking $z = c$, we see that $x = c - 2, y = 8 - 2c, z = c$ constitute the general solution of the given system, where c is an arbitrary constant.

Example 12: Apply the test of rank to examine if the following equations are consistent :

$$2x - y + 3z = 8, -x + 2y + z = 4, 3x + y - 4z = 0$$

and if consistent, find the complete solution.

(Meerut 2007)

Solution : The given system of equations is equivalent to the single matrix equation

$$\mathbf{AX} = \begin{bmatrix} 2 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \\ 0 \end{bmatrix} = \mathbf{B}.$$

The augmented matrix

$$[\mathbf{A} \ \mathbf{B}] = \begin{bmatrix} 2 & -1 & 3 & : & 8 \\ -1 & 2 & 1 & : & 4 \\ 3 & 1 & -4 & : & 0 \end{bmatrix}.$$

We shall reduce the augmented matrix to Echelon form by applying elementary row transformations only. Applying $R_1 \leftrightarrow R_2$, we get

$$\begin{aligned} [\mathbf{A} \ \mathbf{B}] &\sim \begin{bmatrix} -1 & 2 & 1 & : & 4 \\ 2 & -1 & 3 & : & 8 \\ 3 & 1 & -4 & : & 0 \end{bmatrix} \\ &\sim \begin{bmatrix} -1 & 2 & 1 & : & 4 \\ 0 & 3 & 5 & : & 16 \\ 0 & 7 & -1 & : & 12 \end{bmatrix}, \text{ by } R_2 \rightarrow R_2 + 2R_1, R_3 \rightarrow R_3 + 3R_1 \\ &\sim \begin{bmatrix} -1 & 2 & 1 & : & 4 \\ 0 & 3 & 5 & : & 16 \\ 0 & 21 & -3 & : & 36 \end{bmatrix}, \text{ by } R_3 \rightarrow 3R_3 \\ &\sim \begin{bmatrix} -1 & 2 & 1 & : & 4 \\ 0 & 3 & 5 & : & 16 \\ 0 & 0 & -38 & : & -76 \end{bmatrix}, \text{ by } R_3 \rightarrow R_3 - 7R_2 \\ &\sim \begin{bmatrix} -1 & 2 & 1 & : & 4 \\ 0 & 3 & 5 & : & 16 \\ 0 & 0 & 1 & : & 2 \end{bmatrix}, \text{ by } R_3 \rightarrow -\frac{1}{38} R_3. \end{aligned}$$

Above is the Echelon form of the matrix $[A \ B]$. We have the rank $[A \ B] =$ the number of non-zero rows in this Echelon form $= 3$.

By the same transformations, we get $A \sim \begin{bmatrix} -1 & 2 & 1 \\ 0 & 3 & 5 \\ 0 & 0 & 1 \end{bmatrix}$.

Obviously rank $A = 3$. Since rank $A =$ rank $[A \ B]$, therefore the given equations are consistent. Here the number of unknowns is 3. Since rank A is equal to the number of unknowns, therefore the given equations have a unique solution. We see that the given equations are equivalent to the matrix equation

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 3 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 16 \\ 2 \end{bmatrix}.$$

This matrix equation is equivalent to the equations

$$-x + 2y + z = 4, 3y + 5z = 16, z = 2.$$

These give $z = 2, y = 2, x = 2$.

Example 13: Show that the equations

$$x + 2y - z = 3, 3x - y + 2z = 1, 2x - 2y + 3z = 2, x - y + z = -1$$

are consistent and solve them.

(Meerut 2006B, 09; Rohilkhand 06; Garhwal 15)

Solution: The given system of equations is equivalent to the single matrix equation

$$AX = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix} = B.$$

The augmented matrix

$$[A \ B] = \begin{bmatrix} 1 & 2 & -1 & \vdots & 3 \\ 3 & -1 & 2 & \vdots & 1 \\ 2 & -2 & 3 & \vdots & 2 \\ 1 & -1 & 1 & \vdots & -1 \end{bmatrix}.$$

Performing $R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - R_1$, we get

$$\begin{aligned}
 [\mathbf{A} \ \mathbf{B}] &\sim \begin{bmatrix} 1 & 2 & -1 & \vdots & 3 \\ 0 & -7 & 5 & \vdots & -8 \\ 0 & -6 & 5 & \vdots & -4 \\ 0 & -3 & 2 & \vdots & -4 \end{bmatrix} \\
 &\sim \begin{bmatrix} 1 & 2 & -1 & \vdots & 3 \\ 0 & -1 & 0 & \vdots & -4 \\ 0 & -6 & 5 & \vdots & -4 \\ 0 & -3 & 2 & \vdots & -4 \end{bmatrix}, \text{ by } R_2 \rightarrow R_2 - R_3 \\
 &\sim \begin{bmatrix} 1 & 2 & -1 & \vdots & 3 \\ 0 & -1 & 0 & \vdots & -4 \\ 0 & 0 & 5 & \vdots & 20 \\ 0 & 0 & 2 & \vdots & 8 \end{bmatrix}, \text{ by } R_3 \rightarrow R_3 - 6R_2, R_4 \rightarrow R_4 - 3R_2 \\
 &\sim \begin{bmatrix} 1 & 2 & -1 & \vdots & 3 \\ 0 & -1 & 0 & \vdots & -4 \\ 0 & 0 & 1 & \vdots & 4 \\ 0 & 0 & 1 & \vdots & 4 \end{bmatrix}, \text{ by } R_3 \rightarrow \frac{1}{5}R_3, R_4 \rightarrow \frac{1}{2}R_4 \\
 &\sim \begin{bmatrix} 1 & 2 & -1 & \vdots & 3 \\ 0 & -1 & 0 & \vdots & -4 \\ 0 & 0 & 1 & \vdots & 4 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}, \text{ by } R_4 \rightarrow R_4 - R_3.
 \end{aligned}$$

Thus the matrix $[\mathbf{A} \ \mathbf{B}]$ has been reduced to Echelon form. We have $\text{rank} [\mathbf{A} \ \mathbf{B}] =$ the number of non-zero rows in this Echelon form $= 3$. Also

$$\mathbf{A} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

We have $\text{rank} \ \mathbf{A} = 3$. Since $\text{rank} [\mathbf{A} \ \mathbf{B}] = \text{rank} \ \mathbf{A}$, therefore the given equations are consistent. Since $\text{rank} \ \mathbf{A} = 3 =$ the number of unknowns, therefore the given equations have unique solution. The given equations are equivalent to the equations

$$x + 2y - z = 3, -y = -4, z = 4.$$

These give $z = 4, y = 4, x = -1$.

Example 14: State the conditions under which a system of non-homogeneous equations will have (i) no solution (ii) a unique solution (iii) infinity of solutions. (Lucknow 2008)

Solution : Let $\mathbf{AX} = \mathbf{B}$ be a system of linear non-homogeneous equations, where $\mathbf{A}$, $\mathbf{X}$, $\mathbf{B}$ are $m \times n$, $n \times 1$, $m \times 1$ matrices respectively.

- (i) These equations will have no solution if the coefficient matrix $\mathbf{A}$ and the augmented matrix $[\mathbf{A} \ \mathbf{B}]$ are not of the same rank.
- (ii) These equations will possess a unique solution if the matrices $\mathbf{A}$ and $[\mathbf{A} \ \mathbf{B}]$ are of the same rank and the rank is equal to the number of variables. In particular if $\mathbf{A}$ is a square matrix, these equations will possess a unique solution if and only if the matrix $\mathbf{A}$ is non-singular.
- (iii) These equations will have infinity of solutions if the matrices $\mathbf{A}$ and $[\mathbf{A} \ \mathbf{B}]$ are of the same rank and the rank is less than the number of variables.

Example 15: By the use of matrices, solve the equations :

$$x + y + z = 9, \quad 2x + 5y + 7z = 52, \quad 2x + y - z = 0.$$

(Bundelkhand 2011; Avadh 10; Rohilkhand 10)

Solution : The given system of equations is equivalent to the single matrix equation

$$\mathbf{AX} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix} = \mathbf{B}.$$

The augmented matrix

$$\begin{aligned} [\mathbf{A} \ \mathbf{B}] &= \begin{bmatrix} 1 & 1 & 1 & \vdots & 9 \\ 2 & 5 & 7 & \vdots & 52 \\ 2 & 1 & -1 & \vdots & 0 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 9 \\ 0 & 3 & 5 & \vdots & 34 \\ 0 & -1 & -3 & \vdots & -18 \end{bmatrix}, \text{ by } R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 2R_1 \\ &\sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 9 \\ 0 & -1 & -3 & \vdots & -18 \\ 0 & 3 & 5 & \vdots & 34 \end{bmatrix}, \text{ by } R_2 \leftrightarrow R_3 \end{aligned}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 9 \\ 0 & -1 & -3 & \vdots & -18 \\ 0 & 0 & -4 & \vdots & -20 \end{bmatrix}, \text{ by } R_3 \rightarrow R_3 + 3R_2.$$

Above is the Echelon form of the matrix $[\mathbf{A} \ \mathbf{B}]$. We have $\text{rank} [\mathbf{A} \ \mathbf{B}] =$ the number of non-zero rows in this Echelon form $= 3$.

Also by the same E -row transformations, we get

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -3 \\ 0 & 0 & -4 \end{bmatrix}.$$

$\therefore \text{rank } \mathbf{A} = 3$.

Since $\text{rank } \mathbf{A} = \text{rank} [\mathbf{A} \ \mathbf{B}]$, therefore the given equations are consistent. Also $\text{rank } \mathbf{A} = 3$ and the number of unknowns is also 3. Hence the given equations will have a unique solution. To find the solution we see that the given system of equations is equivalent to the matrix equation

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -3 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ -18 \\ -20 \end{bmatrix}.$$

This matrix equation is equivalent to the system of equations

$$x + y + z = 9, -y - 3z = -18, -4z = -20.$$

Solving these, we get $z = 5, y = 3, x = 1$.

Example 16: Investigate for what values of λ, μ the simultaneous equations

$$x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu$$

have (i) no solution, (ii) a unique solution, (iii) an infinite number of solutions.

(Meerut 2006, 09B; Garhwal 06, 11; Bundelkhand 09; Rohilkhand 07; Kanpur 09)

Solution : The matrix form of the given system of equations is

$$\mathbf{AX} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix} = \mathbf{B}.$$

The augmented matrix $[\mathbf{A} \ \mathbf{B}] = \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 1 & 2 & 3 & \vdots & 10 \\ 1 & 2 & \lambda & \vdots & \mu \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 0 & 1 & 2 & \vdots & 4 \\ 0 & 1 & \lambda - 1 & \vdots & \mu - 6 \end{bmatrix} \text{ by } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & \vdots & 6 \\ 0 & 1 & 2 & \vdots & 4 \\ 0 & 0 & \lambda - 3 & \vdots & \mu - 10 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - R_2.$$

If $\lambda \neq 3$, we have $\text{rank} [\mathbf{A} \ \mathbf{B}] = 3 = \text{rank} \mathbf{A}$.

So in this case the given system of equations is consistent. Since $\text{rank} \mathbf{A} =$ the number of unknowns, therefore the given system of equations possesses a unique solution. Thus if $\lambda \neq 3$, the given system of equations possesses a unique solution for any value of μ .

If $\lambda = 3$ and $\mu \neq 10$, we have $\text{rank} [\mathbf{A} \ \mathbf{B}] = 3$ and $\text{rank} \mathbf{A} = 2$. Thus in this case $\text{rank} [\mathbf{A} \ \mathbf{B}] \neq \text{rank} \mathbf{A}$ and so the given system of equations is inconsistent *i.e.*, possesses no solution.

If $\lambda = 3$ and $\mu = 10$, we have $\text{rank} [\mathbf{A} \ \mathbf{B}] = 2 = \text{rank} \mathbf{A}$.

So in this case the given system of equations is again consistent. Since $\text{rank} \mathbf{A} <$ the number of unknowns, therefore in this case the given system of equations possesses an infinite number of solutions.

Example 17: For what values of η the equations

$$x + y + z = 1, x + 2y + 4z = \eta, x + 4y + 10z = \eta^2,$$

have a solution and solve them completely in each case.

(Meerut 2011; Agra 07; Kanpur 06, 08, 10; Kumaun 10, 13)

Solution : The matrix form of the given system is

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \eta \\ \eta^2 \end{bmatrix}.$$

Performing $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$, we get

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 3 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \eta - 1 \\ \eta^2 - 1 \end{bmatrix}.$$

Performing $R_3 \rightarrow R_3 - 3R_2$, we have

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \eta - 1 \\ \eta^2 - 3\eta + 2 \end{bmatrix} \quad \dots(1)$$

Now the given equations will be consistent if and only if

$$\eta^2 - 3\eta + 2 = 0, \text{ i.e., iff } (\eta - 2)(\eta - 1) = 0, \text{ i.e., iff } \eta = 2 \text{ or } \eta = 1.$$

Case I. If $\eta = 2$, the equation (1) becomes

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

The above system of equations is equivalent to

$$y + 3z = 1, x + y + z = 1.$$

$$\therefore y = 1 - 3z, x = 2z.$$

Thus, $x = 2k, y = 1 - 3k, z = k$ constitute the general solution where k is an arbitrary constant.

Case II. If $\eta = 1$, the equation (1) becomes

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The above system of equations is equivalent to

$$y + 3z = 0, x + y + z = 1.$$

$$\therefore y = -3z, x = 1 + 2z.$$

Thus $x = 1 + 2c, y = -3c, z = c$ constitute the general solution, where c is an arbitrary constant.

Comprehensive Exercise 2

1. Use the test of rank to show that the following equations are not consistent :
 $2x - y + z = 4, 3x - y + z = 6, 4x - y + 2z = 7, -x + y - z = 9.$
(Rohilkhand 2009; Kumaun 11)
2. Show that the equations $-2x + y + z = a, x - 2y + z = b, x + y - 2z = c$ have no solution unless $a + b + c \neq 0$
(Kanpur 2009)
3. Apply the test of rank to examine if the following system of equations is consistent and if consistent, find the complete solution :
 $x + y + z = 6, x + 2y + 3z = 10, x + 2y + 4z = 1.$
4. Use matrix method to solve the equations :
 $2x - y + 3z = 9, x + y + z = 6, x - y + z = 2.$
(Meerut 2005; Avadh 06,11; Garhwal 12)
5. Solve completely the equations
 $x + 2y + 3z = 6, 2x + 4y + z = 7, 3x + 2y + 9z = 4.$
(Kumaun 2014)
6. Show that the equations $x + 2y - 5z = -9, 3x - y + 2z = 5, 2x + 3y - z = 3, 4x - 5y + z = -3$ are consistent and solve the same.
7. Show that the equations
 $x - 3y - 8z + 10 = 0, 3x + y - 4z = 0, 2x + 5y + 6z - 13 = 0$
 are consistent and solve the same. (Bundelkhand 2007)
8. Solve completely the equations $2x + 3y + z = 9, x + 2y + 3z = 6, 3x + y + 2z = 8.$
9. Solve completely the equations
 $x + y + z = 1, x + 2y + 3z = 4, x + 3y + 5z = 7, x + 4y + 7z = 10.$
(Lucknow 2009)
10. Express the following system of equations into the matrix equation form $\mathbf{AX} = \mathbf{B}$:
 $x + 2y + z = -1, 6x + y + z = -4, 2x - 3y - z = 0, -x - 7y - 2z = 7, x - y = 1.$
 Determine if this system of equations is consistent and if so find its solution.
11. Examine if the system of equations :
 $x + y + 4z = 6, 3x + 2y - 2z = 9, 5x + y + 2z = 13$
 is consistent. Find also the solution if it is consistent.
12. Prove, without actually solving, that the following system of equations has a unique solution :
 $5x + 3y + 14z = 4, y + 2z = 1, x - y + 2z = 0.$

13. For what values of the parameter λ will the following equations fail to have a unique solution

$$3x - y + \lambda z = 1, 2x + y + z = 2, x + 2y - \lambda z = -1?$$

Will the equations have any solutions for these values of λ ?

(Garhwal 2008)

14. Solve the equations

$$\lambda x + 2y - 2z - 1 = 0, 4x + 2\lambda y - z - 2 = 0, 6x + 6y + \lambda z - 3 = 0,$$

considering specially the case when $\lambda = 2$.

15. Discuss for all values of λ , the system of equations

$$x + y + 4z = 6, x + 2y - 2z = 6, \lambda x + y + z = 6,$$

as regards existence and nature of solutions.

16. Solve the following equations by matrix method :

$$5x + 3y + 7z = 4, 3x + 26y + 2z = 9, 7x + 2y + 10z = 5.$$

17. Solve the following system of linear equations by matrix method :

$$5x - 6y + 4z = 15, 7x + 4y - 3z = 19, 2x + y + 6z = 46.$$

18. Solve the following equations by matrix method :

$$x - y + 2z = 4, 3x + y + 4z = 6, x + y + z = 1.$$

(Garhwal 2014)

19. Solve the following equations by matrix method :

$$x - 2y + 3z = 6, 3x + y - 4z = -7, 5x - 3y + 2z = 5. \quad (\text{Meerut 2010; Garhwal 13})$$

Answers 2

3. $x = -7, y = 22, z = -9$
4. $x = 1, y = 2, z = 3$
5. $x = 1, y = 1, z = 1$
6. $x = \frac{1}{2}, y = \frac{3}{2}, z = \frac{5}{2}$
7. $x = -1 + 2c, y = 3 - 2c, z = c$, where c is arbitrary
8. $x = \frac{35}{18}, y = \frac{29}{18}, z = \frac{5}{18}$
9. $x = c - 2, y = 3 - 2c, z = c$
10. Consistent; $x = -1, y = -2, z = 4$

11. Consistent ; $x = 2, y = 2, z = \frac{1}{2}$
13. $\lambda \neq \frac{7}{2}$, the solution is unique; $\lambda = -\frac{7}{2}$, no solution
14. In case $\lambda = 2$, the general solution of the given system of equations is given by
 $x = \frac{1}{2} - c, y = c, z = 0$
15. $\lambda \neq \frac{7}{10}$, the solution is unique; $\lambda = -\frac{7}{10}$, no solution
16. $x = \frac{7}{11} - \frac{16}{11}c, y = \frac{3}{11} + \frac{1}{11}c, z = c$
17. $x = 3, y = 4, z = 6$
18. $x = \frac{5}{2} - \frac{3}{2}c, y = -\frac{3}{2} + \frac{1}{2}c, z = c$
19. $x = \frac{-8}{7} + \frac{5}{7}c, y = -\frac{25}{7} + \frac{13}{7}c, z = c$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. A set of r n -vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_r$ is said to be linearly independent if every relation of the type $k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 + \dots + k_r \mathbf{X}_r = \mathbf{O}$ implies
- (a) $k_1 + k_2 + \dots + k_r = 0$
- (b) $k_1 = k_2 = \dots = k_r = 0$
- (c) $k_1 + k_2 + \dots + k_r = 1$
- (d) none of these
2. Consider the system of equations

$$a_1x + b_1y + c_1z = 0, a_2x + b_2y + c_2z = 0, a_3x + b_3y + c_3z = 0.$$

If $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$, then the system has

- (a) more than two solutions
(b) only trivial solution
(c) no solution
(d) none of these
3. The system of linear equations $x + y + z = 2$, $2x + y - z = 3$, $3x + 2y + kz = 4$ has a unique solution if
- (a) $k \neq 0$
(b) $-1 < k < 1$
(c) $-2 < k < 2$
(d) $k = 0$
4. The system of equations $x + 2y + 3z = 1$, $2x + y + 3z = 2$, $5x + y + 9z = 4$ has
- (a) only one solution
(b) infinitely many solutions
(c) no solution
(d) none of these

(Meerut 2003)

Fill in the Blank(s)

Fill in the blanks ".....," so that the following statements are complete and correct.

1. The set consisting only of the zero vector, $\mathbf{O}$, is linearly
 2. The set of three vectors $\mathbf{X}_1 = (1, 0, 0)$, $\mathbf{X}_2 = (0, 1, 0)$, $\mathbf{X}_3 = (0, 0, 1)$ is linearly
 3. The number of linearly independent solutions of the system of m homogeneous linear equations in n variables, $\mathbf{AX} = \mathbf{O}$, is, where r is the rank of the matrix $\mathbf{A}$.
 4. When the system of equations has one or more solutions, the equations are said to be, otherwise they are said to be
 5. The system of m non-homogeneous linear equations in n unknowns, $\mathbf{AX} = \mathbf{B}$, is if and only if the coefficient matrix $\mathbf{A}$ and the augmented matrix $[\mathbf{A} \ \mathbf{B}]$ are of the same rank.
(Meerut 2001)
 6. If $\mathbf{A}$ be an n -rowed non-singular matrix, $\mathbf{X}$ be an $n \times 1$ matrix, $\mathbf{B}$ be an $n \times 1$ matrix, the system of equations $\mathbf{AX} = \mathbf{B}$, has a solution.
 7. If the number of equations is less than the number of unknowns, the equations will always have an number of solutions, provided they are consistent.
-

True or False

Write 'T' for true and 'F' for false statement.

1. A set of r n -vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_r$, is said to be linearly dependent if there exist r scalars $k_1, k_2, \dots, k_r$, not all zero, such that $k_1\mathbf{X}_1 + k_2\mathbf{X}_2 + \dots + k_r\mathbf{X}_r = \mathbf{O}$, where $\mathbf{O}$ denotes the n -vector whose components are all zero.
2. If a set of vectors is linearly independent, then at least one member of the set can be expressed as a linear combination of the remaining members.
3. The zero solution *i.e.*, $x = y = z = 0$ is the only solution of the system of equations

$$x + 2y + 3z = 0, 3x + 4y + 4z = 0, 7x + 10y + 12z = 0.$$
4. Let $\mathbf{AX} = \mathbf{B}$ be a system of m non-homogeneous linear equations in n unknowns. These equations will have no solution if the coefficient matrix $\mathbf{A}$ and the augmented matrix $[\mathbf{A} \ \mathbf{B}]$ are not of the same rank.
5. The system of m non-homogeneous linear equations in n unknowns, $\mathbf{AX} = \mathbf{B}$ will possess a unique solution if the matrices $\mathbf{A}$ and $[\mathbf{A} \ \mathbf{B}]$ are of the same rank and the rank is less than the number of variables.

Answers

Multiple Choice Questions

- | | | | |
|--------|--------|--------|--------|
| 1. (b) | 2. (a) | 3. (a) | 4. (a) |
|--------|--------|--------|--------|

Fill in the Blank(s)

- | | |
|---------------|-----------------------------|
| 1. dependent | 2. independent |
| 3. $(n - r)$ | 4. consistent; inconsistent |
| 5. consistent | 6. unique |
| 7. infinite | |

True or False

- | | | |
|------|------|------|
| 1. T | 2. F | 3. T |
| 4. T | 5. F | |

